

Categorical Automata Theory

Conor Sheridan

July 15, 2026

Contents

I	Residual Behaviour and Operational Algebra	5
1	Internal Categories and Mealy Morphisms in the Bicategory of Spans	6
1.1	Orientation	6
1.2	Spans, monads, and internal categories	7
1.2.1	Two running specializations	13
1.3	Mealy morphisms	16
1.3.1	The one-object specialization	21
1.3.2	The stateless specialization	24
1.4	Mealy simulations	28
1.4.1	From strict carrier maps to output-comparison cells	28
1.4.2	The output-comparison monad	32
1.4.3	The input-processing monad and Kleisli algebras	37
1.4.4	One-object simulations	47
1.4.5	Stateless simulations	49
1.5	The bicategory of Mealy morphisms	52
1.6	Examples	73
1.7	Chapter summary	82
2	Behaviour, Fibrations, and the Categorical Myhill–Nerode Theorem	84
2.1	Fibred orientation and standing assumptions	84
2.2	Residual syntax and the behaviour fibration	87
2.2.1	Residual syntax as a split codomain opfibration	87
2.2.2	The behaviour fibration	97
2.3	The cleavage Mealy morphism	110
2.3.1	Internal derivative and anchor-output maps	110
2.3.2	The canonical behaviour carrier	112
2.3.3	The cleavage Mealy morphism	112
2.3.4	Derivative-closed subobjects	113
2.4	Mealy actions and residual semantics	116
2.4.1	The execution category	116
2.4.2	The output functor	117
2.4.3	Residual orbit functors	118
2.4.4	Residual behaviour of a state	120
2.4.5	The behaviour assignment	120
2.5	Terminal residual semantics	124
2.5.1	Strict semantic homomorphisms	124

2.5.2	Terminality of the canonical behaviour machine	125
2.5.3	Behavioural separatedness	127
2.6	Simulations and induced behaviour comparisons	127
2.6.1	Simulation-induced vertical behaviour comparisons	127
2.7	Specifications and derivative images	128
2.7.1	Specifications	128
2.7.2	Formal derivative orbits	129
2.7.3	Derivative images	130
2.7.4	Derivative closure	131
2.8	The Categorical Myhill–Nerode Theorem	133
2.8.1	The Nerode relation	134
2.8.2	The Nerode quotient	134
2.8.3	The minimal residual realization	135
2.8.4	Realizations and reachability	135
2.8.5	Semantic quotient and minimality	143
2.8.6	The categorical theorem	145
2.8.7	Finite-state corollary	146
2.9	Generated residual behaviour categories	148
2.9.1	Derivative-closed full internal subcategories	148
2.9.2	The generated residual behaviour category	148
2.9.3	The syntactic residual comparison category	149
2.9.4	The full residual image	150
2.10	The one-object specialization	151
2.10.1	Residual right-Cayley syntax	151
2.10.2	Residual behaviours and comparisons	152
2.10.3	The canonical residual-cocycle machine	152
2.10.4	Residual semantics of a one-object Mealy morphism	153
2.10.5	Specifications, Nerode quotients, and generated comparisons	153
2.11	The stateless specialization	155
2.11.1	Slice-restricted residual semantics	155
2.11.2	Simulations and residual natural transformations	156
2.11.3	Stateless specifications and their Nerode quotients	157
2.12	Examples	162
2.13	Chapter summary	166
3	Coequational Subobjects and Birkhoff Theorems	168
3.1	Overview and standing assumptions	168
3.1.1	Orientation	168
3.1.2	Ambient assumptions	169
3.1.3	Strict Mealy morphisms and terminal semantics	170
3.2	Local coequational theories	177
3.2.1	Canonical derivative structure	177
3.2.2	Behavioural predicates and residual invariance	178
3.2.3	Generated closure, safety core, and the adjoint triple	179
3.2.4	Coequational subobjects	183
3.2.5	Models of local coequational theories	186
3.2.6	Restricted canonical subobjects	188
3.3	Semantic images and regular-image structure	189

3.3.1	Regular-epimorphic descent for subobjects	189
3.3.2	Regular images of strict homomorphisms	190
3.3.3	Semantic images	197
3.3.4	Meets and coproducts	199
3.4	Universal realization	201
3.4.1	The behaviour–realization correspondence	201
3.4.2	The universal property	202
3.4.3	Unit, counit, and immediate consequences	205
3.4.4	Relation with minimal residual realization	208
3.5	The local coequational Birkhoff theorem	208
3.5.1	Closure properties of model classes	208
3.5.2	Joins of local theories	211
3.5.3	The theorem	214
3.6	Structured output and algebraic closure	218
3.6.1	Structured output categories	218
3.6.2	Pointwise operations on behaviours	218
3.6.3	T-coequational theories	224
3.6.4	Semantic images and structured operations	230
3.6.5	T-coequational closure	231
3.6.6	The structured Birkhoff theorem	233
3.7	Global input reindexing	236
3.7.1	Reindexing residual behaviours along input functors	236
3.7.2	Input pullback of Mealy morphisms	239
3.8	Global coequational theories	248
3.8.1	Global theories and global models	248
3.8.2	Global closure	249
3.9	The global coequational Birkhoff theorem	250
3.9.1	The theorem	250
3.10	One-object specialization	253
3.11	Summary	256
4	Operational Equations, Syntactic Transition–Output Quotients, and Algebraic Eilenberg Duality	257
4.1	Overview and standing assumptions	257
4.1.1	Orientation	257
4.1.2	Standing assumptions and notation	259
4.2	Mealy morphisms as bicategorical representations	261
4.2.1	Internal categories as monads in spans	261
4.2.2	The output-comparison Kleisli monad	264
4.2.3	The internal Kleisli endomorphism category	266
4.2.4	Mealy actions as internal representations	274
4.3	Operational equations and behavioural satisfaction	277
4.3.1	Parallel pairs and operation congruences	277
4.3.2	Satisfaction of operational equations	279
4.3.3	The one-step equation predicate	280
4.3.4	The largest invariant theory satisfying a congruence	283
4.4	Faithful operational images	285
4.4.1	Identity-on-objects quotients and regular images	285

4.4.2	The transition–output image	289
4.4.3	Descent of operational equality	290
4.4.4	Recognition and faithful recognition	291
4.5	Local algebraic Eilenberg duality	294
4.5.1	Semantic closure operators	294
4.5.2	Complete-lattice identities	296
4.5.3	The local algebraic Eilenberg theorem	298
4.5.4	Subdirect operational representation	299
4.6	Specifications and syntactic transition–output categories	301
4.6.1	Generated derivative theories	301
4.6.2	Regular presentations of operational equations	305
4.6.3	Context expansion	306
4.6.4	The syntactic universal property	307
4.6.5	Two-stage operational Myhill–Nerode theory	308
4.6.6	A finiteness criterion	311
4.7	Structured output and automatic algebraic closure	312
4.7.1	Pointwise output operations	312
4.7.2	Recognized theories are automatically structured	314
4.8	Globalization as an indexed operational polarity	316
4.8.1	Residual reindexing	316
4.8.2	Operational inverse image	321
4.8.3	Raw pullback and semantic completion	322
4.8.4	Lax functoriality	323
4.8.5	Saturation-adjusted reindexing	325
4.8.6	Global theories and global congruence systems	325
4.9	One-object and stateless specializations	328
4.9.1	One-object input categories	328
4.9.2	Arrow monoids and execution monoids	329
4.9.3	Residual cocycles	330
4.9.4	Stateless functors	330
4.10	Boolean languages and classical syntactic monoids	332
4.10.1	Boolean residual behaviour	333
4.10.2	Recovery of the syntactic monoid	334
4.10.3	Finite Boolean saturation and right quotients	335
4.10.4	Residual reindexing as inverse morphic preimage	338
4.11	Compatibility with the classical Eilenberg correspondence	339
4.11.1	Varieties of regular languages and pseudovarieties	339
4.11.2	Syntactic separation	340
4.11.3	Finite syntactic generation	341
4.11.4	Recognition through divisors of products	343
4.11.5	The classical correspondence	346
4.12	Synthesis: state minimization and operation minimization	350

Part I

Residual Behaviour and Operational Algebra

Chapter 1

Internal Categories and Mealy Morphisms in the Bicategory of Spans

1.1 Orientation

Let $\mathcal{E}$ be a finitely complete category. Since $\mathcal{E}$ has pullbacks, one may form the bicategory

$$\mathbf{Span}(\mathcal{E}),$$

whose objects are those of $\mathcal{E}$, whose 1-cells are spans, and whose 2-cells are maps of spans. In the formal theory of monads, internal categories in $\mathcal{E}$ may be identified with monads in this bicategory. More explicitly, an endospan

$$C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0$$

equipped with unit and multiplication as a monad in $\mathbf{Span}(\mathcal{E})$ is precisely an internal category in $\mathcal{E}$ [2, 24, 13, 14].

The morphisms produced by this bicategorical identification require some care. We use the orientation in which a morphism from a monad A on A_0 to a monad B on B_0 , carried by a 1-cell

$$P : A_0 \rightarrow B_0,$$

has structure 2-cell

$$\Phi : P \circ A \Rightarrow B \circ P.$$

This orientation is commonly called a monad opmorphism, or a colax monad morphism, although the terminology varies between sources. We shall call it the *Mealy orientation* and shall always display the structure cell explicitly.

In $\mathbf{Span}(\mathcal{E})$, the carrier P is a span

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0.$$

By contrast, an internal functor

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

has object part a single morphism

$$F_0 : A_0 \rightarrow B_0.$$

Thus the Mealy-oriented monad-morphism calculus and the ordinary functorial calculus for internal categories produce different kinds of morphism.

The present chapter takes the span-carried morphisms themselves as the object of study. We retain the span

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

as part of the structure and read it as carrying state. The map p records the input interface of a state, and q records its output interface. The oriented monad-morphism structure then supplies a transition-output operation. From this point of view, the span calculus suggests a typed internal form of automata theory: internal categories provide the input and output type systems, while monad morphisms in spans behave as stateful internal functors.

This interpretation is close in spirit to Paré's Mealy morphisms of enriched categories. Paré introduces Mealy morphisms as morphisms lying between enriched functors and profunctors, with the construction motivated by bicategorical considerations and by the classical Mealy-machine pattern from automata theory [19]. The present chapter develops an internal-categorical version of that idea in the bicategory of spans.

1.2 Spans, monads, and internal categories

A span from X to Y is written

$$X \xleftarrow{p} S \xrightarrow{q} Y.$$

The map $p : S \rightarrow X$ is its left leg, and $q : S \rightarrow Y$ is its right leg. We often display this as the diagram

$$\begin{array}{ccc} & S & \\ p \swarrow & & \searrow q \\ X & & Y. \end{array}$$

When the structure maps are determined by context, we suppress them in pullback notation. For example, if

$$A_1 \xrightarrow{t_A} A_0 \quad \text{and} \quad P \xrightarrow{p} A_0$$

are fixed, we write

$$A_1 \times_{A_0} P$$

for

$$A_1 \times_{t_A, A_0, p} P.$$

Thus $A_1 \times_{A_0} P$ is the pullback object in the square

$$\begin{array}{ccc} A_1 \times_{A_0} P & \xrightarrow{\pi_P} & P \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

When several maps into the same object occur simultaneously, the relevant structure maps are displayed explicitly.

Let

$$X \xleftarrow{p} S \xrightarrow{q} Y$$

and

$$Y \xleftarrow{r} T \xrightarrow{s} Z$$

be spans in $\mathcal{E}$. Their composite is obtained by forming the pullback of $q : S \rightarrow Y$ and $r : T \rightarrow Y$:

$$\begin{array}{ccccc} S \times_Y T & \xrightarrow{\pi_T} & T & \xrightarrow{s} & Z \\ \pi_S \downarrow & \lrcorner & \downarrow r & & \\ S & \xrightarrow{q} & Y & & \\ p \downarrow & & & & \\ X & & & & \end{array}$$

We write this composite as $T \circ S$, so S is performed first and T second.

A map of spans

$$\alpha : \left(X \xleftarrow{p} S \xrightarrow{q} Y \right) \Rightarrow \left(X \xleftarrow{p'} S' \xrightarrow{q'} Y \right)$$

is a morphism $\alpha : S \rightarrow S'$ in $\mathcal{E}$ such that

$$p'\alpha = p, \quad q'\alpha = q.$$

Equivalently, it is a commuting diagram

$$\begin{array}{ccccc} & & S & & \\ & p \swarrow & & \searrow q & \\ X & & & & Y \\ & p' \swarrow & \alpha \downarrow & \searrow q' & \\ & & S' & & \end{array}$$

where the two triangular regions commute.

Definition 1.2.1 (Bicategory of spans). *The **bicategory of spans** in $\mathcal{E}$, denoted*

$$\mathbf{Span}(\mathcal{E}),$$

has

1. *objects the objects of $\mathcal{E}$;*
2. *1-cells $X \rightarrow Y$ the spans*
3. *2-cells the maps of spans.*

The identity 1-cell on X is

$$X \xleftarrow{1_X} X \xrightarrow{1_X} X.$$

Horizontal composition is given by the pullback construction above.

Proposition 1.2.1. *If $\mathcal{E}$ has pullbacks, then the data of Definition 1.2.1 form a bicategory.*

Proof.

1. **Composition of 2-cells.** Vertical composition is inherited from $\mathcal{E}$: a vertical composite of maps of spans is the composite of their apex maps. The defining equations with the left and right legs are preserved by ordinary composition in $\mathcal{E}$.

Horizontal composition is induced by pullback. If

$$\alpha : S \rightarrow S', \quad \beta : T \rightarrow T'$$

are maps of spans over a common middle object Y , then the universal property of the pullback gives a unique map

$$\alpha \times_Y \beta : S \times_Y T \rightarrow S' \times_Y T'$$

whose projections are $\alpha\pi_S$ and $\beta\pi_T$. This is the horizontal composite of α and β .

2. **Associators and unitors.** For composable spans with apexes S, T, U , the two possible composite apexes

$$(S \times_Y T) \times_Z U \quad \text{and} \quad S \times_Y (T \times_Z U)$$

both represent the object of triples (s, t, u) satisfying the two matching equations. The associator is the canonical isomorphism between these two pullback presentations, uniquely determined by its projections to S, T, U .

The left and right unitors are the canonical pullback isomorphisms

$$X \times_X S \cong S, \quad S \times_Y Y \cong S.$$

3. **Coherence.** The interchange law, the pentagon identity, and the triangle identity compare canonical maps between objects obtained from the same iterated pullback data. Since all such maps are uniquely determined by their projections to the original apexes, the relevant diagrams commute. Therefore the span data form a bicategory.

□

Lemma 1.2.1 (Component calculus for span maps). *Let*

$$X \leftarrow S \rightarrow Y, \quad X \leftarrow T \rightarrow Y$$

be spans in $\mathcal{E}$. A map of spans $S \rightarrow T$ is determined by its apex map, and an equality of maps of spans is an equality of apex maps.

Moreover, whenever the codomain apex is a pullback, a map into it is determined by its projections. Thus the generalized-element equations used below are equivalently morphism equations out of the pullback object that expresses the displayed typing constraints.

Remark 1.2.1. *The construction of $\mathbf{Span}(\mathcal{E})$ is one of the basic examples in bicategory theory following Bénabou's introduction of bicategories [2].*

Let C_0 be an object of $\mathcal{E}$. The hom-category

$$\mathbf{Span}(\mathcal{E})(C_0, C_0)$$

is a monoidal category. Its objects are endospans

$$C_0 \xleftarrow{s} S \xrightarrow{t} C_0,$$

its morphisms are maps of spans, its tensor product is horizontal composition of spans, and its tensor unit is the identity span

$$C_0 \xleftarrow{1_{C_0}} C_0 \xrightarrow{1_{C_0}} C_0.$$

We write this monoidal category as

$$\text{EndSpan}_{\mathcal{E}}(C_0) := \mathbf{Span}(\mathcal{E})(C_0, C_0).$$

With our convention for categorical order, if

$$C_0 \xleftarrow{s} S \xrightarrow{t} C_0, \quad C_0 \xleftarrow{s'} T \xrightarrow{t'} C_0,$$

then

$$S \otimes T := T \circ S$$

has apex

$$S \times_{t, C_0, s'} T.$$

Thus $S \otimes T$ denotes first S , then T :

$$\begin{array}{ccc} \begin{array}{c} S \\ \swarrow s \quad \searrow t \\ C_0 \quad \quad C_0 \end{array} & \begin{array}{c} T \\ \swarrow s' \quad \searrow t' \\ C_0 \quad \quad C_0 \end{array} & \rightsquigarrow \quad \begin{array}{c} S \times_{C_0} T \\ \swarrow s\pi_S \quad \searrow t'\pi_T \\ C_0 \quad \quad C_0 \end{array} \end{array}$$

Remark 1.2.2. In any bicategory $\mathcal{B}$, a monad on an object X is equivalently a monoid object in the monoidal category

$$\mathcal{B}(X, X),$$

where the monoidal product is horizontal composition and the unit object is 1_X . Hence a monad on C_0 in $\mathbf{Span}(\mathcal{E})$ is equivalently a monoid object in

$$\text{EndSpan}_{\mathcal{E}}(C_0).$$

This is a basic instance of the formal theory of monads in bicategories [24].

Let

$$C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0$$

be an endospan. Composing this span with itself requires the pullback of

$$t : C_1 \rightarrow C_0 \quad \text{and} \quad s : C_1 \rightarrow C_0.$$

We denote this pullback by

$$C_2 := C_1 \times_{t, C_0, s} C_1.$$

Thus C_2 is the object of composable pairs (a, b) satisfying

$$t(a) = s(b).$$

The defining pullback is

$$\begin{array}{ccc} C_2 & \xrightarrow{\pi_2} & C_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow s \\ C_1 & \xrightarrow{t} & C_0. \end{array}$$

The composite endospan is

$$C_0 \xleftarrow{s\pi_1} C_2 \xrightarrow{t\pi_2} C_0.$$

A monoid structure on the endospan

$$C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0$$

consists of a unit map of spans

$$\eta : \left(C_0 \xleftarrow{1_{C_0}} C_0 \xrightarrow{1_{C_0}} C_0 \right) \Rightarrow \left(C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0 \right)$$

and a multiplication map of spans

$$\mu : \left(C_0 \xleftarrow{s\pi_1} C_2 \xrightarrow{t\pi_2} C_0 \right) \Rightarrow \left(C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0 \right).$$

Equivalently, the unit is a map

$$e : C_0 \rightarrow C_1$$

such that

$$se = 1_{C_0}, \quad te = 1_{C_0},$$

as in

$$\begin{array}{ccccc} & & C_0 & & \\ & \swarrow 1_{C_0} & \downarrow e & \searrow 1_{C_0} & \\ C_0 & \xleftarrow{s} & C_1 & \xrightarrow{t} & C_0, \end{array}$$

and the multiplication is a map

$$m : C_2 \rightarrow C_1$$

such that

$$sm = s\pi_1, \quad tm = t\pi_2,$$

as in

$$\begin{array}{ccccc} & & C_2 & & \\ & \swarrow s\pi_1 & \downarrow m & \searrow t\pi_2 & \\ C_0 & \xleftarrow{s} & C_1 & \xrightarrow{t} & C_0. \end{array}$$

In generalized-element notation, if

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

then

$$m(a, b) = b \circ a : u \rightarrow w.$$

Thus the two projections from C_2 are read as

$$u \xrightarrow{a} v \xrightarrow{b} w.$$

Definition 1.2.2 (Internal category). An *internal category* in $\mathcal{E}$ is a tuple

$$\mathbb{C} = (C_0, C_1, s, t, e, m)$$

where

$$s, t : C_1 \rightarrow C_0, \quad e : C_0 \rightarrow C_1, \quad m : C_1 \times_{t, C_0, s} C_1 \rightarrow C_1,$$

satisfying the source-target equations

$$se = 1_{C_0}, \quad te = 1_{C_0},$$

$$sm = s\pi_1, \quad tm = t\pi_2,$$

the identity equations

$$m \circ \langle es, 1_{C_1} \rangle = 1_{C_1}, \quad m \circ \langle 1_{C_1}, et \rangle = 1_{C_1},$$

and the associativity equation

$$m \circ \langle m \circ \langle \pi_1, \pi_2 \rangle, \pi_3 \rangle = m \circ \langle \pi_1, m \circ \langle \pi_2, \pi_3 \rangle \rangle$$

as maps

$$C_3 \rightarrow C_1,$$

where

$$C_3 := C_1 \times_{t, C_0, s} C_1 \times_{t, C_0, s} C_1$$

is the object of composable triples, with projections

$$\pi_1, \pi_2, \pi_3 : C_3 \rightarrow C_1.$$

All the displayed pairings are the unique maps into

$$C_1 \times_{t, C_0, s} C_1$$

induced by its pullback universal property.

Theorem 1.2.1 (Endospan monoids are internal categories). Let $\mathcal{E}$ be a category with pullbacks. To give a monoid object in

$$\text{EndSpan}_{\mathcal{E}}(C_0)$$

is equivalently to give an internal category in $\mathcal{E}$ with object-of-objects C_0 .

Proof.

1. **Unpack the monoid data.** A monoid object in $\text{EndSpan}_{\mathcal{E}}(C_0)$ is an endospan

$$C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0$$

together with a unit map of spans $e : C_0 \rightarrow C_1$ and a multiplication map of spans $m : C_1 \times_{t, C_0, s} C_1 \rightarrow C_1$. The fact that these are maps of spans is exactly the source-target data

$$se = 1_{C_0}, \quad te = 1_{C_0}, \quad sm = s\pi_1, \quad tm = t\pi_2.$$

2. **Identify the axioms.** The monoid unit axioms say that

$$C_1 \xrightarrow{\langle es, 1 \rangle} C_1 \times_{C_0} C_1 \xrightarrow{m} C_1, \quad C_1 \xrightarrow{\langle 1, et \rangle} C_1 \times_{C_0} C_1 \xrightarrow{m} C_1$$

are both 1_{C_1} . These are precisely the internal identity laws. The monoid associativity axiom is the equality of maps $C_3 \rightarrow C_1$ which, on a composable triple

$$u \xrightarrow{a} v \xrightarrow{b} w \xrightarrow{c} z,$$

says

$$c \circ (b \circ a) = (c \circ b) \circ a.$$

Thus the monoid axioms are exactly the internal-category identity and associativity axioms.

3. **Reverse the construction.** Conversely, an internal category (C_0, C_1, s, t, e, m) determines the endospan

$$C_0 \xleftarrow{s} C_1 \xrightarrow{t} C_0$$

with unit e and multiplication m . The internal-category axioms are exactly the monoid axioms just identified. Hence the two structures are equivalent. $\square$

Corollary 1.2.1 (Monads in spans are internal categories). *Let $\mathcal{E}$ be a category with pullbacks. To give a monad in $\mathbf{Span}(\mathcal{E})$ on an object C_0 is equivalently to give an internal category in $\mathcal{E}$ with object-of-objects C_0 .*

Proof.

1. **Move from monads to monoids in an endohom.** By Remark 1.2.2, monads on C_0 in $\mathbf{Span}(\mathcal{E})$ are monoid objects in

$$\mathbf{EndSpan}_{\mathcal{E}}(C_0) = \mathbf{Span}(\mathcal{E})(C_0, C_0).$$

2. **Apply the endospan-monoid identification.** By Theorem 1.2.1, monoid objects in this endohom category are exactly internal categories in $\mathcal{E}$ whose object-of-objects is C_0 . This proves the claim. $\square$

Remark 1.2.3. *The identification of internal categories with monads in spans is a standard feature of the formal theory of internal categories [24, 13, 14]. The present chapter uses this identification as the source of the stateful morphisms introduced below.*

1.2.1 Two running specializations

Two special cases will be carried through the chapter.

First, since $\mathcal{E}$ is finitely complete, it has a terminal object 1 and finite products. With the standing choice of 1 , binary products are realized as pullbacks over 1 :

$$\begin{array}{ccc} X \times Y & \longrightarrow & Y \\ \downarrow & \lrcorner & \downarrow ! \\ X & \longrightarrow & 1. \end{array}$$

A one-object internal category means an internal category whose object-of-objects is 1. Such an internal category is equivalently a monoid object in $\mathcal{E}$.

We shall use the following order convention for this specialization. If

$$M = (M, e_M, \cdot_M)$$

is a monoid object and multiplication is written by juxtaposition, then the associated one-object internal category

$$\mathbb{M} = (1, M, !, !, e_M, m_{\mathbb{M}})$$

has internal-category composition

$$m_{\mathbb{M}} = \cdot_M \circ \tau_{M,M},$$

where $\tau_{M,M} : M \times M \rightarrow M \times M$ is the symmetry. Equivalently, for a composable pair (a, b) ,

$$m_{\mathbb{M}}(a, b) = ba.$$

Thus, in the one-object category,

$$b \circ a = ba.$$

This convention is chosen so that the target-to-source execution law below becomes the usual right-action law

$$x \cdot (mn) = (x \cdot m) \cdot n.$$

Thus the one-object specialization of the theory below is a theory of internal monoid-indexed Mealy systems.

Second, a morphism

$$F_0 : A_0 \rightarrow B_0$$

determines a canonical span

$$\begin{array}{ccc} & A_0 & \\ 1_{A_0} \swarrow & & \searrow F_0 \\ A_0 & & B_0. \end{array}$$

It also determines the reverse span

$$\begin{array}{ccc} & A_0 & \\ F_0 \swarrow & & \searrow 1_{A_0} \\ B_0 & & A_0. \end{array}$$

The map-representable part of the theory recovers internal functors. We denote these two spans by

$$F_{0!} := \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right)$$

and

$$F_0^* := \left(B_0 \xleftarrow{F_0} A_0 \xrightarrow{1_{A_0}} A_0 \right),$$

respectively.

Definition 1.2.3 (Map-representable span). *A span*

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

*is called **map-representable**, or a **map span**, if it is isomorphic, as a span, to a span of the form*

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0$$

for some map $F_0 : A_0 \rightarrow B_0$. Equivalently, the span is map-representable iff p is an isomorphism. In that case it is isomorphic to

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{qp^{-1}} B_0.$$

The isomorphism of spans is displayed by

$$\begin{array}{ccccc} & & P & & \\ & \swarrow p & \downarrow \cong p & \searrow q & \\ A_0 & \xleftarrow{1_{A_0}} & A_0 & \xrightarrow{qp^{-1}} & B_0. \end{array}$$

Proposition 1.2.2 (Canonical adjunction associated to a map). *There is a canonical adjunction*

$$F_{0!} \dashv F_0^*$$

in $\mathbf{Span}(\mathcal{E})$.

Proof.

1. **Compute the composite $F_0^* \circ F_{0!}$.** The composite

$$F_0^* \circ F_{0!} : A_0 \rightarrow A_0$$

has apex the pullback of F_0 with itself:

$$\begin{array}{ccccc} A_0 \times_{B_0} A_0 & \xrightarrow{\pi_2} & A_0 & & \\ \pi_1 \downarrow & \lrcorner & \downarrow F_0 & & \\ A_0 & \xrightarrow{F_0} & B_0. & & \end{array}$$

Hence

$$F_0^* \circ F_{0!} = \left(A_0 \xleftarrow{\pi_1} A_0 \times_{B_0} A_0 \xrightarrow{\pi_2} A_0 \right).$$

2. **Define the unit.** The unit

$$\eta_F : 1_{A_0} \Rightarrow F_0^* \circ F_{0!}$$

is induced by the diagonal map

$$\Delta : A_0 \rightarrow A_0 \times_{B_0} A_0.$$

The defining diagram is

$$\begin{array}{ccccc} & & A_0 & & \\ & \swarrow 1_{A_0} & \downarrow \Delta & \searrow 1_{A_0} & \\ A_0 & \xleftarrow{\pi_1} & A_0 \times_{B_0} A_0 & \xrightarrow{\pi_2} & A_0. \end{array}$$

3. **Compute the composite** $F_{0!} \circ F_0^*$. The composite

$$F_{0!} \circ F_0^* : B_0 \rightarrow B_0$$

has apex canonically isomorphic to A_0 , since the relevant pullback is

$$\begin{array}{ccc} A_0 \times_{A_0} A_0 & \longrightarrow & A_0 \\ \downarrow & \lrcorner & \downarrow 1_{A_0} \\ A_0 & \xrightarrow{1_{A_0}} & A_0. \end{array}$$

Under this identification,

$$F_{0!} \circ F_0^* = \left(B_0 \xleftarrow{F_0} A_0 \xrightarrow{F_0} B_0 \right).$$

4. **Define the counit.** The counit

$$\varepsilon_F : F_{0!} \circ F_0^* \Rightarrow 1_{B_0}$$

is induced by the map

$$F_0 : A_0 \rightarrow B_0.$$

It is displayed as

$$\begin{array}{ccccc} & & A_0 & & \\ & F_0 \swarrow & \downarrow F_0 & \searrow F_0 & \\ B_0 & \xleftarrow{1_{B_0}} & B_0 & \xrightarrow{1_{B_0}} & B_0. \end{array}$$

5. **Verify the triangle identities.** The two triangle identities reduce to the equations defining the diagonal and the terminal pullback isomorphisms

$$A_0 \times_{A_0} A_0 \cong A_0.$$

Equivalently, both composites are canonical pullback-induced maps with the same projections. By uniqueness in the relevant pullback universal properties, the triangle identities hold.

□

1.3 Mealy morphisms

Let

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A)$$

and

$$\mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B)$$

be internal categories in $\mathcal{E}$, regarded as monads in $\mathbf{Span}(\mathcal{E})$:

$$\begin{array}{ccc} & A_1 & \\ s_A \swarrow & & \searrow t_A \\ A_0 & & A_0 \end{array} \quad \begin{array}{ccc} & B_1 & \\ s_B \swarrow & & \searrow t_B \\ B_0 & & B_0. \end{array}$$

Let

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

be a span:

$$\begin{array}{ccc} & P & \\ p \swarrow & & \searrow q \\ A_0 & & B_0. \end{array}$$

There are two composite spans associated with P , A , and B .

First, the composite $P \circ A$ is obtained by pulling back

$$t_A : A_1 \rightarrow A_0 \quad \text{and} \quad p : P \rightarrow A_0.$$

Thus

$$\begin{array}{ccc} A_1 \times_{t_A, A_0, p} P & \xrightarrow{\pi_P} & P \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p \\ A_1 & \xrightarrow{t_A} & A_0, \end{array}$$

and

$$P \circ A = \left(A_0 \xleftarrow{s_A \pi_{A_1}} A_1 \times_{t_A, A_0, p} P \xrightarrow{q \pi_P} B_0 \right).$$

Second, the composite $B \circ P$ is obtained by pulling back

$$q : P \rightarrow B_0 \quad \text{and} \quad s_B : B_1 \rightarrow B_0.$$

Thus

$$\begin{array}{ccc} P \times_{q, B_0, s_B} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_P \downarrow & \lrcorner & \downarrow s_B \\ P & \xrightarrow{q} & B_0, \end{array}$$

and

$$B \circ P = \left(A_0 \xleftarrow{p \pi_P} P \times_{q, B_0, s_B} B_1 \xrightarrow{t_B \pi_{B_1}} B_0 \right).$$

Definition 1.3.1 (Mealy morphism). *Let $\mathbb{A}$ and $\mathbb{B}$ be internal categories, regarded as monads in $\text{Span}(\mathcal{E})$. A **Mealy morphism***

$$(P, \Phi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

consists of a span

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

and a 2-cell in the Mealy orientation

$$\Phi : P \circ A \Rightarrow B \circ P$$

satisfying the unit and multiplication compatibility axioms

$$\Phi \circ (P\eta_A) = \eta_B P, \tag{1.1}$$

$$\Phi \circ (P\mu_A) = (\mu_B P) \circ (B\Phi) \circ (\Phi A). \tag{1.2}$$

The first equation is an equality of 2-cells

$$P \Rightarrow B \circ P,$$

and the second is an equality of 2-cells

$$P \circ A \circ A \Rightarrow B \circ P.$$

The equations are read with the canonical associativity and unit isomorphisms of $\mathbf{Span}(\mathcal{E})$ inserted where necessary.

In terminology that calls a structure cell

$$P \circ A \Rightarrow B \circ P$$

a monad opmorphism or colax monad morphism, a Mealy morphism is precisely such a morphism carried by a span.

The 2-cell

$$\Phi : P \circ A \Rightarrow B \circ P$$

is a map

$$\Phi : A_1 \times_{t_A, A_0, p} P \longrightarrow P \times_{q, B_0, s_B} B_1$$

making the following span-map diagram commute:

$$\begin{array}{ccccc} & & A_1 \times_{A_0} P & & \\ & \swarrow s_A \pi_{A_1} & \downarrow \Phi & \searrow q \pi_P & \\ A_0 & \xleftarrow{p \pi_P} & P \times_{B_0} B_1 & \xrightarrow{t_B \pi_{B_1}} & B_0. \end{array}$$

Write

$$\Phi(a, x) = (\delta(a, x), \omega(a, x)).$$

Then the pullback condition says

$$s_B(\omega(a, x)) = q(\delta(a, x)),$$

and the two span-compatibility equations say

$$p(\delta(a, x)) = s_A(a), \quad t_B(\omega(a, x)) = q(x).$$

Equivalently, a Mealy morphism consists of maps

$$\delta : A_1 \times_{t_A, A_0, p} P \rightarrow P, \quad \omega : A_1 \times_{t_A, A_0, p} P \rightarrow B_1,$$

satisfying

$$p\delta = s_A \pi_{A_1}, \tag{1.3}$$

$$s_B \omega = q\delta, \tag{1.4}$$

$$t_B \omega = q\pi_P. \tag{1.5}$$

These equations are summarized by the diagram

$$\begin{array}{ccccc}
 A_1 \times_{A_0} P & \xrightarrow{\delta} & P & & A_1 \times_{A_0} P & \xrightarrow{\omega} & B_1 & & A_1 \times_{A_0} P & \xrightarrow{\omega} & B_1 \\
 \pi_{A_1} \downarrow & & \downarrow p & & \delta \downarrow & & \downarrow s_B & & \pi_P \downarrow & & \downarrow t_B \\
 A_1 & \xrightarrow{s_A} & A_0 & & P & \xrightarrow{q} & B_0 & & P & \xrightarrow{q} & B_0.
 \end{array}$$

More transparently, an input arrow

$$u \xrightarrow{a} v$$

admissible at a state x with $p(x) = v$ updates the state to one over u , while producing an output arrow

$$q(\delta(a, x)) \xrightarrow{\omega(a, x)} q(x).$$

Thus the state update and the output arrow follow the target-to-source direction imposed by the structure cell

$$\Phi : P \circ A \Rightarrow B \circ P.$$

Proposition 1.3.1 (The Mealy laws). *For a Mealy morphism*

$$(P, \delta, \omega) : \mathbb{A} \rightsquigarrow \mathbb{B},$$

the abstract unit and multiplication axioms (1.1) and (1.2) are equivalent to the following component equations.

The unit law is

$$\delta(e_A(px), x) = x, \tag{1.6}$$

$$\omega(e_A(px), x) = e_B(qx). \tag{1.7}$$

The multiplication law is

$$\delta(b \circ a, x) = \delta(a, \delta(b, x)), \tag{1.8}$$

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)), \tag{1.9}$$

for internally composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

and states $x \in P$ satisfying $p(x) = w$.

Proof.

1. **Unpack the unit input.** The unit of $\mathbb{A}$ induces a map

$$P \rightarrow A_1 \times_{A_0} P, \quad x \mapsto (e_A(px), x).$$

This is the map in the pullback diagram

$$\begin{array}{ccc}
 P & \xrightarrow{\langle e_{AP}, 1_P \rangle} & A_1 \times_{t_A, A_0, p} P \\
 \searrow 1_P & & \swarrow \pi_P \\
 & P & \\
 \swarrow e_{AP} & \downarrow p & \searrow \pi_{A_1} \\
 & A_0 &
 \end{array}$$

where $t_A e_A p = p$.

2. **Compare the two unit composites.** The unit axiom says that

$$P \xrightarrow{(e_{AP}, 1_P)} A_1 \times_{A_0} P \xrightarrow{\Phi} P \times_{B_0} B_1$$

equals

$$P \xrightarrow{(1_P, e_B q)} P \times_{B_0} B_1.$$

In diagram form:

$$\begin{array}{ccc} P & \xrightarrow{(e_{AP}, 1_P)} & A_1 \times_{A_0} P \\ & \searrow (1_P, e_B q) & \downarrow \Phi \\ & & P \times_{B_0} B_1. \end{array}$$

Comparing the two components of the pullback $P \times_{B_0} B_1$ gives

$$\delta(e_A(px), x) = x, \quad \omega(e_A(px), x) = e_B(qx).$$

3. **Identify the multiplication domain.** For the multiplication axiom, the relevant generalized elements are triples

$$(a, b, x)$$

satisfying

$$t_A(a) = s_A(b), \quad t_A(b) = p(x).$$

They are displayed as

$$u \xrightarrow{a} v \xrightarrow{b} w \quad \text{with} \quad p(x) = w.$$

The corresponding pullback object is

$$A_1 \times_{t_A, A_0, s_A} A_1 \times_{t_A, A_0, p} P.$$

4. **Compute the left-hand side.** The left-hand side of (1.2) first composes a and b in $\mathbb{A}$, then applies Φ :

$$(a, b, x) \longmapsto (b \circ a, x) \longmapsto (\delta(b \circ a, x), \omega(b \circ a, x)).$$

5. **Compute the right-hand side.** The right-hand side first applies Φ to (b, x) :

$$(b, x) \longmapsto (\delta(b, x), \omega(b, x)).$$

Since

$$p(\delta(b, x)) = s_A(b) = t_A(a),$$

the pair $(a, \delta(b, x))$ is admissible. Applying Φ again gives

$$(a, \delta(b, x)) \longmapsto (\delta(a, \delta(b, x)), \omega(a, \delta(b, x))).$$

The two produced $\mathbb{B}$ -arrows are composable:

$$q(\delta(a, \delta(b, x))) \xrightarrow{\omega(a, \delta(b, x))} q(\delta(b, x)) \xrightarrow{\omega(b, x)} q(x).$$

Hence their composite is

$$\omega(b, x) \circ \omega(a, \delta(b, x)).$$

6. **Compare components.** Equality of the two maps into $P \times_{B_0} B_1$ is equality of the two P -components and the two B_1 -components. Therefore the multiplication axiom is equivalent to

$$\delta(b \circ a, x) = \delta(a, \delta(b, x)),$$

and

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

□

Equivalently, using $m_A(a, b) = b \circ a$, the multiplication laws may be written

$$\delta(m_A(a, b), x) = \delta(a, \delta(b, x)), \quad (1.10)$$

$$\omega(m_A(a, b), x) = m_B(\omega(a, \delta(b, x)), \omega(b, x)). \quad (1.11)$$

Remark 1.3.1 (Contravariant input processing). *The input is processed target-to-source. If*

$$a : u \rightarrow v$$

and $p(x) = v$, then

$$p(\delta(a, x)) = u.$$

Operationally, a Mealy morphism may be regarded as a right-action-like contravariant input process on the state span, together with an output cocycle valued in the output category.

1.3.1 The one-object specialization

Let M and N be monoid objects in $\mathcal{E}$. We regard them as one-object internal categories

$$\mathbb{M} = (1, M, !, !, e_M, m_{\mathbb{M}}), \quad \mathbb{N} = (1, N, !, !, e_N, m_{\mathbb{N}}),$$

where

$$m_{\mathbb{M}}(a, b) = ba, \quad m_{\mathbb{N}}(\alpha, \beta) = \beta\alpha.$$

Equivalently, define

$$\rho := \delta \circ \tau_{P, M} : P \times M \rightarrow P,$$

where

$$\tau_{P, M} : P \times M \rightarrow M \times P$$

is the symmetry. We write

$$x \cdot m := \rho(x, m) = \delta(m, x).$$

The state-transition unit and multiplication laws are then precisely the internal right-action laws

$$x \cdot 1 = x, \quad x \cdot (mn) = (x \cdot m) \cdot n.$$

Equivalently, as morphism equations,

$$\rho \circ \langle 1_P, e_M ! \rangle = 1_P$$

and

$$\rho \circ (\rho \times 1_M) = \rho \circ (1_P \times \cdot_M)$$

as maps

$$P \times M \times M \rightarrow P.$$

The output map ω is the corresponding internal N -valued output cocycle:

$$\omega(1, x) = 1, \quad \omega(mn, x) = \omega(m, x)\omega(n, x \cdot m).$$

The sequential execution picture is

$$x \xrightarrow{m/\omega(m, x)} x \cdot m \xrightarrow{n/\omega(n, x \cdot m)} (x \cdot m) \cdot n.$$

Proposition 1.3.2 (One-object Mealy morphisms). *Let M and N be monoid objects in $\mathcal{E}$, regarded as one-object internal categories by the convention*

$$b \circ a = ba.$$

To give a one-object Mealy morphism

$$M \rightsquigarrow N$$

is equivalent to give an internal right M -object

$$\rho : P \times M \rightarrow P$$

together with a map

$$\omega : M \times P \rightarrow N$$

satisfying

$$\omega(1, x) = 1, \quad \omega(mn, x) = \omega(m, x)\omega(n, x \cdot m),$$

where

$$x \cdot m := \rho(x, m).$$

Proof.

1. **Identify the carrier and transition map.** In the one-object case, the carrier span is simply an object P , and the pullbacks defining $P \circ M$ and $N \circ P$ reduce to products. A Mealy structure therefore consists of maps

$$\delta : M \times P \rightarrow P, \quad \omega : M \times P \rightarrow N.$$

Using the cartesian symmetry, define

$$\rho := \delta \circ \tau_{P, M} : P \times M \rightarrow P.$$

Conversely,

$$\delta = \rho \circ \tau_{M, P}.$$

Hence specifying δ is equivalent to specifying the conventionally typed operation ρ .

2. **Identify the right-action laws.** Proposition 1.3.1, together with the convention $b \circ a = ba$, gives

$$\delta(1, x) = x$$

and

$$\delta(mn, x) = \delta(n, \delta(m, x)).$$

In terms of

$$x \cdot m := \rho(x, m) = \delta(m, x),$$

these equations become

$$x \cdot 1 = x$$

and

$$x \cdot (mn) = (x \cdot m) \cdot n.$$

Thus ρ is an internal right action of M on P .

3. Identify the output-cocycle laws. The Mealy unit law for the output is

$$\omega(1, x) = 1.$$

Its multiplication law is

$$\omega(mn, x) = \omega(m, x)\omega(n, \delta(m, x)),$$

which, using

$$\delta(m, x) = x \cdot m,$$

is exactly

$$\omega(mn, x) = \omega(m, x)\omega(n, x \cdot m).$$

4. Reverse the construction. Conversely, let

$$\rho : P \times M \rightarrow P$$

be an internal right action, and let

$$\omega : M \times P \rightarrow N$$

satisfy the displayed unit and cocycle equations. Define

$$\delta := \rho \circ \tau_{M, P} : M \times P \rightarrow P.$$

The right-action equations give the transition unit and multiplication laws, while the equations for ω give the output unit and multiplication laws. Hence (P, δ, ω) determines a one-object Mealy morphism.

□

Remark 1.3.2 (Set-based one-object systems). *When $\mathcal{E} = \mathbf{Set}$, Proposition 1.3.2 says that a one-object Mealy morphism is a set with a right action of the input monoid, together with an output cocycle valued in the output monoid. The set-theoretic reading is therefore a specialization of the internal statement, not part of the definition.*

1.3.2 The stateless specialization

A Mealy morphism

$$(P, \Phi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

has a state object P . The map-representable case is obtained by taking the canonical carrier

$$F_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right).$$

The state is then the current input object:

$$\begin{array}{ccc} & A_0 & \\ 1_{A_0} \swarrow & & \searrow F_0 \\ A_0 & & B_0. \end{array}$$

Let

$$F_0 : A_0 \rightarrow B_0$$

be a morphism in $\mathcal{E}$. A Mealy structure on $F_{0!}$ is a 2-cell

$$\Phi : F_{0!} \circ A \Rightarrow B \circ F_{0!}.$$

First,

$$F_{0!} \circ A \cong \left(A_0 \xleftarrow{s_A} A_1 \xrightarrow{F_0 t_A} B_0 \right).$$

This comes from the pullback

$$\begin{array}{ccccc} A_1 \times_{t_A, A_0, 1_{A_0}} A_0 & \longrightarrow & A_0 \\ \downarrow & \lrcorner & \downarrow 1_{A_0} \\ A_1 & \xrightarrow{t_A} & A_0, \end{array}$$

which is canonically isomorphic to A_1 .

Second,

$$B \circ F_{0!} = \left(A_0 \xleftarrow{\pi_1} A_0 \times_{F_0, B_0, s_B} B_1 \xrightarrow{t_B \pi_2} B_0 \right),$$

where

$$\begin{array}{ccccc} A_0 \times_{F_0, B_0, s_B} B_1 & \xrightarrow{\pi_2} & B_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow s_B \\ A_0 & \xrightarrow{F_0} & B_0. \end{array}$$

Thus Φ is equivalently a map

$$\Phi : A_1 \rightarrow A_0 \times_{B_0} B_1.$$

Write

$$\Phi(a) = (\delta(a), F_1(a)).$$

The condition that $\Phi(a)$ lies in the pullback says

$$F_0(\delta(a)) = s_B(F_1(a)).$$

The condition that Φ is a map of spans says

$$\delta(a) = s_A(a), \quad t_B(F_1(a)) = F_0(t_A(a)).$$

Therefore

$$s_B F_1 = F_0 s_A, \quad t_B F_1 = F_0 t_A.$$

The arrow part is summarized by

$$\begin{array}{ccc} A_1 & \xrightarrow{F_1} & B_1 \\ (s_A, t_A) \downarrow & & \downarrow (s_B, t_B) \\ A_0 \times A_0 & \xrightarrow{F_0 \times F_0} & B_0 \times B_0. \end{array}$$

Definition 1.3.2 (Internal functor). *Let*

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A)$$

and

$$\mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B)$$

be internal categories in $\mathcal{E}$. An **internal functor**

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

consists of maps

$$F_0 : A_0 \rightarrow B_0, \quad F_1 : A_1 \rightarrow B_1,$$

such that

$$s_B F_1 = F_0 s_A, \tag{1.12}$$

$$t_B F_1 = F_0 t_A, \tag{1.13}$$

$$F_1 e_A = e_B F_0, \tag{1.14}$$

$$F_1 m_A = m_B F_1^{(2)}. \tag{1.15}$$

Here

$$F_1^{(2)} : A_1 \times_{t_A, A_0, s_A} A_1 \longrightarrow B_1 \times_{t_B, B_0, s_B} B_1$$

is the unique map induced by the pullback defining the object of composable pairs in $\mathbb{B}$, with projections

$$\pi_1 F_1^{(2)} = F_1 \pi_1, \quad \pi_2 F_1^{(2)} = F_1 \pi_2.$$

It is well-defined because, on

$$A_1 \times_{t_A, A_0, s_A} A_1,$$

one has

$$t_B F_1 \pi_1 = F_0 t_A \pi_1 = F_0 s_A \pi_2 = s_B F_1 \pi_2.$$

The source-target equations say that

$$\begin{array}{ccc} A_1 & \xrightarrow{F_1} & B_1 \\ s_A \downarrow & & \downarrow s_B \\ A_0 & \xrightarrow{F_0} & B_0 \end{array} \quad \begin{array}{ccc} A_1 & \xrightarrow{F_1} & B_1 \\ t_A \downarrow & & \downarrow t_B \\ A_0 & \xrightarrow{F_0} & B_0 \end{array}$$

commute, while the identity and composition equations say that

$$\begin{array}{ccc} A_0 & \xrightarrow{F_0} & B_0 \\ e_A \downarrow & & \downarrow e_B \\ A_1 & \xrightarrow{F_1} & B_1 \end{array}$$

commutes and that F_1 preserves composition in the sense that the diagram

$$\begin{array}{ccc} A_1 \times_{t_A, A_0, s_A} A_1 & \xrightarrow{F_1^{(2)}} & B_1 \times_{t_B, B_0, s_B} B_1 \\ m_A \downarrow & & \downarrow m_B \\ A_1 & \xrightarrow{F_1} & B_1 \end{array}$$

commutes.

Theorem 1.3.1 (Stateless Mealy morphisms are internal functors). *Let*

$$\mathbb{A}, \mathbb{B}$$

be internal categories in $\mathcal{E}$, and let

$$F_0 : A_0 \rightarrow B_0$$

be a morphism in $\mathcal{E}$. To give an internal functor

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

with object map F_0 is equivalently to give a Mealy morphism

$$(F_0!, \Phi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

whose carrier is the canonical map-representable span

$$F_0! = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right).$$

Proof.

1. **Identify the structure cell.** By the calculation preceding the theorem, a 2-cell

$$\Phi : F_0! \circ A \Rightarrow B \circ F_0!$$

is equivalently a map

$$\Phi : A_1 \rightarrow A_0 \times_{F_0, B_0, s_B} B_1.$$

Writing $\Phi(a) = (s_A(a), F_1(a))$, the span-map and pullback conditions are exactly

$$s_B F_1 = F_0 s_A, \quad t_B F_1 = F_0 t_A.$$

2. **Identify the laws.** Under this identification, Proposition 1.3.1 says that the Mealy unit law is

$$F_1 e_A = e_B F_0,$$

and the Mealy multiplication law is

$$F_1(b \circ a) = F_1(b) \circ F_1(a),$$

equivalently

$$F_1 m_A = m_B F_1^{(2)}.$$

Hence a stateless Mealy structure on $F_0!$ is exactly an internal functor with object map F_0 .

3. Construct the inverse assignment. Conversely, an internal functor (F_0, F_1) defines

$$\Phi(a) = (s_A(a), F_1(a)).$$

The source-target equations for F_1 make this a map of spans, and identity and composition preservation are exactly the Mealy unit and multiplication laws. The two constructions are inverse.

□

Corollary 1.3.1 (Map-representable Mealy morphisms). *Let*

$$(P, \Phi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be a Mealy morphism whose carrier span

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

is map-representable. Put

$$F_0 := qp^{-1} : A_0 \rightarrow B_0.$$

Then the span isomorphism

$$\sigma := p : \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right) \xrightarrow{\cong} \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right) = F_0!$$

transports Φ to a Mealy structure

$$\Phi^F : F_0! \circ A \Rightarrow B \circ F_0!.$$

Consequently, (P, Φ) determines an internal functor

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

with object map F_0 .

Proof.

- 1. Construct the carrier-span isomorphism.** Since the carrier is map-representable, the left leg

$$p : P \rightarrow A_0$$

is an isomorphism. Put

$$F_0 = qp^{-1}.$$

The map

$$\sigma := p : P \rightarrow A_0$$

is then a map of spans

$$\sigma : \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right) \Longrightarrow \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right),$$

because

$$1_{A_0}\sigma = p$$

and

$$F_0\sigma = qp^{-1}p = q.$$

Its inverse as a span map is $p^{-1} : A_0 \rightarrow P$.

2. **Transport the Mealy structure.** Define

$$\Phi^F := (B\sigma) \circ \Phi \circ (\sigma^{-1}A).$$

Explicitly, this is the composite

$$F_{0!} \circ A \xRightarrow{\sigma^{-1}A} P \circ A \xRightarrow{\Phi} B \circ P \xRightarrow{B\sigma} B \circ F_{0!}.$$

Since σ is invertible, conjugating the unit and multiplication equations for Φ by σ gives the unit and multiplication equations for Φ^F . Hence

$$(F_{0!}, \Phi^F) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

is a Mealy morphism.

3. **Apply the stateless correspondence.** By Theorem 1.3.1, a Mealy structure on the canonical carrier $F_{0!}$ is equivalently an internal functor

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

whose object map is F_0 . Therefore the original map-representable Mealy morphism determines such an internal functor.

□

1.4 Mealy simulations

1.4.1 From strict carrier maps to output-comparison cells

The previous section identified the 1-cells. The next question is how to compare two such oriented structures.

Let

$$(P, \Phi), (Q, \Psi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be Mealy morphisms, with carrier spans

$$A_0 \xleftarrow{p_P} P \xrightarrow{q_P} B_0, \quad A_0 \xleftarrow{p_Q} Q \xrightarrow{q_Q} B_0.$$

The first comparison available in the span bicategory is a map of carrier spans. In the direction used below, this is a map

$$\theta : Q \rightarrow P$$

satisfying

$$p_P \theta = p_Q, \quad q_P \theta = q_Q.$$

It is displayed as

$$\begin{array}{ccccc} & & Q & & \\ & \swarrow p_Q & \downarrow \theta & \searrow q_Q & \\ A_0 & \xleftarrow{p_P} & P & \xrightarrow{q_P} & B_0. \end{array}$$

Requiring this carrier map to preserve the Mealy structures gives the equations

$$\theta(\delta_Q(a, y)) = \delta_P(a, \theta y), \quad \omega_Q(a, y) = \omega_P(a, \theta y).$$

Thus a strict comparison preserves the input interface, the output interface, the state transition, and the emitted output arrow.

This gives a valid strict output-preserving comparison. Its behaviour in the map-representable case, however, is too rigid. Let

$$F, G : \mathbb{A} \rightarrow \mathbb{B}$$

be internal functors, regarded as Mealy morphisms with canonical carriers

$$F_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right), \quad G_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{G_0} B_0 \right).$$

A strict carrier map

$$G_{0!} \rightarrow F_{0!}$$

is a map

$$\theta : A_0 \rightarrow A_0$$

such that

$$1_{A_0} \theta = 1_{A_0}, \quad F_0 \theta = G_0.$$

Hence

$$\theta = 1_{A_0}, \quad F_0 = G_0.$$

Thus, already at the level of carrier spans, a strict comparison can exist only when the two functors have the same object map.

If the carrier map is required to preserve the Mealy structures strictly, then the output-preservation equation gives, for every arrow $a \in A_1$,

$$G_1(a) = F_1(a).$$

Indeed, the only possible carrier map is 1_{A_0} , and the strict output equation specializes to

$$\omega_G(a, t_A a) = \omega_F(a, t_A a),$$

that is,

$$G_1(a) = F_1(a).$$

Consequently a strict comparison between canonical map-representable Mealy morphisms exists only when

$$F = G,$$

and in that case it is the identity strict comparison.

Strict carrier comparisons therefore do not recover nonidentity internal natural transformations. The component arrow

$$\alpha_x : F_0x \rightarrow G_0x$$

has no place in the strict carrier-map datum, even when the two functors are related by a nontrivial internal natural transformation.

This calculation indicates the required enlargement. A comparison from (P, Φ) to (Q, Ψ) should still send each Q -state to a P -state over the same input interface:

$$\theta : Q \rightarrow P, \quad p_P \theta = p_Q.$$

The equality of output interfaces should be replaced by an arrow in the output category

$$\alpha_y : q_P(\theta y) \rightarrow q_Q(y).$$

In the stateless case this is exactly the shape of a natural-transformation component.

The span calculus already contains the object that stores such data. The composite

$$B \circ P$$

has apex

$$P \times_{q_P, B_0, s_B} B_1$$

and is the span

$$A_0 \xleftarrow{p_P \pi_P} P \times_{q_P, B_0, s_B} B_1 \xrightarrow{t_B \pi_{B_1}} B_0.$$

The defining pullback is

$$\begin{array}{ccc} P \times_{q_P, B_0, s_B} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_P \downarrow & \lrcorner & \downarrow s_B \\ P & \xrightarrow{q_P} & B_0. \end{array}$$

A generalized element of $B \circ P$ is a pair

$$(x, \beta)$$

where

$$x \in P, \quad \beta : q_P(x) \rightarrow b$$

is an arrow of $\mathbb{B}$. Therefore a comparison assigning to a Q -state a P -state together with an output-comparison arrow is exactly a map of spans

$$\Theta : Q \Rightarrow B \circ P.$$

It is displayed as

$$\begin{array}{ccccc} & & Q & & \\ & \swarrow p_Q & \downarrow \Theta & \searrow q_Q & \\ A_0 & \xleftarrow{p_P \pi_P} & P \times_{B_0} B_1 & \xrightarrow{t_B \pi_{B_1}} & B_0. \end{array}$$

Writing

$$\Theta(y) = (\theta(y), \alpha_y),$$

the span-map and pullback conditions give

$$p_P\theta = p_Q, \quad s_B\alpha = q_P\theta, \quad t_B\alpha = q_Q.$$

Thus each state $y \in Q$ is compared with a state $\theta y \in P$ over the same input interface, and the output interfaces are compared by an arrow

$$q_P(\theta y) \xrightarrow{\alpha_y} q_Q(y)$$

in $\mathbb{B}$.

For canonical map-representable carriers, a map of spans

$$G_0! \Rightarrow B \circ F_0!$$

is equivalently a map

$$\alpha : A_0 \rightarrow B_1$$

satisfying

$$s_B\alpha = F_0, \quad t_B\alpha = G_0.$$

These are precisely the component-typing equations for an internal natural transformation

$$\alpha : F \Rightarrow G.$$

The remaining compatibility condition should therefore be the internal naturality equation

$$\alpha_v \circ F_1(a) = G_1(a) \circ \alpha_u$$

for every arrow $a : u \rightarrow v$ of $\mathbb{A}$.

Now notice that on the local span category

$$\mathcal{C}_{A,B} = \mathbf{Span}(\mathcal{E})(A_0, B_0),$$

postcomposition by B defines an endofunctor

$$T_B = B \circ -.$$

Its unit attaches an identity output-comparison arrow, and its multiplication composes output-comparison arrows. Thus T_B is the output-comparison monad. A map

$$Q \rightarrow T_B(P)$$

is exactly the comparison cell just described.

In the case of the source category $\mathbb{A}$, we also have the supplied the input-processing monad

$$R_A = - \circ A.$$

The associativity isomorphisms of the span bicategory identify

$$R_A T_B(P) = (B \circ P) \circ A$$

with

$$T_B R_A(P) = B \circ (P \circ A),$$

and give a distributive law

$$R_A T_B \Rightarrow T_B R_A.$$

Consequently R_A lifts to a monad on the Kleisli category of T_B .

With this organization, a Mealy morphism is an algebra for the lifted input-processing monad, and a Mealy simulation is an oppositely displayed morphism of such algebras. The displayed direction is chosen so that, in the map-representable case, a simulation

$$F_0! \Rightarrow G_0!$$

has component arrows

$$F_0 x \rightarrow G_0 x$$

and hence recovers an internal natural transformation

$$F \Rightarrow G.$$

1.4.2 The output-comparison monad

Fix internal categories

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A), \quad \mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B).$$

Let

$$\mathcal{C}_{A,B} := \mathbf{Span}(\mathcal{E})(A_0, B_0)$$

be the local span category. Its objects are spans

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0,$$

and its morphisms are maps of spans. Thus a morphism

$$f : P \rightarrow Q$$

in $\mathcal{C}_{A,B}$ is a commuting diagram

$$\begin{array}{ccccc} & & P & & \\ & \swarrow p_P & \downarrow f & \searrow q_P & \\ A_0 & \xleftarrow{p_Q} & Q & \xrightarrow{q_Q} & B_0. \end{array}$$

Equivalently,

$$p_Q f = p_P, \quad q_Q f = q_P.$$

Lemma 1.4.1 (Local monads induced by endpoint monads). *Let A be a monad on an object A_0 , and let B be a monad on an object B_0 , in a bicategory $\mathcal{B}$. Write*

$$a_{H,G,F} : (H \circ G) \circ F \Longrightarrow H \circ (G \circ F)$$

for the associator, and write

$$\ell_F : 1 \circ F \Longrightarrow F, \quad r_F : F \circ 1 \Longrightarrow F$$

for the unitors.

On the hom-category

$$\mathcal{B}(A_0, B_0),$$

precomposition by A defines a monad

$$R = - \circ A,$$

and postcomposition by B defines a monad

$$T = B \circ -.$$

For a 1-cell $P : A_0 \rightarrow B_0$, the unit and multiplication of R are

$$\eta_P^R := (P\eta_A) \circ r_P^{-1} : P \Longrightarrow P \circ A,$$

and

$$\mu_P^R := (P\mu_A) \circ a_{P,A,A} : (P \circ A) \circ A \Longrightarrow P \circ A.$$

The unit and multiplication of T are

$$\eta_P^T := (\eta_B P) \circ \ell_P^{-1} : P \Longrightarrow B \circ P,$$

and

$$\mu_P^T := (\mu_B P) \circ a_{B,B,P}^{-1} : B \circ (B \circ P) \Longrightarrow B \circ P.$$

On a 2-cell $f : P \Rightarrow Q$, the two endofunctors act by whiskering:

$$R(f) = fA, \quad T(f) = Bf.$$

Proof. Naturality of the displayed units and multiplications follows from naturality of the bicategorical associators and unitors together with functoriality of whiskering.

For R , the two unit equations reduce, after expanding η^R and μ^R , to the two unit equations for the monad A , with the triangle identity supplying the required cancellation of unitors. The multiplication equation reduces to the multiplication equation for A , with the pentagon identity identifying the two reassociations of

$$((P \circ A) \circ A) \circ A.$$

Dually, the two unit equations for T reduce to the unit equations for B , and the multiplication equation reduces to the multiplication equation for B . The required changes of parentheses are supplied by the triangle and pentagon identities.

Hence $R = - \circ A$ and $T = B \circ -$ are monads on $\mathcal{B}(A_0, B_0)$. $\square$

Definition 1.4.1 (The output-comparison endofunctor). *Define an endofunctor*

$$T_B : \mathcal{C}_{A,B} \rightarrow \mathcal{C}_{A,B}$$

by

$$T_B(P) := B \circ P.$$

Explicitly, if

$$P = \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right),$$

then $T_B(P)$ is obtained from the pullback

$$\begin{array}{ccc} P \times_{q, B_0, s_B} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_P \downarrow & \lrcorner & \downarrow s_B \\ P & \xrightarrow{q} & B_0, \end{array}$$

and is the span

$$T_B(P) = \left(A_0 \xleftarrow{p\pi_P} P \times_{q, B_0, s_B} B_1 \xrightarrow{t_B\pi_{B_1}} B_0 \right).$$

Thus

$$A_0 \xleftarrow{p\pi_P} P \times_{B_0} B_1 \xrightarrow{t_B\pi_{B_1}} B_0$$

is the carrier span $T_B(P)$.

Let

$$f : P \rightarrow Q$$

be a map of carrier spans, so that

$$p_Q f = p_P, \quad q_Q f = q_P.$$

Define

$$T_B(f) : P \times_{q_P, B_0, s_B} B_1 \longrightarrow Q \times_{q_Q, B_0, s_B} B_1$$

to be the unique map induced by the pair

$$f\pi_P : P \times_{q_P, B_0, s_B} B_1 \rightarrow Q, \quad \pi_{B_1} : P \times_{q_P, B_0, s_B} B_1 \rightarrow B_1.$$

It is well-defined because

$$q_Q f\pi_P = q_P \pi_P = s_B \pi_{B_1}.$$

Equivalently, it is characterized by

$$\pi_Q T_B(f) = f\pi_P, \quad \pi_{B_1} T_B(f) = \pi_{B_1}.$$

The defining diagram is

$$\begin{array}{ccccc} P \times_{q_P, B_0, s_B} B_1 & \xrightarrow{T_B(f)} & Q \times_{q_Q, B_0, s_B} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_P \downarrow & & \downarrow \pi_Q & \lrcorner & \downarrow s_B \\ P & \xrightarrow{f} & Q & \xrightarrow{q_Q} & B_0. \end{array}$$

In generalized-element notation,

$$T_B(f)(x, \beta) = (fx, \beta).$$

The equations characterizing $T_B(f)$ immediately imply

$$T_B(1_P) = 1_{T_B(P)}$$

and

$$T_B(gf) = T_B(g)T_B(f),$$

so T_B is an endofunctor.

A generalized element of $T_B(P)$ is a pair

$$(x, \beta)$$

where $x \in P$ and

$$\beta : q(x) \rightarrow b$$

is an arrow of $\mathbb{B}$. Thus $T_B(P)$ freely attaches an output comparison arrow to a state:

$$q(x) \xrightarrow{\beta} b.$$

Proposition 1.4.1 (The output-comparison monad). *The endofunctor*

$$T_B = B \circ -$$

is a monad on $\mathcal{C}_{A,B}$.

Proof.

1. **Use the endpoint monad.** This is Lemma 1.4.1 applied in $\mathbf{Span}(\mathcal{E})$ to the monad B on B_0 . Thus postcomposition by B gives a monad $T_B = B \circ -$ on $\mathcal{C}_{A,B}$.
2. **Unpack the unit.** For a carrier span

$$P = \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right),$$

the unit is the span map

$$\eta_P^T : P \rightarrow T_B(P), \quad x \mapsto (x, e_B(qx)).$$

It is well-defined because $s_B e_B q = q$, and its right leg is $t_B e_B q = q$.

3. **Unpack the multiplication.** A generalized element of $T_B^2(P)$ is a state $x \in P$ together with composable arrows

$$q(x) \xrightarrow{\beta} b \xrightarrow{\gamma} c.$$

The multiplication is

$$\mu_P^T(x, \beta, \gamma) = (x, \gamma \circ \beta),$$

or equivalently $(x, m_B(\beta, \gamma))$.

4. **Monad laws.** The monad unit and associativity axioms are exactly the identity and associativity laws in the internal category $\mathbb{B}$, after whiskering by P . Hence T_B is a monad. □

Definition 1.4.2 (The Kleisli category of output comparisons). *The **output Kleisli category** associated to $\mathbb{B}$ over (A_0, B_0) is the Kleisli category*

$$\mathbf{Kl}(T_B).$$

Its objects are carrier spans

$$A_0 \leftarrow P \rightarrow B_0.$$

A Kleisli arrow

$$Q \rightarrow P$$

is a map in $\mathcal{C}_{A,B}$

$$Q \rightarrow T_B(P) = B \circ P.$$

Unpacked, a Kleisli arrow

$$Q \rightarrow P$$

is a pair of maps

$$\theta : Q \rightarrow P, \quad \alpha : Q \rightarrow B_1,$$

satisfying

$$p_P \theta = p_Q, \tag{1.16}$$

$$s_B \alpha = q_P \theta, \tag{1.17}$$

$$t_B \alpha = q_Q. \tag{1.18}$$

Equivalently, it is the map of spans

$$\begin{array}{ccccc} & & Q & & \\ & p_Q \swarrow & \downarrow \langle \theta, \alpha \rangle & \searrow q_Q & \\ A_0 & \xleftarrow[p_P \pi_P]{} & P \times_{B_0} B_1 & \xrightarrow[t_B \pi_{B_1}]{} & B_0. \end{array}$$

Thus a state $y \in Q$ is sent to a state $\theta y \in P$ over the same input interface, together with an output-comparison arrow

$$q_P(\theta y) \xrightarrow{\alpha_y} q_Q(y).$$

Kleisli composition is output-arrow composition. If

$$(\theta, \alpha) : Q \rightarrow P, \quad (\kappa, \beta) : R \rightarrow Q$$

are Kleisli arrows, then their composite

$$R \rightarrow P$$

is represented by

$$R \xrightarrow{\langle \kappa, \beta \rangle} T_B(Q) \xrightarrow{T_B \langle \theta, \alpha \rangle} T_B^2(P) \xrightarrow{\mu_P^T} T_B(P).$$

In components, it has state map

$$\theta \kappa : R \rightarrow P$$

and comparison component

$$\gamma_z = \beta_z \circ \alpha_{\kappa z}.$$

The composite comparison arrow is the path

$$q_P(\theta \kappa z) \xrightarrow{\alpha_{\kappa z}} q_Q(\kappa z) \xrightarrow{\beta_z} q_R(z).$$

The identity Kleisli arrow on P is

$$x \mapsto (x, e_B(q_P x)).$$

1.4.3 The input-processing monad and Kleisli algebras

The source internal category $\mathbb{A}$ induces a second local monad.

Definition 1.4.3 (The input-processing endofunctor). *Define an endofunctor*

$$R_A : \mathcal{C}_{A,B} \rightarrow \mathcal{C}_{A,B}$$

by

$$R_A(P) := P \circ A.$$

Explicitly, if

$$P = \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right),$$

then $R_A(P)$ is obtained from the pullback

$$\begin{array}{ccc} A_1 \times_{t_A, A_0, p} P & \xrightarrow{\pi_P} & P \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p \\ A_1 & \xrightarrow{t_A} & A_0, \end{array}$$

and is the span

$$R_A(P) = \left(A_0 \xleftarrow{s_A \pi_{A_1}} A_1 \times_{t_A, A_0, p} P \xrightarrow{q \pi_P} B_0 \right).$$

Thus

$$A_0 \xleftarrow{s_A \pi_{A_1}} A_1 \times_{A_0} P \xrightarrow{q \pi_P} B_0$$

is the carrier span $R_A(P)$.

If

$$f : P \rightarrow Q$$

is a map of spans, then

$$R_A(f) : R_A(P) \rightarrow R_A(Q)$$

is induced by

$$(a, x) \mapsto (a, fx).$$

This is well-defined because

$$p_Q(fx) = p_P(x) = t_A(a).$$

A generalized element of $R_A(P)$ is a pair

$$(a, x)$$

where

$$a : u \rightarrow v$$

is an arrow of $\mathbb{A}$ and $p(x) = v$. Thus $R_A(P)$ is the object of states equipped with an admissible input arrow:

$$u \xrightarrow{a} v = p(x).$$

Proposition 1.4.2 (The input-processing monad). *The endofunctor*

$$R_A = - \circ A$$

is a monad on $\mathcal{C}_{A,B}$.

Proof.

1. **Use the endpoint monad.** This is Lemma 1.4.1 applied in $\mathbf{Span}(\mathcal{E})$ to the monad A on A_0 . Thus precomposition by A gives a monad $R_A = - \circ A$ on $\mathcal{C}_{A,B}$.

2. **Unpack the unit.** For a carrier span

$$P = \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right),$$

the unit is the span map

$$\eta_P^R : P \rightarrow R_A(P), \quad x \mapsto (e_A(px), x).$$

It is well-defined because $t_A e_A p = p$, and its left leg is $s_A e_A p = p$.

3. **Unpack the multiplication.** A generalized element of $R_A^2(P)$ is a state $x \in P$ together with composable input arrows

$$u \xrightarrow{a} v \xrightarrow{b} w = p(x).$$

The multiplication is

$$\mu_P^R(a, b, x) = (b \circ a, x),$$

or equivalently $(m_A(a, b), x)$.

4. **Monad laws.** The monad unit and associativity axioms are exactly the identity and associativity laws in the internal category $\mathbb{A}$, after whiskering by P . Hence R_A is a monad.

□

The two local monads are related by span associativity. For every carrier span P , there is a canonical isomorphism

$$R_A T_B(P) = (B \circ P) \circ A \cong B \circ (P \circ A) = T_B R_A(P).$$

On apexes, both sides are canonically isomorphic to the same object of triples

$$(a, x, \beta)$$

with

$$t_A(a) = p(x), \quad s_B(\beta) = q(x).$$

The two pullback presentations are displayed as

$$\begin{array}{ccc} A_1 \times_{A_0} (P \times_{B_0} B_1) & \longrightarrow & P \times_{B_0} B_1 \\ \downarrow & \lrcorner & \downarrow p\pi_P \\ A_1 & \xrightarrow{t_A} & A_0 \end{array} \quad \begin{array}{ccc} (A_1 \times_{A_0} P) \times_{B_0} B_1 & \longrightarrow & B_1 \\ \downarrow & \lrcorner & \downarrow s_B \\ A_1 \times_{A_0} P & \xrightarrow{q\pi_P} & B_0. \end{array}$$

In generalized-element notation, this isomorphism sends

$$(a, (x, \beta))$$

to

$$((a, x), \beta).$$

Lemma 1.4.2 (Endpoint distributive law). *Let A be a monad on A_0 , and let B be a monad on B_0 , in a bicategory $\mathcal{B}$. On the hom-category*

$$\mathcal{B}(A_0, B_0),$$

write

$$R = - \circ A, \quad T = B \circ -$$

for the local monads of Lemma 1.4.1.

There is a natural isomorphism

$$\lambda : RT \Rightarrow TR$$

whose component at a 1-cell $P : A_0 \rightarrow B_0$ is the bicategorical associator

$$\lambda_P := a_{B,P,A} : (B \circ P) \circ A \xrightarrow{\cong} B \circ (P \circ A).$$

This natural isomorphism is a distributive law of R over T . Thus its components satisfy

$$\lambda \circ \eta^R T = T \eta^R, \tag{1.19}$$

$$\lambda \circ R \eta^T = \eta^T R, \tag{1.20}$$

$$\lambda \circ \mu^R T = T \mu^R \circ \lambda R \circ R \lambda, \tag{1.21}$$

$$\lambda \circ R \mu^T = \mu^T R \circ T \lambda \circ \lambda T. \tag{1.22}$$

We use the convention that a distributive law

$$RT \Rightarrow TR$$

lifts R to the Kleisli category of T .

Proof. Naturality of λ is naturality of the associator $a_{B,P,A}$ in the middle variable P .

For the R -unit equation, both sides are canonical composites

$$B \circ P \Longrightarrow B \circ (P \circ A)$$

obtained by inserting the unit of A . After expanding η^R , their equality is the triangle identity together with naturality of the associator.

For the T -unit equation, both sides are canonical composites

$$P \circ A \Longrightarrow B \circ (P \circ A)$$

obtained by inserting the unit of B . Their equality is again the triangle identity together with naturality of the associator.

For the R -multiplication equation, both sides are composites

$$((B \circ P) \circ A) \circ A \Longrightarrow B \circ (P \circ A)$$

which reassociate the four factors and then apply the multiplication of A . Naturality of the associator moves the whiskered multiplication through the reassociation, and the pentagon identity identifies the two possible reassociation composites. This gives

$$\lambda \circ \mu^R T = T \mu^R \circ \lambda R \circ R \lambda.$$

Similarly, the T -multiplication equation compares two composites

$$(B \circ (B \circ P)) \circ A \implies B \circ (P \circ A)$$

which reassociate the four factors and then apply the multiplication of B . Naturality of the associator and the pentagon identity give

$$\lambda \circ R \mu^T = \mu^T R \circ T \lambda \circ \lambda T.$$

Hence $\lambda : RT \Rightarrow TR$ is a distributive law. $\square$

Proposition 1.4.3 (The canonical distributive law). *The associativity constraints of $\mathbf{Span}(\mathcal{E})$ determine a distributive-law isomorphism*

$$\lambda : R_A T_B \Rightarrow T_B R_A$$

of the input-processing monad over the output-comparison monad.

Proof.

1. **Define the component.** By Lemma 1.4.2, the component at a carrier span P is the span associator

$$\lambda_P : (B \circ P) \circ A \xrightarrow{\cong} B \circ (P \circ A).$$

On apexes, both sides are canonically pullback presentations of the object of triples (a, x, β) satisfying

$$t_A(a) = p(x), \quad s_B(\beta) = q(x),$$

and λ_P sends

$$(a, (x, \beta)) \longmapsto ((a, x), \beta).$$

2. **Check naturality.** If $f : P \rightarrow Q$ is a map of carrier spans, both composites in the naturality square send $(a, (x, \beta))$ to $((a, fx), \beta)$. Hence λ is natural.
3. **Check the distributive-law axioms.** The compatibility with the two units inserts either an identity input arrow or an identity output arrow, and the associator leaves the other datum unchanged. The compatibility with the two multiplications says that reassociation commutes with composition in $\mathbb{A}$ and $\mathbb{B}$. Componentwise, for example, the T_B -multiplication compatibility sends

$$(a, x, \beta, \gamma)$$

on both sides to

$$((a, x), \gamma \circ \beta),$$

while the R_A -multiplication compatibility sends

$$(a, b, x, \beta)$$

on both sides to

$$((b \circ a, x), \beta).$$

These are exactly the distributive-law equations for $R_A T_B \Rightarrow T_B R_A$.

□

Corollary 1.4.1 (The lifted input-processing monad). *The distributive law of Proposition 1.4.3 lifts R_A to a monad*

$$\overline{R}_A$$

on the Kleisli category

$$\mathbf{Kl}(T_B).$$

Proof.

1. **Apply the Kleisli lifting theorem.** By the standard Kleisli lifting theorem for distributive laws of monads, a distributive law $RT \Rightarrow TR$ lifts R to a monad on $\mathbf{Kl}(T)$. Applying this to $R_A = - \circ A$, $T_B = B \circ -$, and Proposition 1.4.3 gives the lifted monad $\overline{R}_A$.
2. **Record the formula.** On objects,

$$\overline{R}_A(P) = R_A(P).$$

If a Kleisli arrow $f : P \rightarrow Q$ is represented in $\mathcal{C}_{A,B}$ by a map $f : P \rightarrow T_B(Q)$, then $\overline{R}_A(f)$ is represented by

$$R_A(P) \xrightarrow{R_A f} R_A T_B(Q) \xrightarrow{\lambda_Q} T_B R_A(Q).$$

Thus, if $f(x) = (\theta x, \alpha_x)$, then

$$(a, x) \mapsto ((a, \theta x), \alpha_x).$$

□

Theorem 1.4.1 (Mealy morphisms as Kleisli algebras). *Let $\mathbb{A}$ and $\mathbb{B}$ be internal categories in $\mathcal{E}$. To give a Mealy morphism*

$$\mathbb{A} \rightsquigarrow \mathbb{B}$$

is equivalently to give an algebra for the lifted monad

$$\overline{R}_A$$

on the Kleisli category

$$\mathbf{Kl}(T_B).$$

Proof.

1. **Unpack the algebra structure.** An algebra for $\overline{R}_A$ on an object P of $\mathbf{Kl}(T_B)$ is a Kleisli arrow

$$\Phi : \overline{R}_A(P) \rightarrow P.$$

Since

$$\overline{R}_A(P) = R_A(P),$$

such a Kleisli arrow is represented in $\mathcal{C}_{A,B}$ by a map

$$\Phi : R_A(P) \rightarrow T_B(P).$$

Using

$$R_A(P) = P \circ A, \quad T_B(P) = B \circ P,$$

this is exactly a 2-cell

$$\Phi : P \circ A \Rightarrow B \circ P$$

in $\mathbf{Span}(\mathcal{E})$.

2. **Identify the component data.** On apexes, Φ is a map

$$A_1 \times_{t_A, A_0, p} P \longrightarrow P \times_{q, B_0, s_B} B_1.$$

Write

$$\Phi(a, x) = (\delta(a, x), \omega(a, x)).$$

The condition that this is a map of spans is exactly

$$p\delta = s_A\pi_{A_1}, \quad s_B\omega = q\delta, \quad t_B\omega = q\pi_P.$$

Thus the algebra structure map is precisely the transition-output datum of a Mealy morphism.

3. **Compute the algebra unit law.** The unit of the lifted monad $\bar{R}_A$ is the image in $\text{Kl}(T_B)$ of the unit

$$\eta_P^R : P \rightarrow R_A(P).$$

As a map in $\mathcal{C}_{A,B}$, this Kleisli arrow is represented by

$$P \xrightarrow{\eta_P^R} R_A(P) \xrightarrow{\eta_{R_A(P)}^T} T_B R_A(P).$$

The algebra unit equation in $\text{Kl}(T_B)$ says that the Kleisli composite of this arrow with Φ is the identity Kleisli arrow on P . Expanding Kleisli composition and using the unit laws of T_B , this equation reduces to

$$\Phi \circ \eta_P^R = \eta_P^T$$

as maps

$$P \rightarrow T_B(P).$$

Under the endpoint descriptions of the two monads,

$$\eta_P^R = P\eta_A, \quad \eta_P^T = \eta_B P.$$

Hence the algebra unit law is exactly

$$\Phi \circ (P\eta_A) = \eta_B P,$$

which is the abstract Mealy unit law (1.1).

4. **Compute the algebra multiplication law.** The multiplication of the lifted monad is the image in $\text{Kl}(T_B)$ of

$$\mu_P^R : R_A^2(P) \rightarrow R_A(P).$$

The algebra multiplication equation in the Kleisli category compares the two Kleisli arrows

$$R_A^2(P) \rightarrow P$$

obtained by first using μ_P^R , or by first applying the lifted endofunctor to Φ .

Expanding both Kleisli composites gives the equality

$$\Phi \circ \mu_P^R = \mu_P^T \circ T_B(\Phi) \circ \lambda_P \circ R_A(\Phi)$$

as maps

$$R_A^2(P) \rightarrow T_B(P).$$

Here

$$R_A(\Phi) : R_A^2(P) \rightarrow R_A T_B(P)$$

is followed by the distributive-law component

$$\lambda_P : R_A T_B(P) \rightarrow T_B R_A(P),$$

then by

$$T_B(\Phi) : T_B R_A(P) \rightarrow T_B^2(P),$$

and finally by

$$\mu_P^T : T_B^2(P) \rightarrow T_B(P).$$

Under the endpoint descriptions,

$$\mu_P^R = P\mu_A, \quad \mu_P^T = \mu_B P,$$

$$R_A(\Phi) = \Phi A, \quad T_B(\Phi) = B\Phi,$$

while λ_P is the canonical reassociation

$$(B \circ P) \circ A \xrightarrow{\cong} B \circ (P \circ A).$$

Therefore the algebra multiplication law is exactly

$$\Phi \circ (P\mu_A) = (\mu_B P) \circ (B\Phi) \circ \lambda_P \circ (\Phi A).$$

With the canonical associativity constraints suppressed, this is

$$\Phi \circ (P\mu_A) = (\mu_B P) \circ (B\Phi) \circ (\Phi A),$$

namely the abstract Mealy multiplication law (1.2).

5. **Conclude.** An $\overline{R}_A$ -algebra therefore consists exactly of a carrier span P , a structure cell

$$\Phi : P \circ A \Rightarrow B \circ P,$$

and the Mealy unit and multiplication equations. These are precisely the data and laws of a Mealy morphism

$$\mathbb{A} \rightsquigarrow \mathbb{B}.$$

The two constructions are inverse.

□

Let

$$(P, \Phi), (Q, \Psi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be Mealy morphisms. By Theorem 1.4.1, they are algebras for $\overline{R}_A$ in $\mathbf{Kl}(T_B)$. The simulation direction used in this chapter is the direction that recovers ordinary internal natural transformations in the stateless case.

The 2-cells used here are not the strict 2-cells between oriented monad morphisms in $\mathbf{Span}(\mathcal{E})$. They are Kleisli output-comparison cells for the monad $B \circ -$, with the direction reversed so that the stateless case gives ordinary internal natural transformations.

Definition 1.4.4 (Mealy simulation). *Let*

$$(P, \Phi), (Q, \Psi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

*be Mealy morphisms. A **Mealy simulation***

$$(P, \Phi) \Rightarrow (Q, \Psi)$$

is a morphism

$$(Q, \Psi) \rightarrow (P, \Phi)$$

of $\bar{R}_A$ -algebras in the Kleisli category

$$\mathbf{Kl}(T_B).$$

Equivalently, it is a map in the local span category

$$\Theta : Q \rightarrow T_B(P),$$

or, in bicategorical notation, a map of spans

$$\Theta : Q \Rightarrow B \circ P,$$

satisfying the algebra-morphism equation

$$\mu_P^T \circ T_B(\Phi) \circ \lambda_P \circ R_A(\Theta) = \mu_P^T \circ T_B(\Theta) \circ \Psi. \quad (1.23)$$

Both sides of this equation are maps

$$R_A(Q) \rightarrow T_B(P)$$

in $\mathcal{C}_{A,B}$.

In the span-composition notation of the chapter, equation (1.23) is

$$(\mu_B P) \circ (B\Phi) \circ \lambda_P \circ (\Theta A) = (\mu_B P) \circ (B\Theta) \circ \Psi. \quad (1.24)$$

Here

$$\lambda_P : (B \circ P) \circ A \xrightarrow{\cong} B \circ (P \circ A)$$

is the canonical distributive-law component induced by the associativity constraint of $\mathbf{Span}(\mathcal{E})$.

Writing

$$\Theta(y) = (\theta(y), \alpha(y)),$$

the underlying comparison data are maps

$$\theta : Q \rightarrow P, \quad \alpha : Q \rightarrow B_1,$$

satisfying

$$p_P \theta = p_Q, \quad (1.25)$$

$$s_B \alpha = q_P \theta, \quad (1.26)$$

$$t_B \alpha = q_Q. \quad (1.27)$$

Thus each state $y \in Q$ is compared with a state $\theta y \in P$ over the same input interface, together with an output-comparison arrow

$$\alpha_y : q_P(\theta y) \rightarrow q_Q(y)$$

in $\mathbb{B}$.

Proposition 1.4.4 (The simulation laws). *The abstract compatibility equation (1.24) is equivalent to the following component equations:*

$$\theta(\delta_Q(a, y)) = \delta_P(a, \theta y), \quad (1.28)$$

$$\alpha_y \circ \omega_P(a, \theta y) = \omega_Q(a, y) \circ \alpha_{\delta_Q(a, y)}. \quad (1.29)$$

Proof.

1. **Fix the relevant input data.** The relevant generalized elements are pairs

$$(a, y)$$

satisfying

$$t_A(a) = p_Q(y).$$

Thus $a : u \rightarrow v$ is an input arrow with $p_Q(y) = v$.

2. **Use preservation of the input interface.** Since

$$p_P \theta = p_Q,$$

the pair

$$(a, \theta y)$$

is admissible for P :

$$t_A(a) = p_Q(y) = p_P(\theta y).$$

Hence P can execute the same input arrow a at the compared state θy .

3. **Compute the left-hand side of the abstract equation.** The left-hand side of (1.24) first compares y with θy , and then executes P along a . The state part is

$$\delta_P(a, \theta y).$$

The output arrows form the composite

$$q_P(\delta_P(a, \theta y)) \xrightarrow{\omega_P(a, \theta y)} q_P(\theta y) \xrightarrow{\alpha_y} q_Q(y).$$

Therefore the output component is

$$\alpha_y \circ \omega_P(a, \theta y).$$

4. **Compute the right-hand side of the abstract equation.** The right-hand side first executes Q along a :

$$y \mapsto \delta_Q(a, y),$$

and emits

$$\omega_Q(a, y) : q_Q(\delta_Q(a, y)) \rightarrow q_Q(y).$$

It then compares the updated Q -state with P , giving

$$\theta(\delta_Q(a, y))$$

and

$$\alpha_{\delta_Q(a,y)} : q_P(\theta(\delta_Q(a,y))) \rightarrow q_Q(\delta_Q(a,y)).$$

The output arrows form the composite

$$q_P(\theta(\delta_Q(a,y))) \xrightarrow{\alpha_{\delta_Q(a,y)}} q_Q(\delta_Q(a,y)) \xrightarrow{\omega_Q(a,y)} q_Q(y).$$

Therefore the output component is

$$\omega_Q(a,y) \circ \alpha_{\delta_Q(a,y)}.$$

5. Compare the state components. Equality of the state components gives

$$\theta(\delta_Q(a,y)) = \delta_P(a, \theta y).$$

6. Compare the output components. Equality of the output components gives

$$\alpha_y \circ \omega_P(a, \theta y) = \omega_Q(a, y) \circ \alpha_{\delta_Q(a,y)}.$$

Hence the abstract algebra-morphism condition is equivalent to the two displayed simulation laws.

□

Remark 1.4.1 (Strict simulations). *Suppose*

$$\theta : Q \rightarrow P$$

satisfies

$$p_P \theta = p_Q, \quad q_P \theta = q_Q,$$

and

$$\theta(\delta_Q(a,y)) = \delta_P(a, \theta y), \quad \omega_Q(a,y) = \omega_P(a, \theta y).$$

Then θ defines a Mealy simulation in the sense of Definition 1.4.4 by taking

$$\alpha = e_B q_Q.$$

Conversely, if a Mealy simulation has comparison component

$$\alpha = e_B q_Q,$$

then the equation

$$s_B \alpha = q_P \theta$$

forces

$$q_P \theta = q_Q,$$

and the simulation laws reduce exactly to strict preservation of state transition and output:

$$\theta(\delta_Q(a,y)) = \delta_P(a, \theta y), \quad \omega_Q(a,y) = \omega_P(a, \theta y).$$

1.4.4 One-object simulations

Let

$$M, \quad N$$

be monoid objects in $\mathcal{E}$, regarded as one-object internal categories by the convention

$$b \circ a = ba$$

from Subsection 1.2.1. Let

$$(P, \omega_P), \quad (Q, \omega_Q)$$

be one-object Mealy morphisms

$$M \rightsquigarrow N.$$

Thus P and Q are internal right M -objects, and

$$\omega_P : M \times P \rightarrow N, \quad \omega_Q : M \times Q \rightarrow N$$

are output cocycles.

A Mealy simulation

$$P \Rightarrow Q$$

consists of maps

$$\theta : Q \rightarrow P, \quad \alpha : Q \rightarrow N.$$

The transition equation says that θ is M -equivariant:

$$\theta(y \cdot m) = \theta(y) \cdot m.$$

The output equation says

$$\alpha(y) \omega_P(m, \theta y) = \omega_Q(m, y) \alpha(y \cdot m).$$

The corresponding output-comparison diagram is

$$\begin{array}{ccc} & \xrightarrow{\omega_P(m, \theta y)} & \\ \alpha_{y \cdot m} \downarrow & & \downarrow \alpha_y \\ & \xrightarrow{\omega_Q(m, y)} & \end{array}$$

where the labels multiply according to the one-object convention. Thus one-object Mealy simulations compare output cocycles by a state-dependent cochain. When the output monoid is a group, this becomes the usual nonabelian coboundary relation. The strict output-preserving simulations are the special case

$$\alpha(y) = 1$$

for all y .

Proposition 1.4.5 (One-object Mealy simulations). *Let*

$$P, Q : M \rightsquigarrow N$$

be one-object Mealy morphisms internal to $\mathcal{E}$. A Mealy simulation

$$P \Rightarrow Q$$

is equivalently an M -equivariant map

$$\theta : Q \rightarrow P$$

together with a map

$$\alpha : Q \rightarrow N$$

satisfying

$$\alpha(y) \omega_P(m, \theta y) = \omega_Q(m, y) \alpha(y \cdot m)$$

as a morphism equation $M \times Q \rightarrow N$.

Proof.

1. **Specialize the comparison data.** In the one-object case all interface equations are vacuous. Thus a simulation consists of maps

$$\theta : Q \rightarrow P, \quad \alpha : Q \rightarrow N.$$

2. **Specialize the simulation laws.** Proposition 1.4.4 gives

$$\theta(y \cdot m) = \theta(y) \cdot m,$$

so θ is M -equivariant. Under the one-object convention $b \circ a = ba$, the output equation becomes

$$\alpha(y) \omega_P(m, \theta y) = \omega_Q(m, y) \alpha(y \cdot m).$$

3. **Reverse the specialization.** Conversely, an M -equivariant map θ and a map α satisfying the displayed equation determine exactly the component data and laws of a one-object Mealy simulation.

□

The stateless one-object case is obtained by taking $P = 1$. A Mealy morphism

$$M \rightsquigarrow N$$

then reduces to a monoid morphism

$$h : M \rightarrow N.$$

A simulation between stateless one-object Mealy morphisms

$$h, k : M \rightarrow N$$

is a global element

$$\alpha : 1 \rightarrow N$$

satisfying

$$\alpha h(m) = k(m) \alpha$$

as a morphism equation $M \rightarrow N$. This is the component form of

$$\alpha \circ h(m) = k(m) \circ \alpha$$

under the one-object convention $b \circ a = ba$. In **Set**, this is precisely a natural transformation between the corresponding one-object functors.

1.4.5 Stateless simulations

Let

$$F, G : \mathbb{A} \rightarrow \mathbb{B}$$

be internal functors, regarded as Mealy morphisms carried by the canonical map-representable spans

$$F_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right), \quad G_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{G_0} B_0 \right).$$

A Mealy simulation

$$F_{0!} \Rightarrow G_{0!}$$

is, by definition, a map of spans

$$G_{0!} \Rightarrow B \circ F_{0!}$$

satisfying the simulation law.

The composite

$$B \circ F_{0!} : A_0 \rightarrow B_0$$

has apex

$$A_0 \times_{F_0, B_0, s_B} B_1$$

and is the span

$$A_0 \xleftarrow{\pi_1} A_0 \times_{F_0, B_0, s_B} B_1 \xrightarrow{t_B \pi_2} B_0.$$

The defining pullback is

$$\begin{array}{ccc} A_0 \times_{F_0, B_0, s_B} B_1 & \xrightarrow{\pi_2} & B_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow s_B \\ A_0 & \xrightarrow{F_0} & B_0. \end{array}$$

A map of spans

$$\bar{\alpha} : G_{0!} \Rightarrow B \circ F_{0!}$$

is therefore equivalently a map

$$\bar{\alpha} : A_0 \rightarrow A_0 \times_{F_0, B_0, s_B} B_1$$

such that

$$\pi_1 \bar{\alpha} = 1_{A_0}, \quad t_B \pi_2 \bar{\alpha} = G_0.$$

The span-map diagram is

$$\begin{array}{ccccc} & & A_0 & & \\ & \swarrow 1_{A_0} & \downarrow \bar{\alpha} & \searrow G_0 & \\ A_0 & \xleftarrow{\pi_1} & A_0 \times_{B_0} B_1 & \xrightarrow{t_B \pi_2} & B_0. \end{array}$$

Writing

$$\alpha := \pi_2 \bar{\alpha},$$

the pullback condition gives

$$s_B \alpha = F_0,$$

and the right-leg condition gives

$$t_B \alpha = G_0.$$

Thus a stateless Mealy simulation has components

$$\alpha_x : F_0 x \rightarrow G_0 x$$

in $\mathbb{B}$, displayed as

$$F_0 x \xrightarrow{\alpha_x} G_0 x.$$

Definition 1.4.5 (Internal natural transformation). *Let*

$$F, G : \mathbb{A} \rightarrow \mathbb{B}$$

*be internal functors. An **internal natural transformation***

$$\alpha : F \Rightarrow G$$

is a map

$$\alpha : A_0 \rightarrow B_1$$

satisfying

$$s_B \alpha = F_0, \quad t_B \alpha = G_0,$$

and

$$m_B \langle F_1, \alpha t_A \rangle = m_B \langle \alpha s_A, G_1 \rangle.$$

Here

$$\langle F_1, \alpha t_A \rangle : A_1 \rightarrow B_1 \times_{t_B, B_0, s_B} B_1$$

is well-defined because

$$t_B F_1 = F_0 t_A = s_B \alpha t_A,$$

and

$$\langle \alpha s_A, G_1 \rangle : A_1 \rightarrow B_1 \times_{t_B, B_0, s_B} B_1$$

is well-defined because

$$t_B \alpha s_A = G_0 s_A = s_B G_1.$$

In generalized-element notation, for an arrow $a : u \rightarrow v$, this equation is

$$\alpha_v \circ F_1(a) = G_1(a) \circ \alpha_u.$$

The naturality square is

$$\begin{array}{ccc} F_0 u & \xrightarrow{F_1(a)} & F_0 v \\ \alpha_u \downarrow & & \downarrow \alpha_v \\ G_0 u & \xrightarrow{G_1(a)} & G_0 v. \end{array}$$

Theorem 1.4.2 (Stateless Mealy simulations are internal natural transformations). *Let*

$$F, G : \mathbb{A} \rightarrow \mathbb{B}$$

be internal functors, regarded as canonical map-representable Mealy morphisms. To give a Mealy simulation

$$F_{0!} \Rightarrow G_{0!}$$

is equivalently to give an internal natural transformation

$$F \Rightarrow G.$$

Proof.

1. **Identify the component map.** A Mealy simulation

$$F_{0!} \Rightarrow G_{0!}$$

is a map of spans

$$G_{0!} \Rightarrow B \circ F_{0!}.$$

By the calculation preceding the theorem, this is equivalently a map

$$\alpha : A_0 \rightarrow B_1$$

satisfying

$$s_B \alpha = F_0, \quad t_B \alpha = G_0.$$

These are exactly the source and target equations for the components of an internal natural transformation.

2. **Identify naturality.** For the canonical Mealy structures associated to F and G , an input arrow $a : u \rightarrow v$ emits $F_1(a) : F_0 u \rightarrow F_0 v$ and $G_1(a) : G_0 u \rightarrow G_0 v$, respectively. Proposition 1.4.4 therefore specializes to

$$\alpha_v \circ F_1(a) = G_1(a) \circ \alpha_u,$$

which is precisely internal naturality.

3. **Reverse the construction.** Conversely, an internal natural transformation $\alpha : F \Rightarrow G$ defines the span map

$$\bar{\alpha} = \langle 1_{A_0}, \alpha \rangle : A_0 \rightarrow A_0 \times_{F_0, B_0, s_B} B_1,$$

and the naturality equation is exactly the Mealy simulation law. Hence the two notions are equivalent.

□

1.5 The bicategory of Mealy morphisms

Mealy morphisms compose by composition of oriented monad morphisms in spans. We first give the explicit formula.

Let

$$(P, \delta_P, \omega_P) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

and

$$(Q, \delta_Q, \omega_Q) : \mathbb{B} \rightsquigarrow \mathbb{C}$$

be Mealy morphisms, with underlying spans

$$A_0 \xleftarrow{p_P} P \xrightarrow{q_P} B_0, \quad B_0 \xleftarrow{p_Q} Q \xrightarrow{q_Q} C_0.$$

Their composite carrier is the composite span

$$Q \circ P : A_0 \xleftarrow{p_P \pi_P} P \times_{q_P, B_0, p_Q} Q \xrightarrow{q_Q \pi_Q} C_0.$$

The defining pullback is

$$\begin{array}{ccc} P \times_{q_P, B_0, p_Q} Q & \xrightarrow{\pi_Q} & Q \\ \pi_P \downarrow & \lrcorner & \downarrow p_Q \\ P & \xrightarrow{q_P} & B_0. \end{array}$$

Operationally, a composite state is a pair

$$(x, y)$$

satisfying

$$q_P(x) = p_Q(y).$$

This is the pipeline diagram

$$A_0 \xleftarrow{p_P} P \xrightarrow{q_P} B_0 \xleftarrow{p_Q} Q \xrightarrow{q_Q} C_0.$$

Given an input arrow

$$a : u \rightarrow v$$

of $\mathbb{A}$ with

$$p_P(x) = v,$$

the first machine updates x to

$$\delta_P(a, x)$$

and emits a $\mathbb{B}$ -arrow

$$\omega_P(a, x) : q_P(\delta_P(a, x)) \rightarrow q_P(x).$$

Since

$$q_P(x) = p_Q(y),$$

this emitted arrow is admissible input for the second machine at y . The second machine updates y to

$$\delta_Q(\omega_P(a, x), y)$$

and emits

$$\omega_Q(\omega_P(a, x), y).$$

Thus the composite transition and output are

$$\delta_{QP}(a, (x, y)) = (\delta_P(a, x), \delta_Q(\omega_P(a, x), y)),$$

and

$$\omega_{QP}(a, (x, y)) = \omega_Q(\omega_P(a, x), y).$$

The execution picture is

$$x \xrightarrow{a/\omega_P(a, x)} \delta_P(a, x)$$

$$\xrightarrow[\omega_P(a, x)/\omega_Q(\omega_P(a, x), y)]{y} \delta_Q(\omega_P(a, x), y).$$

Proposition 1.5.1 (Pipeline composition). *The formulas above define a Mealy morphism*

$$Q \circ P : \mathbb{A} \rightsquigarrow \mathbb{C}.$$

Proof.

1. **Check that the transition lands in the pullback carrier.** We must show that the two updated states have matching middle interface. By the typing equations for Q ,

$$p_Q(\delta_Q(\omega_P(a, x), y)) = s_B(\omega_P(a, x)).$$

By the typing equations for P ,

$$s_B(\omega_P(a, x)) = q_P(\delta_P(a, x)).$$

Hence

$$p_Q(\delta_Q(\omega_P(a, x), y)) = q_P(\delta_P(a, x)),$$

so

$$(\delta_P(a, x), \delta_Q(\omega_P(a, x), y))$$

lies in

$$P \times_{B_0} Q.$$

2. **Check the output typing.** The output of the composite is

$$\omega_Q(\omega_P(a, x), y).$$

Since Q is a Mealy morphism,

$$s_C(\omega_Q(\omega_P(a, x), y)) = q_Q(\delta_Q(\omega_P(a, x), y)),$$

and

$$t_C(\omega_Q(\omega_P(a, x), y)) = q_Q(y).$$

Thus the output arrow has the required type

$$q_Q(\delta_Q(\omega_P(a, x), y)) \rightarrow q_Q(y).$$

3. **Verify the unit law.** Let $x \in P$, $y \in Q$, and suppose $q_P(x) = p_Q(y)$. For the identity input arrow $e_A(p_P x)$, the unit law for P gives

$$\delta_P(e_A(p_P x), x) = x, \quad \omega_P(e_A(p_P x), x) = e_B(q_P x).$$

The unit law for Q then gives

$$\delta_Q(e_B(q_P x), y) = y, \quad \omega_Q(e_B(q_P x), y) = e_C(q_Q y).$$

Hence

$$\delta_{QP}(e_A(p_P x), (x, y)) = (x, y),$$

and

$$\omega_{QP}(e_A(p_P x), (x, y)) = e_C(q_Q y).$$

4. **Prepare the multiplication calculation.** Let

$$a : u \rightarrow v, \quad b : v \rightarrow w, \quad p_P(x) = w.$$

Put

$$\beta_b := \omega_P(b, x), \quad \beta_a := \omega_P(a, \delta_P(b, x)).$$

Then

$$\beta_a : q_P(\delta_P(a, \delta_P(b, x))) \rightarrow q_P(\delta_P(b, x)),$$

and

$$\beta_b : q_P(\delta_P(b, x)) \rightarrow q_P(x).$$

By the Mealy multiplication law for P ,

$$\omega_P(b \circ a, x) = \beta_b \circ \beta_a.$$

5. **Verify the state part of multiplication.** The state part of executing the composite input $b \circ a$ is

$$\delta_{QP}(b \circ a, (x, y)) = (\delta_P(b \circ a, x), \delta_Q(\omega_P(b \circ a, x), y)).$$

By the Mealy multiplication law for P ,

$$\delta_P(b \circ a, x) = \delta_P(a, \delta_P(b, x)).$$

By the Mealy multiplication law for Q , applied to the composite

$$\beta_b \circ \beta_a,$$

we have

$$\delta_Q(\beta_b \circ \beta_a, y) = \delta_Q(\beta_a, \delta_Q(\beta_b, y)).$$

This is exactly the Q -state obtained by first executing b , then executing a , in the composite machine.

6. **Verify the output part of multiplication.** The output for the composite input is

$$\omega_{QP}(b \circ a, (x, y)) = \omega_Q(\omega_P(b \circ a, x), y).$$

Since

$$\omega_P(b \circ a, x) = \beta_b \circ \beta_a,$$

the Mealy multiplication law for Q gives

$$\omega_Q(\beta_b \circ \beta_a, y) = \omega_Q(\beta_b, y) \circ \omega_Q(\beta_a, \delta_Q(\beta_b, y)).$$

This is precisely the composite of the outputs produced by first executing b , then executing a , in the pipeline. Hence the multiplication law holds. □

The identity Mealy morphism on $\mathbb{A}$ has underlying identity span

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{1_{A_0}} A_0.$$

At a state $v \in A_0$, an arrow

$$a : u \rightarrow v$$

is processed by

$$v \xrightarrow{a/a} u.$$

Thus the identity Mealy morphism has

$$\delta(a, v) = s_A(a), \quad \omega(a, v) = a.$$

The corresponding diagram is

$$u \xrightarrow{a} v \quad \rightsquigarrow \quad v \xrightarrow{a/a} u.$$

Recall that the displayed simulation direction is opposite to the underlying Kleisli algebra-morphism direction: a simulation

$$P \Rightarrow Q$$

has underlying state map

$$Q \rightarrow P.$$

Proposition 1.5.2 (Vertical composition of simulations). *Mealy simulations compose vertically. If*

$$(\theta, \alpha) : P \Rightarrow Q$$

and

$$(\kappa, \beta) : Q \Rightarrow R$$

are simulations between Mealy morphisms

$$P, Q, R : \mathbb{A} \rightsquigarrow \mathbb{B},$$

where

$$\theta : Q \rightarrow P, \quad \kappa : R \rightarrow Q,$$

then their vertical composite

$$(\kappa, \beta) \circ (\theta, \alpha) : P \Rightarrow R$$

has state map

$$\theta\kappa : R \rightarrow P$$

and comparison component

$$\gamma_z = \beta_z \circ \alpha_{\kappa z}.$$

Proof.

1. **Use composition of Kleisli algebra morphisms.** A simulation

$$(\theta, \alpha) : P \Rightarrow Q$$

is, by definition, a morphism of $\overline{R}_A$ -algebras

$$(Q, \Psi) \rightarrow (P, \Phi)$$

in $\text{Kl}(T_B)$. Similarly,

$$(\kappa, \beta) : Q \Rightarrow R$$

is a morphism of algebras

$$(R, \Xi) \rightarrow (Q, \Psi).$$

Their composite in the algebra category is therefore a morphism

$$(R, \Xi) \rightarrow (P, \Phi),$$

which, under the displayed simulation convention, is a simulation

$$P \Rightarrow R.$$

2. **Unpack the Kleisli composite.** The underlying Kleisli arrows are represented by

$$R \xrightarrow{(\kappa, \beta)} T_B(Q)$$

and

$$Q \xrightarrow{(\theta, \alpha)} T_B(P).$$

Their Kleisli composite has state map

$$\theta\kappa : R \rightarrow P.$$

At a state $z \in R$, its comparison component is the composite

$$q_P(\theta\kappa z) \xrightarrow{\alpha_{\kappa z}} q_Q(\kappa z) \xrightarrow{\beta_z} q_R(z),$$

namely

$$\gamma_z = \beta_z \circ \alpha_{\kappa z}.$$

3. **Conclude.** Morphisms of $\overline{R}_A$ -algebras are closed under composition. Hence the pair

$$(\theta\kappa, \gamma)$$

is again a Mealy simulation, and it is the vertical composite

$$(\kappa, \beta) \circ (\theta, \alpha).$$

□

Proposition 1.5.3 (Horizontal composition of simulations). *Let*

$$P, P' : \mathbb{A} \rightsquigarrow \mathbb{B}, \quad Q, Q' : \mathbb{B} \rightsquigarrow \mathbb{C}$$

be Mealy morphisms. Suppose

$$(\theta, \alpha) : P \Rightarrow P', \quad (\kappa, \beta) : Q \Rightarrow Q'$$

are Mealy simulations, so that

$$\theta : P' \rightarrow P, \quad \kappa : Q' \rightarrow Q.$$

Their horizontal composite

$$(\kappa, \beta) * (\theta, \alpha) : Q \circ P \Rightarrow Q' \circ P'$$

has state map

$$P' \times_{B_0} Q' \longrightarrow P \times_{B_0} Q$$

given by

$$(x', y') \longmapsto (\theta x', \delta_Q(\alpha_{x'}, \kappa y')),$$

and comparison component

$$\beta_{y'} \circ \omega_Q(\alpha_{x'}, \kappa y').$$

Proof.

1. **Check that the state map is defined.** Let $(x', y') \in P' \times_{B_0} Q'$, so

$$q_{P'}(x') = p_{Q'}(y').$$

Since $p_Q \kappa = p_{Q'}$, the comparison arrow

$$\alpha_{x'} : q_P(\theta x') \rightarrow q_{P'}(x')$$

is admissible input for Q at $\kappa y'$. Hence $\delta_Q(\alpha_{x'}, \kappa y')$ is defined, and its input interface is

$$p_Q \delta_Q(\alpha_{x'}, \kappa y') = s_B \alpha_{x'} = q_P \theta x'.$$

Thus

$$(\theta x', \delta_Q(\alpha_{x'}, \kappa y'))$$

lies in $P \times_{B_0} Q$.

2. **Check the comparison arrow.** The output of Q while processing $\alpha_{x'}$ at $\kappa y'$ is

$$\omega_Q(\alpha_{x'}, \kappa y') : q_Q \delta_Q(\alpha_{x'}, \kappa y') \rightarrow q_Q \kappa y'.$$

Composing with

$$\beta_{y'} : q_Q \kappa y' \rightarrow q_{Q'} y'$$

gives the required comparison arrow

$$\beta_{y'} \circ \omega_Q(\alpha_{x'}, \kappa y') : q_Q \delta_Q(\alpha_{x'}, \kappa y') \rightarrow q_{Q'} y'.$$

3. **Verify the simulation law.** Let $a : u \rightarrow v$ be an input arrow for $\mathbb{A}$, and let $p_{P'}(x') = v$. Put

$$x'_1 := \delta_{P'}(a, x'), \quad \varphi := \omega_{P'}(a, x'), \quad y'_1 := \delta_{Q'}(\varphi, y').$$

The state equation for (θ, α) gives

$$\theta x'_1 = \delta_P(a, \theta x').$$

The output equation for (θ, α) gives

$$\alpha_{x'} \circ \omega_P(a, \theta x') = \varphi \circ \alpha_{x'_1}.$$

Applying the Mealy multiplication law for Q to this equality and then the state equation for (κ, β) gives

$$\delta_Q(\omega_P(a, \theta x'), \delta_Q(\alpha_{x'}, \kappa y')) = \delta_Q(\alpha_{x'_1}, \kappa y'_1).$$

Hence the state part of the simulation law holds.

For the output part, set

$$\zeta_{(x', y')} := \beta_{y'} \circ \omega_Q(\alpha_{x'}, \kappa y').$$

The required equation is obtained by pasting the output square for (θ, α) , the Mealy multiplication law for Q , and the output square for (κ, β) . Explicitly, the left-hand output composite reduces to

$$\begin{aligned} & \beta_{y'} \circ \omega_Q(\alpha_{x'} \circ \omega_P(a, \theta x'), \kappa y') \\ &= \beta_{y'} \circ \omega_Q(\varphi \circ \alpha_{x'_1}, \kappa y') \\ &= \omega_{Q'}(\varphi, y') \circ \beta_{y'_1} \circ \omega_Q(\alpha_{x'_1}, \kappa y'_1), \end{aligned}$$

which is

$$\omega_{Q' \circ P'}(a, (x', y')) \circ \zeta_{(x'_1, y'_1)}.$$

This is exactly the simulation output equation. □

Proposition 1.5.4 (Functoriality of horizontal composition of simulations). *Let*

$$P, P', P'' : \mathbb{A} \rightsquigarrow \mathbb{B}, \quad Q, Q', Q'' : \mathbb{B} \rightsquigarrow \mathbb{C}$$

be Mealy morphisms. Suppose we are given vertically composable Mealy simulations

$$(\theta, \alpha) : P \Rightarrow P', \quad (\theta', \alpha') : P' \Rightarrow P'',$$

and

$$(\kappa, \beta) : Q \Rightarrow Q', \quad (\kappa', \beta') : Q' \Rightarrow Q''.$$

Thus

$$\theta : P' \rightarrow P, \quad \theta' : P'' \rightarrow P', \quad \kappa : Q' \rightarrow Q, \quad \kappa' : Q'' \rightarrow Q'$$

are the underlying state maps. Horizontal composition of simulations preserves identities and vertical composites. Hence, for every triple of internal categories

$$\mathbb{A}, \mathbb{B}, \mathbb{C},$$

pipeline composition defines a functor

$$\mathbf{Mly}(\mathcal{E})(\mathbb{B}, \mathbb{C}) \times \mathbf{Mly}(\mathcal{E})(\mathbb{A}, \mathbb{B}) \longrightarrow \mathbf{Mly}(\mathcal{E})(\mathbb{A}, \mathbb{C}).$$

Proof.

1. **Identities.** If both simulations are identity simulations, then

$$\theta = 1_P, \quad \alpha_x = e_B(q_P x), \quad \kappa = 1_Q, \quad \beta_y = e_C(q_Q y).$$

The formula of Proposition 1.5.3 sends $(x, y) \in P \times_{B_0} Q$ to

$$(x, \delta_Q(e_B(q_P x), y)) = (x, y)$$

by the Mealy unit law for Q . Its comparison component is

$$e_C(q_Q y) \circ \omega_Q(e_B(q_P x), y) = e_C(q_Q y).$$

Hence horizontal composition preserves identity simulations.

2. **Vertical composites and their horizontal composite.** The vertical composites

$$(\theta', \alpha') \circ (\theta, \alpha) : P \Rightarrow P''$$

and

$$(\kappa', \beta') \circ (\kappa, \beta) : Q \Rightarrow Q''$$

have state maps

$$\theta\theta' : P'' \rightarrow P, \quad \kappa\kappa' : Q'' \rightarrow Q,$$

and comparison components

$$\begin{aligned} \gamma_{x''} &:= \alpha'_{x''} \circ \alpha_{\theta' x''}, \\ \eta_{y''} &:= \beta'_{y''} \circ \beta_{\kappa' y''}. \end{aligned}$$

Horizontally composing these vertical composites gives the state map

$$(x'', y'') \longmapsto (\theta\theta' x'', \delta_Q(\gamma_{x''}, \kappa\kappa' y''))$$

and comparison component

$$\eta_{y''} \circ \omega_Q(\gamma_{x''}, \kappa\kappa' y'').$$

3. **Horizontally compose first.** Horizontally composing

$$(\theta, \alpha) : P \Rightarrow P'$$

with

$$(\kappa, \beta) : Q \Rightarrow Q'$$

gives a simulation

$$Q \circ P \Rightarrow Q' \circ P'$$

whose state map is

$$(x', y') \longmapsto (\theta x', \delta_Q(\alpha_{x'}, \kappa y'))$$

and whose comparison component is

$$\beta_{y'} \circ \omega_Q(\alpha_{x'}, \kappa y').$$

Similarly, horizontally composing

$$(\theta', \alpha') : P' \Rightarrow P''$$

with

$$(\kappa', \beta') : Q' \Rightarrow Q''$$

gives a simulation

$$Q' \circ P' \Rightarrow Q'' \circ P''$$

whose state map is

$$(x'', y'') \longmapsto (\theta' x'', \delta_{Q'}(\alpha'_{x''}, \kappa' y''))$$

and whose comparison component is

$$\beta'_{y''} \circ \omega_{Q'}(\alpha'_{x''}, \kappa' y'').$$

Put

$$\bar{y} := \delta_{Q'}(\alpha'_{x''}, \kappa' y'').$$

4. **Compare the state maps.** Vertically composing the two horizontally composed simulations gives the state

$$(\theta\theta' x'', \delta_Q(\alpha_{\theta' x''}, \kappa \bar{y})).$$

The transition equation for the simulation $(\kappa, \beta) : Q \Rightarrow Q'$, applied to the input arrow $\alpha'_{x''}$, gives

$$\kappa \bar{y} = \delta_Q(\alpha'_{x''}, \kappa \kappa' y'').$$

Therefore the second state component is

$$\delta_Q(\alpha_{\theta' x''}, \delta_Q(\alpha'_{x''}, \kappa \kappa' y'')).$$

By the Mealy multiplication law for Q ,

$$\begin{aligned} & \delta_Q(\alpha_{\theta' x''}, \delta_Q(\alpha'_{x''}, \kappa \kappa' y'')) \\ &= \delta_Q(\alpha'_{x''} \circ \alpha_{\theta' x''}, \kappa \kappa' y'') \\ &= \delta_Q(\gamma_{x''}, \kappa \kappa' y''). \end{aligned}$$

Thus the two constructions have the same state map.

5. **Compare the output components.** The output-comparison component obtained by horizontally composing first and then vertically composing is

$$\beta'_{y''} \circ \omega_{Q'}(\alpha'_{x''}, \kappa' y'') \circ \beta_{\bar{y}} \circ \omega_Q(\alpha_{\theta' x''}, \kappa \bar{y}).$$

The output equation for the simulation $(\kappa, \beta) : Q \Rightarrow Q'$, applied to $\alpha'_{x''}$ at the state $\kappa' y''$, is

$$\beta_{\kappa' y''} \circ \omega_Q(\alpha'_{x''}, \kappa \kappa' y'') = \omega_{Q'}(\alpha'_{x''}, \kappa' y'') \circ \beta_{\bar{y}}.$$

Using this equation and the previously established identity

$$\kappa \bar{y} = \delta_Q(\alpha'_{x''}, \kappa \kappa' y''),$$

the output component becomes

$$\begin{aligned} & \beta'_{y''} \circ \beta_{\kappa' y''} \circ \omega_Q(\alpha'_{x''}, \kappa \kappa' y'') \circ \omega_Q(\alpha_{\theta' x''}, \delta_Q(\alpha'_{x''}, \kappa \kappa' y'')) \\ &= \eta_{y''} \circ \omega_Q(\alpha'_{x''} \circ \alpha_{\theta' x''}, \kappa \kappa' y'') \\ &= \eta_{y''} \circ \omega_Q(\gamma_{x''}, \kappa \kappa' y''), \end{aligned}$$

where the first equality uses the Mealy output multiplication law for Q .

This is exactly the comparison component obtained by first taking the vertical composites and then horizontally composing them.

6. **Conclusion.** Horizontal composition preserves identity simulations and vertical composites. Hence, for every triple of internal categories

$$\mathbb{A}, \mathbb{B}, \mathbb{C},$$

pipeline composition defines a functor

$$\mathbf{Mly}(\mathcal{E})(\mathbb{B}, \mathbb{C}) \times \mathbf{Mly}(\mathcal{E})(\mathbb{A}, \mathbb{B}) \longrightarrow \mathbf{Mly}(\mathcal{E})(\mathbb{A}, \mathbb{C}).$$

□

Lemma 1.5.1 (Span coherence induces strict Mealy coherence). *The canonical associativity and unit isomorphisms for pullback composition determine invertible strict Mealy simulations.*

More precisely, because the displayed direction of a Mealy simulation is opposite to the direction of its underlying state map, the state map of each Mealy coherence simulation is the inverse of the apex map of the corresponding coherence 2-cell in $\mathbf{Span}(\mathcal{E})$.

Proof.

1. **Associators.** Let

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}, \quad Q : \mathbb{B} \rightsquigarrow \mathbb{C}, \quad R : \mathbb{C} \rightsquigarrow \mathbb{D}$$

be composable Mealy morphisms.

The carrier of

$$(R \circ Q) \circ P$$

is

$$P \times_{B_0} (Q \times_{C_0} R),$$

whereas the carrier of

$$R \circ (Q \circ P)$$

is

$$(P \times_{B_0} Q) \times_{C_0} R.$$

Both are iterated-pullback presentations of the object of triples

$$(x, y, z)$$

satisfying

$$q_P(x) = p_Q(y), \quad q_Q(y) = p_R(z).$$

The ordinary span associator

$$(R \circ Q) \circ P \Longrightarrow R \circ (Q \circ P)$$

has apex map

$$P \times_{B_0} (Q \times_{C_0} R) \longrightarrow (P \times_{B_0} Q) \times_{C_0} R,$$

characterized in generalized-element notation by

$$(x, (y, z)) \longmapsto ((x, y), z).$$

We define the Mealy associator

$$a_{R,Q,P} : (R \circ Q) \circ P \Rightarrow R \circ (Q \circ P)$$

using the inverse pullback isomorphism as its state map:

$$\begin{aligned} \hat{a}_{P,Q,R} : (P \times_{B_0} Q) \times_{C_0} R &\xrightarrow{\cong} P \times_{B_0} (Q \times_{C_0} R), \\ \hat{a}_{P,Q,R}((x, y), z) &= (x, (y, z)). \end{aligned}$$

This direction is forced by the convention that a simulation from a source Mealy morphism to a target Mealy morphism has a state map from the target carrier to the source carrier.

Under the canonical identification of both carriers with the same triple-pullback object, the two parenthesized pipelines execute P , then Q , then R . They therefore have the same transition map and emit the same output arrow. Consequently $\hat{a}_{P,Q,R}$, equipped with the identity output comparison, is a strict Mealy simulation.

The ordinary span-associator apex map gives the state map of the inverse Mealy simulation

$$R \circ (Q \circ P) \Rightarrow (R \circ Q) \circ P.$$

Hence the Mealy associator is invertible.

2. Left unitors. Let

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

The carrier of

$$1_{\mathbb{B}} \circ P$$

is

$$P \times_{q_P, B_0, 1_{B_0}} B_0.$$

The ordinary span left unitor

$$1_{B_0} \circ P \Longrightarrow P$$

has apex map the canonical projection

$$P \times_{q_P, B_0, 1_{B_0}} B_0 \rightarrow P.$$

The Mealy left unitor is displayed in the same direction,

$$\ell_P : 1_{\mathbb{B}} \circ P \Rightarrow P,$$

but its state map must run from the target carrier to the source carrier. We therefore use the inverse pullback isomorphism

$$\widehat{\ell}_P : P \longrightarrow P \times_{q_P, B_0, 1_{B_0}} B_0, \quad x \longmapsto (x, q_P x).$$

After P executes an input arrow a at x , it updates to $\delta_P(a, x)$ and emits

$$\omega_P(a, x) : q_P(\delta_P(a, x)) \rightarrow q_P(x).$$

The identity Mealy morphism on $\mathbb{B}$ processes this emitted arrow at the object-state $q_P(x)$, updates that state to $q_P(\delta_P(a, x))$, and emits the same arrow. Thus the updated composite state is

$$(\delta_P(a, x), q_P(\delta_P(a, x))) = \widehat{\ell}_P(\delta_P(a, x)),$$

and the composite output is $\omega_P(a, x)$.

Hence $\widehat{\ell}_P$, equipped with the identity output comparison, is a strict Mealy simulation. The canonical projection

$$P \times_{q_P, B_0, 1_{B_0}} B_0 \rightarrow P$$

gives the state map of its inverse strict simulation

$$P \Rightarrow 1_{\mathbb{B}} \circ P.$$

3. Right unitors. The carrier of

$$P \circ 1_{\mathbb{A}}$$

is

$$A_0 \times_{1_{A_0}, A_0, p_P} P.$$

The ordinary span right unitor

$$P \circ 1_{A_0} \Longrightarrow P$$

has apex map the canonical projection

$$A_0 \times_{1_{A_0}, A_0, p_P} P \rightarrow P.$$

The Mealy right unitor

$$r_P : P \circ 1_{\mathbb{A}} \Rightarrow P$$

therefore uses the inverse pullback isomorphism as its state map:

$$\widehat{r}_P : P \longrightarrow A_0 \times_{1_{A_0}, A_0, p_P} P, \quad x \longmapsto (p_P x, x).$$

Let

$$a : u \rightarrow v$$

be an input arrow with $p_P(x) = v$. The identity Mealy morphism on $\mathbb{A}$ updates its object-state from v to u and emits a . The machine P then processes a at x , updating to $\delta_P(a, x)$ and emitting $\omega_P(a, x)$. Since

$$p_P(\delta_P(a, x)) = u,$$

the updated composite state is

$$(u, \delta_P(a, x)) = \widehat{r}_P(\delta_P(a, x)).$$

Thus $\widehat{r}_P$ strictly preserves both transition and output.

Hence $\widehat{r}_P$, equipped with the identity output comparison, is a strict Mealy simulation. The canonical projection

$$A_0 \times_{1_{A_0}, A_0, p_P} P \rightarrow P$$

gives the state map of its inverse strict simulation

$$P \Rightarrow P \circ 1_{\mathbb{A}}.$$

4. **Invertibility.** In each case, the state maps of the two displayed simulations are inverse pullback isomorphisms, and every output-comparison component is an identity arrow. Hence the associators and unitors are invertible strict Mealy simulations.

□

Lemma 1.5.2 (Naturality of pipeline coherence). *The associativity and unit simulations of Lemma 1.5.1 are natural with respect to Mealy simulations.*

Proof.

1. **Set up the associator naturality calculation.** Let

$$(\theta, \alpha) : P \Rightarrow P', \quad (\kappa, \beta) : Q \Rightarrow Q', \quad (\ell, \chi) : R \Rightarrow R'$$

be simulations between composable Mealy morphisms

$$P, P' : \mathbb{A} \rightsquigarrow \mathbb{B},$$

$$Q, Q' : \mathbb{B} \rightsquigarrow \mathbb{C},$$

and

$$R, R' : \mathbb{C} \rightsquigarrow \mathbb{D}.$$

Thus the state maps have directions

$$\theta : P' \rightarrow P, \quad \kappa : Q' \rightarrow Q, \quad \ell : R' \rightarrow R.$$

Let

$$(x', y', z')$$

be a state of the target triple pipeline. Put

$$\gamma_{x', y'} := \omega_Q(\alpha_{x'}, \kappa y')$$

and

$$\zeta_{x', y'} := \beta_{y'} \circ \gamma_{x', y'}.$$

Here $\gamma_{x', y'}$ and $\beta_{y'}$ are composable arrows in $\mathbb{C}$.

2. **Compare the state components for the associator.** After identifying the two parenthesized carrier objects with the same triple pullback, one route around the associator naturality square gives the state

$$(\theta x', \delta_Q(\alpha_{x'}, \kappa y'), \delta_R(\zeta_{x', y'}, \ell z')).$$

The other route gives

$$(\theta x', \delta_Q(\alpha_{x'}, \kappa y'), \delta_R(\gamma_{x', y'}, \delta_R(\beta_{y'}, \ell z'))).$$

By the state multiplication law for R ,

$$\delta_R(\beta_{y'} \circ \gamma_{x', y'}, \ell z') = \delta_R(\gamma_{x', y'}, \delta_R(\beta_{y'}, \ell z')).$$

Since

$$\zeta_{x', y'} = \beta_{y'} \circ \gamma_{x', y'},$$

the two state components agree.

3. **Compare the output components for the associator.** The first route has output-comparison component

$$\chi_{z'} \circ \omega_R(\zeta_{x', y'}, \ell z').$$

The second route has output-comparison component

$$\chi_{z'} \circ \omega_R(\beta_{y'}, \ell z') \circ \omega_R(\gamma_{x', y'}, \delta_R(\beta_{y'}, \ell z')).$$

By the output multiplication law for R ,

$$\omega_R(\beta_{y'} \circ \gamma_{x', y'}, \ell z') = \omega_R(\beta_{y'}, \ell z') \circ \omega_R(\gamma_{x', y'}, \delta_R(\beta_{y'}, \ell z')).$$

Therefore the two output-comparison components agree. This proves naturality of the associator.

4. **Naturality of the left unitor.** Let

$$(\theta, \alpha) : P \Rightarrow P'$$

be a Mealy simulation between morphisms

$$P, P' : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

Thus

$$\theta : P' \rightarrow P$$

and

$$\alpha_{x'} : q_P(\theta x') \rightarrow q_{P'}(x').$$

Under the canonical pullback presentation, a state of

$$1_{\mathbb{B}} \circ P'$$

has the form

$$(x', q_{P'} x').$$

Horizontally composing (θ, α) with the identity simulation on $1_{\mathbb{B}}$ sends this state to

$$(\theta x', \delta_{1_{\mathbb{B}}}(\alpha_{x'}, q_{P'} x')).$$

The identity Mealy morphism on $\mathbb{B}$ satisfies

$$\delta_{1_{\mathbb{B}}}(\beta, t_B \beta) = s_B \beta, \quad \omega_{1_{\mathbb{B}}}(\beta, t_B \beta) = \beta.$$

Since

$$s_B \alpha_{x'} = q_P(\theta x'), \quad t_B \alpha_{x'} = q_{P'}(x'),$$

the resulting state is

$$(\theta x', q_P \theta x').$$

Its output-comparison component is

$$e_B(q_P x') \circ \omega_{1_{\mathbb{B}}}(\alpha_{x'}, q_{P'} x') = \alpha_{x'}.$$

The other route around the left-unitor naturality square first applies the simulation (θ, α) and then the left-unitor simulation for P . It has the same state map

$$x' \mapsto (\theta x', q_P \theta x')$$

and the same output-comparison component $\alpha_{x'}$. Hence the left unitor is natural.

5. Naturality of the right unitor. Under the canonical pullback presentation, a state of

$$P' \circ 1_{\mathbb{A}}$$

has the form

$$(p_{P'} x', x').$$

Horizontally composing the identity simulation on $1_{\mathbb{A}}$ with (θ, α) sends this state to

$$(p_{P'} x', \delta_P(e_A(p_{P'} x'), \theta x')).$$

Since

$$p_P \theta = p_{P'},$$

the Mealy unit law for P gives

$$\delta_P(e_A(p_{P'} x'), \theta x') = \theta x'.$$

Thus the resulting state is

$$(p_P \theta x', \theta x').$$

Its output-comparison component is

$$\alpha_{x'} \circ \omega_P(e_A(p_{P'} x'), \theta x') = \alpha_{x'}.$$

The other route around the right-unitor naturality square first applies (θ, α) and then the right-unitor simulation for P . It has the same state map

$$x' \mapsto (p_P \theta x', \theta x')$$

and the same output-comparison component $\alpha_{x'}$. Hence the right unitor is natural.

□

Theorem 1.5.1 (The Mealy bicategory). *If $\mathcal{E}$ has pullbacks, then there is a bicategory*

$$\mathbf{Mly}(\mathcal{E})$$

whose objects are internal categories in $\mathcal{E}$, whose 1-cells are Mealy morphisms, and whose 2-cells are Mealy simulations.

Proof.

1. **Objects and hom-categories.** The objects are internal categories in $\mathcal{E}$, equivalently monads in $\mathbf{Span}(\mathcal{E})$.

For internal categories $\mathbb{A}$ and $\mathbb{B}$, the objects of

$$\mathbf{Mly}(\mathcal{E})(\mathbb{A}, \mathbb{B})$$

are Mealy morphisms

$$\mathbb{A} \rightsquigarrow \mathbb{B}.$$

By Theorem 1.4.1, these are algebras for the lifted monad

$$\overline{R}_A$$

in $\mathbf{Kl}(T_B)$.

A Mealy simulation

$$(P, \Phi) \Rightarrow (Q, \Psi)$$

is, by definition, an algebra morphism

$$(Q, \Psi) \rightarrow (P, \Phi).$$

Hence the hom-category is the opposite of the corresponding algebra category. Its identity morphisms and vertical composition are therefore inherited from the algebra category, with vertical composition given explicitly by Proposition 1.5.2.

2. **Horizontal composition.** Horizontal composition of 1-cells is pipeline composition. For

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}, \quad Q : \mathbb{B} \rightsquigarrow \mathbb{C},$$

the composite carrier is

$$A_0 \xleftarrow{p_P \pi_P} P \times_{q_P, B_0, p_Q} Q \xrightarrow{q_Q \pi_Q} C_0,$$

with transition and output maps

$$\delta_{QP}(a, (x, y)) = (\delta_P(a, x), \delta_Q(\omega_P(a, x), y))$$

and

$$\omega_{QP}(a, (x, y)) = \omega_Q(\omega_P(a, x), y).$$

Proposition 1.5.1 proves that these data satisfy the Mealy laws.

Horizontal composition of 2-cells is the construction of Proposition 1.5.3. Proposition 1.5.4 proves that this operation preserves identity simulations and vertical composites. Thus pipeline composition defines a functor on each pair of hom-categories.

3. **Identity 1-cells.** The identity 1-cell on an internal category $\mathbb{A}$ has carrier

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{1_{A_0}} A_0.$$

Its transition and output maps are

$$\delta(a, v) = s_A(a), \quad \omega(a, v) = a,$$

where necessarily $v = t_A(a)$.

For the identity input at v , one has

$$\delta(e_A(v), v) = s_A e_A(v) = v$$

and

$$\omega(e_A(v), v) = e_A(v),$$

so the Mealy unit law holds.

Now let

$$a : u \rightarrow v, \quad b : v \rightarrow w$$

be composable arrows. The transition multiplication law is

$$\begin{aligned} \delta(b \circ a, w) &= s_A(b \circ a) \\ &= u \\ &= s_A(a) \\ &= \delta(a, s_A(b)) \\ &= \delta(a, \delta(b, w)). \end{aligned}$$

The output multiplication law is

$$\begin{aligned} \omega(b \circ a, w) &= b \circ a \\ &= \omega(b, w) \circ \omega(a, \delta(b, w)). \end{aligned}$$

Thus these formulas define the identity Mealy morphism on $\mathbb{A}$.

4. **Associators and unitors.** The associators and unitors are obtained from the canonical associativity and unit isomorphisms for pullback composition. Because the displayed direction of a Mealy simulation is opposite to the direction of its underlying state map, their state maps are the inverse apex maps of the corresponding span-coherence 2-cells.

By Lemma 1.5.1, these maps, equipped with identity output comparisons, are invertible strict Mealy simulations. By Lemma 1.5.2, they are natural with respect to arbitrary Mealy simulations.

5. **Interchange.** The interchange law is precisely the functoriality of horizontal composition established in Proposition 1.5.4.

6. **Pentagon and triangle coherence.** The state maps of the associators and unitors are canonical isomorphisms between iterated pullback presentations. Their horizontal whiskerings by identity simulations again have the corresponding canonical pullback maps as their state components.

Indeed, the comparison component of an identity simulation is an identity arrow. In the horizontal-composition formula of Proposition 1.5.3, the additional transition and output terms introduced by whiskering therefore reduce by the Mealy unit laws:

$$\delta(e(-), x) = x, \quad \omega(e(-), x) = e(-).$$

Consequently, whiskering an associator or unitor coherence simulation by an identity simulation again produces a strict simulation with identity output-comparison component.

Every route in the pentagon therefore determines a canonical map from the carrier of the final parenthesization to the carrier of the initial parenthesization. The maps obtained along the two routes have the same projections to the four original carrier objects, and hence are equal by the universal properties of the relevant iterated pullbacks.

The same argument applies to the two routes in the triangle: both state maps have the same projections to the original carrier objects, and hence coincide.

Every output-comparison component occurring in the pentagon and triangle is an identity arrow. Their vertical composites are therefore identity arrows as well. Hence the pentagon and triangle identities hold as equations of Mealy simulations.

7. **Conclusion.** The preceding data provide hom-categories, functorial horizontal composition, identity 1-cells, natural invertible associators and unitors, and the required interchange and coherence equations. They therefore define a bicategory

$$\mathbf{Mly}(\mathcal{E})$$

whose objects are internal categories, whose 1-cells are Mealy morphisms, and whose 2-cells are Mealy simulations. □

Corollary 1.5.1 (The functorial sub-bicategory). *Let*

$$\mathbf{Mly}_{\text{map}}(\mathcal{E})$$

denote the locally full sub-bicategory of

$$\mathbf{Mly}(\mathcal{E})$$

having all internal categories as objects and the map-representable Mealy morphisms as 1-cells. Then

$$\mathbf{Mly}_{\text{map}}(\mathcal{E})$$

is biequivalent to the ordinary 2-category

$$\mathbf{Cat}(\mathcal{E})$$

of internal categories, internal functors, and internal natural transformations.

More precisely, after restricting to the canonical carriers

$$F_{0!} = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right)$$

and transporting pipeline composition along the canonical pullback isomorphisms, one obtains the usual strict 2-categorical composition of internal functors and internal natural transformations.

Proof.

1. **Map-representable carriers form a sub-bicategory.** Identity carriers are map-representable, since the identity carrier on $\mathbb{A}$ is

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{1_{A_0}} A_0.$$

It remains to check closure under pipeline composition. Let

$$A_0 \xleftarrow{p_P} P \xrightarrow{q_P} B_0, \quad B_0 \xleftarrow{p_Q} Q \xrightarrow{q_Q} C_0$$

be map-representable carriers. Thus p_P and p_Q are isomorphisms. Their composite carrier has apex

$$P \times_{q_P, B_0, p_Q} Q$$

and left leg

$$p_P \pi_P : P \times_{q_P, B_0, p_Q} Q \rightarrow A_0.$$

In the defining pullback square

$$\begin{array}{ccc} P \times_{q_P, B_0, p_Q} Q & \xrightarrow{\pi_Q} & Q \\ \pi_P \downarrow & \lrcorner & \downarrow p_Q \\ P & \xrightarrow{q_P} & B_0, \end{array}$$

the projection π_P is a pullback of the isomorphism p_Q , and is therefore an isomorphism. Hence $p_P \pi_P$ is an isomorphism, so the composite carrier is map-representable.

Since the sub-bicategory is locally full, it contains every Mealy simulation between its 1-cells. Therefore

$$\mathbf{Mly}_{\text{map}}(\mathcal{E})$$

is well-defined.

2. **Canonical map carriers correspond to internal functors.** By Theorem 1.3.1, for every morphism

$$F_0 : A_0 \rightarrow B_0,$$

Mealy structures on the canonical carrier

$$F_0! = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right)$$

are in bijective correspondence with internal functors

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

having object map F_0 .

3. **Every map-representable carrier is isomorphic to a canonical carrier in the Mealy hom-category.** Let

$$(P, \Phi) : \mathbb{A} \rightsquigarrow \mathbb{B}$$

have map-representable carrier

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0.$$

Then p is an isomorphism. Put

$$F_0 := qp^{-1} : A_0 \rightarrow B_0.$$

By Corollary 1.3.1, the carrier isomorphism

$$p : \left(A_0 \xleftarrow{p} P \xrightarrow{q} B_0 \right) \xrightarrow{\cong} F_0!$$

transports Φ to the Mealy structure

$$\Phi^F := (Bp) \circ \Phi \circ (p^{-1}A)$$

on $F_0!$.

We first construct a strict simulation

$$(P, \Phi) \Rightarrow (F_0!, \Phi^F).$$

Its state map is

$$p^{-1} : A_0 \rightarrow P.$$

This is a map of carrier spans because

$$pp^{-1} = 1_{A_0}$$

and

$$qp^{-1} = F_0.$$

Its output-comparison component is therefore the identity arrow

$$e_B F_0 : A_0 \rightarrow B_1.$$

The strict compatibility equation for this state map is

$$(Bp^{-1}) \circ \Phi^F = \Phi \circ (p^{-1}A).$$

Indeed,

$$\begin{aligned} (Bp^{-1}) \circ \Phi^F &= (Bp^{-1}) \circ (Bp) \circ \Phi \circ (p^{-1}A) \\ &= \Phi \circ (p^{-1}A). \end{aligned}$$

Here and below the canonical associativity constraints are suppressed.

Conversely, define a strict simulation

$$(F_0!, \Phi^F) \Rightarrow (P, \Phi)$$

with state map

$$p : P \rightarrow A_0.$$

Its output-comparison component is the identity arrow

$$e_B q : P \rightarrow B_1.$$

The required strict compatibility equation is

$$(Bp) \circ \Phi = \Phi^F \circ (pA).$$

This follows from

$$\begin{aligned} \Phi^F \circ (pA) &= (Bp) \circ \Phi \circ (p^{-1}A) \circ (pA) \\ &= (Bp) \circ \Phi. \end{aligned}$$

The two state maps p and p^{-1} are inverse. Since both simulations have identity output-comparison components, their vertical composites also have identity output comparisons. Hence the two strict simulations are mutually inverse.

Therefore every map-representable Mealy morphism is isomorphic in its Mealy hom-category to a canonical one arising from an internal functor.

4. **Simulations correspond to internal natural transformations.** By Theorem 1.4.2, simulations

$$F_{0!} \Rightarrow G_{0!}$$

between canonical map-representable Mealy morphisms are exactly internal natural transformations

$$F \Rightarrow G.$$

Thus the correspondence between internal functors and canonical map-representable Mealy morphisms is locally an isomorphism of categories, and the inclusion of canonical carriers into all map-representable carriers is locally essentially surjective.

5. **Compare horizontal composition.** Let

$$F : \mathbb{A} \rightarrow \mathbb{B}, \quad G : \mathbb{B} \rightarrow \mathbb{C}$$

be internal functors. The pipeline composite of their canonical carriers has apex

$$A_0 \times_{F_0, B_0, 1_{B_0}} B_0,$$

which is canonically isomorphic to A_0 . Under this pullback isomorphism, the composite carrier is identified with

$$(G_0 F_0)! = \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{G_0 F_0} C_0 \right).$$

The pipeline transition-output structure transports to the arrow map

$$G_1 F_1 : A_1 \rightarrow C_1.$$

Hence pipeline composition agrees with ordinary functor composition up to the canonical carrier isomorphism.

Under the same identifications, the horizontal-composition formula for simulations becomes the usual horizontal pasting formula for internal natural transformations. The coherence isomorphisms are inherited from the canonical pullback associators and unitors.

6. **Establish the biequivalence.** The assignment

$$F \mapsto F_{0!}$$

together with the identification of internal natural transformations with stateless simulations is locally fully faithful on canonical carriers. Every map-representable 1-cell is isomorphic to one in its image by the strict simulations constructed above. The canonical pullback isomorphisms supply the pseudofunctorial composition and unit constraints. Therefore

$$\mathbf{Cat}(\mathcal{E}) \simeq \mathbf{Mly}_{\text{map}}(\mathcal{E})$$

as bicategories.

7. **Recover the strict 2-category.** Restrict to canonical carriers $F_{0!}$ and define their composition by transporting pipeline composition along the canonical isomorphism

$$G_{0!} \circ F_{0!} \cong (G_0 F_0)_{!}.$$

Define the transported composite to be the canonical carrier of the ordinary composite internal functor GF , and perform the analogous transport for simulations.

With this transported composition, identities and associativity are those of ordinary internal functors and internal natural transformations, and hence hold strictly. The resulting strict 2-category is the usual

$$\mathbf{Cat}(\mathcal{E}).$$

□

1.6 Examples

Unless otherwise stated, the examples in this section are taken in $\mathcal{E} = \mathbf{Set}$. Thus internal categories are small categories. Accordingly, all categories appearing below are understood to be small. In particular, when using categories such as finite sets and injections, we use a fixed small skeleton.

The first example records ordinary deterministic Mealy machines as a special case. The following examples exhibit genuinely many-object, stateful instances of the internal-category construction.

Example 1.6.1 (Classical deterministic Mealy machines). *Let Σ be an input alphabet and Γ an output alphabet. Let*

$$\Sigma^*, \quad \Gamma^*$$

be the free monoids. Regard them as one-object categories using the one-object convention fixed earlier:

$$b \circ a = ba.$$

A deterministic Mealy machine consists of a state set S , a transition function

$$\tau : S \times \Sigma \rightarrow S,$$

and an output function

$$o : S \times \Sigma \rightarrow \Gamma.$$

It extends uniquely to functions

$$S \times \Sigma^* \rightarrow S, \quad \Sigma^* \times S \rightarrow \Gamma^*,$$

written

$$s \cdot w, \quad \omega(w, s),$$

such that

$$s \cdot \varepsilon = s, \quad \omega(\varepsilon, s) = \varepsilon,$$

and

$$s \cdot (uv) = (s \cdot u) \cdot v, \quad \omega(uv, s) = \omega(u, s)\omega(v, s \cdot u).$$

The sequential execution diagram is

$$s \xrightarrow{u/\omega(u, s)} s \cdot u \xrightarrow{v/\omega(v, s \cdot u)} s \cdot (uv).$$

These are precisely the one-object Mealy laws for a Mealy morphism

$$\Sigma^* \rightsquigarrow \Gamma^*.$$

There is one minor convention to record. A general one-object Mealy morphism

$$\Sigma^* \rightsquigarrow \Gamma^*$$

allows the output on a generator

$$a \in \Sigma$$

to be an arbitrary word

$$\omega(a, s) \in \Gamma^*.$$

A classical letter-output Mealy machine is the subclass satisfying

$$\omega(a, s) \in \Gamma \subseteq \Gamma^*$$

for every $a \in \Sigma$ and state $s \in S$.

Thus ordinary deterministic Mealy machines are recovered by passing from alphabets to their free monoids and imposing the generator-level single-letter output condition. The many-object examples below should be read as typed and stateful generalizations of this one-object recovery.

Example 1.6.2 (The base- r carry machine). Let $r \geq 2$. Let the input monoid be

$$M = (\mathbb{N}, +),$$

let the state set be

$$P = \{0, 1, \dots, r-1\},$$

and define a right action by

$$x \cdot n = (x + n) \bmod r.$$

Let the output monoid also be

$$N = (\mathbb{N}, +).$$

Define

$$\omega(n, x) = \left\lfloor \frac{x + n}{r} \right\rfloor.$$

Then the additive output cocycle law becomes the carry identity

$$\left\lfloor \frac{x + m + n}{r} \right\rfloor = \left\lfloor \frac{x + m}{r} \right\rfloor + \left\lfloor \frac{(x + m) \bmod r + n}{r} \right\rfloor.$$

Thus a two-step addition has the form

$$x \xrightarrow{m/\lfloor (x+m)/r \rfloor} (x+m) \bmod r \xrightarrow{n/\lfloor ((x+m) \bmod r + n)/r \rfloor} (x+m+n) \bmod r.$$

Therefore this data defines a Mealy morphism

$$(\mathbb{N}, +) \rightsquigarrow (\mathbb{N}, +).$$

The state records a residue modulo r , and the output records the carry produced by adding the input.

Example 1.6.3 (Typed Mealy machines from path categories). Let G and H be small directed graphs, and let

$$\mathbb{A} = \mathbf{Path}(G), \quad \mathbb{B} = \mathbf{Path}(H)$$

be their free categories. The vertices of G may be read as input modes, and the edges of G as typed input symbols. Similarly, the vertices of H may be read as output modes, and the edges of H as typed output symbols.

A Mealy morphism

$$\mathbf{Path}(G) \rightsquigarrow \mathbf{Path}(H)$$

assigns to each state $x \in P$ an input vertex

$$p(x) \in G_0$$

and an output vertex

$$q(x) \in H_0.$$

If

$$\alpha : u \rightarrow v$$

is a path in G and $p(x) = v$, then execution along α produces a new state

$$\delta(\alpha, x)$$

over the source vertex u , together with an output path

$$\omega(\alpha, x) : q(\delta(\alpha, x)) \rightarrow q(x)$$

in H . The typing diagram is

$$u \xrightarrow{\alpha} v = p(x) \quad \rightsquigarrow \quad q(\delta(\alpha, x)) \xrightarrow{\omega(\alpha, x)} q(x).$$

The multiplication law says that the output path for a concatenated input path is obtained by concatenating the output paths produced by the sequential executions:

$$u \xrightarrow{\alpha} v \xrightarrow{\beta} w = p(x) \quad \rightsquigarrow \quad q(\delta(\alpha, \delta(\beta, x))) \xrightarrow{\omega(\alpha, \delta(\beta, x))} q(\delta(\beta, x)) \xrightarrow{\omega(\beta, x)} q(x).$$

Thus a Mealy morphism between path categories is a typed path-transducer: it reads input paths backwards relative to the chosen orientation and emits output paths in the output graph.

When each of G and H has one vertex, the standard path-composition convention yields the opposite free monoids relative to the one-object convention fixed in Subsection 1.2.1. More precisely,

if Σ and Γ are the respective edge sets, then the one-vertex path categories are the one-object categories associated to

$$(\Sigma^*)^{\text{op}} \quad \text{and} \quad (\Gamma^*)^{\text{op}}.$$

Word reversal gives monoid isomorphisms

$$\text{rev} : \Sigma^* \xrightarrow{\cong} (\Sigma^*)^{\text{op}}, \quad \text{rev} : \Gamma^* \xrightarrow{\cong} (\Gamma^*)^{\text{op}}.$$

Hence, after transport along word reversal, the one-vertex case recovers the ordinary word-monoid example.

In the many-object case, the construction records source and target modes explicitly. This is the first genuinely internal-category extension of ordinary Mealy machines: inputs and outputs are no longer words in a single alphabet, but typed paths whose endpoints matter.

A Mealy simulation between two such typed path-transducers assigns to each state of the second transducer a state of the first with the same input vertex, together with a path in H comparing the two output vertices. The simulation law says that these comparison paths commute with the output paths emitted during execution.

Example 1.6.4 (Context-restriction and register-allocation machines). *This example gives a deterministic computer-science instance whose input and output categories are genuinely many-object and whose carrier is genuinely stateful.*

Fix two countably infinite sets

$$\text{Var} \quad \text{and} \quad \text{Reg},$$

thought of as source variables and register names. Let

$$\mathbf{FinSub}(\text{Var})$$

be the small poset category whose objects are finite subsets

$$\Gamma \subseteq \text{Var},$$

and whose arrows are inclusions

$$a : \Gamma' \hookrightarrow \Gamma.$$

Similarly, let

$$\mathbf{FinSub}(\text{Reg})$$

be the small poset category of finite subsets of registers and inclusions.

Take

$$\mathbb{A} = \mathbf{FinSub}(\text{Var}), \quad \mathbb{B} = \mathbf{FinSub}(\text{Reg}).$$

An object Γ of $\mathbb{A}$ is read as a finite source context, for example a set of currently live source variables. An arrow

$$a : \Gamma' \hookrightarrow \Gamma$$

is a context inclusion. With the orientation of this chapter, processing this arrow target-to-source restricts from the larger live context Γ to the smaller live context Γ' :

$$\Gamma' \xleftarrow{a} \Gamma.$$

Define a state to be a bijective allocation

$$\rho : \Gamma \cong R$$

from a finite source context $\Gamma \subseteq \mathbf{Var}$ to a finite register context $R \subseteq \mathbf{Reg}$. Let P be the set of triples

$$(\Gamma, R, \rho)$$

where

$$\Gamma \subseteq \mathbf{Var}, \quad R \subseteq \mathbf{Reg}$$

are finite subsets and

$$\rho : \Gamma \cong R$$

is a bijection. Define

$$p(\Gamma, R, \rho) = \Gamma, \quad q(\Gamma, R, \rho) = R.$$

Thus the carrier span records both the source context and the currently chosen register context:

$$\begin{array}{ccc} & P & \\ p \swarrow & & \searrow q \\ \mathbf{FinSub}(\mathbf{Var})_0 & & \mathbf{FinSub}(\mathbf{Reg})_0. \end{array}$$

Now let

$$a : \Gamma' \hookrightarrow \Gamma$$

be an input arrow and let

$$x = (\Gamma, R, \rho)$$

be a current allocation state. Define

$$R' := \rho(\Gamma') \subseteq R.$$

Since the output category has finite subsets of $\mathbf{Reg}$ as literal objects, R' is an object of $\mathbb{B}$, and the inclusion

$$R' \hookrightarrow R$$

is a literal arrow of $\mathbb{B}$. Define the updated allocation by restriction:

$$\rho|_{\Gamma'} : \Gamma' \cong R'.$$

Thus set

$$\delta(a, (\Gamma, R, \rho)) := (\Gamma', R', \rho|_{\Gamma'}),$$

and define the output arrow

$$\omega(a, (\Gamma, R, \rho)) : R' \hookrightarrow R$$

to be the inclusion of finite register subsets.

The execution square is

$$\begin{array}{ccc} \Gamma' & \xrightarrow{a} & \Gamma \\ \rho|_{\Gamma'} \downarrow \sim & & \sim \downarrow \rho \\ R' & \xrightarrow{\omega(a, (\Gamma, R, \rho))} & R. \end{array}$$

Thus execution restricts a source allocation along a context inclusion and emits the corresponding inclusion of register contexts.

The unit law is immediate. Restricting along the identity inclusion

$$1_\Gamma : \Gamma \hookrightarrow \Gamma$$

leaves the allocation unchanged and emits the identity inclusion:

$$\delta(1_\Gamma, (\Gamma, R, \rho)) = (\Gamma, R, \rho), \quad \omega(1_\Gamma, (\Gamma, R, \rho)) = 1_R.$$

The multiplication law says that restriction along a composite inclusion is the same as restriction in two stages. If

$$\Gamma'' \xrightarrow{b} \Gamma' \xrightarrow{a} \Gamma$$

are inclusions, and if

$$x = (\Gamma, R, \rho),$$

then

$$\delta(a \circ b, x) = \delta(b, \delta(a, x)).$$

Indeed, both sides are the allocation

$$\rho|_{\Gamma''} : \Gamma'' \cong \rho(\Gamma'').$$

The output arrows also compose strictly. Put

$$R' = \rho(\Gamma'), \quad R'' = \rho(\Gamma'').$$

Then

$$R'' \subseteq R' \subseteq R,$$

and

$$\omega(a \circ b, x) = \omega(a, x) \circ \omega(b, \delta(a, x))$$

as inclusions

$$R'' \hookrightarrow R.$$

The two-stage square is

$$\begin{array}{ccccc} \Gamma'' & \xrightarrow{b} & \Gamma' & \xrightarrow{a} & \Gamma \\ \rho|_{\Gamma''} \downarrow \sim & & \rho|_{\Gamma'} \downarrow \sim & & \sim \downarrow \rho \\ R'' & \hookrightarrow & R' & \hookrightarrow & R. \end{array}$$

Therefore this data defines a Mealy morphism

$$\mathbf{FinSub}(\mathbf{Var}) \rightsquigarrow \mathbf{FinSub}(\mathbf{Reg}).$$

This example is not map-representable. For a fixed source context Γ , there may be many allocations

$$\rho : \Gamma \cong R$$

with different choices of register subsets $R \subseteq \mathbf{Reg}$, or different bijections to the same register subset. Hence the state is not determined by the input interface. The example models deterministic bookkeeping for environment restriction, live-variable restriction, or register-allocation restriction, without adding nondeterministic or probabilistic effects.

Example 1.6.5 (Posets as internal categories). *Let A and B be small posets. Every small poset may be regarded as a small category with a unique arrow*

$$u \rightarrow v$$

precisely when

$$u \leq v.$$

Thus a Mealy morphism between posets A and B consists of a state set P equipped with interface maps

$$p : P \rightarrow A, \quad q : P \rightarrow B,$$

together with a transition operation defined whenever

$$u \leq v \quad \text{and} \quad p(x) = v.$$

The transition produces a new state

$$\delta(u \leq v, x)$$

with

$$p(\delta(u \leq v, x)) = u,$$

and an output inequality

$$q(\delta(u \leq v, x)) \leq q(x)$$

in B . The operational picture is

$$u \xrightarrow{\leq} v = p(x) \quad \rightsquigarrow \quad q(\delta(u \leq v, x)) \xrightarrow{\leq} q(x).$$

The unit law says that the reflexive inequality $v \leq v$ preserves the state and emits the reflexive inequality $q(x) \leq q(x)$. The multiplication law says that for

$$u \leq v \leq w$$

and $p(x) = w$, executing along the composite inequality

$$u \leq w$$

is the same as first executing along $v \leq w$ and then along $u \leq v$. Thus a Mealy morphism between posets is a stateful refinement process that moves backwards along input refinements while emitting corresponding output refinements.

A Mealy simulation

$$P \Rightarrow Q$$

between such Mealy morphisms consists of a function

$$\theta : Q \rightarrow P$$

over A , together with inequalities

$$q_P(\theta y) \leq q_Q(y)$$

in B , compatible with execution. The comparison inequality is

$$q_P(\theta y) \xrightarrow{\leq} q_Q(y).$$

The simulation law says that the comparison inequalities commute with the output inequalities emitted by the two machines.

A map-representable Mealy morphism between posets is just a monotone map. Indeed, if the carrier is

$$A \xleftarrow{1_A} A \xrightarrow{F_0} B,$$

then the output inequality associated to $u \leq v$ is forced to be

$$F_0 u \leq F_0 v.$$

A Mealy simulation between map-representable carriers is a pointwise inequality

$$F_0 x \leq G_0 x$$

natural with respect to the order. This is the posetal form of an internal natural transformation.

Example 1.6.6 (Action groupoids). Let a group G act on a set X on the right, and let a group H act on a set Y on the right. The corresponding action groupoids

$$X//G, \quad Y//H$$

are small categories. Their arrows are

$$(x, g) : x \rightarrow xg, \quad (y, h) : y \rightarrow yh.$$

Composition is given by

$$(xg, h) \circ (x, g) = (x, gh),$$

and similarly for $Y//H$.

A Mealy morphism

$$X//G \rightsquigarrow Y//H$$

has a state set P with interface maps

$$p : P \rightarrow X, \quad q : P \rightarrow Y.$$

With the orientation of this chapter, an input arrow

$$(x, g) : x \rightarrow xg$$

is processed target-to-source: a state $z \in P$ with

$$p(z) = xg$$

is updated to a state

$$\delta((x, g), z)$$

with

$$p(\delta((x, g), z)) = x.$$

The output is an arrow in the action groupoid $Y//H$. Therefore it is determined by an element

$$h((x, g), z) \in H$$

such that

$$q(\delta((x, g), z)) h((x, g), z) = q(z).$$

The execution diagram is

$$x \xrightarrow{g} xg = p(z) \quad \rightsquigarrow \quad q(\delta((x, g), z)) \xrightarrow{h((x, g), z)} q(z).$$

A Mealy simulation between action-groupoid Mealy morphisms assigns to each state of the second system a state of the first system over the same input object, together with an H -label comparing the corresponding output objects. If this comparison label is $\alpha_z \in H$, then it satisfies

$$q_P(\theta z)\alpha_z = q_Q(z).$$

The simulation law is the groupoid-valued naturality equation saying that comparison before execution and comparison after execution yield the same arrow in the output action groupoid.

With the standard composition convention for right-action groupoids, the one-object case corresponds to the opposite group relative to the one-object convention

$$b \circ a = ba$$

fixed in Subsection 1.2.1. Inversion gives a group isomorphism

$$G \xrightarrow{\cong} G^{\text{op}}, \quad g \mapsto g^{-1},$$

and similarly for H . After transport along these isomorphisms, the one-object case is the right-action-plus-cocycle theory of Subsection 1.3.1.

Thus this example is the multi-object groupoid analogue of that one-object description. The state update is a contravariant action along input groupoid arrows, the emitted H -label is a groupoid-valued cocycle, and simulations compare such cocycles by coherent state-dependent output arrows.

Example 1.6.7 (Group-valued coboundary outputs). Let M be a monoid acting on the right on a set P , and let G be a group. Given a function

$$\lambda : P \rightarrow G,$$

define

$$\omega(m, x) = \lambda(x)^{-1} \lambda(x \cdot m).$$

Then

$$\begin{aligned} \omega(m, x) \omega(n, x \cdot m) &= \lambda(x)^{-1} \lambda(x \cdot m) \lambda(x \cdot m)^{-1} \lambda((x \cdot m) \cdot n) \\ &= \lambda(x)^{-1} \lambda((x \cdot m) \cdot n) \\ &= \lambda(x)^{-1} \lambda(x \cdot (mn)) \\ &= \omega(mn, x). \end{aligned}$$

Hence

$$\omega(mn, x) = \omega(m, x) \omega(n, x \cdot m),$$

so ω is a valid output cocycle.

The cancellation in this calculation occurs entirely in the output group:

$$\lambda(x \cdot m) \lambda(x \cdot m)^{-1} = 1_G.$$

Thus, when G is regarded as a one-object category, all the displayed group elements are arrows at its unique object, and the cocycle equation is an equality of composites of such arrows.

This example shows that, for group-valued outputs, the cocycle terminology is literal: the output law is a nonabelian cocycle equation over a right action.

The same form describes simulations in the one-object group-valued case. If two G -valued cocycles differ by a state-dependent coboundary, then they are connected by a Mealy simulation. For example, when $P = Q$ and $\theta = 1_P$, the simulation equation reads

$$\alpha(x) \omega_P(m, x) = \omega_Q(m, x) \alpha(x \cdot m).$$

Equivalently,

$$\omega_Q(m, x) = \alpha(x) \omega_P(m, x) \alpha(x \cdot m)^{-1}.$$

Taking

$$\omega_P(m, x) = 1_G$$

and

$$\alpha(x) = \lambda(x)^{-1},$$

one obtains

$$\begin{aligned} \omega_Q(m, x) &= \lambda(x)^{-1} (\lambda(x \cdot m)^{-1})^{-1} \\ &= \lambda(x)^{-1} \lambda(x \cdot m). \end{aligned}$$

Thus the coboundary cocycle

$$\omega_Q(m, x) = \lambda(x)^{-1} \lambda(x \cdot m)$$

is connected to the trivial cocycle by a simulation from the trivial cocycle to the coboundary cocycle.

1.7 Chapter summary

The constructions of the chapter may be summarized as follows:

structure	interpretation
internal category	monad in $\mathbf{Span}(\mathcal{E})$
Mealy morphism	oriented monad morphism $PA \Rightarrow BP$
$T_B = B \circ -$	output-comparison monad
$R_A = - \circ A$	input-processing monad
Mealy morphism	$\overline{R}_A$ -algebra in $\mathbf{Kl}(T_B)$
Mealy simulation	oppositely directed Kleisli algebra morphism
strict Mealy simulation	identity-output-comparison special case
one-object Mealy morphism	internal right monoid action with output cocycle
one-object Mealy simulation	equivariant map with coherent cocycle comparison
map-representable Mealy morphism	internal functor, up to canonical carrier isomorphism
map-representable Mealy simulation	internal natural transformation, up to canonical carrier isomorphism.

The main structural result is the bicategory

$$\mathbf{Mly}(\mathcal{E}),$$

whose objects are internal categories, whose 1-cells are Mealy morphisms, and whose 2-cells are Mealy simulations. The one-object specialization is internal under the standing finite-limit assumptions: it

recovers right actions of monoid objects equipped with output cocycles. In **Set**, the free word-monoid case recovers classical deterministic Mealy machines after imposing the usual single-letter output condition on generators. The map-representable specialization recovers the ordinary 2-category of internal categories, internal functors, and internal natural transformations up to the canonical pullback isomorphisms inherent in the span bicategory.

Chapter 2

Behaviour, Fibrations, and the Categorical Myhill–Nerode Theorem

2.1 Fibred orientation and standing assumptions

The Myhill–Nerode theorem characterizes regular behaviour by residuals. For a language, the residuals are the quotients of the language by input words. For a Mealy morphism between internal categories, the residual inputs over an object are organized by a slice internal category. This chapter develops the corresponding residual semantics for the Mealy morphisms of Chapter 1. The language-theoretic residual quotient goes back to Myhill and Nerode, and the derivative viewpoint is closely related to Brzozowski’s calculus of regular-expression derivatives [15, 18, 4]. The categorical formulation below is also related in spirit to the final-semantics and realization-theoretic approaches to automata [7, 20, 21].

The input internal category $\mathbb{A}$ determines a residual syntax opfibration

$$\mathrm{Res}_{\mathbb{A}} \longrightarrow \mathbb{A}.$$

The fibre over an object $v \in A_0$ is the residual slice

$$\mathbb{A}/v.$$

An input arrow

$$a : u \rightarrow v$$

acts on residual syntax by cocartesian transport, given by postcomposition:

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

Thus residual syntax is covariantly indexed by $\mathbb{A}$.

The behaviour assignment is contravariant in residual syntax. A residual behaviour over v is an internal functor

$$H : \mathbb{A}/v \rightarrow \mathbb{B}.$$

Reindexing along

$$a : u \rightarrow v$$

is precomposition with residual transport:

$$a^*H := H \circ a_*.$$

This reindexing is the derivative of H by a :

$$\partial_a H := a^*H = H \circ a_*.$$

Hence the behaviour fibration is the contravariantly indexed internal category

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}} \longrightarrow \mathbb{A}$$

whose fibre over v is

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}(v) = [\mathbb{A}/v, \mathbb{B}].$$

The behaviour fibration carries a canonical Mealy morphism. Its transition is cartesian reindexing,

$$H \longmapsto a^*H = \partial_a H,$$

and its output is the anchor comparison obtained from the residual morphism

$$a_*1_u = a \longrightarrow 1_v$$

in the slice $\mathbb{A}/v$.

Assumption 2.1.1. *Throughout this chapter, $\mathcal{E}$ is a Barr-exact locally cartesian closed category with chosen pullbacks. Thus $\mathcal{E}$ has finite limits, dependent products in slices, stable regular epi–mono factorizations, and effective quotients of internal equivalence relations.*

Notation 2.1.1. *Let*

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A), \quad \mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B)$$

be internal categories in $\mathcal{E}$. We use the composition convention of Chapter 1: if

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

then

$$m_A(a, b) = b \circ a.$$

Thus a Mealy state over w processes the input $b \circ a$ by first processing b , then processing a . We write

$$1_v := e_A(v)$$

for the identity arrow at v , and similarly in $\mathbb{B}$.

Notation 2.1.2. *Generalized-element notation abbreviates morphism equations in $\mathcal{E}$. When variables such as a, r, x, H appear below, the corresponding formal assertion is the equality of maps out of the pullback object expressing the displayed typing constraints.*

Notation 2.1.3. *A Mealy morphism*

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

has carrier span

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

and structure maps

$$\delta : A_1 \times_{t_A, A_0, p} P \rightarrow P, \quad \omega : A_1 \times_{t_A, A_0, p} P \rightarrow B_1.$$

For $a : u \rightarrow v$ and $p(x) = v$, the input is processed target-to-source:

$$p(\delta(a, x)) = u,$$

and the output arrow is

$$\omega(a, x) : q(\delta(a, x)) \rightarrow q(x).$$

The Mealy unit laws are

$$\delta(1_{p(x)}, x) = x, \quad \omega(1_{p(x)}, x) = 1_{q(x)},$$

and the Mealy multiplication laws are

$$\delta(b \circ a, x) = \delta(a, \delta(b, x)),$$

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

Notation 2.1.4. *A Mealy simulation*

$$(\theta, \alpha) : P \Rightarrow Q$$

has underlying state map in the opposite direction

$$\theta : Q \rightarrow P,$$

and an output-comparison component

$$\alpha : Q \rightarrow B_1$$

satisfying

$$p_P \theta = p_Q, \quad s_B \alpha = q_P \theta, \quad t_B \alpha = q_Q.$$

Thus

$$\alpha_y : q_P(\theta y) \rightarrow q_Q(y).$$

The simulation laws are

$$\theta(\delta_Q(a, y)) = \delta_P(a, \theta y),$$

and

$$\alpha_y \circ \omega_P(a, \theta y) = \omega_Q(a, y) \circ \alpha_{\delta_Q(a, y)}.$$

Remark 2.1.1. *A displayed simulation*

$$P \Rightarrow Q$$

has state map $Q \rightarrow P$. The strict semantic homomorphisms used in Section 2.5 have covariantly directed state maps $P \rightarrow Q$.

2.2 Residual syntax and the behaviour fibration

The residual syntax of an input category is the indexed-category presentation of the codomain opfibration. We use the split cleavage given by postcomposition.

2.2.1 Residual syntax as a split codomain opfibration

Definition 2.2.1 (Residual syntax). *The **vertical residual category** of $\mathbb{A}$ is the internal category*

$$\mathbb{R}_{\mathbb{A}}$$

in the slice $\mathcal{E}/A_0$ defined as follows.

Its object object is

$$(\mathbb{R}_{\mathbb{A}})_0 := \left(A_1 \xrightarrow{t_A} A_0 \right).$$

Its arrow object is

$$(\mathbb{R}_{\mathbb{A}})_1 := \left(T_A \xrightarrow{t_A \pi_2} A_0 \right),$$

where

$$T_A := A_1 \times_{t_A, A_0, s_A} A_1.$$

Thus a generalized element of T_A is a composable pair

$$z \xrightarrow{c} w \xrightarrow{r} v.$$

The source and target maps are

$$s_{\mathbb{R}} := m_A : T_A \rightarrow A_1, \quad t_{\mathbb{R}} := \pi_2 : T_A \rightarrow A_1.$$

In generalized-element notation,

$$s_{\mathbb{R}}(c, r) = r \circ c, \quad t_{\mathbb{R}}(c, r) = r.$$

Hence (c, r) represents the residual arrow

$$r \circ c \longrightarrow r.$$

The identity map is

$$e_{\mathbb{R}} := \langle e_A s_A, 1_{A_1} \rangle : A_1 \rightarrow T_A,$$

so that

$$e_{\mathbb{R}}(r) = (1_{s_A(r)}, r).$$

Composition is induced by composition in $\mathbb{A}$. On a residual triple

$$y \xrightarrow{d} z \xrightarrow{c} w \xrightarrow{r} v,$$

it is

$$(c, r) \circ (d, r \circ c) = (c \circ d, r).$$

Residual postcomposition is represented internally as follows. Pull back $\mathbb{R}_{\mathbb{A}}$ along

$$s_A, t_A : A_1 \rightarrow A_0.$$

There is an internal functor in $\mathcal{E}/A_1$

$$\mathbf{post} : s_A^* \mathbb{R}_{\mathbb{A}} \longrightarrow t_A^* \mathbb{R}_{\mathbb{A}}.$$

Its object map is

$$(a, r) \longmapsto (a, a \circ r),$$

where

$$r : w \rightarrow s_A(a),$$

and its arrow map is

$$(a, c, r) \longmapsto (a, c, a \circ r).$$

Fibrewise, over an arrow

$$a : u \rightarrow v,$$

this internal functor is the postcomposition functor

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

The pair

$$(\mathbb{R}_{\mathbb{A}}, \mathbf{post})$$

is called the **residual syntax** of $\mathbb{A}$.

The defining pullback for the residual arrow object is

$$\begin{array}{ccc} T_A & \xrightarrow{\pi_2} & A_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow s_A \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

A generalized element is a pair

$$(c, r)$$

with

$$z \xrightarrow{c} w \xrightarrow{r} v.$$

It represents the residual triangle

$$z \xrightarrow{c} w \xrightarrow{r} v. \quad \text{with a curved arrow } r \circ c \text{ from } z \text{ to } v.$$

Proposition 2.2.1 (Residual fibres). *The data defining $\mathbb{R}_{\mathbb{A}}$ form an internal category in*

$$\mathcal{E}/A_0.$$

Its fibre over an object v of $\mathbb{A}$ is the slice internal category

$$\mathbb{A}/v.$$

Proof.

1. Verify the structure maps lie over A_0 .

Since

$$t_A m_A = t_A \pi_2$$

on T_A , the source map

$$s_{\mathbb{R}} = m_A$$

is a morphism in $\mathcal{E}/A_0$. The target map

$$t_{\mathbb{R}} = \pi_2$$

is also a morphism over A_0 .

The identity map lies over A_0 because

$$t_A e_A s_A = s_A$$

and

$$t_A \pi_2 e_{\mathbb{R}} = t_A.$$

2. Verify source and target of identities.

For $r : w \rightarrow v$,

$$s_{\mathbb{R}} e_{\mathbb{R}}(r) = r \circ 1_w = r,$$

and

$$t_{\mathbb{R}} e_{\mathbb{R}}(r) = r.$$

3. Verify composition.

A composable pair of residual arrows is represented by a diagram

$$y \xrightarrow{d} z \xrightarrow{c} w \xrightarrow{r} v.$$

The two residual arrows are

$$r \circ c \circ d \longrightarrow r \circ c$$

and

$$r \circ c \longrightarrow r.$$

Their composite is represented by

$$(c \circ d, r) : r \circ (c \circ d) \longrightarrow r.$$

Associativity in $\mathbb{A}$ identifies

$$r \circ (c \circ d) = (r \circ c) \circ d,$$

so this has the required source.

4. Verify the category laws.

The two unit equations reduce to the unit equations in $\mathbb{A}$. The associativity equation reduces to associativity in $\mathbb{A}$ on a string

$$x \longrightarrow y \longrightarrow z \longrightarrow w \longrightarrow v.$$

Therefore $\mathbb{R}_{\mathbb{A}}$ is an internal category in $\mathcal{E}/A_0$.

5. **Identify the fibres.**

Pulling $\mathbb{R}_{\mathbb{A}}$ back to an object v gives objects

$$r : w \rightarrow v$$

and arrows

$$r \circ c \rightarrow r$$

represented by commuting triangles over v . These are exactly the objects and arrows of

$$\mathbb{A}/v.$$

□

Definition 2.2.2 (Cocartesian residual transport). *For an arrow*

$$a : u \rightarrow v$$

*of $\mathbb{A}$, the **cocartesian residual transport functor***

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v$$

is the fibre of

$$\text{post} : s_A^* \mathbb{R}_{\mathbb{A}} \longrightarrow t_A^* \mathbb{R}_{\mathbb{A}}$$

over a .

Explicitly,

$$a_*(r) = a \circ r$$

for $r : w \rightarrow u$, and

$$a_*(c, r) = (c, a \circ r)$$

for a residual arrow

$$r \circ c \rightarrow r.$$

The chosen cocartesian residual square is

$$\begin{array}{ccc} w & \xlongequal{\quad} & w \\ r \downarrow & & \downarrow a \circ r \\ u & \xrightarrow{\quad a \quad} & v. \end{array}$$

Proposition 2.2.2 (Split opfibration laws). *Residual transport satisfies*

$$(1_v)_* = 1_{\mathbb{A}/v},$$

and, for composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

$$(b \circ a)_* = b_* \circ a_*.$$

These equations hold on both residual objects and residual arrows.

Proof. On a residual object $r : z \rightarrow u$,

$$(1_u)_*(r) = 1_u \circ r = r.$$

For composable a and b ,

$$(b \circ a)_*(r) = (b \circ a) \circ r = b \circ (a \circ r) = b_*(a_*(r)).$$

The corresponding equations on residual arrows follow from the same associativity equation in $\mathbb{A}$, with the residual triangle map left unchanged. Hence the equations hold as equations of internal functors. $\square$

Theorem 2.2.1 (Total residual syntax and the codomain opfibration). *Let*

$$T_A = A_1 \times_{t_A, A_0, s_A} A_1$$

be the object of composable pairs in $\mathbb{A}$, and define

$$\text{Sq}_A := T_A \times_{m_A, A_1, m_A} T_A.$$

Write

$$\lambda, \rho : \text{Sq}_A \rightrightarrows T_A$$

for the two pullback projections, and put

$$\lambda_i := \pi_i \lambda, \quad \rho_i := \pi_i \rho \quad (i = 1, 2).$$

A generalized element of Sq_A is written

$$((c, r'), (r, a)),$$

where

$$r : w \rightarrow u, \quad r' : w' \rightarrow v, \quad c : w \rightarrow w', \quad a : u \rightarrow v,$$

and the defining pullback equation is

$$r' \circ c = a \circ r.$$

There is an internal category

$$\text{Res}_{\mathbb{A}} := \mathbb{A}^{\rightarrow}$$

whose object object is A_1 , whose arrow object is Sq_A , and whose source and target maps are

$$s_{\text{Res}} := \rho_1 : \text{Sq}_A \rightarrow A_1,$$

$$t_{\text{Res}} := \lambda_2 : \text{Sq}_A \rightarrow A_1.$$

Thus the commutative square

$$((c, r'), (r, a))$$

is an arrow

$$r \longrightarrow r'$$

in $\text{Res}_{\mathbb{A}}$.

The identity at $r : w \rightarrow u$ is the square

$$e_{\text{Res}}(r) = ((1_w, r), (r, 1_u)).$$

In morphism notation,

$$e_{\text{Res}} = \langle \langle e_A s_A, 1_{A_1} \rangle, \langle 1_{A_1}, e_A t_A \rangle \rangle.$$

Composition is induced by horizontal and vertical composition in $\mathbb{A}$. Explicitly, for composable squares

$$((c, r'), (r, a)) : r \rightarrow r',$$

and

$$((d, r''), (r', b)) : r' \rightarrow r'',$$

their composite is

$$((d \circ c, r''), (r, b \circ a)).$$

The maps

$$\text{cod}_0 := t_A : A_1 \rightarrow A_0$$

and

$$\text{cod}_1 := \rho_2 : \text{Sq}_A \rightarrow A_1$$

define an internal functor

$$\text{cod} : \text{Res}_{\mathbb{A}} \longrightarrow \mathbb{A}.$$

This internal functor is a split internal opfibration. Its chosen cocartesian lift of

$$a : u \rightarrow v$$

at an object

$$r : w \rightarrow u$$

is the commutative square

$$\chi(r, a) := ((1_w, a \circ r), (r, a)) : r \longrightarrow a \circ r,$$

lying over a .

The vertical internal category of

$$\text{cod} : \text{Res}_{\mathbb{A}} \rightarrow \mathbb{A}$$

is canonically isomorphic, as an internal category in $\mathcal{E}/A_0$, to

$$\mathbb{R}_{\mathbb{A}}.$$

Under this identification, the transport induced by the chosen cocartesian lifts is exactly the residual postcomposition functor

$$\text{post} : s_A^* \mathbb{R}_{\mathbb{A}} \longrightarrow t_A^* \mathbb{R}_{\mathbb{A}}.$$

Fibrewise, the induced transport along

$$a : u \rightarrow v$$

is

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

Proof.

1. Construct the internal arrow category.

By definition,

$$\mathrm{Sq}_A = T_A \times_{m_A, A_1, m_A} T_A.$$

Hence a generalized element

$$((c, r'), (r, a))$$

consists of two composable pairs satisfying

$$m_A(c, r') = m_A(r, a).$$

In categorical notation this is

$$r' \circ c = a \circ r,$$

so the data are precisely a commutative square in $\mathbb{A}$.

The identity assignment is well-defined because

$$r \circ 1_{s_A(r)} = r$$

and

$$1_{t_A(r)} \circ r = r.$$

To check that composition is well-defined, suppose

$$r' \circ c = a \circ r$$

and

$$r'' \circ d = b \circ r'.$$

Then

$$\begin{aligned} r'' \circ (d \circ c) &= (r'' \circ d) \circ c \\ &= (b \circ r') \circ c \\ &= b \circ (r' \circ c) \\ &= b \circ (a \circ r) \\ &= (b \circ a) \circ r. \end{aligned}$$

Thus

$$((d \circ c, r''), (r, b \circ a))$$

is again an element of Sq_A .

The source and target of the composite are r and r'' . The unit and associativity equations follow componentwise from the unit and associativity equations in $\mathbb{A}$. Therefore

$$\mathrm{Res}_{\mathbb{A}}$$

is an internal category.

2. Construct the codomain functor.

For a commutative square

$$((c, r'), (r, a)),$$

the pair $(r, a) \in T_A$ gives

$$t_A(r) = s_A(a).$$

Moreover,

$$t_A(r') = t_A(r' \circ c) = t_A(a \circ r) = t_A(a).$$

Hence

$$s_A \text{cod}_1 = \text{cod}_0 s_{\text{Res}},$$

and

$$t_A \text{cod}_1 = \text{cod}_0 t_{\text{Res}}.$$

The bottom edge of the identity square at r is $1_{t_A(r)}$, and the bottom edge of the composite of squares over a and b is $b \circ a$. Therefore

$$\text{cod}$$

preserves identities and composition.

3. Define the chosen cocartesian lifts.

Let

$$r : w \rightarrow u, \quad a : u \rightarrow v.$$

Define

$$\chi(r, a) = ((1_w, a \circ r), (r, a)).$$

This lies in Sq_A because

$$(a \circ r) \circ 1_w = a \circ r.$$

Its source is r , its target is $a \circ r$, and its image under cod is a .

Internally, the chosen-lift map is

$$\chi : T_A \rightarrow \text{Sq}_A$$

given by

$$\chi = \langle \langle e_A s_A \pi_1, m_A \rangle, 1_{T_A} \rangle,$$

where a generalized element of T_A is written (r, a) .

4. Verify the cocartesian universal property.

Let

$$a : u \rightarrow v, \quad b : v \rightarrow z,$$

and suppose that

$$g : r \rightarrow s$$

is a square lying over $b \circ a$. Write

$$g = ((d, s), (r, b \circ a)).$$

Its commutativity equation is

$$s \circ d = (b \circ a) \circ r.$$

Define

$$h := ((d, s), (a \circ r, b)).$$

This is a commutative square because

$$\begin{aligned} s \circ d &= (b \circ a) \circ r \\ &= b \circ (a \circ r). \end{aligned}$$

It is an arrow

$$h : a \circ r \rightarrow s$$

lying over b .

By the composition formula in Res_A ,

$$h \circ \chi(r, a) = g.$$

This factorization is unique. Indeed, composition with $\chi(r, a)$ leaves the top edge and the target vertical edge unchanged. Thus any square over b whose composite with $\chi(r, a)$ is g must have top edge d , target edge s , source edge $a \circ r$, and bottom edge b , and is therefore equal to h .

By Notation 2.1.2, this generalized-element calculation is the equality of the corresponding internal morphisms from the pullback object classifying such factorization problems. Hence $\chi(r, a)$ is internally cocartesian.

5. Verify splitness.

For an identity arrow,

$$\chi(r, 1_u) = ((1_w, r), (r, 1_u)) = 1_r.$$

For composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow z,$$

one has

$$\chi(a \circ r, b) \circ \chi(r, a) = \chi(r, b \circ a).$$

Both sides are the square with top edge 1_w , source edge r , target edge

$$b \circ (a \circ r) = (b \circ a) \circ r,$$

and bottom edge $b \circ a$.

Therefore the chosen cocartesian lifts preserve identities and composition strictly.

6. Identify the vertical category.

A vertical arrow is a commutative square whose bottom edge is an identity. Given a composable pair

$$(c, r) \in T_A,$$

define the vertical square

$$\Phi(c, r) := ((c, r), (r \circ c, 1_{t_A(r)})).$$

Its source object is $r \circ c$, its target object is r , and it lies over $1_{t_A(r)}$.

Conversely, let

$$((c, r'), (r, 1_v))$$

be a vertical square. Its commutativity equation gives

$$r' \circ c = 1_v \circ r = r.$$

It is therefore uniquely determined by the composable pair

$$(c, r').$$

These two assignments are mutually inverse. Under this identification, a vertical arrow is precisely

$$r \circ c \longrightarrow r,$$

its identity is

$$(1_{s_A(r)}, r),$$

and its composition is

$$(c, r) \circ (d, r \circ c) = (c \circ d, r).$$

These are exactly the structure maps of

$$\mathbb{R}_{\mathbb{A}}.$$

Hence the vertical category of cod is canonically isomorphic to $\mathbb{R}_{\mathbb{A}}$ in $\mathcal{E}/A_0$.

7. Identify the induced transport.

Let

$$a : u \rightarrow v.$$

The chosen cocartesian lift sends a residual object

$$r : w \rightarrow u$$

to

$$a \circ r : w \rightarrow v.$$

A vertical residual arrow represented by

$$(c, r) : r \circ c \longrightarrow r$$

is transported to the vertical square represented by

$$(c, a \circ r) : (a \circ r) \circ c \longrightarrow a \circ r.$$

By associativity,

$$(a \circ r) \circ c = a \circ (r \circ c).$$

Thus the induced transport is exactly

$$r \longmapsto a \circ r, \quad (c, r) \longmapsto (c, a \circ r),$$

which is the internal functor

$$\mathbf{post} : s_A^* \mathbb{R}_{\mathbb{A}} \longrightarrow t_A^* \mathbb{R}_{\mathbb{A}}.$$

Fibrewise, this is

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

□

2.2.2 The behaviour fibration

Behaviour is obtained from residual syntax by forming the internal functor category from the vertical residual category into the constant internal category $\mathbb{B}$, both regarded as internal categories in the slice $\mathcal{E}/A_0$.

Notation 2.2.1 (The vertical residual category). *Let*

$$\mathbb{R}_{\mathbb{A}}$$

denote the internal category in $\mathcal{E}/A_0$ constructed in Proposition 2.2.1. Thus its object object is

$$A_1 \xrightarrow{t_A} A_0,$$

and its arrow object is

$$T_A = A_1 \times_{t_A, A_0, s_A} A_1 \xrightarrow{t_A \pi_2} A_0.$$

Its fibre over $v \in A_0$ is the slice internal category

$$\mathbb{A}/v.$$

Definition 2.2.3 (Constant target category over the residual base). *Let*

$$\Delta_{A_0} \mathbb{B}$$

denote the constant internal category in $\mathcal{E}/A_0$ whose object and arrow objects are

$$A_0 \times B_0 \longrightarrow A_0, \quad A_0 \times B_1 \longrightarrow A_0,$$

with structure maps

$$1_{A_0} \times s_B, \quad 1_{A_0} \times t_B, \quad 1_{A_0} \times e_B, \quad 1_{A_0} \times m_B.$$

Lemma 2.2.1 (Base change for internal functor categories). *Let*

$$f : X \rightarrow Y$$

be a morphism in a locally cartesian closed category $\mathcal{E}$, and let $\mathbb{C}, \mathbb{D}$ be internal categories in

$$\mathcal{E}/Y.$$

There is a canonical isomorphism of internal categories in $\mathcal{E}/X$

$$f^*[\mathbb{C}, \mathbb{D}]_{\mathcal{E}/Y} \cong [f^*\mathbb{C}, f^*\mathbb{D}]_{\mathcal{E}/X}.$$

Under this isomorphism, the pullback of the evaluation functor agrees with the evaluation functor of the pulled-back internal categories.

Proof. The strict internal functor category

$$[\mathbb{C}, \mathbb{D}]_{\mathcal{E}/Y}$$

is constructed from finite limits and exponentials in the slice $\mathcal{E}/Y$.

Reindexing

$$f^* : \mathcal{E}/Y \rightarrow \mathcal{E}/X$$

preserves finite limits. Since $\mathcal{E}$ is locally cartesian closed, reindexing also preserves slice exponentials. Concretely, this follows by expressing an exponential as a dependent product along a projection and applying the Beck–Chevalley isomorphism for the corresponding pullback square.

Applying f^* to the finite-limit and exponential presentation of the object of internal functors therefore gives the corresponding presentation of

$$[f^*\mathbb{C}, f^*\mathbb{D}]_{\mathcal{E}/X}.$$

The same argument applies to the object of internal natural transformations and to the source, target, identity, and composition maps.

The evaluation functor is induced by the evaluation maps of the exponentials used in the construction. These evaluation maps are preserved by the same base-change isomorphisms. Hence the stated isomorphism is compatible with evaluation. $\square$

Theorem 2.2.2 (Internal category of residual behaviours). *The strict internal functor category*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} := [\mathbb{R}_{\mathbb{A}}, \Delta_{A_0}\mathbb{B}]_{\mathcal{E}/A_0}$$

exists as an internal category in

$$\mathcal{E}/A_0.$$

Write its object and arrow objects as

$$p_{\mathbf{Beh}} : \mathbf{Beh}_0 \rightarrow A_0,$$

and

$$p_{\mathbf{Beh}, 1} : \mathbf{Beh}_1 \rightarrow A_0.$$

Its fibre over an object v of $\mathbb{A}$ is canonically the strict internal functor category

$$[\mathbb{A}/v, \mathbb{B}].$$

Proof. Since $\mathcal{E}$ is locally cartesian closed, the slice

$$\mathcal{E}/A_0$$

is cartesian closed and finitely complete.

In any finitely complete cartesian closed category, the strict internal functor category between two internal categories can be constructed from exponentials and equalizers. Its object object is the subobject of the object of pairs

$$(H_0, H_1)$$

of maps

$$H_0 : (\mathbb{R}_{\mathbb{A}})_0 \rightarrow (\Delta_{A_0}\mathbb{B})_0,$$

$$H_1 : (\mathbb{R}_{\mathbb{A}})_1 \rightarrow (\Delta_{A_0}\mathbb{B})_1$$

satisfying the source, target, identity, and composition equations.

Its arrow object is the object of triples

$$(H, K, \eta)$$

in which H and K are such internal functors and

$$\eta : H \Rightarrow K$$

is an internal natural transformation. The component-typing and naturality equations are again imposed by finite limits.

This constructs

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

in $\mathcal{E}/A_0$.

To identify its fibres, pull the construction back along an object

$$v : 1 \rightarrow A_0$$

or, more generally, along a generalized element of A_0 . By Lemma 2.2.1,

$$v^*[\mathbb{R}_{\mathbb{A}}, \Delta_{A_0} \mathbb{B}]_{\mathcal{E}/A_0} \cong [v^* \mathbb{R}_{\mathbb{A}}, v^* \Delta_{A_0} \mathbb{B}].$$

The pulled-back residual category is

$$v^* \mathbb{R}_{\mathbb{A}} = \mathbb{A}/v,$$

and the pulled-back constant target category is $\mathbb{B}$. Therefore the fibre is

$$[\mathbb{A}/v, \mathbb{B}].$$

□

Definition 2.2.4 (Behaviour comparison). *Let*

$$H, K : \mathbb{A}/v \rightarrow \mathbb{B}$$

*be residual behaviours over the same object v . A **behaviour comparison***

$$\eta : H \Rightarrow K$$

is an internal natural transformation.

Its component at a residual object

$$r : u \rightarrow v$$

is an arrow

$$\eta_r : H(r) \rightarrow K(r)$$

satisfying, for every residual triangle

$$w \xrightarrow{c} u \xrightarrow{r} v,$$

the naturality equation

$$\eta_r \circ H(c, r) = K(c, r) \circ \eta_{r \circ c}.$$

The naturality square is

$$\begin{array}{ccc} H(r \circ c) & \xrightarrow{H(c, r)} & H(r) \\ \eta_{r \circ c} \downarrow & & \downarrow \eta_r \\ K(r \circ c) & \xrightarrow{K(c, r)} & K(r). \end{array}$$

Theorem 2.2.3 (Object of behaviour comparisons). *There is an object*

$$\mathbf{Beh}_1 \rightarrow A_0$$

representing triples

$$(H, K, \eta),$$

where H and K are residual behaviours over the same object of $\mathbb{A}$, and

$$\eta : H \Rightarrow K$$

is a behaviour comparison.

The source and target maps

$$s_{\mathbf{Beh}}, t_{\mathbf{Beh}} : \mathbf{Beh}_1 \rightrightarrows \mathbf{Beh}_0$$

send

$$\eta : H \Rightarrow K$$

to H and K , respectively, and factor through

$$\mathbf{Beh}_0 \times_{A_0} \mathbf{Beh}_0.$$

Proof. The arrow object of the internal functor category

$$[\mathbb{R}_{\mathbb{A}}, \Delta_{A_0} \mathbb{B}]_{\mathcal{E}/A_0}$$

is the object of internal natural transformations.

Over a pair

$$(H, K) \in \mathbf{Beh}_0 \times_{A_0} \mathbf{Beh}_0,$$

a component assignment is a family of arrows of $\mathbb{B}$

$$\eta_r : H(r) \rightarrow K(r)$$

indexed by residual objects r . Such assignments are represented by exponentiation in the slice. Their source and target equations, and the naturality equation indexed by the residual triangle object T_A , are imposed by pullbacks and equalizers. This gives the representing object $\mathbf{Beh}_1$. $\square$

Theorem 2.2.4 (Vertical behaviour category). *The data*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} = (\mathbf{Beh}_0, \mathbf{Beh}_1, s_{\mathbf{Beh}}, t_{\mathbf{Beh}}, e_{\mathbf{Beh}}, m_{\mathbf{Beh}})$$

form an internal category in the slice

$$\mathcal{E}/A_0.$$

In particular, every vertical behaviour comparison has source and target in the same fibre over A_0 .

Proof. Identities and composition are those of the internal functor category

$$[\mathbb{R}_{\mathbb{A}}, \Delta_{A_0} \mathbb{B}]_{\mathcal{E}/A_0}.$$

Fibrewise, they are identity natural transformations and vertical composition of natural transformations. Their laws hold pointwise in $\mathbb{B}$, and hence hold internally in $\mathcal{E}/A_0$. $\square$

Theorem 2.2.5 (Parametrized precomposition). *Residual postcomposition induces an internal functor in $\mathcal{E}/A_1$*

$$\mathbf{pre}_A : t_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \longrightarrow s_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}.$$

Fibrewise, over an arrow

$$a : u \rightarrow v,$$

this functor is precomposition with

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

Thus it sends

$$H : \mathbb{A}/v \rightarrow \mathbb{B}$$

to

$$H \circ a_* : \mathbb{A}/u \rightarrow \mathbb{B},$$

and sends a natural transformation

$$\eta : H \Rightarrow K$$

to

$$\eta * a_* : H \circ a_* \Rightarrow K \circ a_*.$$

Proof. By Lemma 2.2.1, there are canonical isomorphisms of internal categories in $\mathcal{E}/A_1$

$$t_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \cong [t_A^* \mathbb{R}_{\mathbb{A}}, \Delta_{A_1} \mathbb{B}]_{\mathcal{E}/A_1},$$

and

$$s_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \cong [s_A^* \mathbb{R}_{\mathbb{A}}, \Delta_{A_1} \mathbb{B}]_{\mathcal{E}/A_1}.$$

Residual syntax supplies an internal functor

$$\mathbf{post} : s_A^* \mathbb{R}_{\mathbb{A}} \longrightarrow t_A^* \mathbb{R}_{\mathbb{A}}.$$

Precomposition with $\mathbf{post}$ defines an internal functor

$$\mathbf{post}^* : [t_A^* \mathbb{R}_{\mathbb{A}}, \Delta_{A_1} \mathbb{B}]_{\mathcal{E}/A_1} \longrightarrow [s_A^* \mathbb{R}_{\mathbb{A}}, \Delta_{A_1} \mathbb{B}]_{\mathcal{E}/A_1}.$$

On objects it sends an internal functor H to

$$H \circ \mathbf{post},$$

and on arrows it sends an internal natural transformation η to its whiskering

$$\eta * \mathbf{post}.$$

Transporting this internal functor across the two canonical base-change isomorphisms defines

$$\mathbf{pre}_A.$$

Equivalently, $\mathbf{pre}_A$ is the unique internal functor whose pullback along any generalized arrow of $\mathbb{A}$ classifies precomposition by the corresponding residual postcomposition functor.

Over a generalized arrow

$$a : u \rightarrow v,$$

the pullback of $\mathbf{post}$ is

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

Consequently, the pullback of pre_A sends

$$H \longmapsto H \circ a_*,$$

and

$$\eta \longmapsto \eta * a_*,$$

as claimed.

Since internal precomposition is defined by the representing property of the strict internal functor category, equal residual transport functors induce equal maps on both the object and arrow objects of the behaviour category. This observation will be used in Proposition 2.2.3. $\square$

Definition 2.2.5 (Behaviour indexed category). *The **behaviour indexed category** is the vertical behaviour category*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \longrightarrow A_0$$

together with the parametrized reindexing functor

$$\text{pre}_A : t_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \longrightarrow s_A^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}.$$

The object map of pre_A , followed by projection from the pullback over s_A , is denoted

$$\partial_0 : A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0 \longrightarrow \mathbf{Beh}_0.$$

The arrow map is denoted

$$\partial_1 : A_1 \times_{t_A, A_0, p_{\mathbf{Beh}} s_{\mathbf{Beh}}} \mathbf{Beh}_1 \longrightarrow \mathbf{Beh}_1.$$

We also write

$$\partial := \partial_0.$$

These maps satisfy

$$p_{\mathbf{Beh}} \partial_0 = s_A \pi_{A_1},$$

and

$$s_{\mathbf{Beh}} \partial_1 = \partial_0(1_{A_1} \times s_{\mathbf{Beh}}),$$

$$t_{\mathbf{Beh}} \partial_1 = \partial_0(1_{A_1} \times t_{\mathbf{Beh}}).$$

Definition 2.2.6 (Derivative). *Let*

$$a : u \rightarrow v$$

and let

$$H : \mathbb{A}/v \rightarrow \mathbb{B}$$

*be a residual behaviour. The **derivative** of H by a is*

$$\partial_a H := H \circ a_*.$$

Thus, for a residual object

$$r : w \rightarrow u,$$

$$(\partial_a H)(r) = H(a \circ r).$$

Definition 2.2.7 (Derivative of a comparison). *Let*

$$\eta : H \Rightarrow K$$

*be a behaviour comparison over v . Its **derivative** by $a : u \rightarrow v$ is*

$$\partial_a \eta := \eta * a_* : \partial_a H \Rightarrow \partial_a K.$$

Proposition 2.2.3 (Split fibration laws for derivatives). *Derivatives satisfy*

$$\partial_{1_v} H = H,$$

and, for composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

$$\partial_{b \circ a} H = \partial_a(\partial_b H).$$

The same equations hold for behaviour comparisons.

Derivative reindexing also preserves vertical identities and vertical composition:

$$\partial_a(1_H) = 1_{\partial_a H},$$

and

$$\partial_a(\mu \circ \eta) = \partial_a \mu \circ \partial_a \eta.$$

Proof. Residual transport satisfies the strict internal-functor equations

$$(1_v)_* = 1_{\mathbb{A}/v},$$

and

$$(b \circ a)_* = b_* \circ a_*.$$

Therefore, on residual behaviours,

$$\begin{aligned} \partial_{1_v} H &= H \circ (1_v)_* \\ &= H \circ 1_{\mathbb{A}/v} \\ &= H, \end{aligned}$$

and

$$\begin{aligned} \partial_{b \circ a} H &= H \circ (b \circ a)_* \\ &= H \circ (b_* \circ a_*) \\ &= (H \circ b_*) \circ a_* \\ &= \partial_a(\partial_b H). \end{aligned}$$

These are equalities of internal morphisms, rather than merely fibrewise canonical isomorphisms. Indeed, the two sides classify the same parametrized internal functor on residual objects and residual arrows. Since the object of residual behaviours represents such parametrized internal functors, equality of the classified functors gives equality of the corresponding maps into **Beh**₀.

The same argument applies to behaviour comparisons. Explicitly,

$$\partial_{1_v} \eta = \eta * (1_v)_* = \eta,$$

and

$$\begin{aligned}\partial_{b \circ a} \eta &= \eta * (b \circ a)_* \\ &= \eta * (b_* \circ a_*) \\ &= (\eta * b_*) * a_* \\ &= \partial_a(\partial_b \eta).\end{aligned}$$

Representability of the object of internal natural transformations makes these equalities of maps into $\mathbf{Beh}_1$.

Finally, $\mathbf{pre}_A$ is an internal functor. It therefore preserves identity arrows and vertical composition:

$$\partial_a(1_H) = 1_{\partial_a H},$$

and

$$\partial_a(\mu \circ \eta) = \partial_a \mu \circ \partial_a \eta.$$

□

Theorem 2.2.6 (Internal evaluation of residual behaviours). *The internal functor category*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} = [\mathbb{R}_{\mathbb{A}}, \Delta_{A_0} \mathbb{B}]_{\mathcal{E}/A_0}$$

carries a canonical internal evaluation functor

$$\text{ev} : \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \times_{A_0} \mathbb{R}_{\mathbb{A}} \longrightarrow \Delta_{A_0} \mathbb{B}.$$

Write its object and arrow maps, after suppressing the evident A_0 -coordinates, as

$$\text{ev}_0 : \mathbf{Beh}_0 \times_{A_0} A_1 \longrightarrow B_0,$$

and

$$\text{ev}_1 : \mathbf{Beh}_1 \times_{A_0} T_A \longrightarrow B_1.$$

In generalized-element notation,

$$\text{ev}_0(H, r) = H(r).$$

If

$$\eta : H \Rightarrow K$$

and

$$\xi : r \rightarrow r'$$

are arrows in the two factors, then

$$\text{ev}_1(\eta, \xi) : H(r) \rightarrow K(r')$$

is the common composite

$$K(\xi) \circ \eta_r = \eta_{r'} \circ H(\xi).$$

Proof. This is the evaluation functor associated to the internal hom

$$[\mathbb{R}_{\mathbb{A}}, \Delta_{A_0} \mathbb{B}]_{\mathcal{E}/A_0}.$$

Its object map is exponential evaluation. Its arrow map is induced by evaluation of the component of an internal natural transformation followed by evaluation of the relevant residual arrow.

The equality

$$K(\xi) \circ \eta_r = \eta_{r'} \circ H(\xi)$$

is precisely the internal naturality equation for η . Preservation of identities and composition is the standard evaluation-functor calculation in the internal functor category. □

Definition 2.2.8 (Anchor evaluation). *The identity residual section is*

$$e_A : A_0 \rightarrow A_1.$$

It selects

$$1_v : v \rightarrow v$$

in the residual fibre over v .

*The **anchor map** is*

$$q_{\mathbf{Beh}} := \text{ev}_0 \circ \langle 1_{\mathbf{Beh}_0}, e_A p_{\mathbf{Beh}} \rangle : \mathbf{Beh}_0 \rightarrow B_0.$$

Thus, for

$$\begin{aligned} H : \mathbb{A}/v &\rightarrow \mathbb{B}, \\ q_{\mathbf{Beh}}(H) &= H(1_v). \end{aligned}$$

Thus the object of residual behaviours carries the anchor span

$$\begin{array}{ccc} & \mathbf{Beh}_0 & \\ p_{\mathbf{Beh}} \swarrow & & \searrow q_{\mathbf{Beh}} \\ A_0 & & B_0. \end{array}$$

Definition 2.2.9 (Total behaviour category). *Put*

$$X_{\mathbf{Beh}} := A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

A generalized element is a pair

$$(a, H), \quad a : u \rightarrow v, \quad H : \mathbb{A}/v \rightarrow \mathbb{B}.$$

Define the total arrow object

$$\mathbf{Beh}_1^{\text{tot}} := X_{\mathbf{Beh}} \times_{\partial, \mathbf{Beh}_0, t_{\mathbf{Beh}}} \mathbf{Beh}_1.$$

Thus a generalized total arrow is a triple

$$(a, H, \eta)$$

where

$$\eta : K \Rightarrow \partial_a H$$

is a vertical behaviour comparison over u .

Its source and target are

$$s_{\text{tot}}(a, H, \eta) := s_{\mathbf{Beh}}(\eta) = K,$$

$$t_{\text{tot}}(a, H, \eta) := H.$$

Theorem 2.2.7 (The total behaviour fibration). *The object*

$$\mathbf{Beh}_0$$

and the arrow object

$$\mathbf{Beh}_1^{\text{tot}}$$

of Definition 2.2.9 form an internal category

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{tot}}.$$

The maps

$$p_{\mathbf{Beh}} : \mathbf{Beh}_0 \rightarrow A_0$$

and

$$p_{\mathbf{Beh}, 1}^{\text{tot}} : \mathbf{Beh}_1^{\text{tot}} \rightarrow A_1, \quad (a, H, \eta) \mapsto a,$$

define an internal functor

$$\pi_{\mathbf{Beh}} : \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{tot}} \rightarrow \mathbb{A}.$$

This internal functor is a split internal fibration. Its chosen cartesian lift of

$$a : u \rightarrow v$$

at a behaviour H over v is

$$\chi(a, H) := (a, H, 1_{\partial_a H}) : \partial_a H \rightarrow H.$$

Proof.

1. Define identities.

Let H lie over v . Define

$$1_H^{\text{tot}} := (1_v, H, 1_H).$$

This is well-defined because

$$\partial_{1_v} H = H.$$

2. Define composition.

Suppose

$$(a, H, \eta) : K \rightarrow H$$

lies over

$$a : u \rightarrow v,$$

so that

$$\eta : K \Rightarrow \partial_a H,$$

and suppose

$$(b, L, \mu) : H \rightarrow L$$

lies over

$$b : v \rightarrow w,$$

so that

$$\mu : H \Rightarrow \partial_b L.$$

Define

$$(b, L, \mu) \circ (a, H, \eta) := (b \circ a, L, (\partial_a \mu) \circ \eta).$$

This is well-typed because

$$\partial_a \mu : \partial_a H \Rightarrow \partial_a \partial_b L = \partial_{b \circ a} L.$$

3. Verify the category laws.

The left and right unit equations follow from

$$\partial_{1_v} = 1$$

and preservation of identity comparisons by derivative reindexing.

For associativity, let a third total arrow be given. Expanding both parenthesizations reduces their comparison components to the same composite by:

$$\partial_a(\nu \circ \mu) = \partial_a \nu \circ \partial_a \mu,$$

$$\partial_a \partial_b = \partial_{b \circ a},$$

and associativity of vertical composition in $\mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{vert}}$.

Hence the displayed data form an internal category.

4. Verify that the projection is an internal functor.

The identity total arrow at H lies over $1_{p_{\mathbf{Beh}} H}$. The composite of arrows over a and b lies over $b \circ a$. Therefore

$$\pi_{\mathbf{Beh}}$$

preserves identities and composition.

5. Define the cleavage.

Put

$$X_{\mathbf{Beh}} := A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

Define

$$\chi : X_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_1^{\text{tot}}$$

by

$$\chi(a, H) = (a, H, 1_{\partial_a H}).$$

Its target is H , its source is $\partial_a H$, and its image under the projection functor is a .

6. Form the internal object of composite-factorization problems.

Let

$$A_2 := A_1 \times_{t_A, A_0, s_A} A_1.$$

We write a generalized element as

$$(c, a),$$

where

$$c : w \rightarrow u, \quad a : u \rightarrow v.$$

Thus

$$m_A(c, a) = a \circ c.$$

Let $\mathcal{L}$ be the pullback

$$\mathcal{L} := A_2 \times_{m_A, A_1, p_{\mathbf{Beh}, 1}^{\text{tot}}} \mathbf{Beh}_1^{\text{tot}}.$$

A generalized element of $\mathcal{L}$ is a triple

$$(c, a, g),$$

where $g : K \rightarrow H$ is a total arrow lying over

$$a \circ c.$$

Next put

$$Y := A_2 \times_{t_A \pi_2, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

A generalized element of Y is

$$(c, a, H)$$

with H lying over $t_A(a)$.

There is a map

$$\vartheta : Y \rightarrow \mathbf{Beh}_0 \times_{p_{\mathbf{Beh}}, A_0, t_A} A_1$$

defined by

$$\vartheta(c, a, H) = (\partial_a H, c).$$

This is well-defined because

$$p_{\mathbf{Beh}}(\partial_a H) = s_A(a) = t_A(c).$$

Define

$$\mathcal{R} := Y \times_{\vartheta, \mathbf{Beh}_0 \times_{A_0} A_1, (t_{\text{tot}}, p_{\mathbf{Beh}, 1}^{\text{tot}})} \mathbf{Beh}_1^{\text{tot}}.$$

A generalized element of $\mathcal{R}$ is a quadruple

$$(c, a, H, h),$$

where

$$h : K \rightarrow \partial_a H$$

is a total arrow lying over c .

7. Construct the cartesian-factorization isomorphism.

Composition with the chosen lift defines a map

$$\Gamma : \mathcal{R} \rightarrow \mathcal{L}$$

by

$$\Gamma(c, a, H, h) = (c, a, \chi(a, H) \circ h).$$

To compute this map, write

$$h = (c, \partial_a H, \nu),$$

where

$$\nu : K \Rightarrow \partial_c \partial_a H = \partial_{a \circ c} H.$$

Since

$$\chi(a, H) = (a, H, 1_{\partial_a H}),$$

the composition formula gives

$$\chi(a, H) \circ h = (a \circ c, H, \nu).$$

Conversely, let

$$g = (a \circ c, H, \nu)$$

represent an element of $\mathcal{L}$. Define

$$\Gamma^{-1}(c, a, g) := (c, a, H, (c, \partial_a H, \nu)).$$

This is well-defined because

$$\partial_c \partial_a H = \partial_{a \circ c} H.$$

The two constructions leave the vertical comparison ν unchanged. Therefore

$$\Gamma^{-1}\Gamma = 1_{\mathcal{R}}, \quad \Gamma\Gamma^{-1} = 1_{\mathcal{L}}.$$

Hence

$$\Gamma : \mathcal{R} \xrightarrow{\cong} \mathcal{L}$$

is an isomorphism.

This is the internal cartesian universal property: every total arrow

$$g : K \rightarrow H$$

lying over $a \circ c$ factors uniquely as

$$K \xrightarrow{h} \partial_a H \xrightarrow{\chi(a, H)} H,$$

where h lies over c .

8. Verify splitness.

For an identity arrow,

$$\chi(1_v, H) = (1_v, H, 1_H) = 1_H^{\text{tot}}.$$

For composable

$$c : w \rightarrow u, \quad a : u \rightarrow v,$$

one has

$$\chi(a \circ c, H) = \chi(a, H) \circ \chi(c, \partial_a H).$$

Indeed, both sides are the total arrow

$$\partial_{a \circ c} H \rightarrow H$$

over $a \circ c$ with identity vertical comparison, using

$$\partial_c \partial_a H = \partial_{a \circ c} H.$$

Thus the chosen cartesian lifts are strictly compatible with identities and composition, and

$$\pi_{\mathbf{Beh}}$$

is a split internal fibration.

□

Remark 2.2.1. *The internal category*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

contains only comparisons inside a fixed residual fibre. The total category

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{tot}}$$

also contains arrows over non-identity arrows of $\mathbb{A}$.

An arrow over

$$a : u \rightarrow v$$

from a behaviour K over u to a behaviour H over v is a vertical comparison

$$K \Rightarrow \partial_a H.$$

The canonical behaviour Mealy morphism uses the chosen cartesian cleavage of this total fibration.

Remark 2.2.2 (Equality of behaviours). *The object $\mathbf{Beh}_0$ represents residual functors themselves. Thus equality in $\mathbf{Beh}_0$ is equality of represented residual functors, not isomorphism of residual functors and not equivalence inside the vertical behaviour category.*

2.3 The cleavage Mealy morphism

The behaviour indexed category has a split cleavage given by derivative reindexing. Anchor evaluation turns this cleavage into a Mealy morphism.

2.3.1 Internal derivative and anchor-output maps

Lemma 2.3.1 (Internal derivative and anchor-output maps). *Put*

$$X_{\mathbf{Beh}} := A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0,$$

and write its projections as

$$\pi_A : X_{\mathbf{Beh}} \rightarrow A_1, \quad \pi_H : X_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0.$$

The derivative map

$$\partial : X_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0$$

satisfies

$$p_{\mathbf{Beh}} \partial = s_A \pi_A.$$

Define

$$\sigma : A_1 \rightarrow T_A$$

by

$$\sigma := \langle 1_{A_1}, e_A t_A \rangle.$$

Then $\sigma(a)$ represents the residual arrow

$$a \longrightarrow 1_{t_A a}.$$

The anchor-output map is

$$\text{out}_{\mathbf{Beh}} := \text{ev}_1 \circ \langle e_{\mathbf{Beh}} \pi_H, \sigma \pi_A \rangle : X_{\mathbf{Beh}} \rightarrow B_1.$$

It satisfies

$$s_B \text{out}_{\mathbf{Beh}} = q_{\mathbf{Beh}} \partial,$$

and

$$t_B \text{out}_{\mathbf{Beh}} = q_{\mathbf{Beh}} \pi_H.$$

Proof. The derivative equation

$$p_{\mathbf{Beh}} \partial = s_A \pi_A$$

is the base equation for parametrized precomposition.

To define σ , observe that

$$t_A = s_A e_A t_A.$$

Hence the pair

$$(1_{A_1}, e_A t_A)$$

induces a map into

$$T_A = A_1 \times_{t_A, A_0, s_A} A_1.$$

For an arrow $a : u \rightarrow v$,

$$\sigma(a) = (a, 1_v).$$

Its source and target in the residual category are

$$1_v \circ a = a, \quad 1_v,$$

so it represents

$$a \rightarrow 1_v.$$

The pair

$$(e_{\mathbf{Beh}} \pi_H, \sigma \pi_A)$$

is a map into the arrow object of

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \times_{A_0} \mathbb{R}_{\mathbb{A}}.$$

Its first component is the identity natural transformation on H , while its second component is the residual arrow

$$a \rightarrow 1_v.$$

Applying the arrow part of evaluation gives

$$\text{out}_{\mathbf{Beh}}(a, H) = H(a \rightarrow 1_v) : H(a) \rightarrow H(1_v).$$

The target is therefore

$$H(1_v) = q_{\mathbf{Beh}}(H),$$

proving

$$t_B \text{out}_{\mathbf{Beh}} = q_{\mathbf{Beh}} \pi_H.$$

For the source, let $a : u \rightarrow v$. By definition of derivative,

$$(\partial_a H)(1_u) = H(a_* 1_u).$$

Residual postcomposition gives

$$a_* 1_u = a \circ 1_u = a.$$

Consequently,

$$H(a) = (\partial_a H)(1_u) = q_{\mathbf{Beh}}(\partial_a H).$$

Hence

$$s_B \text{out}_{\mathbf{Beh}} = q_{\mathbf{Beh}} \partial.$$

□

2.3.2 The canonical behaviour carrier

Definition 2.3.1 (Canonical behaviour carrier). *The **canonical behaviour carrier** is the anchor span*

$$A_0 \xleftarrow{p_{\mathbf{Beh}}} \mathbf{Beh}_0 \xrightarrow{q_{\mathbf{Beh}}} B_0.$$

Definition 2.3.2 (Canonical transition and output). *The **canonical transition** is derivative reindexing:*

$$\delta_{\mathbf{Beh}} := \partial.$$

*The **canonical output** is the anchor comparison:*

$$\omega_{\mathbf{Beh}} := \text{out}_{\mathbf{Beh}}.$$

In generalized-element notation,

$$\delta_{\mathbf{Beh}}(a, H) = \partial_a H,$$

and

$$\omega_{\mathbf{Beh}}(a, H) = H(a, 1_v) : H(a) \rightarrow H(1_v),$$

where $a : u \rightarrow v$ and $H : \mathbb{A}/v \rightarrow \mathbb{B}$.

The defining residual triangle for the output is

$$u \xrightarrow{a} v \xrightarrow{1_v} v.$$

$\begin{array}{c} \text{ } \\ \text{ } \end{array}$
 $\begin{array}{c} a \\ \text{ } \end{array}$

2.3.3 The cleavage Mealy morphism

Theorem 2.3.1 (Cleavage Mealy morphism of the behaviour fibration). *The anchor span, together with derivative transition and anchor-comparison output, defines a Mealy morphism*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

Proof.

1. **Check typing.** The typing equations are exactly the two parts of Lemma 2.3.1.
2. **Verify the transition laws.** The transition unit and multiplication laws are the split fibration laws:

$$\partial_{1_v} H = H, \quad \partial_{b \circ a} H = \partial_a(\partial_b H).$$

The defining pullback is

$$\begin{array}{ccc} D(S) & \xrightarrow{\pi_S} & S \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p_S \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

Thus a generalized element (a, H) , with

$$a : u \rightarrow p_S(H),$$

has Z -coordinate

$$(u, q_{\mathbf{Beh}}(\partial_a i H)).$$

This coordinate is the input interface and anchor of the derivative behaviour.

Definition 2.3.4 (Derivative-closed subobject). *A subobject*

$$i : S \hookrightarrow \mathbf{Beh}_0$$

*in $\mathcal{E}/Z$ is **derivative-closed** if the derivative map*

$$\partial(1_{A_1} \times i) : D(S) \rightarrow \mathbf{Beh}_0$$

factors through i in $\mathcal{E}/Z$. Equivalently, there is a unique map

$$\bar{\partial}_S : D(S) \rightarrow S$$

over Z such that

$$i\bar{\partial}_S = \partial(1_{A_1} \times i).$$

The factorization condition is displayed by

$$\begin{array}{ccc} D(S) & \xrightarrow{\partial(1_{A_1} \times i)} & \mathbf{Beh}_0 \\ & \searrow \bar{\partial}_S & \nearrow i \\ & S & \end{array} \quad \text{in } \mathcal{E}/Z.$$

Lemma 2.3.2 (Restriction to derivative-closed subobjects). *Let*

$$i : S \hookrightarrow \mathbf{Beh}_0$$

be derivative-closed. Then the canonical behaviour Mealy morphism restricts to a Mealy morphism carried by

$$A_0 \xleftarrow{p_{\mathbf{Beh}}^i} S \xrightarrow{q_{\mathbf{Beh}}^i} B_0.$$

Proof. Write

$$p_S := p_{\mathbf{Beh}}^i, \quad q_S := q_{\mathbf{Beh}}^i.$$

Derivative-closedness supplies a unique map

$$\bar{\partial}_S : A_1 \times_{t_A, A_0, p_S} S \rightarrow S$$

satisfying

$$i\bar{\partial}_S = \partial(1_{A_1} \times i).$$

Define the restricted transition by

$$\delta_S := \bar{\partial}_S.$$

Define the restricted output by

$$\omega_S := \omega_{\mathbf{Beh}} \circ (1_{A_1} \times i).$$

Thus, in generalized-element notation,

$$\delta_S(a, s) = \bar{\partial}_S(a, s),$$

and

$$\omega_S(a, s) = \omega_{\mathbf{Beh}}(a, i(s)).$$

We first verify the typing equations. Since $\bar{\partial}_S$ is a morphism over the derivative-induced Z -structure,

$$p_S(\delta_S(a, s)) = s_A(a).$$

Moreover,

$$\begin{aligned} s_B\omega_S(a, s) &= s_B\omega_{\mathbf{Beh}}(a, i(s)) \\ &= q_{\mathbf{Beh}}(\partial_a i(s)) \\ &= q_{\mathbf{Beh}}(i(\delta_S(a, s))) \\ &= q_S(\delta_S(a, s)), \end{aligned}$$

while

$$\begin{aligned} t_B\omega_S(a, s) &= t_B\omega_{\mathbf{Beh}}(a, i(s)) \\ &= q_{\mathbf{Beh}}(i(s)) \\ &= q_S(s). \end{aligned}$$

For the transition unit law, compose with the monomorphism i :

$$\begin{aligned} i(\delta_S(1_{p_S(s)}, s)) &= \partial_{1_{p_S(s)}} i(s) \\ &= i(s). \end{aligned}$$

Hence

$$\delta_S(1_{p_S(s)}, s) = s.$$

For the transition multiplication law, let

$$a : u \rightarrow v, \quad b : v \rightarrow w, \quad p_S(s) = w.$$

After composing with i , one has

$$\begin{aligned} i(\delta_S(b \circ a, s)) &= \partial_{b \circ a} i(s) \\ &= \partial_a(\partial_b i(s)) \\ &= \partial_a(i(\delta_S(b, s))) \\ &= i(\delta_S(a, \delta_S(b, s))). \end{aligned}$$

Since i is monic,

$$\delta_S(b \circ a, s) = \delta_S(a, \delta_S(b, s)).$$

The output unit law follows directly by restricting the ambient output unit law:

$$\begin{aligned}\omega_S(1_{p_S(s)}, s) &= \omega_{\mathbf{Beh}}(1_{p_S(s)}, i(s)) \\ &= 1_{q_{\mathbf{Beh}}(i(s))} \\ &= 1_{q_S(s)}.\end{aligned}$$

For the output multiplication law,

$$\begin{aligned}\omega_S(b \circ a, s) &= \omega_{\mathbf{Beh}}(b \circ a, i(s)) \\ &= \omega_{\mathbf{Beh}}(b, i(s)) \circ \omega_{\mathbf{Beh}}(a, \partial_b i(s)) \\ &= \omega_{\mathbf{Beh}}(b, i(s)) \circ \omega_{\mathbf{Beh}}(a, i(\delta_S(b, s))) \\ &= \omega_S(b, s) \circ \omega_S(a, \delta_S(b, s)).\end{aligned}$$

Thus the restricted transition and output satisfy all the Mealy laws. $\square$

2.4 Mealy actions and residual semantics

An arbitrary Mealy morphism determines a residual semantics by sending each state to its residual orbit. This construction is the categorical analogue of sending a state of a deterministic automaton to the language, or output behaviour, observed from that state. Related formulations appear in the coalgebraic account of final semantics and minimization [20, 21].

2.4.1 The execution category

The transition part of a Mealy morphism is a contravariant action of $\mathbb{A}$, equivalently a covariant action of $\mathbb{A}^{\text{op}}$. We use the corresponding category of executions, whose arrows are oriented in the direction in which states evolve.

Definition 2.4.1 (Execution category). *Let*

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

*be a Mealy morphism. Its **execution category***

$$\int_{\mathbb{A}} P$$

has object object P and arrow object

$$A_1 \times_{t_A, A_0, P} P.$$

The arrow

$$(a, x)$$

has source and target

$$s_{\int P}(a, x) = x, \quad t_{\int P}(a, x) = \delta(a, x).$$

The identity at x is

$$(1_{p(x)}, x).$$

If

$$(a, x) : x \rightarrow \delta(a, x), \quad (c, \delta(a, x)) : \delta(a, x) \rightarrow \delta(c, \delta(a, x))$$

are composable, where

$$c : w \rightarrow u, \quad a : u \rightarrow v,$$

then their composite is

$$(a \circ c, x) : x \rightarrow \delta(a \circ c, x).$$

Here $a \circ c$ denotes the ordinary categorical composite

$$w \xrightarrow{c} u \xrightarrow{a} v,$$

so, in the notation of Chapter 1,

$$a \circ c = m_A(c, a).$$

The transition square for an execution arrow is

$$\begin{array}{ccc} q(\delta(a, x)) & \xrightarrow{\omega(a, x)} & q(x) \\ \uparrow q & & \uparrow q \\ \delta(a, x) & \xleftarrow{(a, x)} & x \end{array}$$

where the horizontal arrow on the bottom is in the execution category and the top arrow is its emitted $\mathbb{B}$ -output.

Proposition 2.4.1 (Execution category). *The data of Definition 2.4.1 define an internal category*

$$\int_{\mathbb{A}} P.$$

Proof.

1. **Use the Mealy laws for δ .** The identity and composition maps are well-defined by the Mealy unit and multiplication laws for δ . The identity and associativity laws are inherited from the identity and associativity laws in $\mathbb{A}$, using the Mealy laws for δ .
2. **Display the execution order.** The execution category composes in the ordinary categorical order of $\mathbb{A}$, while execution itself processes inputs target-to-source. The composite execution square is

$$\begin{array}{ccccc} x & \xrightarrow{(a, x)} & \delta(a, x) & \xrightarrow{(c, \delta(a, x))} & \delta(c, \delta(a, x)) \\ & \searrow & & \nearrow & \\ & & & & \end{array} \quad \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array} \quad \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array}$$

$(a \circ c, x)$

□

2.4.2 The output functor

Proposition 2.4.2 (Output functor). *The output map of P is equivalently an internal functor*

$$\Omega_P : \int_{\mathbb{A}} P \rightarrow \mathbb{B}^{\text{op}}$$

with object map

$$q : P \rightarrow B_0.$$

It sends an execution arrow

$$(a, x) : x \rightarrow \delta(a, x)$$

to

$$\omega(a, x)^{\text{op}} : q(x) \rightarrow q(\delta(a, x))$$

in $\mathbb{B}^{\text{op}}$.

Proof.

1. **Check arrow typing.** The Mealy typing equations give

$$\omega(a, x) : q(\delta(a, x)) \rightarrow q(x)$$

in $\mathbb{B}$, hence an arrow

$$q(x) \rightarrow q(\delta(a, x))$$

in $\mathbb{B}^{\text{op}}$.

2. **Check identities.** The Mealy output unit law gives identity preservation.
3. **Check composition.** For composition, consider composable execution arrows

$$(a, x) : x \rightarrow \delta(a, x), \quad (c, \delta(a, x)) : \delta(a, x) \rightarrow \delta(c, \delta(a, x)).$$

Their composite is

$$(a \circ c, x).$$

The output of the composite is

$$\omega(a \circ c, x) = \omega(a, x) \circ \omega(c, \delta(a, x))$$

in $\mathbb{B}$. Passing to $\mathbb{B}^{\text{op}}$, this is exactly composition preservation:

$$\omega(c, \delta(a, x))^{\text{op}} \circ \omega(a, x)^{\text{op}} = \omega(a \circ c, x)^{\text{op}}.$$

4. **Display the opposite composition square.** The relevant square in $\mathbb{B}^{\text{op}}$ is

$$q(x) \xrightarrow{\omega(a, x)^{\text{op}}} q(\delta(a, x)) \xrightarrow{\omega(c, \delta(a, x))^{\text{op}}} q(\delta(c, \delta(a, x)))$$

□

2.4.3 Residual orbit functors

Definition 2.4.2 (Residual orbit). *Let $x \in P$ with $p(x) = v$. The **residual orbit** of x is the functor*

$$\rho_x : \mathbb{A}/v \rightarrow \left(\int_{\mathbb{A}} P \right)^{\text{op}}$$

defined as follows. On a residual object

$$r : u \rightarrow v,$$

it is

$$\rho_x(r) = \delta(r, x).$$

2.4.4 Residual behaviour of a state

Definition 2.4.3 (Residual behaviour of a state). *The **residual behaviour** of a state $x \in P$ is the composite*

$$H_x := \Omega_P^{\text{op}} \circ \rho_x : \mathbb{A}/p(x) \rightarrow \mathbb{B}.$$

Thus

$$H_x(r) = q(\delta(r, x)).$$

For a residual triangle

$$w \xrightarrow{a} u \xrightarrow{r} p(x),$$

the arrow value is

$$H_x(a, r) = \omega(a, \delta(r, x)) : q(\delta(r \circ a, x)) \rightarrow q(\delta(r, x)).$$

The residual behaviour is the composite of the orbit and output functor:

$$\begin{array}{ccc} \mathbb{A}/p(x) & \xrightarrow{\rho_x} & (\int_{\mathbb{A}} P)^{\text{op}} \xrightarrow{\Omega_P^{\text{op}}} \mathbb{B}. \\ & \searrow H_x & \nearrow \end{array}$$

Proposition 2.4.4. *For every $x \in P$, the assignment H_x is a residual behaviour. Moreover,*

$$H_x(1_{p(x)}) = q(x).$$

Proof.

1. **Use functoriality of the composite.** The first claim follows because H_x is the composite of two functors.
2. **Evaluate at the anchor.** The second follows from the Mealy unit law:

$$H_x(1_{p(x)}) = q(\delta(1_{p(x)}, x)) = q(x).$$

□

2.4.5 The behaviour assignment

Definition 2.4.4 (Behaviour assignment). *For a Mealy morphism P , the **behaviour assignment** is the map*

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

classified by the residual behaviours

$$x \longmapsto H_x.$$

Lemma 2.4.1 (Internal construction of β_P). *The assignment*

$$x \longmapsto H_x$$

defines a morphism

$$\beta_P : P \rightarrow \mathbf{Beh}_0.$$

Proof. Pull back the vertical residual category along

$$p : P \rightarrow A_0.$$

This gives an internal category

$$p^* \mathbb{R}_{\mathbb{A}}$$

in the slice

$$\mathcal{E}/P.$$

Its object object is

$$P \times_{p, A_0, t_A} A_1 \longrightarrow P,$$

whose generalized elements are pairs

$$(x, r), \quad r : u \rightarrow p(x).$$

Its arrow object is

$$P \times_{p, A_0, t_A \pi_2} T_A \longrightarrow P,$$

whose generalized elements are triples

$$(x, a, r), \quad w \xrightarrow{a} u \xrightarrow{r} p(x).$$

Such a triple represents the residual arrow

$$r \circ a \longrightarrow r.$$

Put

$$\Delta_P \mathbb{B} := p^* \Delta_{A_0} \mathbb{B}.$$

This is the constant internal category $\mathbb{B}$ in $\mathcal{E}/P$. Its object and arrow objects are

$$P \times B_0 \longrightarrow P, \quad P \times B_1 \longrightarrow P.$$

Define an object map in $\mathcal{E}/P$

$$\tilde{H}_{P,0} : P \times_{p, A_0, t_A} A_1 \longrightarrow P \times B_0$$

by

$$\tilde{H}_{P,0}(x, r) := (x, q(\delta(r, x))).$$

Define an arrow map in $\mathcal{E}/P$

$$\tilde{H}_{P,1} : P \times_{p, A_0, t_A \pi_2} T_A \longrightarrow P \times B_1$$

by

$$\tilde{H}_{P,1}(x, a, r) := (x, \omega(a, \delta(r, x))).$$

Both maps preserve the displayed P -coordinate and are therefore morphisms in $\mathcal{E}/P$.

We verify that they define an internal functor

$$\tilde{H}_P : p^* \mathbb{R}_{\mathbb{A}} \longrightarrow \Delta_P \mathbb{B}.$$

First, the source of the arrow

$$\omega(a, \delta(r, x))$$

is

$$\begin{aligned} s_B \omega(a, \delta(r, x)) &= q(\delta(a, \delta(r, x))) \\ &= q(\delta(r \circ a, x)), \end{aligned}$$

using the Mealy multiplication law for δ . This is the B_0 -coordinate of

$$\tilde{H}_{P,0}(x, r \circ a).$$

Its target is

$$t_B \omega(a, \delta(r, x)) = q(\delta(r, x)),$$

which is the B_0 -coordinate of

$$\tilde{H}_{P,0}(x, r).$$

Hence the source and target equations hold in $\mathcal{E}/P$.

Preservation of identities follows from the Mealy output unit law. The identity residual arrow at

$$r : u \rightarrow p(x)$$

is represented by

$$(1_u, r).$$

Therefore

$$\begin{aligned} \tilde{H}_{P,1}(x, 1_u, r) &= (x, \omega(1_u, \delta(r, x))) \\ &= (x, 1_{q(\delta(r, x))}), \end{aligned}$$

which is the identity at

$$\tilde{H}_{P,0}(x, r).$$

To verify preservation of composition, consider a residual string

$$z \xrightarrow{b} w \xrightarrow{a} u \xrightarrow{r} p(x).$$

The two residual arrows are

$$r \circ a \circ b \longrightarrow r \circ a$$

and

$$r \circ a \longrightarrow r,$$

and their composite is represented by

$$(a \circ b, r) : r \circ (a \circ b) \longrightarrow r.$$

The Mealy output multiplication law gives

$$\begin{aligned} \omega(a \circ b, \delta(r, x)) &= \omega(a, \delta(r, x)) \circ \omega(b, \delta(a, \delta(r, x))) \\ &= \omega(a, \delta(r, x)) \circ \omega(b, \delta(r \circ a, x)). \end{aligned}$$

Hence

$$\tilde{H}_{P,1}(x, a \circ b, r) = \tilde{H}_{P,1}(x, a, r) \circ \tilde{H}_{P,1}(x, b, r \circ a).$$

Thus

$$\tilde{H}_P : p^* \mathbb{R}_A \longrightarrow \Delta_P \mathbb{B}$$

is an internal functor in $\mathcal{E}/P$.

By Lemma 2.2.1, there is a canonical isomorphism of internal categories in $\mathcal{E}/P$

$$p^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \cong [p^* \mathbb{R}_{\mathbb{A}}, \Delta_P \mathbb{B}]_{\mathcal{E}/P}.$$

The internal functor $\tilde{H}_P$ therefore determines a section of the object object

$$p^* \mathbf{Beh}_0 \rightarrow P.$$

Equivalently, it determines a unique morphism

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

over A_0 .

Fibrewise, this morphism sends a state x to the residual behaviour

$$H_x : \mathbb{A}/p(x) \rightarrow \mathbb{B}$$

whose object and arrow assignments are

$$H_x(r) = q(\delta(r, x)),$$

and

$$H_x(a, r) = \omega(a, \delta(r, x)).$$

□

Proposition 2.4.5 (Semantic homomorphism). *The map*

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

lies over

$$Z = A_0 \times B_0.$$

It satisfies

$$\beta_P(\delta_P(a, x)) = \partial_a(\beta_P(x)),$$

and

$$\omega_P(a, x) = \omega_{\mathbf{Beh}}(a, \beta_P(x)).$$

Therefore

$$\beta_P : P \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a strict semantic Mealy homomorphism.

Proof.

1. **Check the map over Z .** Since H_x lies over $p(x)$ and satisfies

$$H_x(1_{p(x)}) = q(x),$$

the map β_P lies over Z . This is the commutativity of

$$\begin{array}{ccc} P & \xrightarrow{\beta_P} & \mathbf{Beh}_0 \\ & \searrow (p, q) & \downarrow (p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) \\ & & A_0 \times B_0. \end{array}$$

2. **Check transition preservation.** For $a : u \rightarrow v$ and $p(x) = v$, the residual-orbit identity gives

$$\rho_{\delta(a,x)} = \rho_x \circ a_*.$$

Therefore

$$\beta_P(\delta(a,x)) = H_{\delta(a,x)} = H_x \circ a_* = \partial_a H_x = \partial_a(\beta_P(x)).$$

3. **Check output preservation.** Finally,

$$\omega_{\mathbf{Beh}}(a, \beta_P(x)) = H_x(a, 1_v) = \omega_P(a, \delta_P(1_v, x)) = \omega_P(a, x).$$

□

2.5 Terminal residual semantics

The canonical behaviour machine is terminal among strict semantic Mealy homomorphisms. The terminality is formulated in an ordinary category whose morphisms preserve transition and output strictly. This section is in the same general line as universal-realization and final-semantics formulations of automata behaviour [7, 20, 21].

2.5.1 Strict semantic homomorphisms

Definition 2.5.1 (Strict semantic Mealy homomorphism). *Let*

$$P, Q : \mathbb{A} \rightsquigarrow \mathbb{B}$$

*be Mealy morphisms. A **strict semantic Mealy homomorphism***

$$f : P \rightarrow Q$$

is a map over

$$Z = A_0 \times B_0,$$

so that

$$p_Q f = p_P, \quad q_Q f = q_P,$$

satisfying

$$f(\delta_P(a, x)) = \delta_Q(a, f(x)),$$

and

$$\omega_P(a, x) = \omega_Q(a, f(x)).$$

The strict preservation equations are displayed by

$$\begin{array}{ccc} x & \xrightarrow{a, \omega_P(a,x)} & \delta_P(a, x) \\ f \downarrow & & \downarrow f \\ f(x) & \xrightarrow{a, \omega_Q(a, f(x))} & \delta_Q(a, f(x)). \end{array}$$

Lemma 2.5.1. *Strict semantic Mealy homomorphisms are closed under identities and composition.*

Proof.

1. **Check identities and composition.** This is immediate from the defining preservation equations.

□

Notation 2.5.1. Write

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

for the ordinary category whose objects are Mealy morphisms

$$\mathbb{A} \rightsquigarrow \mathbb{B}$$

and whose morphisms are strict semantic Mealy homomorphisms.

Remark 2.5.1. The terminality statement below is a statement in the ordinary category

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

The bicategory $\mathbf{Mly}(\mathcal{E})$ uses Mealy simulations as 2-cells, and those simulations have oppositely directed state maps.

2.5.2 Terminality of the canonical behaviour machine

Lemma 2.5.2. For the canonical behaviour Mealy morphism,

$$\beta_{\mathbf{Beh}^{\text{can}}} = 1_{\mathbf{Beh}_0}.$$

If

$$i : S \hookrightarrow \mathbf{Beh}_0$$

is derivative-closed and the canonical behaviour machine is restricted to S , then its behaviour assignment is the inclusion

$$S \hookrightarrow \mathbf{Beh}_0.$$

Proof.

1. **Compute the residual behaviour of a canonical state.** Let $H : \mathbb{A}/v \rightarrow \mathbb{B}$ be a state of the canonical behaviour machine. Its residual behaviour sends

$$r : u \rightarrow v$$

to

$$q_{\mathbf{Beh}}(\partial_r H) = H(r),$$

and sends a residual triangle

$$w \xrightarrow{a} u \xrightarrow{r} v$$

to

$$\omega_{\mathbf{Beh}}(a, \partial_r H) = H(a, r).$$

2. **Conclude identity of the behaviour map.** Thus the residual behaviour determined by H is H . The restricted statement follows by the same calculation after composing with the subobject inclusion.

□

Theorem 2.5.1 (Terminal residual semantics). *The canonical behaviour Mealy morphism*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is terminal in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

For every Mealy morphism P , the unique strict semantic homomorphism

$$P \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is

$$\beta_P : x \mapsto H_x.$$

Proof.

1. **Establish existence.** Existence is Proposition 2.4.5.

2. **Set up uniqueness.** For uniqueness, let

$$f : P \rightarrow \mathbf{Beh}_0$$

be a strict semantic Mealy homomorphism. We show $f = \beta_P$.

3. **Recover the object assignment.** Let $x \in P$, and let $r : u \rightarrow p(x)$ be a residual input. Since f preserves transitions,

$$f(\delta_P(r, x)) = \partial_r f(x).$$

Since f lies over Z ,

$$q_{\mathbf{Beh}}(f(\delta_P(r, x))) = q(\delta_P(r, x)).$$

But

$$q_{\mathbf{Beh}}(\partial_r f(x)) = (f(x))(r).$$

Hence

$$(f(x))(r) = q(\delta_P(r, x)) = H_x(r).$$

4. **Recover the arrow assignment.** Now let

$$w \xrightarrow{a} u \xrightarrow{r} p(x)$$

be a residual triangle. Strict output preservation at the state

$$\delta_P(r, x)$$

gives

$$\omega_P(a, \delta_P(r, x)) = \omega_{\mathbf{Beh}}(a, f(\delta_P(r, x))).$$

Using transition preservation,

$$f(\delta_P(r, x)) = \partial_r f(x).$$

Therefore

$$\omega_{\mathbf{Beh}}(a, \partial_r f(x)) = (\partial_r f(x))(a, 1_u) = (f(x))(a, r).$$

Hence

$$(f(x))(a, r) = \omega_P(a, \delta_P(r, x)) = H_x(a, r).$$

Thus $f(x) = H_x$ for every x , so $f = \beta_P$.

5. **Display the reconstruction diagrams.** The proof reconstructs the object and arrow assignments of $f(x)$ by the two diagrams

$$\begin{array}{ccc}
 \delta_P(r, x) & \xrightarrow{f} & \partial_r f(x) \\
 \downarrow q & & \downarrow q_{\mathbf{Beh}} \\
 q(\delta_P(r, x)) & \xlongequal{\quad} & (f(x))(r)
 \end{array}
 \qquad
 \begin{array}{ccc}
 q(\delta_P(r \circ a, x)) & \xrightarrow{\omega_P(a, \delta_P(r, x))} & q(\delta_P(r, x)) \\
 \parallel & & \parallel \\
 (f(x))(r \circ a) & \xrightarrow{(f(x))(a, r)} & (f(x))(r).
 \end{array}$$

□

2.5.3 Behavioural separatedness

Definition 2.5.2 (Behavioural separatedness). *A Mealy morphism*

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

is behaviourally separated if

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

is monic.

2.6 Simulations and induced behaviour comparisons

Simulations compare machines by output-comparison arrows. Under residual semantics, every Mealy simulation induces a vertical natural transformation between the corresponding residual behaviours.

2.6.1 Simulation-induced vertical behaviour comparisons

Theorem 2.6.1 (Simulation-induced vertical behaviour comparison). *Let*

$$(\theta, \alpha) : P \Rightarrow Q$$

be a Mealy simulation. For each $y \in Q$, the family

$$(\hat{\alpha}_y)_r := \alpha_{\delta_Q(r, y)}$$

defines a natural transformation

$$\hat{\alpha}_y : H_{\theta y}^P \Rightarrow H_y^Q.$$

Equivalently, the simulation induces a morphism

$$Q \rightarrow \mathbf{Beh}_1$$

whose source is

$$\beta_P \theta$$

and whose target is

$$\beta_Q.$$

Proof.

1. **Type the components.** The state law gives

$$\theta(\delta_Q(r, y)) = \delta_P(r, \theta y).$$

Hence

$$\alpha_{\delta_Q(r, y)} : q_P(\delta_P(r, \theta y)) \rightarrow q_Q(\delta_Q(r, y)),$$

which is exactly a component

$$H_{\theta y}^P(r) \rightarrow H_y^Q(r).$$

2. **Check naturality.** For a residual triangle

$$w \xrightarrow{a} u \xrightarrow{r} p_Q(y),$$

naturality requires

$$(\hat{\alpha}_y)_r \circ H_{\theta y}^P(a, r) = H_y^Q(a, r) \circ (\hat{\alpha}_y)_{r \circ a}.$$

This is the simulation output law applied to the state

$$\delta_Q(r, y).$$

3. **Display the naturality square.** The square is

$$\begin{array}{ccc} H_{\theta y}^P(r \circ a) & \xrightarrow{H_{\theta y}^P(a, r)} & H_{\theta y}^P(r) \\ \alpha_{\delta_Q(r \circ a, y)} \downarrow & & \downarrow \alpha_{\delta_Q(r, y)} \\ H_y^Q(r \circ a) & \xrightarrow{H_y^Q(a, r)} & H_y^Q(r). \end{array}$$

□

Remark 2.6.1. *The construction in Theorem 2.6.1 is a semantic consequence of a state-level simulation. A vertical behaviour comparison compares the residual behaviours after applying β_P and β_Q . A Mealy simulation also contains the state-level comparison data and satisfies the transition law before residual semantics is applied.*

2.7 Specifications and derivative images

A specification is a family of behaviours. Its formal derivative orbit is obtained by acting on that family by all residual input arrows through the cleavage of the behaviour fibration.

2.7.1 Specifications

Definition 2.7.1 (Behaviour specification). *A **behaviour specification** is a morphism*

$$k : K \rightarrow \mathbf{Beh}_0.$$

Write

$$p_K := p_{\mathbf{Beh}} k.$$

We regard K as an object of

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0,$$

through the composite

$$K \xrightarrow{k} \mathbf{Beh}_0 \xrightarrow{(p_{\mathbf{Beh}}, q_{\mathbf{Beh}})} Z.$$

2.7.2 Formal derivative orbits

Definition 2.7.2 (Formal derivative orbit). *The **formal derivative orbit object** of K is*

$$D_K := A_1 \times_{t_A, A_0, p_K} K.$$

A generalized element is a pair

$$(a, \kappa),$$

where

$$a : u \rightarrow p_K(\kappa).$$

Its defining pullback is

$$\begin{array}{ccc} D_K & \xrightarrow{\pi_K} & K \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p_K \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

Definition 2.7.3 (Residual evaluation). *The **residual evaluation map** is*

$$d_K : D_K \rightarrow \mathbf{Beh}_0$$

defined by

$$d_K(a, \kappa) = \partial_a k(\kappa).$$

Equivalently,

$$d_K = \partial \circ (1_{A_1} \times k).$$

The object D_K is regarded as an object over

$$Z = A_0 \times B_0$$

through the composite

$$D_K \xrightarrow{d_K} \mathbf{Beh}_0 \xrightarrow{(p_{\mathbf{Beh}}, q_{\mathbf{Beh}})} Z.$$

Thus the Z -coordinate of (a, κ) is the input interface and anchor of the derivative behaviour

$$\partial_a k(\kappa).$$

We write

$$p_{D_K} := p_{\mathbf{Beh}} d_K.$$

For a generalized element (a, κ) , with

$$a : u \rightarrow p_K(\kappa),$$

one has

$$p_{D_K}(a, \kappa) = u = s_A(a).$$

Definition 2.7.4 (Unit inclusion of syntax). *The **unit inclusion of syntax***

$$\iota_K : K \rightarrow D_K$$

is

$$\kappa \longmapsto (1_{p_K(\kappa)}, \kappa).$$

It satisfies

$$d_K \iota_K = k.$$

2.7.3 Derivative images

Lemma 2.7.1 (Regular-epi descent for subobjects). *Let $\mathcal{E}$ be regular. If*

$$e : X \twoheadrightarrow Y$$

is a regular epimorphism and

$$m : S \hookrightarrow Z$$

is a subobject, then a map

$$g : Y \rightarrow Z$$

factors through m whenever

$$ge : X \rightarrow Z$$

factors through m .

Proof.

1. **Use the kernel pair of the regular epimorphism.** Suppose $ge = mh$. Let

$$K_e \rightrightarrows X$$

be the kernel pair of e . Since e is the coequalizer of its kernel pair,

$$ge\pi_1 = ge\pi_2.$$

Hence

$$mh\pi_1 = mh\pi_2.$$

2. **Descend through the monomorphism.** Since m is monic,

$$h\pi_1 = h\pi_2.$$

Thus h descends uniquely through e , giving

$$\bar{h} : Y \rightarrow S$$

with

$$\bar{h}e = h.$$

3. **Recover the factorization of g .** Since e is epi,

$$m\bar{h} = g.$$

The descent diagram is

$$\begin{array}{ccccc}
 K_e & \xrightleftharpoons[\pi_2]{\pi_1} & X & \xrightarrow{e} & Y \\
 & & \downarrow h & \swarrow \bar{h} & \downarrow g \\
 & & S & \xrightarrow{m} & Z
 \end{array}$$

□

Since $\mathcal{E}$ is regular, the map

$$d_K : D_K \rightarrow \mathbf{Beh}_0$$

has a regular epi–mono factorization in the slice

$$\mathcal{E}/Z.$$

Write it as

$$D_K \xrightarrow{e_K} \mathrm{Der}_0(K) \xrightarrow{j_K} \mathbf{Beh}_0,$$

where e_K is a regular epimorphism and j_K is monic in $\mathcal{E}/Z$:

$$\begin{array}{ccc} D_K & \xrightarrow{d_K} & \mathbf{Beh}_0 \\ & \searrow e_K & \nearrow j_K \\ & \mathrm{Der}_0(K) & \end{array} \quad \text{in } \mathcal{E}/Z.$$

Definition 2.7.5 (Derivative image). *The object*

$$\mathrm{Der}_0(K)$$

*is the **derivative image**, or **object-level residual closure**, of K .*

2.7.4 Derivative closure

Theorem 2.7.1 (Derivative closure). *The subobject*

$$j_K : \mathrm{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

is the smallest derivative-closed subobject of $\mathbf{Beh}_0$ in $\mathcal{E}/Z$ containing the image of

$$k : K \rightarrow \mathbf{Beh}_0.$$

Proof. Let

$$D_K \xrightarrow{e_K} \mathrm{Der}_0(K) \xrightarrow{j_K} \mathbf{Beh}_0$$

be the regular epi–mono factorization of d_K in $\mathcal{E}/Z$.

1. The image contains the specified behaviours.

The unit inclusion

$$\iota_K : K \rightarrow D_K$$

satisfies

$$d_K \iota_K = k.$$

Since $d_K = j_K e_K$, one has

$$k = j_K e_K \iota_K.$$

Hence the image of k is contained in the subobject j_K .

2. Form the derivative domain of the image.

Let

$$D(\text{Der}_0(K)) = A_1 \times_{t_A, A_0, p_{\text{Der}}} \text{Der}_0(K),$$

where

$$p_{\text{Der}} := p_{\mathbf{Beh}} j_K.$$

Consider the derivative composite

$$g := \partial(1_{A_1} \times j_K) : D(\text{Der}_0(K)) \rightarrow \mathbf{Beh}_0.$$

Pull back the underlying regular epimorphism

$$e_K : D_K \twoheadrightarrow \text{Der}_0(K)$$

along the projection

$$D(\text{Der}_0(K)) \rightarrow \text{Der}_0(K)$$

in $\mathcal{E}$. Stability of regular epimorphisms under pullback gives a regular epimorphism

$$e'_K : \tilde{D}_K \twoheadrightarrow D(\text{Der}_0(K)).$$

By the universal property of the relevant pullbacks, there is a canonical isomorphism

$$\tilde{D}_K \cong A_1 \times_{t_A, A_0, p_{D_K}} D_K,$$

where

$$p_{D_K} := p_{\mathbf{Beh}} d_K.$$

We use this canonical identification below.

3. Define residual concatenation on the cover.

Define

$$\chi_K : A_1 \times_{t_A, A_0, p_{D_K}} D_K \rightarrow D_K$$

by

$$\chi_K(c, (a, \kappa)) := (a \circ c, \kappa).$$

In morphism notation, its first component is induced by the multiplication

$$m_A : A_1 \times_{t_A, A_0, s_A} A_1 \rightarrow A_1,$$

since

$$a \circ c = m_A(c, a),$$

and its second component is κ .

4. Compute the derivative on the regular-epimorphic cover.

The split derivative law gives the morphism equation

$$g e'_K = j_K e_K \chi_K.$$

Indeed, in generalized-element notation,

$$\begin{aligned}
 ge'_K(c, a, \kappa) &= \partial_c(j_K e_K(a, \kappa)) \\
 &= \partial_c(d_K(a, \kappa)) \\
 &= \partial_c(\partial_a k(\kappa)) \\
 &= \partial_{a \circ c} k(\kappa) \\
 &= d_K(a \circ c, \kappa) \\
 &= j_K e_K \chi_K(c, a, \kappa).
 \end{aligned}$$

Therefore ge'_K factors through j_K .

5. Descend the factorization.

Since e'_K is a regular epimorphism and j_K is monic, Lemma 2.7.1 gives a unique map

$$\bar{\partial}_K : D(\text{Der}_0(K)) \rightarrow \text{Der}_0(K)$$

such that

$$j_K \bar{\partial}_K = g.$$

This map lies over Z : after composing with the monomorphism j_K , both composites to Z agree, and j_K is a morphism over Z . Thus j_K is derivative-closed.

6. Prove minimality.

Let

$$i : S \hookrightarrow \mathbf{Beh}_0$$

be derivative-closed and suppose that k factors through i :

$$k = i \bar{k}.$$

Derivative-closedness gives a factorization

$$\partial(1_{A_1} \times i) = i \bar{\partial}_S.$$

Consequently,

$$\begin{aligned}
 d_K &= \partial(1_{A_1} \times k) \\
 &= \partial(1_{A_1} \times i)(1_{A_1} \times \bar{k}) \\
 &= i \bar{\partial}_S(1_{A_1} \times \bar{k}).
 \end{aligned}$$

Hence d_K factors through i . Since j_K is the regular image of d_K , the image subobject j_K factors through i .

Therefore j_K is the smallest derivative-closed subobject containing the image of k .

□

2.8 The Categorical Myhill–Nerode Theorem

We now assemble the equality-level Nerode relation, the derivative image, the minimal residual realization, and the reachability–separatedness criterion. The result is a categorical Myhill–Nerode theorem for the residual semantics developed above. Related categorical minimization theorems for automata and languages appear in functorial and topos-theoretic settings [5, 9]. The theorem below is stated for Mealy morphisms between internal categories and uses the equality of represented residual behaviours.

2.8.1 The Nerode relation

Definition 2.8.1 (Nerode relation). *The **Nerode relation***

$$\equiv_K \rightrightarrows D_K$$

is the kernel pair of

$$d_K : D_K \rightarrow \mathbf{Beh}_0$$

in the slice

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0.$$

Thus

$$(a, \kappa) \equiv_K (a', \kappa')$$

precisely when

$$\partial_a k(\kappa) = \partial_{a'} k(\kappa')$$

as represented residual behaviours.

The kernel-pair square is

$$\begin{array}{ccc} \equiv_K & \xrightarrow{\pi_2} & D_K \\ \pi_1 \downarrow & \lrcorner & \downarrow d_K \\ D_K & \xrightarrow{d_K} & \mathbf{Beh}_0. \end{array}$$

Remark 2.8.1. *This is an equality-level Nerode relation. It identifies formal residual derivatives exactly when they determine the same represented residual functor, including both object and arrow assignments. It does not quotient by isomorphism, by behaviour comparison, or by equivalence inside the vertical behaviour category.*

2.8.2 The Nerode quotient

Theorem 2.8.1 (Categorical Myhill–Nerode quotient). *The quotient*

$$D_K / \equiv_K$$

exists and is effective in

$$\mathcal{E}/Z.$$

Moreover, there is a canonical isomorphism

$$D_K / \equiv_K \cong \mathrm{Der}_0(K)$$

in $\mathcal{E}/Z$.

Proof.

1. **Use exactness of the slice.** Since $\mathcal{E}$ is Barr-exact, the slice $\mathcal{E}/Z$ is Barr-exact. Hence the kernel pair of d_K has an effective quotient.
2. **Identify the quotient with the image.** In a Barr-exact category, the quotient of the kernel pair of a morphism is its regular image. Therefore

$$D_K / \equiv_K \cong \mathrm{Im}(d_K) = \mathrm{Der}_0(K).$$

3. **Display the comparison.** The comparison is represented by the diagram

$$\begin{array}{ccc}
 \equiv_K & \xRightarrow{\quad} & D_K \xrightarrow{\quad} D_K / \equiv_K \\
 & & \searrow e_K \quad \downarrow \cong \\
 & & \text{Der}_0(K).
 \end{array}$$

□

2.8.3 The minimal residual realization

Definition 2.8.2 (Minimal residual realization). *The **minimal residual realization** of a specification*

$$k : K \rightarrow \mathbf{Beh}_0$$

is the restriction of the canonical behaviour Mealy morphism to the derivative-closed image

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0.$$

It is denoted

$$\text{Min}(K) : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

The minimal realization carries a canonical realization map

$$i_{\min} : K \rightarrow \text{Der}_0(K)$$

defined by

$$i_{\min} := e_K \iota_K.$$

It satisfies

$$j_K i_{\min} = j_K e_K \iota_K = d_K \iota_K = k.$$

2.8.4 Realizations and reachability

Definition 2.8.3 (Realization). *A **realization** of a behaviour specification*

$$k : K \rightarrow \mathbf{Beh}_0$$

is a Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

together with a map

$$i : K \rightarrow P$$

such that

$$\beta_P i = k.$$

Remark 2.8.2. *Since β_P and k both lie over $Z = A_0 \times B_0$, the equation*

$$\beta_P i = k$$

implies that i is a map over Z .

Definition 2.8.4 (Reachability map). *Given a realization*

$$i : K \rightarrow P,$$

*the **reachability map***

$$\rho_{P,K} : D_K \rightarrow P$$

is defined by

$$\rho_{P,K}(a, \kappa) = \delta_P(a, i(\kappa)).$$

Lemma 2.8.1. *The reachability map satisfies*

$$\beta_P \rho_{P,K} = d_K.$$

Consequently, $\rho_{P,K}$ is a map over Z , where D_K is viewed over Z through d_K .

Proof.

1. **Compute the behaviour of a reachable state.** For $(a, \kappa) \in D_K$,

$$\begin{aligned} \beta_P(\rho_{P,K}(a, \kappa)) &= \beta_P(\delta_P(a, i(\kappa))) \\ &= \partial_a(\beta_P(i(\kappa))) \\ &= \partial_a k(\kappa) \\ &= d_K(a, \kappa). \end{aligned}$$

2. **Display the commutative triangle.** This is the commutativity of

$$\begin{array}{ccc} D_K & \xrightarrow{\rho_{P,K}} & P \\ & \searrow d_K & \downarrow \beta_P \\ & & \mathbf{Beh}_0. \end{array}$$

□

Definition 2.8.5 (Transition-closed subobject). *Let*

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be a Mealy morphism, and let

$$m : S \hookrightarrow P$$

be a subobject in $\mathcal{E}/Z$, where $Z = A_0 \times B_0$. Write

$$p_S := p_P m.$$

*The **transition domain** of S is*

$$\mathsf{T}(S) := A_1 \times_{t_A, A_0, p_S} S.$$

We regard $\mathsf{T}(S)$ as an object of $\mathcal{E}/Z$ through the composite

$$\mathsf{T}(S) \xrightarrow{1_{A_1} \times m} A_1 \times_{t_A, A_0, p_P} P \xrightarrow{\delta_P} P \xrightarrow{(p_P, q_P)} Z.$$

We say that S is **transition-closed** if the transition map

$$\delta_P(1_{A_1} \times m) : T(S) \rightarrow P$$

factors through m in $\mathcal{E}/Z$. Equivalently, there is a unique map

$$\bar{\delta}_S : T(S) \rightarrow S$$

over Z such that

$$m\bar{\delta}_S = \delta_P(1_{A_1} \times m).$$

The factorization condition is

$$\begin{array}{ccc} T(S) & \xrightarrow{\delta_P(1_{A_1} \times m)} & P \\ & \searrow \bar{\delta}_S & \nearrow m \\ & S & \end{array} \quad \text{in } \mathcal{E}/Z.$$

Lemma 2.8.2 (Restriction to transition-closed subobjects). *Let*

$$m : S \hookrightarrow P$$

be a transition-closed subobject of a Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

in $\mathcal{E}/Z$. Then S , with the restricted carrier

$$A_0 \xleftarrow{p_P m} S \xrightarrow{q_P m} B_0,$$

carries a canonical restricted Mealy morphism structure.

Its transition is the factorization

$$\bar{\delta}_S : A_1 \times_{t_A, A_0, p_P m} S \rightarrow S$$

supplied by transition-closedness, and its output is the restriction of

$$\omega_P : A_1 \times_{t_A, A_0, p_P} P \rightarrow B_1$$

along

$$1_{A_1} \times m.$$

Proof. Write

$$p_S := p_P m, \quad q_S := q_P m.$$

Transition-closedness supplies a unique map

$$\bar{\delta}_S : A_1 \times_{t_A, A_0, p_S} S \rightarrow S$$

satisfying

$$m\bar{\delta}_S = \delta_P(1_{A_1} \times m).$$

Define

$$\delta_S := \bar{\delta}_S.$$

Define the restricted output by

$$\omega_S := \omega_P \circ (1_{A_1} \times m).$$

In generalized-element notation,

$$\delta_S(a, s) = \bar{\delta}_S(a, s),$$

and

$$\omega_S(a, s) = \omega_P(a, m(s)).$$

Since $\bar{\delta}_S$ is a morphism over the transition-induced Z -structure,

$$p_S(\delta_S(a, s)) = s_A(a).$$

The output source equation is

$$\begin{aligned} s_B \omega_S(a, s) &= s_B \omega_P(a, m(s)) \\ &= q_P(\delta_P(a, m(s))) \\ &= q_P(m(\delta_S(a, s))) \\ &= q_S(\delta_S(a, s)). \end{aligned}$$

The output target equation is

$$\begin{aligned} t_B \omega_S(a, s) &= t_B \omega_P(a, m(s)) \\ &= q_P(m(s)) \\ &= q_S(s). \end{aligned}$$

For the transition unit law, compose with m :

$$\begin{aligned} m(\delta_S(1_{p_S(s)}, s)) &= \delta_P(1_{p_P(m(s))}, m(s)) \\ &= m(s). \end{aligned}$$

Since m is monic,

$$\delta_S(1_{p_S(s)}, s) = s.$$

For the transition multiplication law, let

$$a : u \rightarrow v, \quad b : v \rightarrow w, \quad p_S(s) = w.$$

After composing with m ,

$$\begin{aligned} m(\delta_S(b \circ a, s)) &= \delta_P(b \circ a, m(s)) \\ &= \delta_P(a, \delta_P(b, m(s))) \\ &= \delta_P(a, m(\delta_S(b, s))) \\ &= m(\delta_S(a, \delta_S(b, s))). \end{aligned}$$

Hence, by monicity,

$$\delta_S(b \circ a, s) = \delta_S(a, \delta_S(b, s)).$$

The output unit law is the restriction of the output unit law of P :

$$\begin{aligned} \omega_S(1_{p_S(s)}, s) &= \omega_P(1_{p_P(m(s))}, m(s)) \\ &= 1_{q_P(m(s))} \\ &= 1_{q_S(s)}. \end{aligned}$$

For the output multiplication law,

$$\begin{aligned}
 \omega_S(b \circ a, s) &= \omega_P(b \circ a, m(s)) \\
 &= \omega_P(b, m(s)) \circ \omega_P(a, \delta_P(b, m(s))) \\
 &= \omega_P(b, m(s)) \circ \omega_P(a, m(\delta_S(b, s))) \\
 &= \omega_S(b, s) \circ \omega_S(a, \delta_S(b, s)).
 \end{aligned}$$

Therefore the restricted carrier, transition, and output define a Mealy morphism

$$\mathbb{A} \rightsquigarrow \mathbb{B}.$$

By construction, the inclusion

$$m : S \rightarrow P$$

preserves transition and output strictly. □

Definition 2.8.6 (Reachable part). *The **reachable part***

$$\text{Reach}_0(P, K) \hookrightarrow P$$

is the regular image of

$$\rho_{P,K} : D_K \rightarrow P$$

in the slice

$$\mathcal{E}/Z.$$

Proposition 2.8.1 (Reachable part). *Let*

$$i : K \rightarrow P$$

be a realization of

$$k : K \rightarrow \mathbf{Beh}_0.$$

Let

$$D_K \xrightarrow{e_R} \text{Reach}_0(P, K) \xrightarrow{m_R} P$$

be the regular epi–mono factorization of the reachability map

$$\rho_{P,K} : D_K \rightarrow P$$

in $\mathcal{E}/Z$.

Then:

1. *the subobject*

$$m_R : \text{Reach}_0(P, K) \hookrightarrow P$$

is transition-closed;

2. *the restricted Mealy structure makes m_R a strict semantic Mealy homomorphism;*

3. *the map*

$$i_R := e_R \iota_K : K \rightarrow \text{Reach}_0(P, K)$$

is a realization of k ;

4. the realization i_R is reachable;
5. there is a regular epimorphic strict semantic homomorphism

$$h : \text{Reach}_0(P, K) \rightarrow \text{Der}_0(K)$$

satisfying

$$j_K h = \beta_P m_R.$$

Proof. 1. **Prove transition-closedness.**

Put

$$p_{\text{Reach}} := p_P m_R.$$

Consider the transition composite

$$g : A_1 \times_{t_A, A_0, p_{\text{Reach}}} \text{Reach}_0(P, K) \rightarrow P$$

defined by

$$g := \delta_P(1_{A_1} \times m_R).$$

Pull back the underlying regular epimorphism

$$e_R : D_K \twoheadrightarrow \text{Reach}_0(P, K)$$

along the projection

$$A_1 \times_{t_A, A_0, p_{\text{Reach}}} \text{Reach}_0(P, K) \rightarrow \text{Reach}_0(P, K)$$

in $\mathcal{E}$. This gives a regular epimorphism

$$e'_R : \tilde{D}_R \twoheadrightarrow A_1 \times_{t_A, A_0, p_{\text{Reach}}} \text{Reach}_0(P, K).$$

There is a canonical pullback isomorphism

$$\tilde{D}_R \cong A_1 \times_{t_A, A_0, p_{D_K}} D_K.$$

We use this identification below.

Define

$$\chi_{P,K} : A_1 \times_{t_A, A_0, p_{D_K}} D_K \rightarrow D_K$$

by

$$\chi_{P,K}(c, (a, \kappa)) = (a \circ c, \kappa).$$

The Mealy multiplication law for δ_P gives

$$ge'_R = m_R e_R \chi_{P,K}.$$

In generalized-element notation,

$$\begin{aligned} ge'_R(c, a, \kappa) &= \delta_P(c, m_R e_R(a, \kappa)) \\ &= \delta_P(c, \rho_{P,K}(a, \kappa)) \\ &= \delta_P(c, \delta_P(a, i(\kappa))) \\ &= \delta_P(a \circ c, i(\kappa)) \\ &= \rho_{P,K}(a \circ c, \kappa) \\ &= m_R e_R \chi_{P,K}(c, a, \kappa). \end{aligned}$$

Hence ge'_R factors through m_R . Since e'_R is a regular epimorphism and m_R is monic, Lemma 2.7.1 supplies a unique map

$$\bar{\delta}_R : A_1 \times_{t_{A,A_0,p_{\text{Reach}}}} \text{Reach}_0(P, K) \rightarrow \text{Reach}_0(P, K)$$

such that

$$m_R \bar{\delta}_R = g.$$

Thus m_R is transition-closed.

2. Restrict the Mealy structure.

By Lemma 2.8.2, the reachable part carries the restricted Mealy structure. Its transition is $\bar{\delta}_R$, and its output is the restriction of ω_P . By construction,

$$m_R : \text{Reach}_0(P, K) \rightarrow P$$

preserves transition and output strictly and lies over Z . Hence it is a strict semantic Mealy homomorphism.

3. Construct the realization map.

Define

$$i_R := e_R \iota_K.$$

Since

$$m_R e_R = \rho_{P,K},$$

the Mealy unit law gives

$$\begin{aligned} m_R i_R &= m_R e_R \iota_K \\ &= \rho_{P,K} \iota_K \\ &= i. \end{aligned}$$

The composite

$$\beta_P m_R : \text{Reach}_0(P, K) \rightarrow \mathbf{Beh}_0$$

is a strict semantic homomorphism. By terminality of the canonical behaviour machine,

$$\beta_{\text{Reach}} = \beta_P m_R.$$

Therefore

$$\begin{aligned} \beta_{\text{Reach}} i_R &= \beta_P m_R e_R \iota_K \\ &= \beta_P \rho_{P,K} \iota_K \\ &= d_K \iota_K \\ &= k. \end{aligned}$$

Thus i_R is a realization of k .

4. Show that the restricted realization is reachable.

Let

$$\rho_{R,K} : D_K \rightarrow \text{Reach}_0(P, K)$$

be the reachability map of the realization i_R . After composing with m_R , one has

$$\begin{aligned} m_R \rho_{R,K}(a, \kappa) &= \delta_P(a, m_R i_R(\kappa)) \\ &= \delta_P(a, i(\kappa)) \\ &= \rho_{P,K}(a, \kappa) \\ &= m_R e_R(a, \kappa). \end{aligned}$$

Since m_R is monic,

$$\rho_{R,K} = e_R.$$

The map e_R is a regular epimorphism, so its image is all of $\text{Reach}_0(P, K)$. Hence i_R is reachable.

5. Construct the quotient map to the derivative image.

Let

$$D_K \xrightarrow{e_K} \text{Der}_0(K) \xrightarrow{j_K} \mathbf{Beh}_0$$

be the image factorization of d_K . By Lemma 2.8.1,

$$\beta_P m_R e_R = \beta_P \rho_{P,K} = d_K = j_K e_K.$$

Let

$$R_{e_R} \rightrightarrows D_K$$

be the kernel pair of e_R , with projections π_1, π_2 . Then

$$\begin{aligned} j_K e_K \pi_1 &= \beta_P m_R e_R \pi_1 \\ &= \beta_P m_R e_R \pi_2 \\ &= j_K e_K \pi_2. \end{aligned}$$

Since j_K is monic,

$$e_K \pi_1 = e_K \pi_2.$$

Since e_R is the coequalizer of its kernel pair, there is a unique map

$$h : \text{Reach}_0(P, K) \rightarrow \text{Der}_0(K)$$

such that

$$h e_R = e_K.$$

Since both $j_K h$ and $\beta_P m_R$ agree after precomposition with the epimorphism e_R ,

$$j_K h = \beta_P m_R.$$

Factor h as

$$\text{Reach}_0(P, K) \twoheadrightarrow I \rightarrow \text{Der}_0(K)$$

in $\mathcal{E}/Z$. The regular epimorphism

$$e_K = h e_R$$

factors through the displayed monomorphism $I \rightarrow \text{Der}_0(K)$. Since regular epimorphisms are extremal, that monomorphism is an isomorphism. Hence h is a regular epimorphism.

Finally, h is a strict semantic homomorphism because

$$j_K h = \beta_P m_R,$$

the right-hand side is strict, and j_K is the strict inclusion of a derivative-closed canonical submachine. □

Definition 2.8.7 (Reachable realization). *A realization $i : K \rightarrow P$ is **reachable** if*

$$\text{Reach}_0(P, K) \hookrightarrow P$$

is an isomorphism.

2.8.5 Semantic quotient and minimality

Theorem 2.8.2 (Semantic quotient theorem). *Let*

$$i : K \rightarrow P$$

be a realization of

$$k : K \rightarrow \mathbf{Beh}_0.$$

The behaviour assignment restricts on the reachable part to a strict semantic homomorphism

$$\bar{\beta}_{P,K} : \text{Reach}_0(P, K) \rightarrow \text{Min}(K).$$

This map is a regular epimorphism in $\mathcal{E}/Z$. If P is reachable, then

$$\bar{\beta}_{P,K} : P \twoheadrightarrow \text{Min}(K)$$

is a regular epimorphic strict semantic homomorphism.

Proof.

1. **Define the restricted semantic map.** By Proposition 2.8.1, there is a regular epimorphism

$$h : \text{Reach}_0(P, K) \twoheadrightarrow \text{Der}_0(K)$$

satisfying

$$j_K h = \beta_P m_R.$$

Define

$$\bar{\beta}_{P,K} := h.$$

2. **Restrict the Mealy structures.** Since the reachable part is transition-closed and $\text{Der}_0(K)$ is derivative-closed, the source carries its restricted Mealy structure by Lemma 2.8.2, and the target carries its restricted Mealy structure by Lemma 2.3.2.
3. **Check strict preservation.** The map $\bar{\beta}_{P,K}$ is a strict semantic homomorphism because it is the restriction of the strict semantic homomorphism

$$\beta_P : P \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

For example, transition preservation follows after composing with j_K : both sides become the corresponding transition-preservation equation for β_P . Since j_K is monic, the equality holds in $\text{Der}_0(K)$. The output equation is inherited directly from the restricted output structure.

4. **Conclude regular epimorphy.** Its regular epimorphy is the regular epimorphy of h . If P is reachable, then $\text{Reach}_0(P, K) \cong P$, giving the final statement. $\square$

Definition 2.8.8 (Separated realization). *A realization $i : K \rightarrow P$ is **behaviourally separated** if the Mealy morphism P is behaviourally separated, equivalently if*

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

is monic.

Theorem 2.8.3 (Minimal realization theorem). *If a realization*

$$i : K \rightarrow P$$

is reachable and behaviourally separated, then the canonical quotient map

$$\bar{\beta}_{P,K} : P \rightarrow \text{Min}(K)$$

is an isomorphism of Mealy morphisms. Moreover,

$$\bar{\beta}_{P,K} i = i_{\min}.$$

Hence every reachable separated realization of K is isomorphic to

$$\text{Min}(K)$$

as a realization.

Proof.

1. **Use reachability to get a regular epimorphism.** By Theorem 2.8.2, reachability gives a regular epimorphic strict semantic homomorphism

$$\bar{\beta}_{P,K} : P \twoheadrightarrow \text{Min}(K).$$

2. **Use separatedness to get a monomorphism.** Let

$$j_K : \text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

be the image inclusion. Then

$$j_K \bar{\beta}_{P,K} = \beta_P.$$

Since P is behaviourally separated, β_P is monic. Since j_K is monic, $\bar{\beta}_{P,K}$ is monic. A morphism that is both regular epimorphic and monic in a regular category is an isomorphism.

3. **Promote the isomorphism to Mealy morphisms.** Since $\bar{\beta}_{P,K}$ is a strict semantic homomorphism and an isomorphism over Z , its inverse preserves transitions and outputs by applying the inverse to the strict preservation equations. Hence it is an isomorphism of Mealy morphisms.

4. **Compare realization maps.** Finally,

$$j_K \bar{\beta}_{P,K} i = \beta_P i = k = j_K i_{\min}.$$

Since j_K is monic,

$$\bar{\beta}_{P,K} i = i_{\min}.$$

$\square$

2.8.6 The categorical theorem

Theorem 2.8.4 (Categorical Myhill–Nerode theorem). *Let*

$$k : K \rightarrow \mathbf{Beh}_0$$

be a behaviour specification. Then:

1. *the equality-level Nerode quotient*

$$D_K / \equiv_K$$

exists effectively in

$$\mathcal{E} / (A_0 \times B_0);$$

2. *there is a canonical isomorphism*

$$D_K / \equiv_K \cong \text{Der}_0(K);$$

3. *under this isomorphism, the quotient carries the restricted canonical Mealy structure;*

4. *this restricted Mealy morphism is the minimal residual realization*

$$\text{Min}(K);$$

5. *every realization $i : K \rightarrow P$ has a reachable part mapping by a regular epimorphic strict semantic homomorphism onto*

$$\text{Min}(K);$$

6. *every reachable behaviourally separated realization of K is isomorphic to*

$$\text{Min}(K)$$

as a realization.

Proof.

1. **Apply the quotient theorem.** The first two claims are Theorem 2.8.1.
2. **Restrict the canonical Mealy structure.** By Theorem 2.7.1,

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

is derivative-closed. Therefore, by Lemma 2.3.2, the canonical behaviour Mealy morphism restricts to $\text{Der}_0(K)$. This restricted Mealy morphism is $\text{Min}(K)$ by Definition 2.8.2.

3. **Apply the semantic quotient and minimality theorems.** The regular epimorphic quotient from the reachable part of any realization onto $\text{Min}(K)$ is Theorem 2.8.2. The uniqueness of reachable separated realizations is Theorem 2.8.3.

□

2.8.7 Finite-state corollary

Definition 2.8.9 (Finite-state structure). A **finite-state structure** on $\mathcal{E}$ is a choice, for every $Z \in \mathcal{E}$, of a replete full subcategory

$$\mathbf{Fin}_Z \subseteq \mathcal{E}/Z$$

closed under:

1. isomorphisms;
2. subobjects;
3. regular images;
4. effective quotients of internal equivalence relations.

For the remainder of this subsection, put

$$Z = A_0 \times B_0.$$

Definition 2.8.10 (Finite-state realization). A realization with carrier

$$P \rightarrow Z$$

is **finite-state** if

$$P \rightarrow Z$$

lies in

$$\mathbf{Fin}_Z.$$

Corollary 2.8.1 (Finite-state Myhill–Nerode theorem). Let

$$k : K \rightarrow \mathbf{Beh}_0$$

be a behaviour specification. The following are equivalent:

1. K has a finite-state realization.
2. K has a finite-state reachable behaviourally separated realization.
3. The carrier

$$\mathbf{Der}_0(K) \rightarrow Z$$

of

$$\mathbf{Min}(K)$$

lies in $\mathbf{Fin}_Z$.

4. The Nerode quotient

$$D_K / \equiv_K$$

lies in $\mathbf{Fin}_Z$.

Proof.

1. **Derive finiteness of the minimal carrier from a finite realization.** Suppose K has a finite-state realization $i : K \rightarrow P$. The reachable part

$$\text{Reach}_0(P, K) \hookrightarrow P$$

is a subobject of P in $\mathcal{E}/Z$, hence lies in Fin_Z . By Theorem 2.8.2, there is a regular epimorphism

$$\text{Reach}_0(P, K) \twoheadrightarrow \text{Min}(K)$$

in $\mathcal{E}/Z$. Closure under regular images gives

$$\text{Der}_0(K) \in \text{Fin}_Z.$$

Thus (1) implies (3).

2. **Show the minimal realization is reachable and separated.** If

$$\text{Der}_0(K) \in \text{Fin}_Z,$$

then $\text{Min}(K)$ is finite-state. Its realization map is

$$i_{\min} : K \rightarrow \text{Der}_0(K).$$

The reachability map of this realization has image

$$\text{Der}_0(K),$$

because its composite with the inclusion

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

is

$$(a, \kappa) \mapsto \partial_a k(\kappa) = d_K(a, \kappa).$$

Hence $\text{Min}(K)$ is reachable. It is behaviourally separated because, by Lemma 2.5.2, its behaviour map is the monomorphism

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0.$$

Thus (3) implies (2).

3. **Conclude the remaining implications.** The implication (2) implies (1) is immediate. Finally, Theorem 2.8.1 gives

$$D_K / \equiv_K \cong \text{Der}_0(K)$$

in $\mathcal{E}/Z$, so (3) and (4) are equivalent. □

Remark 2.8.3. *In the proof above, finite realizability implies finiteness of the minimal carrier using closure under subobjects and regular images. Closure under effective quotients ensures that, when a formal derivative object lies in the finite-state structure, its Nerode quotient remains in the same finite-state structure.*

Remark 2.8.4. *For finite categorical semantics, one may also require the arrow object*

$$\mathbf{Der}(K)_1$$

to lie in the chosen finite class over

$$\text{Der}_0(K) \times_{A_0} \text{Der}_0(K).$$

This records finiteness of comparison structure as well as finiteness of residual states.

2.9 Generated residual behaviour categories

The minimal Mealy carrier records the object image of the derivative orbit. The vertical behaviour category also records behaviour comparisons between generated residual behaviours.

2.9.1 Derivative-closed full internal subcategories

Definition 2.9.1 (Derivative-closed full internal subcategory). *A full internal subcategory*

$$\mathbb{S} \subseteq \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

is derivative-closed if its object subobject

$$S_0 \hookrightarrow \mathbf{Beh}_0$$

is derivative-closed in the sense of Definition 2.3.4.

2.9.2 The generated residual behaviour category

Definition 2.9.2 (Generated residual behaviour category). *The generated residual behaviour category of a specification*

$$k : K \rightarrow \mathbf{Beh}_0$$

is the full internal subcategory

$$\mathbf{Der}(K) \subseteq \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

on the object subobject

$$\mathbf{Der}_0(K) \hookrightarrow \mathbf{Beh}_0.$$

Thus

$$\mathbf{Der}(K)_0 = \mathbf{Der}_0(K),$$

and

$$\mathbf{Der}(K)_1 = \mathbf{Der}_0(K) \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} \mathbf{Der}_0(K).$$

The arrow object is defined by the pullback rectangle

$$\begin{array}{ccc} \mathbf{Der}(K)_1 & \xrightarrow{\quad} & \mathbf{Beh}_1 \\ \downarrow & \lrcorner & \downarrow (s, t) \\ \mathbf{Der}_0(K) \times_{A_0} \mathbf{Der}_0(K) & \xrightarrow{(j_K, j_K)} & \mathbf{Beh}_0 \times_{A_0} \mathbf{Beh}_0. \end{array}$$

Theorem 2.9.1 (Generated residual behaviour category). *The internal category*

$$\mathbf{Der}(K)$$

is the smallest full derivative-closed internal subcategory of

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

containing the specified behaviours.

Proof.

1. **Use the object-level closure.** By Theorem 2.7.1, the object part

$$\mathbf{Der}_0(K)$$

is the smallest derivative-closed subobject of $\mathbf{Beh}_0$ containing the image of K .

2. **Pass to the full subcategory.** The category $\mathbf{Der}(K)$ is the full internal subcategory on this object subobject. Hence any full derivative-closed internal subcategory containing K contains $\mathbf{Der}_0(K)$ on objects. Since it is full, it contains all arrows between those objects, and therefore contains the full subcategory $\mathbf{Der}(K)$. Thus $\mathbf{Der}(K)$ is the smallest full derivative-closed internal subcategory containing the specified behaviours.

□

2.9.3 The syntactic residual comparison category

Definition 2.9.3 (Syntactic residual comparison category). *The **syntactic residual comparison category** of K is the full inverse-image internal category along the object map d_K :*

$$\mathbf{Syn}(K) := d_K^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}.$$

Thus

$$\mathbf{Syn}(K)_0 = D_K,$$

and

$$\mathbf{Syn}(K)_1 = D_K \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} D_K.$$

A generalized arrow

$$(a, \kappa) \rightarrow (a', \kappa')$$

is a behaviour comparison

$$\partial_a k(\kappa) \Rightarrow \partial_{a'} k(\kappa').$$

The inverse-image square on arrow objects may be displayed fibrewise as

$$\begin{array}{ccc} \mathbf{Syn}(K)_1 & \xrightarrow{\quad} & \mathbf{Beh}_1 \\ \downarrow & \lrcorner & \downarrow (s, t) \\ D_K \times_{A_0} D_K & \xrightarrow{(d_K, d_K)} & \mathbf{Beh}_0 \times_{A_0} \mathbf{Beh}_0. \end{array}$$

The evident projection functor

$$\nu_K : \mathbf{Syn}(K) \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

has object map d_K and sends each arrow, already a behaviour comparison in $\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$, to that same comparison.

Remark 2.9.1. *The comparison-level image considered below is the full subcategory on the object image. It is not the pointwise arrow-image of the residual evaluation functor. The map from $\mathbf{Syn}(K)$ to this full subcategory is regular epimorphic on arrows because $\mathbf{Syn}(K)$ is defined by pulling back the whole vertical behaviour category along d_K .*

2.9.4 The full residual image

Theorem 2.9.2 (Full residual image on the derivative object image). *The residual evaluation functor*

$$\nu_K : \mathbf{Syn}(K) \rightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{vert}}$$

factors as

$$\mathbf{Syn}(K) \twoheadrightarrow \mathbf{Der}(K) \hookrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{vert}},$$

where the first functor is regular epimorphic on objects and arrows, and the second functor is the full inclusion.

Proof.

1. **Identify the object image.** The object map of ν_K is

$$d_K : D_K \rightarrow \mathbf{Beh}_0,$$

whose image is

$$\mathbf{Der}_0(K).$$

2. **Compare the arrow objects.** The arrow object of $\mathbf{Syn}(K)$ is

$$D_K \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} D_K,$$

while the arrow object of $\mathbf{Der}(K)$ is

$$\mathbf{Der}_0(K) \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} \mathbf{Der}_0(K).$$

The induced arrow map factors as

$$\begin{aligned} D_K \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} D_K \\ \twoheadrightarrow \mathbf{Der}_0(K) \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} D_K \\ \twoheadrightarrow \mathbf{Der}_0(K) \times_{\mathbf{Beh}_0, s} \mathbf{Beh}_1 \times_{t, \mathbf{Beh}_0} \mathbf{Der}_0(K). \end{aligned}$$

3. **Use stability of regular epimorphisms.** Each map is a pullback of

$$e_K : D_K \twoheadrightarrow \mathbf{Der}_0(K).$$

Regular epimorphisms are stable under pullback and closed under composition, so the arrow map is a regular epimorphism.

4. **Display the arrow-level image.** The arrow-level image diagram is

$$\begin{array}{ccc} \mathbf{Syn}(K)_1 & \twoheadrightarrow & \mathbf{Der}(K)_1 \\ \downarrow & & \downarrow \\ \mathbf{Beh}_1 & \xlongequal{\quad} & \mathbf{Beh}_1. \end{array}$$

□

2.10 The one-object specialization

Let M and N be monoid objects in $\mathcal{E}$, regarded as one-object internal categories

$$\mathbb{A} = M, \quad \mathbb{B} = N,$$

using the one-object convention of Chapter 1:

$$b \circ a = ba.$$

Thus a Mealy morphism

$$M \rightsquigarrow N$$

is an internal right M -object P , written

$$x \cdot m := \delta(m, x),$$

together with an internal N -valued output cocycle

$$\omega : M \times P \rightarrow N$$

satisfying

$$x \cdot 1 = x, \quad x \cdot (mn) = (x \cdot m) \cdot n,$$

and

$$\omega(1, x) = 1, \quad \omega(mn, x) = \omega(m, x)\omega(n, x \cdot m).$$

2.10.1 Residual right-Cayley syntax

The residual opfibration has a single base object, but its fibre is the internal residual right-Cayley category of M . Its object object is M . An object is written r , and an arrow labelled by m has the form

$$rm \longrightarrow r.$$

Equivalently, the arrow object is $M \times M$, with source and target maps

$$(r, m) \longmapsto rm, \quad (r, m) \longmapsto r.$$

The identity at r is $(r, 1)$. Composition is determined by

$$rmn \xrightarrow{n} rm \xrightarrow{m} r \quad \longmapsto \quad rmn \xrightarrow{mn} r,$$

or, in morphism notation,

$$(rm, n), (r, m) \longmapsto (r, mn).$$

Transport along an input $s \in M$ is postcomposition in the one-object category. Hence

$$s_* r = sr,$$

where sr is the monoid product corresponding to the categorical composite

$$s \circ r.$$

Thus the derivative of a residual behaviour will be given by left translation of the residual-history variable.

2.10.2 Residual behaviours and comparisons

A residual behaviour is an internal functor from the residual right-Cayley category of M to N . Since N has one object, such a functor is equivalently a map

$$\lambda : M \times M \rightarrow N$$

satisfying

$$\lambda(r, 1) = 1, \quad \lambda(r, mn) = \lambda(r, m)\lambda(rm, n).$$

The element $\lambda(r, m)$ is the output attached to the residual arrow

$$rm \longrightarrow r.$$

A behaviour comparison

$$\alpha : \lambda \Rightarrow \mu$$

is an internal natural transformation between the corresponding residual cocycle functors. Equivalently, it is a map

$$\alpha : M \rightarrow N$$

satisfying

$$\alpha_r \lambda(r, m) = \mu(r, m) \alpha_{rm}.$$

This is the naturality square for the residual arrow $rm \rightarrow r$:

$$\begin{array}{ccc} \bullet & \xrightarrow{\lambda(r, m)} & \bullet \\ \alpha_{rm} \downarrow & & \downarrow \alpha_r \\ \bullet & \xrightarrow{\mu(r, m)} & \bullet \end{array}$$

If $N = G$ is an internal group, this may equivalently be written as the nonabelian coboundary formula

$$\mu(r, m) = \alpha_r \lambda(r, m) \alpha_{rm}^{-1}.$$

2.10.3 The canonical residual-cocycle machine

The states of the canonical behaviour Mealy morphism are residual cocycles

$$\lambda : M \times M \rightarrow N.$$

For an input $s \in M$, the transition is derivative reindexing:

$$\lambda \longmapsto \partial_s \lambda, \quad (\partial_s \lambda)(r, m) = \lambda(sr, m).$$

The output on input s is the anchor comparison associated to the residual arrow

$$s \longrightarrow 1$$

in the residual right-Cayley category. Under the one-object identification of $\mathbb{B} = N$, this comparison is represented by

$$\lambda(1, s) \in N.$$

Thus the canonical behaviour Mealy morphism specializes to the residual-cocycle machine

$$\lambda \xrightarrow{s / \lambda(1, s)} \partial_s \lambda.$$

2.10.4 Residual semantics of a one-object Mealy morphism

Let

$$P : M \rightsquigarrow N$$

be a one-object Mealy morphism. Its execution category is the internal action category of the right M -object P . Its objects are states $x \in P$, and an arrow labelled by $m \in M$ has the form

$$x \longrightarrow x \cdot m.$$

The output functor to N^{op} sends this execution arrow to the element

$$\omega(m, x) \in N.$$

The cocycle law says exactly that this assignment preserves composition:

$$x \xrightarrow{m} x \cdot m \xrightarrow{n} (x \cdot m) \cdot n$$

has total output

$$\omega(m, x)\omega(n, x \cdot m) = \omega(mn, x).$$

For a state $x \in P$, the residual behaviour of x is the residual cocycle

$$\lambda_x : M \times M \rightarrow N$$

defined by

$$\lambda_x(r, m) = \omega(m, x \cdot r).$$

Thus $\lambda_x(r, m)$ is the output produced after first moving from x to the residual state $x \cdot r$, and then reading the residual input m . The cocycle equations for ω imply

$$\lambda_x(r, 1) = 1, \quad \lambda_x(r, mn) = \lambda_x(r, m)\lambda_x(rm, n).$$

The behaviour assignment is therefore

$$\beta_P : P \rightarrow \mathbf{Beh}_0, \quad \beta_P(x) = \lambda_x.$$

Terminality of the canonical behaviour machine specializes to the statement that this is the unique strict semantic map from P to the residual-cocycle machine. Behavioural separatedness says that

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

is monic, equivalently that states are separated by their future residual cocycles:

$$\lambda_x = \lambda_y \implies x = y.$$

2.10.5 Specifications, Nerode quotients, and generated comparisons

A behaviour specification

$$k : K \rightarrow \mathbf{Beh}_0$$

is an internal family of residual N -valued cocycles. Write

$$k(\kappa) = \lambda_\kappa.$$

The formal derivative orbit is

$$D_K = M \times K,$$

and the residual evaluation map is

$$d_K : M \times K \rightarrow \mathbf{Beh}_0, \quad d_K(s, \kappa) = \partial_s \lambda_\kappa.$$

Explicitly,

$$(\partial_s \lambda_\kappa)(r, m) = \lambda_\kappa(sr, m).$$

The Nerode relation is the kernel pair of d_K . Hence

$$(s, \kappa) \equiv_K (s', \kappa')$$

precisely when the corresponding residual cocycles are equal:

$$\partial_s \lambda_\kappa = \partial_{s'} \lambda_{\kappa'}.$$

Equivalently,

$$\lambda_\kappa(sr, m) = \lambda_{\kappa'}(s'r, m)$$

as internal residual behaviours.

The Nerode quotient

$$D_K / \equiv_K$$

is canonically the derivative image

$$\mathbf{Der}_0(K) \rightarrow \mathbf{Beh}_0.$$

Thus, in the one-object specialization, the state object of the minimal residual realization is the residual derivative closure of the specified cocycles. Its transition is

$$\lambda \mapsto \partial_s \lambda,$$

and its output on input s is

$$\lambda(1, s) \in N.$$

Hence $\mathbf{Min}(K)$ is the smallest residual-cocycle machine containing the specified cocycles and closed under all residual derivatives.

The generated residual behaviour category

$$\mathbf{Der}(K)$$

is the full residual comparison category on these generated residual cocycles. Its objects are residual derivatives of the specified cocycles. An arrow

$$\lambda \rightarrow \mu$$

is a residual behaviour comparison, equivalently a map

$$\alpha : M \rightarrow N$$

satisfying

$$\alpha_r \lambda(r, m) = \mu(r, m) \alpha_{rm}.$$

If $N = G$ is an internal group, these arrows may be read as nonabelian residual coboundaries:

$$\mu(r, m) = \alpha_r \lambda(r, m) \alpha_{rm}^{-1}.$$

Thus $\mathbf{Der}(K)$ records not only the residual derivatives generated by K , but also all residual coboundary comparisons between them.

In the ordinary external finite-state case, the finite-state theorem says that a finite realization exists precisely when there are only finitely many distinct residual cocycles in $\mathbf{Der}_0(K)$.

2.11 The stateless specialization

Let

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

be an internal functor, regarded as the map-representable Mealy morphism associated to its object map

$$F_0 : A_0 \rightarrow B_0.$$

Its carrier is

$$A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0.$$

Thus a state is simply an input object $v \in A_0$. There is no additional state object beyond the object-of-objects of $\mathbb{A}$.

For an input arrow

$$a : u \rightarrow v,$$

the transition sends the state v to the state u , and the emitted output arrow is

$$F_1(a) : F_0 u \rightarrow F_0 v.$$

Consequently the execution category has the same arrows as $\mathbb{A}$, but read in the target-to-source execution direction. It is canonically identified with

$$\mathbb{A}^{\text{op}},$$

and the output functor is

$$F^{\text{op}} : \mathbb{A}^{\text{op}} \rightarrow \mathbb{B}^{\text{op}}.$$

2.11.1 Slice-restricted residual semantics

For an object $v \in A_0$, the residual behaviour of the stateless state v is the internal functor

$$H_v^F : \mathbb{A}/v \rightarrow \mathbb{B}$$

defined by restriction of F along the domain functor

$$\text{dom}_v : \mathbb{A}/v \rightarrow \mathbb{A}.$$

Thus

$$H_v^F = F \circ \text{dom}_v.$$

Explicitly,

$$H_v^F(r : u \rightarrow v) = F_0(u),$$

and, for a residual triangle

$$w \xrightarrow{a} u \xrightarrow{r} v,$$

one has

$$H_v^F(a, r) = F_1(a) : F_0(w) \rightarrow F_0(u).$$

Hence F determines a canonical map into the behaviour object,

$$J_F : A_0 \rightarrow \mathbf{Beh}_0, \quad J_F(v) = H_v^F = F \circ \text{dom}_v.$$

This is the stateless residual semantics of F . It assigns to an input object v the ordinary slice restriction of F to arrows with codomain v .

For an input arrow $a : u \rightarrow v$, residual derivative is reindexing along postcomposition

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

The defining square is

$$\begin{array}{ccc} \mathbb{A}/u & \xrightarrow{a_*} & \mathbb{A}/v \\ \text{dom}_u \downarrow & & \downarrow \text{dom}_v \\ \mathbb{A} & \xlongequal{\quad} & \mathbb{A}. \end{array}$$

Therefore

$$\partial_a H_v^F = H_v^F \circ a_* = F \circ \text{dom}_v \circ a_* = F \circ \text{dom}_u = H_u^F.$$

Equivalently,

$$\partial_a J_F(v) = J_F(u).$$

The emitted output arrow of the canonical behaviour machine at $J_F(v)$ is the anchor comparison for H_v^F along a . Since H_v^F is the slice restriction of F , this anchor comparison is exactly

$$F_1(a) : F_0 u \rightarrow F_0 v.$$

Thus the canonical behaviour machine contains the stateless submachine

$$H_v^F \xrightarrow{a / F_1(a)} H_u^F.$$

The behaviour assignment of the map-representable Mealy morphism associated to F is precisely

$$\beta_{F_!} = J_F.$$

Moreover,

$$p_{\mathbf{Beh}} J_F = 1_{A_0},$$

so J_F is monic. Hence stateless map-representable machines are behaviourally separated.

For $F = 1_{\mathbb{A}}$, the stateless residual semantics is

$$v \longmapsto \text{dom}_v : \mathbb{A}/v \rightarrow \mathbb{A},$$

so the canonical behaviour machine recovers the codomain-slice geometry of $\mathbb{A}$ itself.

2.11.2 Simulations and residual natural transformations

Let

$$F, G : \mathbb{A} \rightarrow \mathbb{B}$$

be internal functors, regarded as map-representable Mealy morphisms

$$F_!, G_! : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

A Mealy simulation

$$F_! \Rightarrow G_!$$

is exactly an internal natural transformation

$$\alpha : F \Rightarrow G.$$

Thus α consists of a map

$$\alpha : A_0 \rightarrow B_1$$

satisfying

$$s_B \alpha = F_0, \quad t_B \alpha = G_0,$$

together with the internal naturality equation.

The induced behaviour comparison is obtained by restricting this natural transformation to each slice. For each $v \in A_0$, it is a vertical natural transformation

$$\hat{\alpha}_v : H_v^F \Rightarrow H_v^G$$

between

$$H_v^F = F \circ \text{dom}_v, \quad H_v^G = G \circ \text{dom}_v.$$

At an object

$$r : u \rightarrow v$$

of the slice $\mathbb{A}/v$, its component is

$$(\hat{\alpha}_v)_r = \alpha_u : F_0(u) \rightarrow G_0(u).$$

For a residual triangle

$$w \xrightarrow{a} u \xrightarrow{r} v,$$

naturality of $\hat{\alpha}_v$ is the ordinary naturality square for α at $a : w \rightarrow u$:

$$\begin{array}{ccc} F_0(w) & \xrightarrow{F_1(a)} & F_0(u) \\ \alpha_w \downarrow & & \downarrow \alpha_u \\ G_0(w) & \xrightarrow{G_1(a)} & G_0(u). \end{array}$$

Equivalently,

$$\hat{\alpha}_v = \alpha * \text{dom}_v : F \circ \text{dom}_v \Rightarrow G \circ \text{dom}_v.$$

Thus simulation-induced behaviour comparison specializes to ordinary restriction of a natural transformation along slices.

2.11.3 Stateless specifications and their Nerode quotients

Let

$$i : K \rightarrow A_0$$

be a family of specified input objects. The corresponding stateless specification for F is

$$k := J_F i : K \rightarrow \mathbf{Beh}_0.$$

Thus the specified behaviours are

$$H_{i(\kappa)}^F = F \circ \text{dom}_{i(\kappa)}.$$

The formal derivative orbit is

$$D_K = A_1 \times_{t_A, A_0, i} K.$$

A generalized element is an arrow

$$a : u \rightarrow i(\kappa)$$

into one of the specified objects.

The residual evaluation map is

$$d_K : D_K \rightarrow \mathbf{Beh}_0, \quad d_K(a, \kappa) = \partial_a J_F(i(\kappa)).$$

Since

$$\partial_a J_F(i(\kappa)) = J_F(s_A a),$$

one has the factorization

$$d_K = J_F s_A \pi_{A_1}.$$

Put

$$Z := A_0 \times B_0.$$

Regard A_0 as an object of $\mathcal{E}/Z$ through

$$\bar{F} := (1_{A_0}, F_0) : A_0 \rightarrow Z.$$

The map

$$J_F : A_0 \rightarrow \mathbf{Beh}_0$$

is a morphism in $\mathcal{E}/Z$, because

$$p_{\mathbf{Beh}} J_F = 1_{A_0},$$

and

$$q_{\mathbf{Beh}} J_F = F_0.$$

Moreover, J_F is monic, since $p_{\mathbf{Beh}}$ is a left inverse:

$$p_{\mathbf{Beh}} J_F = 1_{A_0}.$$

The object D_K is regarded as an object of $\mathcal{E}/Z$ through

$$(p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) d_K.$$

The map

$$s_A \pi_{A_1} : D_K \rightarrow A_0$$

is a morphism in $\mathcal{E}/Z$. Indeed,

$$\begin{aligned} (1_{A_0}, F_0) s_A \pi_{A_1} &= (p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) J_F s_A \pi_{A_1} \\ &= (p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) d_K. \end{aligned}$$

Take the regular epi–mono factorization of

$$s_A \pi_{A_1} : D_K \rightarrow A_0$$

in the slice $\mathcal{E}/Z$:

$$D_K \xrightarrow{e_S} S \xrightarrow{m_S} A_0.$$

Thus e_S is a regular epimorphism and m_S is monic in $\mathcal{E}/Z$, and

$$m_S e_S = s_A \pi_{A_1}.$$

The composite

$$J_F m_S : S \rightarrow \mathbf{Beh}_0$$

is monic. Moreover,

$$\begin{aligned} d_K &= J_F s_A \pi_{A_1} \\ &= J_F m_S e_S. \end{aligned}$$

Hence

$$D_K \xrightarrow{e_S} S \xrightarrow{J_F m_S} \mathbf{Beh}_0$$

is a regular epi–mono factorization of d_K in $\mathcal{E}/Z$.

Let

$$D_K \xrightarrow{e_K} \mathrm{Der}_0(K) \xrightarrow{j_K} \mathbf{Beh}_0$$

be the regular epi–mono factorization used in Definition 2.7.5. By uniqueness of regular images, there is a unique isomorphism

$$\varphi_K : S \xrightarrow{\cong} \mathrm{Der}_0(K)$$

in $\mathcal{E}/Z$ satisfying

$$\varphi_K e_S = e_K,$$

and

$$j_K \varphi_K = J_F m_S.$$

We henceforth use this canonical isomorphism to identify

$$\mathrm{Der}_0(K) \cong S.$$

Under this identification,

$$e_K = e_S, \quad j_K = J_F m_S.$$

The Nerode relation is the kernel pair of d_K . Since

$$d_K = J_F s_A \pi_{A_1}$$

and J_F is monic, the kernel pair of d_K is the kernel pair of

$$s_A \pi_{A_1}.$$

Since

$$s_A \pi_{A_1} = m_S e_S$$

and m_S is monic, this is in turn the kernel pair of e_S .

Consequently,

$$(a, \kappa) \equiv_K (a', \kappa')$$

if and only if

$$s_A a = s_A a'.$$

Since e_S is the effective quotient of its kernel pair,

$$D_K / \equiv_K \cong S \cong \mathrm{Der}_0(K)$$

in $\mathcal{E}/Z$.

The subobject

$$m_S : S \hookrightarrow A_0$$

is downward closed under domains of arrows. More precisely, there is a unique map

$$\bar{s} : A_1 \times_{t_A, A_0, m_S} S \rightarrow S$$

such that

$$m_S \bar{s} = s_A \pi_{A_1}.$$

To construct this map, transport the derivative-closure transition

$$\bar{\partial}_K : A_1 \times_{t_A, A_0, p_{\text{Der}}} \text{Der}_0(K) \rightarrow \text{Der}_0(K)$$

of Theorem 2.7.1 across the canonical isomorphism

$$\varphi_K : S \cong \text{Der}_0(K).$$

For

$$a : u \rightarrow v, \quad m_S(s) = v,$$

derivative closure gives

$$J_F m_S(\bar{s}(a, s)) = \partial_a J_F(m_S s).$$

But

$$\partial_a J_F(m_S s) = J_F(s_A a).$$

Since J_F is monic,

$$m_S(\bar{s}(a, s)) = s_A a.$$

Thus

$$m_S \bar{s} = s_A \pi_{A_1}.$$

The minimal residual realization generated by the stateless specification has carrier

$$A_0 \xleftarrow{m_S} S \xrightarrow{F_0 m_S} B_0.$$

For an arrow

$$a : u \rightarrow v$$

with v represented by a state $s \in S$, its transition is

$$s \mapsto \bar{s}(a, s),$$

whose image in A_0 is u , and its emitted output is

$$F_1(a) : F_0 u \rightarrow F_0 v.$$

Thus the minimal residual realization is the restriction of the map-representable machine associated to F to the generated downward-closed object subobject

$$m_S : S \hookrightarrow A_0.$$

Its carrier is map-representable in the sense of Chapter 1 if and only if

$$m_S : S \rightarrow A_0$$

is an isomorphism.

Let

$$\mathbb{A}|_S$$

denote the full internal subcategory of $\mathbb{A}$ on S . Its object object is S , and its arrow object is

$$S \times_{m_S, A_0, s_A} A_1 \times_{t_A, A_0, m_S} S.$$

The downward-closure equation

$$m_S \bar{s} = s_A \pi_{A_1}$$

means that every arrow of $\mathbb{A}$ whose target lies in S also has its source in S .

The execution category of the minimal residual realization is canonically

$$(\mathbb{A}|_S)^{\text{op}},$$

and its output functor is the restriction of

$$F^{\text{op}} : \mathbb{A}^{\text{op}} \rightarrow \mathbb{B}^{\text{op}}.$$

The subobject S is the smallest downward-closed subobject of A_0 containing the image of i .

First, identities show that the image of i is contained in S . Indeed, the unit inclusion

$$\iota_K : K \rightarrow D_K$$

satisfies

$$s_A \pi_{A_1} \iota_K = i,$$

and therefore

$$i = m_S e_S \iota_K.$$

Now let

$$n : T \hookrightarrow A_0$$

be any downward-closed subobject through which i factors. For every generalized element

$$(a, \kappa) \in D_K,$$

the target $t_A(a) = i(\kappa)$ lies in T . Downward closure therefore implies that the source $s_A(a)$ also lies in T . Consequently the map

$$s_A \pi_{A_1} : D_K \rightarrow A_0$$

factors through n . Since m_S is its regular image, m_S factors through n . Thus S is the smallest such subobject.

For the full specification

$$i = 1_{A_0} : A_0 \rightarrow A_0,$$

there is a canonical isomorphism

$$D_K \cong A_1,$$

under which the map defining S is

$$s_A : A_1 \rightarrow A_0.$$

Since

$$s_A e_A = 1_{A_0},$$

the map s_A is a split epimorphism, and its regular image is all of A_0 . Therefore

$$S \cong A_0,$$

and the minimal residual realization is the original map-representable machine associated to F .

2.12 Examples

Example 2.12.1 (Classical deterministic Mealy machines). *Let*

$$M = I^*$$

be the free monoid on an input alphabet I , and let N be an output monoid.

The residual syntax fibre is the residual right-Cayley category of I^ . A residual behaviour is a residual cocycle*

$$\lambda : I^* \times I^* \rightarrow N$$

satisfying

$$\lambda(w, 1) = 1,$$

and

$$\lambda(w, uv) = \lambda(w, u)\lambda(wu, v).$$

Since I^ is free, such a cocycle is determined by its values on generators:*

$$\bar{\lambda} : I^* \times I \rightarrow N.$$

Equivalently, define

$$o : I^+ \rightarrow N$$

by

$$o(wi) := \bar{\lambda}(w, i).$$

The associated residual cocycle is

$$\lambda_o(w, i_1 \cdots i_n) = o(wi_1)o(wi_1i_2) \cdots o(wi_1 \cdots i_n),$$

with

$$\lambda_o(w, 1) = 1.$$

The derivative by $w \in I^$ is left translation of the residual history:*

$$(\partial_w o)(u) = o(wu), \quad u \in I^+.$$

The minimal residual realization has state set

$$\{\partial_w o \mid w \in I^*\}.$$

On a generator $i \in I$, its transition is

$$\partial_w o \mapsto \partial_{wi} o,$$

and its output is

$$o(wi) \in N.$$

This describes deterministic Mealy machines whose single-step outputs take values in an arbitrary output monoid N .

To recover the usual letter-output formulation, let O be an output alphabet, take

$$N = O^*,$$

and impose the generator-level condition

$$\bar{\lambda}(w, i) \in O \subseteq O^*$$

for every $w \in I^*$ and $i \in I$. The cumulative output on a word is then the concatenation of the individual output letters emitted during its sequential processing.

In either formulation, the finite-state criterion is

$$\{\partial_w o \mid w \in I^*\}$$

finite. In the letter-output specialization, this is the deterministic Mealy analogue of the residual finiteness criterion in the classical Myhill–Nerode theorem [15, 18, 4].

Example 2.12.2 (Group-valued cumulative output languages). Let G be a group and let

$$\ell : \Sigma^* \rightarrow G$$

be a cumulative output function. It determines an incremental residual cocycle by

$$\lambda_\ell(r, m) = \ell(r)^{-1} \ell(rm).$$

Then

$$\lambda_\ell(r, 1) = 1,$$

and

$$\lambda_\ell(r, mn) = \lambda_\ell(r, m) \lambda_\ell(rm, n).$$

Thus group-valued cumulative output languages fit the present Mealy-type semantics after passing to prefix quotients.

Example 2.12.3 (Base- r carry machine). Let

$$M = N = (\mathbb{N}, +), \quad P = \{0, \dots, r-1\}.$$

Define

$$x \cdot n = (x + n) \bmod r,$$

and

$$\omega(n, x) = \left\lfloor \frac{x + n}{r} \right\rfloor.$$

The behaviour of x is

$$\lambda_x(s, n) = \omega(n, x \cdot s) = \left\lfloor \frac{(x + s) \bmod r + n}{r} \right\rfloor.$$

The derivative closure is

$$\{\lambda_x \mid x \in \{0, \dots, r-1\}\}.$$

The Nerode quotient identifies residual histories with the same carry residue.

Example 2.12.4 (Typed path-category machines). *Let G and H be directed graphs, and consider a Mealy morphism*

$$\mathbf{Path}(G) \rightsquigarrow \mathbf{Path}(H).$$

For a vertex v of G , the residual syntax fibre is

$$\mathbf{Path}(G)/v.$$

A residual behaviour is a functor

$$\mathbf{Path}(G)/v \rightarrow \mathbf{Path}(H).$$

It assigns coherent output paths in H to input paths ending at v . Derivatives are reindexing along residual transport by path concatenation. The categorical Myhill–Nerode object is the derivative image of these path-output behaviours. Simulations induce natural transformations between the corresponding residual path-output functors.

Example 2.12.5 (Context restriction and register allocation). *Consider the register-allocation Mealy morphism whose states are triples*

$$(\Gamma, R, \rho),$$

where Γ is a finite variable context, R is a finite register context, and

$$\rho : \Gamma \cong R$$

is a bijection. For an inclusion

$$\Gamma' \hookrightarrow \Gamma,$$

the residual state is

$$(\Gamma', \rho(\Gamma'), \rho|_{\Gamma'}).$$

The residual behaviour sends the residual object

$$\Gamma' \hookrightarrow \Gamma$$

to the restricted register context

$$\rho(\Gamma') \subseteq R.$$

For a residual triangle

$$\Gamma'' \hookrightarrow \Gamma' \hookrightarrow \Gamma,$$

the behaviour arrow is the inclusion

$$\rho(\Gamma'') \hookrightarrow \rho(\Gamma').$$

The generated residual behaviour category records residual restriction behaviours and coherent register-interface comparisons.

The residual behaviour square is

$$\begin{array}{ccc} \Gamma'' & \hookrightarrow & \Gamma' \\ \rho|_{\Gamma''} \downarrow \sim & & \sim \downarrow \rho|_{\Gamma'} \\ \rho(\Gamma'') & \hookrightarrow & \rho(\Gamma'). \end{array}$$

Example 2.12.6 (Poset-valued systems). *If $\mathbb{A}$ and $\mathbb{B}$ are internal posets, then the residual fibre over v is the downset of objects mapping to v . A residual behaviour over v is a monotone assignment from this residual downset into $\mathbb{B}$. Behaviour comparisons are pointwise inequalities. Thus*

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

is an internal preorder, and $\mathbf{Der}(K)$ is the derivative-closed residual preorder generated by the specification.

Example 2.12.7 (Action groupoids and gauge comparisons). *Let*

$$X//G \quad \text{and} \quad Y//H$$

be action groupoids. The residual syntax fibre over $x \in X$ is the category of arrows ending at x . A residual behaviour records coherent H -transport along residual G -histories. Behaviour comparisons are gauge transformations between such transport assignments. The generated residual behaviour category is the full residual image of the syntactic residual comparison category in the behaviour category.

Example 2.12.8 (Group-valued coboundary outputs). *Let $N = G$ be a group. A behaviour comparison*

$$\alpha : \lambda \Rightarrow \mu$$

between residual cocycles is a family

$$\alpha_r \in G$$

satisfying

$$\alpha_r \lambda(r, m) = \mu(r, m) \alpha_{rm}.$$

Equivalently,

$$\mu(r, m) = \alpha_r \lambda(r, m) \alpha_{rm}^{-1}.$$

Thus the vertical behaviour category is a category of residual cocycles and nonabelian coboundary comparisons.

2.13 Chapter summary

The constructions of this chapter may be summarized as follows:

fibred structure	Mealy-theoretic role
$\text{Res}_{\mathbb{A}} \rightarrow \mathbb{A}$	split codomain opfibration of residual input syntax
$\mathbb{A}/v$	residual histories ending at v
$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v$	cocartesian residual transport by postcomposition
$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}} \rightarrow \mathbb{A}$	behaviour fibration
$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}(v)$	$[\mathbb{A}/v, \mathbb{B}]$
$\partial_a H = a^* H = H \circ a_*$	cartesian derivative reindexing
$\mathbf{Beh}_0$	total object of residual behaviours
$\mathbf{Beh}_1$	vertical behaviour comparisons
$q_{\mathbf{Beh}}(H) = H(1_v)$	anchor evaluation
cleavage of $\mathbf{Beh}$	canonical behaviour Mealy morphism
$\beta_P(x) = \Omega_P^{\text{op}} \rho_x$	classifying map of residual orbit semantics
D_K	formal derivative orbit of a specification
$d_K : D_K \rightarrow \mathbf{Beh}_0$	fibred residual evaluation
$\equiv_K$	kernel-pair Nerode relation
$D_K / \equiv_K \cong \text{Der}_0(K)$	categorical Myhill–Nerode quotient
$\text{Der}_0(K)$	object image of the derivative orbit
$\text{Min}(K)$	minimal residual realization
$\text{Reach}_0(P, K) \rightarrow \text{Min}(K)$	semantic quotient of a reachable realization
$\mathbf{Der}(K)$	generated full residual behaviour category

The residual syntax of the input category is organized by the split codomain opfibration

$$\text{Res}_{\mathbb{A}} \rightarrow \mathbb{A}.$$

Its fibre over v is the residual slice

$$\mathbb{A}/v,$$

and transport along $a : u \rightarrow v$ is postcomposition

$$a_* : \mathbb{A}/u \rightarrow \mathbb{A}/v.$$

Behaviours form the corresponding contravariant behaviour fibration:

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}(v) = [\mathbb{A}/v, \mathbb{B}].$$

Derivatives are cartesian reindexing:

$$\partial_a H = H \circ a_*.$$

The cleavage of the behaviour fibration makes the object of behaviours into the state object of the canonical behaviour Mealy morphism

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

The transition is derivative reindexing, and the output is the anchor comparison

$$H(a, 1_v) : H(a) \rightarrow H(1_v).$$

For any Mealy morphism P , each state x determines an orbit functor

$$\rho_x : \mathbb{A}/p(x) \rightarrow \left(\int_{\mathbb{A}} P \right)^{\text{op}},$$

and composing this orbit with the output functor gives the residual behaviour

$$H_x = \Omega_P^{\text{op}} \rho_x.$$

Thus the semantic map

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

is determined by fibred residual orbit semantics. The canonical behaviour machine is terminal in the ordinary category of strict semantic Mealy homomorphisms.

For a specification

$$k : K \rightarrow \mathbf{Beh}_0,$$

the residual evaluation map

$$d_K : D_K \rightarrow \mathbf{Beh}_0$$

sends a formal derivative (a, κ) to

$$\partial_a k(\kappa).$$

Its kernel pair is the equality-level Nerode relation, and exactness gives

$$D_K / \equiv_K \cong \text{Der}_0(K).$$

The right-hand side is derivative-closed, so the canonical behaviour machine restricts to it. This restricted machine is the minimal residual realization

$$\text{Min}(K).$$

Every reachable realization maps regularly epimorphically onto $\text{Min}(K)$, and every reachable behaviourally separated realization is isomorphic to it. With a finite-state structure, a finite realization exists exactly when the Nerode quotient, equivalently the minimal residual realization, is finite.

The generated residual behaviour category

$$\mathbf{Der}(K)$$

records the vertical behaviour comparisons between generated residual behaviours. It is the full internal subcategory of

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}}$$

on the derivative image

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_0.$$

Thus the categorical Myhill–Nerode theorem has an object-level Mealy component, namely $\text{Min}(K)$, and a comparison-level fibred component, namely $\mathbf{Der}(K)$.

Chapter 3

Coequational Subobjects and Birkhoff Theorems

3.1 Overview and standing assumptions

3.1.1 Orientation

The preceding chapter constructed, for a behaviour specification

$$k : K \rightarrow \mathbf{Beh}_0,$$

its derivative image

$$\mathrm{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

and the corresponding minimal residual Mealy morphism

$$\mathrm{Min}(K) : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

The present chapter reorganizes residual semantics in coequational form. The central semantic object is the canonical behaviour Mealy morphism

$$\mathbf{Beh}_{\mathbb{A},\mathbb{B}}^{\mathrm{can}} : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

By Chapter 2, this object is terminal in the ordinary category

$$\mathbf{Mly}_{\mathbb{A},\mathbb{B}}^{\mathrm{str}}$$

of Mealy morphisms and strict semantic homomorphisms. Hence a local coequational theory is a subobject of this terminal semantic object:

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B}}^{\mathrm{can}}.$$

A Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

models V when its unique semantic map to the terminal object factors through V :

$$\begin{array}{ccc} P & \xrightarrow{\beta_P} & \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}} \\ & \searrow \tilde{\beta}_P & \nearrow j_V \\ & V & \end{array}$$

Equivalently, the carrier map

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

factors through the underlying carrier subobject

$$V \hookrightarrow \mathbf{Beh}_0$$

in the slice over

$$Z := A_0 \times B_0.$$

The basic examples of coequational theories are semantic images. For a Mealy morphism P , its semantic image is the regular image of its terminal semantic map:

$$P \rightarrow \mathbf{Beh}(P) \hookrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}.$$

The first main task of the chapter is to prove that regular images of strict semantic homomorphisms are created by the ambient carrier slice. This makes

$$\mathbf{Beh}(P)$$

again a subobject of the canonical behaviour Mealy morphism.

The chapter proves a coequational Birkhoff theorem. Locally, a replete full class of Mealy morphisms is definable by a local coequational theory exactly when it is closed under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts. The representing coequational theory associated to a class

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbf{A}, \mathbf{B}}^{\text{str}}$$

is the join of the semantic images of its objects:

$$V_{\mathcal{C}} := \bigvee_{P \in \mathcal{C}} \mathbf{Beh}(P) \hookrightarrow \mathbf{Beh}_0.$$

The use of arbitrary joins of subobjects is the point at which this chapter works in a Grothendieck topos. The resulting statement is in the same lineage as Birkhoff and Eilenberg type correspondences for recognizable languages and categorical varieties of languages [6, 1, 22, 23].

3.1.2 Ambient assumptions

Assumption 3.1.1. *Throughout this chapter, $\mathcal{E}$ is a Grothendieck topos with chosen pullbacks. Thus $\mathcal{E}$ is Barr-exact and locally cartesian closed, has stable regular epi-mono factorizations, effective quotients of internal equivalence relations, small coproducts, and complete subobject lattices in every slice. All coproducts and joins below are small, relative to the fixed ambient universe. We use these topos-theoretic properties in the form recorded in [12, 3, 10].*

Internal-language convention. All arguments below written in generalized-element notation are interpreted in the Mitchell–Bénabou internal language of $\mathcal{E}$. Equivalently, each such argument may be read after pullback along an arbitrary generalized element. In particular, assertions of the form

$$x \in X,$$

implications between subobject-membership predicates, and elementwise equalities are shorthand for the corresponding internal statements in $\mathcal{E}$.

The input and output internal categories are

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A), \quad \mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B).$$

When the input and output categories are fixed, we write

$$Z := A_0 \times B_0.$$

Composition in $\mathbb{A}$ is written

$$m_A(a, b) = b \circ a$$

for composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w.$$

Thus a Mealy state over w processes $b \circ a$ by first processing b , then processing a .

Small coproducts include the empty coproduct. Thus every coequationally definable class contains the initial strict Mealy morphism. Dually, the empty join of local coequational theories is the bottom subobject of the canonical behaviour object.

3.1.3 Strict Mealy morphisms and terminal semantics

Throughout, Mealy morphisms

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

are considered as objects of the ordinary category

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

A strict semantic homomorphism

$$f : P \rightarrow Q$$

is a map in the carrier slice

$$\mathcal{E}/Z$$

preserving transition and output on the nose. Thus, writing

$$P = (P, p_P, q_P, \delta_P, \omega_P), \quad Q = (Q, p_Q, q_Q, \delta_Q, \omega_Q),$$

strictness means

$$p_Q f = p_P, \quad q_Q f = q_P,$$

and

$$f \delta_P = \delta_Q(1 \times f), \quad \omega_P = \omega_Q(1 \times f).$$

For every Mealy morphism

$$P = (P, p_P, q_P, \delta_P, \omega_P),$$

write

$$D_P := A_1 \times_{t_A, A_0, p_P} P$$

for its transition domain, and write

$$\lambda_P : D_P \rightarrow A_1, \quad \rho_P : D_P \rightarrow P$$

for the two pullback projections. Thus

$$t_A \lambda_P = p_P \rho_P.$$

The transition and output typing equations may then be written

$$p_P \delta_P = s_A \lambda_P,$$

and

$$s_B \omega_P = q_P \delta_P, \quad t_B \omega_P = q_P \rho_P.$$

For a map

$$f : P \rightarrow Q$$

over A_0 , write

$$\hat{f} : D_P \rightarrow D_Q$$

for the induced map on transition domains, characterized by

$$\lambda_Q \hat{f} = \lambda_P, \quad \rho_Q \hat{f} = f \rho_P.$$

In generalized-element notation,

$$\hat{f}(a, x) = (a, fx).$$

Subobjects, coproducts, and full subcategories below are taken in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

unless explicitly stated otherwise. Regular quotients are understood in the carrier-based sense of the following definition.

Definition 3.1.1 (Regular quotient). *A strict semantic homomorphism*

$$e : P \rightarrow Q$$

*is a **regular quotient** if its underlying carrier map*

$$Ue : UP \rightarrow UQ$$

is a regular epimorphism in

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0.$$

Thus regular-quotient closure throughout this chapter means closure under strict semantic homomorphisms whose carriers are regular epimorphisms in the carrier slice.

A replete full subcategory

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

is said to be **closed under strict homomorphic preimages** if, for every strict semantic homomorphism

$$f : P \rightarrow Q,$$

with

$$Q \in \mathcal{C},$$

one has

$$P \in \mathcal{C}.$$

In particular, this includes closure under strict subobjects, since a strict subobject is a strict semantic homomorphism into its ambient object.

Recall from Chapter 2 that every Mealy morphism P has a residual semantics map

$$\beta_P : P \rightarrow \mathbf{Beh}_0.$$

For $x \in P$, the residual behaviour

$$\beta_P(x) = H_x : \mathbb{A}/p_P(x) \rightarrow \mathbb{B}$$

is given on objects by

$$H_x(r) = q_P(\delta_P(r, x)),$$

and on residual arrows

$$w \xrightarrow{a} u \xrightarrow{r} p_P(x)$$

by

$$H_x(a, r) = \omega_P(a, \delta_P(r, x)).$$

Equivalently,

$$\beta_P : P \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is the unique strict semantic homomorphism from P to the terminal object

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

of $\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$. The use of terminal semantic objects follows the categorical semantics of realization and coalgebraic behaviour maps [7, 21, 11].

Lemma 3.1.1 (Strict maps preserve residual semantics). *Let*

$$f : P \rightarrow Q$$

be a strict semantic homomorphism. Then

$$\beta_Q f = \beta_P.$$

Proof.

1. **Use terminality.** Since f is strict and β_Q is strict, the composite

$$\beta_Q f : P \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a strict semantic homomorphism. The map β_P is also a strict semantic homomorphism from P to the terminal object.

This gives the parallel pair

$$P \begin{array}{c} \xrightarrow{\beta_Q f} \\ \xrightarrow{\beta_P} \end{array} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

2. **Conclude uniqueness.** By terminality of

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

in $\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$, the two maps agree:

$$\beta_Q f = \beta_P.$$

□

Lemma 3.1.2 (Pullbacks and monomorphisms are created by carriers). *The carrier functor*

$$U : \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \longrightarrow \mathcal{E}/Z$$

creates pullbacks.

Consequently, a strict semantic homomorphism

$$f : P \rightarrow Q$$

is monic in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

if and only if its underlying carrier map is monic in

$$\mathcal{E}/Z.$$

Proof. Let

$$f : P \rightarrow R, \quad g : Q \rightarrow R$$

be strict semantic homomorphisms. Form their pullback in the carrier slice:

$$\begin{array}{ccc} X & \xrightarrow{\pi_Q} & Q \\ \pi_P \downarrow & \lrcorner & \downarrow g \\ P & \xrightarrow{f} & R. \end{array}$$

Thus

$$X = P \times_R Q$$

in $\mathcal{E}/Z$. Put

$$p_X := p_P \pi_P = p_Q \pi_Q, \quad q_X := q_P \pi_P = q_Q \pi_Q.$$

Write

$$D_X := A_1 \times_{t_A, A_0, p_X} X$$

and similarly for D_P, D_Q, D_R . The pullback projections induce maps

$$\hat{\pi}_P : D_X \rightarrow D_P, \quad \hat{\pi}_Q : D_X \rightarrow D_Q.$$

Since f and g are strict,

$$\begin{aligned} f\delta_P\hat{\pi}_P &= \delta_R\hat{f}\hat{\pi}_P \\ &= \delta_R\hat{g}\hat{\pi}_Q \\ &= g\delta_Q\hat{\pi}_Q. \end{aligned}$$

The middle equality follows from

$$f\pi_P = g\pi_Q,$$

which implies

$$\hat{f}\hat{\pi}_P = \hat{g}\hat{\pi}_Q.$$

Hence the pair

$$(\delta_P\hat{\pi}_P, \delta_Q\hat{\pi}_Q)$$

factors uniquely through the underlying pullback X in $\mathcal{E}$. Define

$$\delta_X : D_X \rightarrow X$$

by

$$\pi_P\delta_X = \delta_P\hat{\pi}_P, \quad \pi_Q\delta_X = \delta_Q\hat{\pi}_Q.$$

The transition has the required input typing:

$$\begin{aligned} p_X\delta_X &= p_P\pi_P\delta_X \\ &= p_P\delta_P\hat{\pi}_P \\ &= s_A\lambda_P\hat{\pi}_P \\ &= s_A\lambda_X. \end{aligned}$$

Strictness of f and g also gives

$$\begin{aligned} \omega_P\hat{\pi}_P &= \omega_R\hat{f}\hat{\pi}_P \\ &= \omega_R\hat{g}\hat{\pi}_Q \\ &= \omega_Q\hat{\pi}_Q. \end{aligned}$$

Define the common map to be

$$\omega_X : D_X \rightarrow B_1.$$

The output map has the required source typing:

$$\begin{aligned} s_B\omega_X &= s_B\omega_P\hat{\pi}_P \\ &= q_P\delta_P\hat{\pi}_P \\ &= q_P\pi_P\delta_X \\ &= q_X\delta_X. \end{aligned}$$

It also has the required target typing. Writing

$$\rho_X : D_X \rightarrow X$$

for the state projection,

$$\begin{aligned} t_B \omega_X &= t_B \omega_P \widehat{\pi}_P \\ &= q_P \rho_P \widehat{\pi}_P \\ &= q_P \pi_P \rho_X \\ &= q_X \rho_X. \end{aligned}$$

The transition unit law holds after composing with both pullback projections. For example,

$$\begin{aligned} \pi_P \delta_X(1_{p_X x}, x) &= \delta_P(1_{p_P \pi_P x}, \pi_P x) \\ &= \pi_P x. \end{aligned}$$

The analogous equality holds after composing with π_Q . Since the pullback projections are jointly monic,

$$\delta_X(1_{p_X x}, x) = x.$$

For composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

and $x \in X$ over w , one has

$$\begin{aligned} \pi_P \delta_X(b \circ a, x) &= \delta_P(b \circ a, \pi_P x) \\ &= \delta_P(a, \delta_P(b, \pi_P x)) \\ &= \pi_P \delta_X(a, \delta_X(b, x)). \end{aligned}$$

The same equality holds after composing with π_Q . Joint monicity therefore gives

$$\delta_X(b \circ a, x) = \delta_X(a, \delta_X(b, x)).$$

The output unit law follows by restriction:

$$\begin{aligned} \omega_X(1_{p_X x}, x) &= \omega_P(1_{p_P \pi_P x}, \pi_P x) \\ &= 1_{q_P \pi_P x} \\ &= 1_{q_X x}. \end{aligned}$$

Likewise,

$$\begin{aligned} \omega_X(b \circ a, x) &= \omega_P(b \circ a, \pi_P x) \\ &= \omega_P(b, \pi_P x) \circ \omega_P(a, \delta_P(b, \pi_P x)) \\ &= \omega_X(b, x) \circ \omega_X(a, \delta_X(b, x)). \end{aligned}$$

Thus

$$X = (X, p_X, q_X, \delta_X, \omega_X)$$

is a Mealy morphism. By construction, the projections

$$\pi_P : X \rightarrow P, \quad \pi_Q : X \rightarrow Q$$

preserve transition and output strictly.

Now suppose that

$$h : T \rightarrow P, \quad k : T \rightarrow Q$$

are strict semantic homomorphisms satisfying

$$fh = gk.$$

The carrier pullback gives a unique map

$$\ell : T \rightarrow X$$

in $\mathcal{E}/Z$ satisfying

$$\pi_P \ell = h, \quad \pi_Q \ell = k.$$

To prove transition preservation, compose with the pullback projections:

$$\begin{aligned} \pi_P \ell \delta_T &= h \delta_T \\ &= \delta_P \widehat{h} \\ &= \delta_P \widehat{\pi_P \ell} \\ &= \pi_P \delta_X \widehat{\ell}, \end{aligned}$$

and similarly

$$\pi_Q \ell \delta_T = \pi_Q \delta_X \widehat{\ell}.$$

Joint monicity gives

$$\ell \delta_T = \delta_X \widehat{\ell}.$$

Output preservation follows from

$$\begin{aligned} \omega_X \widehat{\ell} &= \omega_P \widehat{\pi_P \ell} \\ &= \omega_P \widehat{h} \\ &= \omega_T. \end{aligned}$$

Hence ℓ is strict. This proves that U creates the displayed pullback.

Suppose now that

$$f : P \rightarrow Q$$

is a strict semantic homomorphism whose carrier is monic in $\mathcal{E}/Z$. Since the carrier functor is faithful, f is monic in the strict Mealy category.

Conversely, suppose that f is monic in the strict Mealy category. Since pullbacks are created by U , the kernel pair of f is

$$P \times_Q P \begin{array}{c} \xrightarrow{r_1} \\ \xrightarrow{r_2} \end{array} P \xrightarrow{f} Q.$$

Since

$$f r_1 = f r_2$$

and f is monic,

$$r_1 = r_2.$$

Therefore the underlying carrier kernel-pair projections agree. Hence the carrier of f is monic in $\mathcal{E}/Z$. $\square$

3.2 Local coequational theories

3.2.1 Canonical derivative structure

The canonical behaviour object

$$\mathbf{Beh}_0$$

carries the canonical behaviour Mealy morphism

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} : \mathbb{A} \rightsquigarrow \mathbb{B}.$$

A state over $v \in A_0$ is a residual behaviour

$$H : \mathbb{A}/v \rightarrow \mathbb{B}.$$

For an arrow

$$a : u \rightarrow v,$$

the canonical transition is the derivative

$$\partial_a H = H \circ a_*,$$

and the canonical output comparison is

$$\omega_{\mathbf{Beh}}(a, H) : q_{\mathbf{Beh}}(\partial_a H) \rightarrow q_{\mathbf{Beh}}(H),$$

the arrow assigned by H to the residual morphism

$$a \rightarrow 1_v$$

in $\mathbb{A}/v$. This derivative notation is compatible with the Brzozowski derivative viewpoint in the one-object free-monoid specialization [4].

The output comparison is displayed by applying H to the residual triangle

$$u \xrightarrow{\quad a \quad} v \xrightarrow{\quad 1_v \quad} v. \quad \rightsquigarrow \quad H(a : u \rightarrow v) \xrightarrow{H(a \rightarrow 1_v)} H(1_v).$$

Equivalently, the transition-output structure is the map of spans

$$\begin{array}{ccccc} A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0 & \xrightarrow{\langle \delta_{\mathbf{Beh}}, \omega_{\mathbf{Beh}} \rangle} & \mathbf{Beh}_0 \times_{q_{\mathbf{Beh}}, B_0, s_B} B_1 & & \\ & \searrow s_A \pi_{A_1} & \swarrow p_{\mathbf{Beh}} \pi_{\mathbf{Beh}} & & \searrow t_B \pi_{B_1} \\ & & A_0 & & B_0. \end{array}$$

Put

$$Z = A_0 \times B_0$$

and write

$$D_{\mathbf{Beh}} := A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

There are two maps

$$\text{pr}_{\partial} : D_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0, \quad \partial : D_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0,$$

defined by

$$\text{pr}_\partial(a, H) = H, \quad \partial(a, H) = \partial_a H.$$

Whenever a derivative domain occurs as a carrier over Z , it is regarded over Z by the derivative state. Thus $D_{\mathbf{Beh}}$ is regarded as an object of $\mathcal{E}/Z$ by the composite

$$\begin{array}{ccc} D_{\mathbf{Beh}} & \xrightarrow{\partial} & \mathbf{Beh}_0 \\ & \searrow (s_A \pi_{A_1}, q_{\mathbf{Beh}} \partial) & \downarrow (p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) \\ & & Z. \end{array}$$

Thus, if

$$a : u \rightarrow v$$

and H lies over v , then (a, H) has Z -coordinate

$$(u, q_{\mathbf{Beh}}(\partial_a H)).$$

With this convention, the derivative map

$$\partial : D_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0$$

is a morphism in $\mathcal{E}/Z$. The state projection

$$\text{pr}_\partial : D_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0$$

is not in general a morphism over Z , since it remembers the interface and anchor coordinates of the original behaviour rather than those of its derivative.

Pullbacks and universal quantification along pr_∂ below are therefore taken in $\mathcal{E}$. Since subobjects of $\mathbf{Beh}_0$ in $\mathcal{E}$ are canonically the same as subobjects of $\mathbf{Beh}_0$ in $\mathcal{E}/Z$, the resulting predicates are again regarded over Z .

3.2.2 Behavioural predicates and residual invariance

Definition 3.2.1 (Behavioural predicate). *A **behavioural predicate** over $(\mathbb{A}, \mathbb{B})$ is a subobject*

$$j_C : C \hookrightarrow \mathbf{Beh}_0$$

in $\mathcal{E}/Z$. Write

$$\text{Pred}_{\mathbb{A}, \mathbb{B}} := \text{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_0)$$

for the poset of behavioural predicates, ordered by inclusion.

A behavioural predicate specifies a class of residual behaviours without yet requiring stability under future input. The residual stability condition is exactly derivative-closedness.

Definition 3.2.2 (Residual invariance). *A behavioural predicate*

$$j_V : V \hookrightarrow \mathbf{Beh}_0$$

*is **residually invariant**, or **derivative-closed**, if the canonical derivative map restricts to a morphism*

$$\delta_V : A_1 \times_{t_A, A_0, p_V} V \rightarrow V$$

in $\mathcal{E}/Z$, equivalently if

$$j_V \delta_V = \partial(1 \times j_V).$$

In generalized-element notation, this says

$$H \in V, \quad a : u \rightarrow p_V(H) \implies \partial_a H \in V.$$

Write

$$\text{Coeq}_{\mathbb{A}, \mathbb{B}} \subseteq \text{Pred}_{\mathbb{A}, \mathbb{B}}$$

for the full subposet of residually invariant predicates, and write

$$I_\partial : \text{Coeq}_{\mathbb{A}, \mathbb{B}} \hookrightarrow \text{Pred}_{\mathbb{A}, \mathbb{B}}$$

for the inclusion.

3.2.3 Generated closure, safety core, and the adjoint triple

There are two canonical ways to replace an arbitrary behavioural predicate by a residually invariant one. The first freely adjoins all residual derivatives; the second retains only those behaviours whose every residual derivative already satisfies the predicate.

Definition 3.2.3 (Generated residual closure). *Let*

$$j_C : C \hookrightarrow \mathbf{Beh}_0$$

be a behavioural predicate. Its derivative domain is

$$D(C) := A_1 \times_{t_A, A_0, p_C} C,$$

regarded over Z by the derivative state. Factor the derivative map in $\mathcal{E}/Z$ as

$$\begin{array}{ccc} D(C) & \xrightarrow{\partial(1 \times j_C)} & \mathbf{Beh}_0 \\ & \searrow e_C \quad \swarrow g_C & \\ & \text{Gen}_\partial(C) & \end{array}$$

with e_C regular epimorphic and g_C monic. The subobject

$$g_C : \text{Gen}_\partial(C) \hookrightarrow \mathbf{Beh}_0$$

*is the **generated residual closure** of C .*

For the specification given by the inclusion

$$j_C : C \rightarrow \mathbf{Beh}_0,$$

this is precisely the derivative image constructed in Chapter 2. The derivative-closure theorem, Theorem 2.7.1, therefore gives the following universal property.

Proposition 3.2.1 (Universal property of generated residual closure). *For every behavioural predicate C , the subobject*

$$\text{Gen}_\partial(C) \hookrightarrow \mathbf{Beh}_0$$

is residually invariant and is the least residually invariant predicate containing C . Consequently, for every $V \in \text{Coeq}_{\mathbb{A}, \mathbb{B}}$,

$$\text{Gen}_\partial(C) \leq V \iff C \leq I_\partial(V).$$

Proof. Apply Theorem 2.7.1 to the behaviour specification

$$j_C : C \rightarrow \mathbf{Beh}_0.$$

Its formal derivative orbit is

$$A_1 \times_{t_A, A_0, p_C} C,$$

and its residual evaluation map is

$$\partial(1 \times j_C) : A_1 \times_{t_A, A_0, p_C} C \rightarrow \mathbf{Beh}_0.$$

By definition, the regular image of this map is

$$\text{Gen}_\partial(C) \hookrightarrow \mathbf{Beh}_0.$$

Theorem 2.7.1 states that this regular image is residually invariant and is the smallest residually invariant subobject containing the image of j_C . Since j_C is itself a monomorphism, its image is exactly the predicate C . Hence

$$\text{Gen}_\partial(C)$$

is the least residually invariant predicate containing C .

Therefore, for every residually invariant predicate V ,

$$\text{Gen}_\partial(C) \leq V$$

if and only if

$$C \leq V.$$

Equivalently,

$$\text{Gen}_\partial(C) \leq V \iff C \leq I_\partial(V).$$

□

The second construction uses universal quantification over all derivatives of a behaviour. Since $\mathcal{E}$ is locally cartesian closed, pullback along the state projection

$$\text{pr}_\partial : D_{\mathbf{Beh}} \rightarrow \mathbf{Beh}_0$$

has a right adjoint on subobject posets,

$$\forall_{\text{pr}_\partial} : \text{Sub}_{\mathcal{E}}(D_{\mathbf{Beh}}) \rightarrow \text{Sub}_{\mathcal{E}}(\mathbf{Beh}_0).$$

Definition 3.2.4 (Residual safety core). *Let*

$$j_C : C \hookrightarrow \mathbf{Beh}_0$$

*be a behavioural predicate. Its **residual safety core** is*

$$\text{Safe}_\partial(C) := \forall_{\text{pr}_\partial} (\partial^* C) \hookrightarrow \mathbf{Beh}_0,$$

where

$$\partial^* C \hookrightarrow D_{\mathbf{Beh}}$$

is the pullback of

$$C \hookrightarrow \mathbf{Beh}_0$$

along the derivative map ∂ .

Internally, the defining predicate is

$$H \in \text{Safe}_\partial(C) \iff \forall a : u \rightarrow p_{\mathbf{Beh}}(H), \partial_a H \in C.$$

Thus $\text{Safe}_\partial(C)$ consists of those behaviours for which every admissible residual continuation satisfies C .

Proposition 3.2.2 (Universal property of the residual safety core). *For every behavioural predicate C , the residual safety core*

$$\text{Safe}_\partial(C) \hookrightarrow \mathbf{Beh}_0$$

is residually invariant and is the greatest residually invariant predicate contained in C .

Consequently, for every

$$V \in \text{Coeq}_{\mathbb{A}, \mathbb{B}},$$

$$I_\partial(V) \leq C \iff V \leq \text{Safe}_\partial(C).$$

Proof.

1. The safety core lies in the original predicate.

The identity map of the input category gives a section

$$\iota_{\mathbf{Beh}} : \mathbf{Beh}_0 \rightarrow D_{\mathbf{Beh}}, \quad H \mapsto (1_{p_{\mathbf{Beh}}(H)}, H),$$

satisfying

$$\text{pr}_\partial \iota_{\mathbf{Beh}} = 1_{\mathbf{Beh}_0},$$

and

$$\partial \iota_{\mathbf{Beh}} = 1_{\mathbf{Beh}_0}.$$

Hence, if every derivative of H lies in C , then in particular the identity derivative of H lies in C . Therefore

$$\text{Safe}_\partial(C) \leq C.$$

2. The safety core is derivative-closed.

Let

$$H \in \text{Safe}_\partial(C), \quad a : u \rightarrow p_{\mathbf{Beh}}(H).$$

To show that

$$\partial_a H \in \text{Safe}_\partial(C),$$

let

$$c : w \rightarrow u.$$

The derivative multiplication law gives

$$\partial_c(\partial_a H) = \partial_{a \circ c} H.$$

Since H lies in the safety core, every derivative of H , including $\partial_{a \circ c} H$, lies in C . Thus every derivative of $\partial_a H$ lies in C , and therefore

$$\partial_a H \in \text{Safe}_\partial(C).$$

Hence $\text{Safe}_\partial(C)$ is residually invariant.

3. Prove maximality.

Let V be residually invariant and suppose

$$V \leq C.$$

If $H \in V$, residual invariance gives

$$\partial_a H \in V$$

for every admissible input arrow a . Since $V \leq C$, every such derivative lies in C . Hence

$$H \in \text{Safe}_\partial(C).$$

Therefore

$$V \leq \text{Safe}_\partial(C).$$

Conversely, if

$$V \leq \text{Safe}_\partial(C),$$

then

$$V \leq C$$

by the first part.

This proves

$$I_\partial(V) \leq C \iff V \leq \text{Safe}_\partial(C),$$

and establishes the claimed maximality.

□

Theorem 3.2.1 (Generation–inclusion–safety adjoint triple). *The generated residual closure and residual safety core define monotone maps*

$$\text{Gen}_\partial, \text{Safe}_\partial : \text{Pred}_{\mathbb{A}, \mathbb{B}} \rightarrow \text{Coeq}_{\mathbb{A}, \mathbb{B}},$$

and there is an adjoint triple of posets

$$\text{Gen}_\partial \dashv I_\partial \dashv \text{Safe}_\partial.$$

Explicitly, for every behavioural predicate C and every residually invariant predicate V ,

$$\text{Gen}_\partial(C) \leq V \iff C \leq I_\partial(V),$$

and

$$I_\partial(V) \leq C \iff V \leq \text{Safe}_\partial(C).$$

Proof. The first adjunction is Proposition 3.2.1, and the second is Proposition 3.2.2. $\square$

Corollary 3.2.1 (Closure and interior). *For every behavioural predicate C ,*

$$\text{Safe}_\partial(C) \leq C \leq \text{Gen}_\partial(C).$$

Moreover, the following are equivalent:

1. C is residually invariant;
2. $\text{Gen}_\partial(C) = C$;
3. $\text{Safe}_\partial(C) = C$.

Thus residually invariant predicates are simultaneously the fixed points of the generated residual closure and of the residual safety interior.

Proof. The two inequalities are the unit of $\text{Gen}_\partial \dashv I_\partial$ and the counit of $I_\partial \dashv \text{Safe}_\partial$. If C is residually invariant, both universal properties force equality. Conversely, either equality exhibits C as an object of $\text{Coeq}_{\mathbb{A}, \mathbb{B}}$. $\square$

3.2.4 Coequational subobjects

Definition 3.2.5 (Local coequational theory and residual behavioural contract). *A **local coequational theory** over $(\mathbb{A}, \mathbb{B})$, equivalently a **residual behavioural contract**, is a residually invariant behavioural predicate*

$$j_V : V \hookrightarrow \mathbf{Beh}_0.$$

The terminology records two aspects of the same structure. The phrase “coequational theory” emphasizes the Birkhoff and Eilenberg developments of this chapter and the next. The phrase “residual behavioural contract” emphasizes that V is a behavioural requirement stable under every future typed input. The following proposition identifies these predicates with strict subobjects of the canonical behaviour machine.

Proposition 3.2.3 (Residual contracts as subobjects of the canonical behaviour machine). *To give a local coequational theory*

$$j_V : V \hookrightarrow \mathbf{Beh}_0$$

is equivalently to give a subobject

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Under this equivalence, the canonical derivative and output maps restrict to

$$\delta_V : A_1 \times_{t_A, A_0, p_V} V \rightarrow V$$

and

$$\omega_V : A_1 \times_{t_A, A_0, p_V} V \rightarrow B_1.$$

Proof. Suppose first that

$$j_V : V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a subobject in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Choose a monic representative j_V . By Lemma 3.1.2, its underlying carrier map

$$j_V : V \hookrightarrow \mathbf{Beh}_0$$

is monic in $\mathcal{E}/Z$.

Since j_V is strict, its transition equation is

$$j_V \delta_V = \delta_{\mathbf{Beh}}(1 \times j_V) = \partial(1 \times j_V).$$

Hence the underlying carrier subobject is derivative-closed. Therefore it is a local coequational theory.

Conversely, suppose

$$j_V : V \hookrightarrow \mathbf{Beh}_0$$

is a derivative-closed subobject in $\mathcal{E}/Z$. Write

$$p_V := p_{\mathbf{Beh}} j_V, \quad q_V := q_{\mathbf{Beh}} j_V.$$

Derivative-closedness supplies a unique map

$$\delta_V : A_1 \times_{t_A, A_0, p_V} V \rightarrow V$$

satisfying

$$j_V \delta_V = \delta_{\mathbf{Beh}}(1 \times j_V).$$

Define

$$\omega_V := \omega_{\mathbf{Beh}}(1 \times j_V).$$

The output typing equations are inherited from the canonical behaviour machine:

$$\begin{aligned} s_B \omega_V &= s_B \omega_{\mathbf{Beh}} \widehat{j_V} \\ &= q_{\mathbf{Beh}} \delta_{\mathbf{Beh}} \widehat{j_V} \\ &= q_{\mathbf{Beh}} j_V \delta_V \\ &= q_V \delta_V. \end{aligned}$$

and

$$\begin{aligned} t_B \omega_V &= t_B \omega_{\mathbf{Beh} \hat{j}_V} \\ &= q_{\mathbf{Beh} \rho_{\mathbf{Beh} \hat{j}_V}} \\ &= q_{\mathbf{Beh} j_V \rho_V} \\ &= q_V \rho_V. \end{aligned}$$

The transition unit law follows from

$$\begin{aligned} j_V \delta_V(1_{p_V H}, H) &= \delta_{\mathbf{Beh}}(1_{p_{\mathbf{Beh} j_V H}}, j_V H) \\ &= j_V H. \end{aligned}$$

Since j_V is monic,

$$\delta_V(1_{p_V H}, H) = H.$$

For composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

and $H \in V$ over w ,

$$\begin{aligned} j_V \delta_V(b \circ a, H) &= \delta_{\mathbf{Beh}}(b \circ a, j_V H) \\ &= \delta_{\mathbf{Beh}}(a, \delta_{\mathbf{Beh}}(b, j_V H)) \\ &= \delta_{\mathbf{Beh}}(a, j_V \delta_V(b, H)) \\ &= j_V \delta_V(a, \delta_V(b, H)). \end{aligned}$$

Monicity gives

$$\delta_V(b \circ a, H) = \delta_V(a, \delta_V(b, H)).$$

The output unit law is obtained by direct restriction:

$$\begin{aligned} \omega_V(1_{p_V H}, H) &= \omega_{\mathbf{Beh}}(1_{p_{\mathbf{Beh} j_V H}}, j_V H) \\ &= 1_{q_{\mathbf{Beh} j_V H}} \\ &= 1_{q_V H}. \end{aligned}$$

The output multiplication law is likewise the restriction of the canonical output law:

$$\begin{aligned} \omega_V(b \circ a, H) &= \omega_{\mathbf{Beh}}(b \circ a, j_V H) \\ &= \omega_{\mathbf{Beh}}(b, j_V H) \circ \omega_{\mathbf{Beh}}(a, \delta_{\mathbf{Beh}}(b, j_V H)) \\ &= \omega_V(b, H) \circ \omega_{\mathbf{Beh}}(a, j_V \delta_V(b, H)) \\ &= \omega_V(b, H) \circ \omega_V(a, \delta_V(b, H)). \end{aligned}$$

Thus V carries a Mealy structure. By construction,

$$j_V : V \rightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}$$

preserves transition and output strictly. Its carrier is monic, so Lemma 3.1.2 implies that it is monic in the strict Mealy category. Hence it is a subobject of the canonical behaviour machine. $\square$

The derivative-closedness condition is the factorization

$$\begin{array}{ccc} A_1 \times_{t_A, A_0, p_V} V & \xrightarrow{\partial(1 \times j_V)} & \mathbf{Beh}_0 \\ & \searrow \delta_V & \nearrow j_V \\ & V & \end{array}$$

in $\mathcal{E}/Z$. Equivalently, the transition-output cell of the canonical machine restricts along j_V :

$$\begin{array}{ccc} A_1 \times_{A_0} V & \xrightarrow{1 \times j_V} & A_1 \times_{A_0} \mathbf{Beh}_0 \\ \langle \delta_V, \omega_V \rangle \downarrow & & \downarrow \langle \delta_{\mathbf{Beh}}, \omega_{\mathbf{Beh}} \rangle \\ V \times_{B_0} B_1 & \xrightarrow{j_V \times 1} & \mathbf{Beh}_0 \times_{B_0} B_1. \end{array}$$

When $V \hookrightarrow \mathbf{Beh}_0$ is a local coequational theory, we write

$$\delta_V(a, H) = \partial_a H$$

and

$$\omega_V(a, H) = \omega_{\mathbf{Beh}}(a, H).$$

3.2.5 Models of local coequational theories

Definition 3.2.6 (Models and contract satisfaction). *Let*

$$j_V : V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

be a local coequational theory, and regard its carrier as the monomorphism

$$j_V : V \hookrightarrow \mathbf{Beh}_0$$

in

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0.$$

A Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

models V , or satisfies the residual behavioural contract V , if its carrier semantic map

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

factors through j_V in the carrier slice $\mathcal{E}/Z$. Equivalently, there is a necessarily unique morphism

$$\bar{\beta}_P : P \rightarrow V$$

in $\mathcal{E}/Z$ such that

$$j_V \bar{\beta}_P = \beta_P.$$

We write

$$P \models V$$

and denote by

$$\text{Mod}(V)$$

the full subcategory of

$$\text{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

on the objects that model V .

Lemma 3.2.1 (Model factorizations are strict). *If $P \models V$, then the unique factorization*

$$\bar{\beta}_P : P \rightarrow V$$

satisfying

$$j_V \bar{\beta}_P = \beta_P$$

is a strict semantic homomorphism.

Proof.

1. **Check the map over Z .** The equation

$$j_V \bar{\beta}_P = \beta_P$$

and the fact that both j_V and β_P are maps over Z imply that $\bar{\beta}_P$ is a map over Z . This is the lower triangle in

$$\begin{array}{ccc}
 P & \xrightarrow{\beta_P} & \mathbf{Beh}_0 \\
 & \searrow \bar{\beta}_P & \nearrow j_V \\
 & V & \\
 (p_P, q_P) \swarrow & \downarrow (p_V, q_V) & \searrow (p_{\mathbf{Beh}}, q_{\mathbf{Beh}}) \\
 & Z &
 \end{array}$$

2. **Check transition preservation.** Both composites

$$j_V \bar{\beta}_P \delta_P \quad \text{and} \quad j_V \delta_V (1 \times \bar{\beta}_P)$$

are equal to the transition-preservation composite for the strict map

$$\beta_P : P \rightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}.$$

The comparison is the outer rectangle

$$\begin{array}{ccccc}
 & & 1 \times \beta_P & & \\
 & \nearrow & & \searrow & \\
 A_1 \times_{A_0} P & \xrightarrow{1 \times \bar{\beta}_P} & A_1 \times_{A_0} V & \xrightarrow{1 \times j_V} & A_1 \times_{A_0} \mathbf{Beh}_0 \\
 \delta_P \downarrow & & \downarrow \delta_V & & \downarrow \delta_{\mathbf{Beh}} \\
 P & \xrightarrow{\bar{\beta}_P} & V & \xrightarrow{j_V} & \mathbf{Beh}_0 \\
 & \searrow \beta_P & & \nearrow & \\
 & & & &
 \end{array}$$

Since j_V is monic, the transition equation holds in V .

3. **Check output preservation.** The output equation for $\bar{\beta}_P$ follows from the output equation for β_P , since ω_V is the restriction of $\omega_{\mathbf{Beh}}$. It is represented by

$$\begin{array}{ccccc}
 A_1 \times_{A_0} P & \xrightarrow{1 \times \bar{\beta}_P} & A_1 \times_{A_0} V & \xrightarrow{1 \times j_V} & A_1 \times_{A_0} \mathbf{Beh}_0 \\
 & \searrow \omega_P & \downarrow \omega_V & \swarrow \omega_{\mathbf{Beh}} & \\
 & & B_1 & &
 \end{array}$$

□

3.2.6 Restricted canonical subobjects

Lemma 3.2.2 (Semantic map of a restricted canonical subobject). *Let*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

be a local coequational theory. Then the residual semantics map

$$\beta_V : V \rightarrow \mathbf{Beh}_0$$

is the inclusion

$$j_V : V \hookrightarrow \mathbf{Beh}_0.$$

Proof.

1. **Compute residual objects.** Let

$$H \in V_v$$

and let

$$r : u \rightarrow v$$

be a residual object. In the restricted canonical object, execution of r from H gives

$$\partial_r H.$$

Its anchor output is

$$q_V(\partial_r H) = H(r).$$

This is the object part of the equality displayed by

$$\begin{array}{ccc} H & \longmapsto & H(1_v) \\ \partial_r \downarrow & & \\ \partial_r H & \longmapsto & H(r). \end{array}$$

2. **Compute residual arrows.** For a residual arrow

$$w \xrightarrow{a} u \xrightarrow{r} v,$$

the output produced by the restricted canonical object at $\partial_r H$ along a is

$$\omega_V(a, \partial_r H).$$

This is the arrow assigned by H to the residual morphism

$$r \circ a \rightarrow r.$$

The comparison is

$$w \begin{array}{c} \xrightarrow{a} \\ \curvearrowright^{r \circ a} \\ \xrightarrow{r} \end{array} u \rightarrow v \quad \rightsquigarrow \quad H(r \circ a) \xrightarrow{H(r \circ a \rightarrow r)} H(r).$$

3. **Conclude equality of represented behaviours.** Thus the residual behaviour computed from H in the restricted canonical object agrees with H on residual objects and residual arrows. Hence

$$\beta_V(H) = H.$$

Therefore

$$\beta_V = j_V.$$

Equivalently, the terminal semantic triangle is

$$\begin{array}{ccc} V & \xrightarrow{\beta_V} & \mathbf{Beh}^{\text{can}} \\ & \searrow & \nearrow j_V \\ & V & \end{array}$$

and the diagonal identity is forced by the previous computation.

□

3.3 Semantic images and regular-image structure

3.3.1 Regular-epimorphic descent for subobjects

The following elementary descent fact will be used repeatedly.

Lemma 3.3.1 (Regular-epimorphic descent for subobject membership). *Let $\mathcal{X}$ be either $\mathcal{E}$ or a slice $\mathcal{E}/S$. Let*

$$e : X \twoheadrightarrow Y$$

be a regular epimorphism in $\mathcal{X}$, and let

$$j : V \hookrightarrow W$$

be a monomorphism in $\mathcal{X}$. If

$$h : Y \rightarrow W$$

is a morphism in $\mathcal{X}$ such that

$$he$$

factors through j , then h factors uniquely through j .

Proof.

1. **Use the kernel pair of the regular epimorphism.** Let

$$R \rightrightarrows X$$

be the kernel pair of e in $\mathcal{X}$. Since e is a regular epimorphism, it is the coequalizer of this kernel pair:

$$R \begin{array}{c} \xrightarrow{r_1} \\ \xrightarrow{r_2} \end{array} X \xrightarrow{e} Y.$$

2. **Descend the factorization.** Suppose

$$he = jg$$

for some $g : X \rightarrow V$. Since

$$her_1 = her_2,$$

we have

$$jgr_1 = jgr_2.$$

Since j is monic,

$$gr_1 = gr_2.$$

Therefore g descends uniquely through e to a map

$$\bar{g} : Y \rightarrow V$$

with

$$\bar{g}e = g.$$

The descent diagram is

$$\begin{array}{ccccc} R & \xrightarrow[r_2]{r_1} & X & \xrightarrow{e} & Y \\ & & & \searrow g & \downarrow \bar{g} \\ & & & & V. \end{array}$$

3. **Compare with h .** Now

$$j\bar{g}e = jg = he.$$

Since e is epic,

$$j\bar{g} = h.$$

Uniqueness follows from monicity of j . The final comparison is

$$\begin{array}{ccc} X & \xrightarrow{e} & Y \\ & \searrow g & \downarrow \bar{g} \\ & & V \\ & \searrow he & \downarrow j \\ & & W. \end{array}$$

□

3.3.2 Regular images of strict homomorphisms

This section proves that regular images of strict semantic homomorphisms are created by the carrier slice.

The following proposition should be read as a carrier-creation statement for regular images. In particular, the regular image factorization of a strict semantic homomorphism is computed in the carrier slice

$$\mathcal{E}/Z$$

and then carries the unique Mealy structure making the two factorization maps strict. Thus the phrase “regular image” below always refers to the regular image in the carrier slice, equipped with this created strict Mealy structure.

Proposition 3.3.1 (Regular images are created by the carrier slice). *Let*

$$f : P \rightarrow Q$$

be a strict semantic homomorphism. Factor its underlying carrier map in $\mathcal{E}/Z$ as

$$P \xrightarrow{e} I \xrightarrow{m} Q,$$

where e is a regular epimorphism and m is a monomorphism.

Then I carries a unique Mealy structure such that

$$e : P \twoheadrightarrow I$$

and

$$m : I \hookrightarrow Q$$

are strict semantic homomorphisms. Moreover:

1. *e is a regular epimorphism in*

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}};$$

2. *m is a monomorphism in*

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}};$$

3. *the factorization*

$$f = me$$

is the regular image factorization of f in the strict Mealy category.

Consequently the carrier functor

$$U : \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \rightarrow \mathcal{E}/Z$$

creates regular image factorizations.

Proof. Put

$$p_I := p_Q m, \quad q_I := q_Q m,$$

and write

$$D_I := A_1 \times_{t_A, A_0, p_I} I.$$

Since regular epimorphisms are stable under pullback, the induced map

$$\widehat{e} : D_P \twoheadrightarrow D_I$$

is a regular epimorphism. In generalized-element notation,

$$\widehat{e}(a, x) = (a, ex).$$

Consider the underlying map in $\mathcal{E}$

$$h_\delta := \delta_Q \widehat{m} : D_I \rightarrow Q.$$

Since $f = me$ and f is strict,

$$\begin{aligned} h_\delta \widehat{e} &= \delta_Q \widehat{m} \widehat{e} \\ &= \delta_Q \widehat{f} \\ &= f \delta_P \\ &= me \delta_P. \end{aligned}$$

Thus $h_\delta \widehat{e}$ factors through the underlying monomorphism

$$m : I \hookrightarrow Q.$$

At this point the transition map on I has not yet been constructed, so D_I is not yet being regarded over Z by a derivative-state coordinate. We therefore apply Lemma 3.3.1 in the underlying category $\mathcal{E}$.

Since $\widehat{e}$ is regular epimorphic in $\mathcal{E}$, m is monic in $\mathcal{E}$, and

$$h_\delta \widehat{e} = m(e \delta_P),$$

there is a unique map

$$\delta_I : D_I \rightarrow I$$

satisfying

$$m \delta_I = \delta_Q \widehat{m}.$$

Define

$$\omega_I := \omega_Q \widehat{m} : D_I \rightarrow B_1.$$

The transition has the required input typing:

$$\begin{aligned} p_I \delta_I &= p_Q m \delta_I \\ &= p_Q \delta_Q \widehat{m} \\ &= s_A \lambda_Q \widehat{m} \\ &= s_A \lambda_I. \end{aligned}$$

The output map has the required source typing:

$$\begin{aligned} s_B \omega_I &= s_B \omega_Q \widehat{m} \\ &= q_Q \delta_Q \widehat{m} \\ &= q_Q m \delta_I \\ &= q_I \delta_I. \end{aligned}$$

It also has the required target typing:

$$\begin{aligned} t_B \omega_I &= t_B \omega_Q \widehat{m} \\ &= q_Q \rho_Q \widehat{m} \\ &= q_Q m \rho_I \\ &= q_I \rho_I. \end{aligned}$$

These equations show that δ_I and ω_I are correctly typed. In particular, D_I may now be regarded over Z by

$$\langle s_A \lambda_I, q_I \delta_I \rangle : D_I \rightarrow A_0 \times B_0.$$

The transition unit law holds after composing with m :

$$\begin{aligned} m\delta_I(1_{p_I z}, z) &= \delta_Q(1_{p_Q m z}, m z) \\ &= m z. \end{aligned}$$

Since m is monic,

$$\delta_I(1_{p_I z}, z) = z.$$

For composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w,$$

and $z \in I$ over w ,

$$\begin{aligned} m\delta_I(b \circ a, z) &= \delta_Q(b \circ a, m z) \\ &= \delta_Q(a, \delta_Q(b, m z)) \\ &= \delta_Q(a, m\delta_I(b, z)) \\ &= m\delta_I(a, \delta_I(b, z)). \end{aligned}$$

Monicity of m gives

$$\delta_I(b \circ a, z) = \delta_I(a, \delta_I(b, z)).$$

The output unit law follows by restriction:

$$\begin{aligned} \omega_I(1_{p_I z}, z) &= \omega_Q(1_{p_Q m z}, m z) \\ &= 1_{q_Q m z} \\ &= 1_{q_I z}. \end{aligned}$$

Similarly,

$$\begin{aligned} \omega_I(b \circ a, z) &= \omega_Q(b \circ a, m z) \\ &= \omega_Q(b, m z) \circ \omega_Q(a, \delta_Q(b, m z)) \\ &= \omega_I(b, z) \circ \omega_Q(a, m\delta_I(b, z)) \\ &= \omega_I(b, z) \circ \omega_I(a, \delta_I(b, z)). \end{aligned}$$

Hence

$$I = (I, p_I, q_I, \delta_I, \omega_I)$$

is a Mealy morphism.

The equations

$$m\delta_I = \delta_Q \hat{m}$$

and

$$\omega_I = \omega_Q \hat{m}$$

show that m is strict.

To show that e is strict, compute

$$\begin{aligned} me\delta_P &= f\delta_P \\ &= \delta_Q \hat{f} \\ &= \delta_Q \hat{m} \hat{e} \\ &= m\delta_I \hat{e}. \end{aligned}$$

Since m is monic,

$$e\delta_P = \delta_I \hat{e}.$$

For the output,

$$\begin{aligned}\omega_I \widehat{e} &= \omega_Q \widehat{m} \widehat{e} \\ &= \omega_Q \widehat{f} \\ &= \omega_P.\end{aligned}$$

Hence e is strict.

Since the carrier of m is monic, Lemma 3.1.2 implies that m is monic in the strict Mealy category.

We next prove that e is a regular epimorphism in the strict Mealy category. Let

$$R_e \rightrightarrows P$$

be its kernel pair there. By Lemma 3.1.2, this kernel pair is created by the carrier functor.

Let

$$h : P \rightarrow X$$

be a strict semantic homomorphism coequalizing the kernel pair of e . Since the carrier map e is the coequalizer of its carrier kernel pair, the underlying carrier map of h descends uniquely to a map

$$\bar{h} : I \rightarrow X$$

in $\mathcal{E}/Z$ satisfying

$$\bar{h}e = h.$$

To prove transition preservation, precompose with the regular epimorphism $\widehat{e}$:

$$\begin{aligned}\bar{h}\delta_I \widehat{e} &= \bar{h}e\delta_P \\ &= h\delta_P \\ &= \delta_X \widehat{h} \\ &= \delta_X \widehat{h} \widehat{e}.\end{aligned}$$

Since $\widehat{e}$ is epic,

$$\bar{h}\delta_I = \delta_X \widehat{h}.$$

Likewise,

$$\begin{aligned}\omega_X \widehat{h} \widehat{e} &= \omega_X \widehat{h} \\ &= \omega_P \\ &= \omega_I \widehat{e}.\end{aligned}$$

Since $\widehat{e}$ is epic,

$$\omega_X \widehat{h} = \omega_I.$$

Thus $\bar{h}$ is strict.

Therefore e is the coequalizer of its kernel pair in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}},$$

and hence is a regular epimorphism there.

Finally, suppose that f factors through a strict monomorphism

$$n : J \hookrightarrow Q :$$

$$f = nh$$

for some strict semantic homomorphism

$$h : P \rightarrow J.$$

Since

$$P \xrightarrow{e} I \xrightarrow{m} Q$$

is the carrier image factorization of f , there is a unique carrier map

$$u : I \rightarrow J$$

in $\mathcal{E}/Z$ satisfying

$$ue = h, \quad nu = m.$$

The map u preserves transition:

$$\begin{aligned} nu\delta_I &= m\delta_I \\ &= \delta_Q \widehat{m} \\ &= \delta_Q \widehat{n} \widehat{u} \\ &= n\delta_J \widehat{u}. \end{aligned}$$

Since the carrier of n is monic,

$$u\delta_I = \delta_J \widehat{u}.$$

It also preserves output:

$$\begin{aligned} \omega_J \widehat{u} &= \omega_Q \widehat{n} \widehat{u} \\ &= \omega_Q \widehat{m} \\ &= \omega_I. \end{aligned}$$

Hence u is strict.

Its uniqueness as a strict map follows from its uniqueness as a carrier map. Therefore $m : I \hookrightarrow Q$ has the universal property of the image of f , and

$$P \xrightarrow{e} I \xrightarrow{m} Q$$

is the regular image factorization in the strict Mealy category.

Finally, the Mealy structure on I is unique. Indeed, strictness of m forces

$$m\delta_I = \delta_Q \widehat{m}$$

and

$$\omega_I = \omega_Q \widehat{m},$$

and monicity of m determines δ_I uniquely. □

Corollary 3.3.1 (Regular epimorphisms are detected by carriers). *Let*

$$e : P \rightarrow Q$$

be a strict semantic homomorphism. Then the following are equivalent:

1. the underlying carrier map

$$Ue : UP \rightarrow UQ$$

is a regular epimorphism in

$$\mathcal{E}/Z;$$

2. the strict semantic homomorphism

$$e : P \rightarrow Q$$

is a regular epimorphism in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Consequently, the regular quotients of Definition 3.1.1 are exactly the categorical regular epimorphisms in the strict Mealy category.

Proof.

1. **Carrier regular epimorphisms are categorical regular epimorphisms.**

Suppose that

$$Ue : UP \rightarrow UQ$$

is a regular epimorphism in $\mathcal{E}/Z$.

Apply Proposition 3.3.1 to e . Since the carrier image of a regular epimorphism is its entire codomain, its carrier regular-image factorization is, up to the unique canonical isomorphism of image representatives,

$$UP \xrightarrow{Ue} UQ \xrightarrow{1_{UQ}} UQ.$$

The proposition creates this factorization in the strict Mealy category and identifies its first map as a categorical regular epimorphism. Hence e is a regular epimorphism in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

2. **Categorical regular epimorphisms have regular-epimorphic carriers.**

Conversely, suppose

$$e : P \rightarrow Q$$

is a regular epimorphism in the strict Mealy category.

Factor its carrier in $\mathcal{E}/Z$ as

$$UP \xrightarrow{Ue'} UI \xrightarrow{Um} UQ,$$

where Ue' is regular epimorphic and Um is monic.

By Proposition 3.3.1, the object I carries a unique Mealy structure for which

$$P \xrightarrow{e'} I \xrightarrow{m} Q$$

is the regular image factorization of e in the strict Mealy category. Thus

$$e = me',$$

with e' a regular epimorphism and m a monomorphism.

Regular epimorphisms are extremal. Since the regular epimorphism e factors through the monomorphism m , the map m is an isomorphism. Therefore its carrier

$$Um : UI \rightarrow UQ$$

is an isomorphism in $\mathcal{E}/Z$.

Since

$$Ue = (Um)(Ue')$$

and Ue' is regular epimorphic, it follows that Ue is a regular epimorphism in $\mathcal{E}/Z$.

□

Remark 3.3.1. *By the preceding corollary, there is no distinction in this chapter between:*

1. *a strict semantic homomorphism whose carrier is a regular epimorphism in $\mathcal{E}/Z$;*
2. *a regular epimorphism in*

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Accordingly, the terms

$$\text{regular quotient} \quad \text{and} \quad \text{regular-epimorphic quotient}$$

may be used interchangeably for strict semantic homomorphisms.

3.3.3 Semantic images

Definition 3.3.1 (Semantic image). *For a Mealy morphism P , its **semantic image** is the regular image factorization of the unique strict map to the terminal object:*

$$P \xrightarrow{e_P} \text{Beh}(P) \xrightarrow{m_P} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

On carriers, this is the regular image factorization of

$$\beta_P : P \rightarrow \mathbf{Beh}_0$$

in $\mathcal{E}/Z$.

Remark 3.3.2 (Image subobjects and chosen representatives). *The notation*

$$\text{Beh}(P)$$

is used primarily for the regular-image subobject

$$\text{Im}(\beta_P) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

regarded as an element of

$$\text{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_0).$$

Accordingly, an equality such as

$$\text{Beh}(P) = V$$

means equality of subobject classes in this subobject poset. If particular regular epi-mono factorizations have been chosen, the corresponding carrier objects need not be literally equal; instead, there is a unique canonical isomorphism between the chosen image representatives commuting with their monomorphisms into $\mathbf{Beh}_0$.

The same convention applies to chosen representatives of derivative images, joins, meets, and other regular-image subobjects appearing below.

The semantic image is displayed by

$$\begin{array}{ccc}
 P & \xrightarrow{\beta_P} & \mathbf{Beh}_0 \\
 e_P \searrow & & \nearrow m_P \\
 & \mathbf{Beh}(P) &
 \end{array}
 \quad \text{in } \mathcal{E}/Z.$$

The same factorization, with the Mealy structure created by Proposition 3.3.1, is

$$\begin{array}{ccc}
 P & \xrightarrow{\beta_P} & \mathbf{Beh}^{\text{can}} \\
 e_P \searrow & & \nearrow m_P \\
 & \mathbf{Beh}(P) &
 \end{array}
 \quad \text{in } \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Corollary 3.3.2 (Semantic images are coequational theories). *For every Mealy morphism P , the semantic image*

$$\mathbf{Beh}(P) \hookrightarrow \mathbf{Beh}^{\text{can}}$$

is a local coequational theory. Moreover,

$$P \twoheadrightarrow \mathbf{Beh}(P)$$

is a regular quotient in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Proof.

1. **Apply image creation.** The semantic image is the regular image of the strict semantic homomorphism

$$\beta_P : P \rightarrow \mathbf{Beh}^{\text{can}}.$$

Proposition 3.3.1 creates this image in the strict Mealy category.

2. **Identify the image as a subobject of the terminal object.** The monic part

$$\mathbf{Beh}(P) \hookrightarrow \mathbf{Beh}^{\text{can}}$$

is a subobject, hence a local coequational theory.

3. **Identify the quotient map.** The regular-epimorphic part

$$P \twoheadrightarrow \mathbf{Beh}(P)$$

is strict by the same proposition.

□

3.3.4 Meets and coproducts

Lemma 3.3.2 (Meets of local theories). *If*

$$S, T \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

are local coequational theories, then their meet

$$S \wedge T \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a local coequational theory.

Its carrier is the pullback subobject

$$S \wedge T = S \times_{\mathbf{Beh}_0} T$$

in $\mathcal{E}/Z$.

Proof. Regard S and T as strict subobjects of the canonical behaviour machine by Proposition 3.2.3. Form the pullback

$$\begin{array}{ccc} S \wedge T & \longrightarrow & T \\ \downarrow & \lrcorner & \downarrow \\ S & \hookrightarrow & \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} \end{array}$$

in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

By Lemma 3.1.2, this pullback is created by the carrier functor. Its carrier is therefore

$$S \times_{\mathbf{Beh}_0} T$$

in $\mathcal{E}/Z$, which is the meet of the two carrier subobjects.

The induced map

$$S \wedge T \longrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is strict and has monic carrier. Hence it is a subobject in the strict Mealy category. By Proposition 3.2.3, its carrier is a derivative-closed behavioural predicate. Therefore it is a local coequational theory.

Explicitly, its transition and output are the restrictions

$$\delta_{S \wedge T}(a, H) = \partial_a H$$

and

$$\omega_{S \wedge T}(a, H) = \omega_{\mathbf{Beh}}(a, H).$$

□

Proposition 3.3.2 (Small coproducts are created by the carrier slice). *A small family*

$$(P_i)_{i \in I}$$

of Mealy morphisms over $(\mathbb{A}, \mathbb{B})$ has a coproduct in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Its carrier is the coproduct

$$\coprod_{i \in I} P_i$$

in the slice

$$\mathcal{E}/Z,$$

and its transition and output maps are induced componentwise.

Proof.

1. **Use preservation of coproducts by pullback.** In a Grothendieck topos, pullback preserves small coproducts in every slice. Hence

$$A_1 \times_{t_A, A_0, P \coprod_i P_i} \coprod_i P_i \cong \coprod_i (A_1 \times_{t_A, A_0, P_{P_i}} P_i).$$

The coproduct of transition domains is displayed by

$$\begin{array}{ccc} A_1 \times_{A_0} P_i & \longrightarrow & A_1 \times_{A_0} \coprod_i P_i \\ \downarrow & & \downarrow \\ P_i & \longrightarrow & \coprod_i P_i. \end{array}$$

2. **Define transition and output componentwise.** The transition and output maps are induced by the component maps

$$\delta_{P_i}, \quad \omega_{P_i}.$$

Equivalently, for each i the following diagrams commute:

$$\begin{array}{ccc} A_1 \times_{A_0} P_i & \longrightarrow & A_1 \times_{A_0} \coprod_i P_i \\ \delta_{P_i} \downarrow & & \downarrow \delta_{\coprod_i P_i} \\ P_i & \longrightarrow & \coprod_i P_i \end{array} \quad \begin{array}{ccc} A_1 \times_{A_0} P_i & \longrightarrow & A_1 \times_{A_0} \coprod_i P_i \\ & \searrow \omega_{P_i} & \downarrow \omega_{\coprod_i P_i} \\ & & B_1. \end{array}$$

3. **Verify the laws.** The Mealy laws hold after restriction to each coproduct summand. Since the summand inclusions are jointly epic, the laws hold on the coproduct.
4. **Check the universal property.** A strict semantic homomorphism

$$\coprod_i P_i \rightarrow Q$$

is equivalently a compatible family of strict semantic homomorphisms

$$P_i \rightarrow Q.$$

This is the universal property of the coproduct in $\mathcal{E}/Z$ together with the componentwise Mealy structure:

$$\begin{array}{ccc} P_i & \longrightarrow & \coprod_i P_i \\ & \searrow f_i & \downarrow f \\ & & Q. \end{array}$$

□

3.4 Universal realization

3.4.1 The behaviour–realization correspondence

The semantic-image construction is the local form of the classical behaviour–realization adjunction. In the terminology of Goguen and Naudé, behaviour and realization form an adjoint situation: a system has a universal behaviour quotient, and a behaviour object has a canonical realization [7, 16, 17]. In the present setting, behaviour objects are local coequational subobjects of the terminal residual semantic Mealy morphism.

Let

$$\mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}$$

denote the poset category of local coequational theories

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

ordered by inclusion as subobjects in

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0.$$

Thus a morphism

$$V \rightarrow W$$

exists precisely when

$$V \leq W.$$

If

$$j_V : V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a local coequational theory, then by Proposition 3.2.3 the carrier V already carries the restricted canonical Mealy structure. Explicitly, its carrier span is

$$A_0 \xleftarrow{p_V} V \xrightarrow{q_V} B_0,$$

its transition is

$$\delta_V(a, H) = \partial_a H,$$

and its output is

$$\omega_V(a, H) = \omega_{\mathbf{Beh}}(a, H).$$

Equivalently, the structure is obtained by restriction along

$$j_V : V \hookrightarrow \mathbf{Beh}_0 :$$

$$\begin{array}{ccccc} A_1 \times_{t_A, A_0, p_V} V & \xrightarrow{1 \times j_V} & A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0 & \xrightarrow{\omega_{\mathbf{Beh}}} & B_1 \\ \delta_V \downarrow & \circlearrowleft & \downarrow \delta_{\mathbf{Beh}} & & \\ V & \xrightarrow{j_V} & \mathbf{Beh}_0 & & \end{array}$$

The output map is

$$\omega_V = \omega_{\mathbf{Beh}}(1 \times j_V).$$

If

$$V \leq W$$

are local coequational theories, then the inclusion

$$V \hookrightarrow W$$

is a strict semantic homomorphism, since both transition and output are inherited from the canonical behaviour machine. Thus local coequational theories embed into

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

by sending each subobject to its restricted canonical Mealy morphism.

On the other side, the assignment

$$P \mapsto \text{Beh}(P)$$

is functorial as a map

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \longrightarrow \mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}.$$

Indeed, if

$$f : P \rightarrow Q$$

is a strict semantic homomorphism, then by Lemma 3.1.1,

$$\beta_Q f = \beta_P.$$

Taking regular images in the carrier slice gives

$$\text{Beh}(P) \leq \text{Beh}(Q),$$

because the regular image of a composite is contained in the regular image of its second factor. The image-containment diagram is

$$\begin{array}{ccccc}
 P & \xrightarrow{f} & Q & \xrightarrow{\beta_Q} & \mathbf{Beh}^{\text{can}} \\
 e_P \downarrow & & e_Q \downarrow & \nearrow m_Q & \\
 \text{Beh}(P) & \dashrightarrow & \text{Beh}(Q) & &
 \end{array}$$

where the dashed arrow exists because

$$m_Q e_Q f = \beta_Q f = \beta_P.$$

Since

$$\mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}$$

is posetal, these inclusions are automatically functorial.

3.4.2 The universal property

Theorem 3.4.1 (Universal realization adjunction). *The semantic-image assignment*

$$\text{Beh} : \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \longrightarrow \mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}$$

is left adjoint to the inclusion of local coequational theories as restricted canonical Mealy morphisms

$$\iota : \mathbf{Coeq}_{\mathbb{A}, \mathbb{B}} \longrightarrow \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Thus there is an adjunction

$$\mathbf{Beh} \dashv \iota.$$

Explicitly, for every Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

and every local coequational theory

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

there is a natural bijection

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}(P, \iota V) \cong \mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}(\mathbf{Beh}(P), V).$$

Equivalently,

$$P \rightarrow V \text{ strictly} \iff \mathbf{Beh}(P) \leq V.$$

Proof. Let

$$j_V : V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be a local coequational theory.

1. **From a strict map into V to semantic containment.** Suppose first that there is a strict semantic homomorphism

$$f : P \rightarrow V.$$

Then

$$j_V f : P \rightarrow \mathbf{Beh}^{\text{can}}$$

is a strict semantic homomorphism. By terminality of

$$\mathbf{Beh}^{\text{can}}$$

in

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}},$$

this composite is the terminal semantic map:

$$j_V f = \beta_P.$$

Hence β_P factors through V . Since

$$\mathbf{Beh}(P)$$

is the regular image of β_P , it follows that

$$\mathbf{Beh}(P) \leq V.$$

This implication is displayed by

$$\begin{array}{ccccc}
 & & \beta_P & & \\
 & \nearrow & & \searrow & \\
 P & \xrightarrow{f} & V & \xrightarrow{j_V} & \mathbf{Beh}^{\text{can}} \\
 \downarrow e_P & \nearrow & \nearrow & & \nearrow \\
 \mathbf{Beh}(P) & & & \nearrow m_P &
 \end{array}$$

2. **From semantic containment to a strict map into V .** Conversely, suppose

$$\text{Beh}(P) \leq V.$$

The semantic image factorization of P is

$$P \xrightarrow{e_P} \text{Beh}(P) \xrightarrow{m_P} \mathbf{Beh}^{\text{can}}.$$

Since

$$\text{Beh}(P) \leq V,$$

the monomorphism m_P factors through j_V :

$$\begin{array}{ccc} \text{Beh}(P) & \xrightarrow{m_P} & \mathbf{Beh}^{\text{can}} \\ & \searrow \bar{m}_P & \nearrow j_V \\ & V & \end{array}$$

Therefore β_P factors through V :

$$P \xrightarrow{e_P} \text{Beh}(P) \xrightarrow{\bar{m}_P} V \xrightarrow{j_V} \mathbf{Beh}^{\text{can}}.$$

β_P (curved arrow from P to $\mathbf{Beh}^{\text{can}}$)

By Lemma 3.2.1, the induced map

$$P \rightarrow V$$

is a strict semantic homomorphism.

3. **Prove uniqueness.** Let

$$f, g : P \rightrightarrows V$$

be strict semantic homomorphisms. Then

$$j_V f \quad \text{and} \quad j_V g$$

are strict semantic homomorphisms from P to

$$\mathbf{Beh}^{\text{can}}.$$

Hence, by terminality,

$$j_V f = \beta_P = j_V g.$$

Since j_V is monic,

$$f = g.$$

4. **Conclude the bijection.** The left-hand hom-set

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}(P, V)$$

is empty or singleton according as

$$\text{Beh}(P) \leq V$$

fails or holds. This is exactly the right-hand hom-set in the poset

$$\mathbf{Coeq}_{\mathbb{A}, \mathbb{B}}.$$

Therefore the required natural bijection holds.

□

3.4.3 Unit, counit, and immediate consequences

Proposition 3.4.1 (Unit and counit). *Under the adjunction of Theorem 3.4.1, the unit at a Mealy morphism P is the semantic quotient*

$$\eta_P : P \twoheadrightarrow \text{Beh}(P).$$

The counit at a local coequational theory V is the canonical isomorphism

$$\text{Beh}(V) \xrightarrow{\cong} V.$$

Proof.

1. **Identify the unit.** The unit at P is the unique strict semantic homomorphism

$$P \rightarrow \text{Beh}(P)$$

corresponding to the identity inclusion

$$\text{Beh}(P) \leq \text{Beh}(P).$$

By construction, this is the regular-epimorphic part of the semantic image factorization:

$$P \xrightarrow{\eta_P} \text{Beh}(P) \xrightarrow{m_P} \mathbf{Beh}^{\text{can}}.$$

β_P

2. **Identify the counit.** Let

$$j_V : V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be a local coequational theory. By Lemma 3.2.2, the semantic map of the restricted canonical object V is its inclusion:

$$\beta_V = j_V.$$

Hence the semantic image of V is exactly V :

$$\text{Beh}(V) \cong V.$$

Diagrammatically:

$$V \xrightarrow{\beta_V = j_V} \text{Beh}(V) \xrightarrow{\quad} \mathbf{Beh}^{\text{can}}.$$

□

Corollary 3.4.1 (Full faithfulness of restricted canonical realization). *The inclusion of*

$$\text{Coeq}_{\mathbb{A}, \mathbb{B}}$$

into

$$\text{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

by restricted canonical Mealy morphisms is fully faithful.

Proof.

1. **Use the standard adjunction criterion.** In an adjunction

$$L \dashv R,$$

the right adjoint R is fully faithful precisely when the counit is an isomorphism.

2. **Apply the counit computation.** Here the counit is an isomorphism by Proposition 3.4.1. Therefore the inclusion of local coequational theories as restricted canonical Mealy morphisms is fully faithful.

□

Corollary 3.4.2 (Terminal realization of a coequational theory). *Let*

$$V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be a local coequational theory. Then V , with its restricted canonical Mealy structure, is terminal in

$$\mathbf{Mod}(V).$$

Proof.

1. **Use the model condition.** If

$$P \models V,$$

then

$$\mathbf{Beh}(P) \leq V.$$

2. **Apply the universal property.** By Theorem 3.4.1, the containment

$$\mathbf{Beh}(P) \leq V$$

is equivalent to the existence of a unique strict semantic homomorphism

$$P \rightarrow V.$$

3. **Identify the unique map.** Hence V is terminal in its model category. The unique map is the factorization of the terminal semantic map through V :

$$\begin{array}{ccc} P & \xrightarrow{\beta_P} & \mathbf{Beh}^{\text{can}} \\ & \searrow \tilde{\beta}_P & \nearrow j_V \\ & V & \end{array}$$

□

Corollary 3.4.3 (Behaviourally separated reflection). *The semantic quotient*

$$\eta_P : P \rightarrow \mathbf{Beh}(P)$$

is an isomorphism if and only if P is behaviourally separated.

Proof. The semantic image factorization is

$$P \xrightarrow{\eta_P} \text{Beh}(P) \xrightarrow{m_P} \mathbf{Beh}^{\text{can}}.$$

β_P (curved arrow from P to $\mathbf{Beh}^{\text{can}}$)

1. Behavioural separatedness implies reflection isomorphism.

Suppose P is behaviourally separated. Then

$$\beta_P = m_P \eta_P$$

is monic.

We claim that η_P is monic. Let

$$f, g : X \rightrightarrows P$$

satisfy

$$\eta_P f = \eta_P g.$$

Then

$$\begin{aligned} \beta_P f &= m_P \eta_P f \\ &= m_P \eta_P g \\ &= \beta_P g. \end{aligned}$$

Since β_P is monic,

$$f = g.$$

Hence η_P is monic.

By construction, η_P is also a regular epimorphism in the strict Mealy category. A morphism that is both a regular epimorphism and a monomorphism is an isomorphism. Therefore

$$\eta_P : P \xrightarrow{\cong} \text{Beh}(P).$$

2. Reflection isomorphism implies behavioural separatedness.

Conversely, suppose

$$\eta_P : P \xrightarrow{\cong} \text{Beh}(P)$$

is an isomorphism. Since

$$m_P : \text{Beh}(P) \hookrightarrow \mathbf{Beh}^{\text{can}}$$

is monic, the composite

$$\beta_P = m_P \eta_P$$

is monic. Therefore P is behaviourally separated.

□

3.4.4 Relation with minimal residual realization

Remark 3.4.1 (Relation with minimal residual realization). *Let*

$$k : K \rightarrow \mathbf{Beh}_0$$

be a behaviour specification. By Theorem 2.7.1, its derivative image

$$\mathbf{Der}_0(K) \hookrightarrow \mathbf{Beh}_0$$

is derivative-closed, hence is a local coequational theory. The minimal residual realization of Chapter 2 is precisely the restricted canonical Mealy morphism on this derivative image:

$$\mathbf{Min}(K) = \mathbf{Der}_0(K)$$

as a restricted canonical Mealy morphism.

If

$$i : K \rightarrow P$$

is a reachable realization of k , then

$$\mathbf{Beh}(P) = \mathbf{Der}_0(K).$$

Under this identification, the unit at P is the regular semantic quotient

$$P \twoheadrightarrow \mathbf{Min}(K)$$

from the categorical Myhill–Nerode theorem. This is the diagram

$$\begin{array}{ccccc} & & \beta_P & & \\ & \nearrow & & \searrow & \\ P & \twoheadrightarrow & \mathbf{Beh}(P) & \hookrightarrow & \mathbf{Beh}^{\text{can}} \\ & & \cong \downarrow & \nearrow j_K & \\ & & \mathbf{Der}_0(K) & & \end{array}$$

and the restricted canonical Mealy morphism on

$$\mathbf{Der}_0(K)$$

is

$$\mathbf{Min}(K).$$

3.5 The local coequational Birkhoff theorem

3.5.1 Closure properties of model classes

Theorem 3.5.1 (Local coequational closure). *Let*

$$V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be a local coequational theory. Then:

1. $\text{Mod}(V)$ is closed under strict homomorphic preimages.
2. $\text{Mod}(V)$ is closed under regular-epimorphic quotients.
3. $\text{Mod}(V)$ is closed under small coproducts.
4. V , regarded as the restricted canonical object, is itself a model of V , and

$$\text{Beh}(V) = V.$$

Consequently the assignment

$$V \mapsto \text{Mod}(V)$$

is monotone and order-reflecting.

Proof. Let

$$j_V : V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

denote the inclusion.

1. **Closure under strict homomorphic preimages.** Let

$$f : P \rightarrow Q$$

be strict and suppose $Q \models V$. Choose

$$\bar{\beta}_Q : Q \rightarrow V$$

with

$$j_V \bar{\beta}_Q = \beta_Q.$$

Then, by Lemma 3.1.1,

$$\beta_P = \beta_Q f = j_V \bar{\beta}_Q f,$$

so $P \models V$. The factorization is

$$\begin{array}{ccccc}
 P & \xrightarrow{f} & Q & \xrightarrow{\beta_Q} & \mathbf{Beh}^{\text{can}} \\
 & \searrow \bar{\beta}_Q f & \downarrow \bar{\beta}_Q & \nearrow j_V & \\
 & & V & &
 \end{array}$$

β_P (curved arrow from P to $\mathbf{Beh}^{\text{can}}$)
 β_Q (curved arrow from Q to $\mathbf{Beh}^{\text{can}}$)

2. **Closure under regular-epimorphic quotients.** Let

$$e : P \twoheadrightarrow Q$$

be a regular-epimorphic strict semantic homomorphism and suppose $P \models V$. Since

$$\beta_P = \beta_Q e,$$

and β_P factors through

$$j_V : V \hookrightarrow \mathbf{Beh}^{\text{can}},$$

Lemma 3.3.1 implies that β_Q factors through j_V . Hence $Q \models V$. Diagrammatically,

$$\begin{array}{ccc}
 P & \xrightarrow{e} & Q \\
 & \searrow \beta_P & \downarrow \beta_Q \\
 & & V \\
 & \searrow \beta_P & \downarrow j_V \\
 & & \mathbf{Beh}^{\text{can}}.
 \end{array}$$

3. **Closure under small coproducts.** Let

$$(P_i)_{i \in I}$$

be a small family of models. For each i , the factorization

$$\bar{\beta}_i : P_i \rightarrow V$$

is strict by Lemma 3.2.1. By the coproduct universal property in $\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$, these strict maps induce a strict map

$$\coprod_i P_i \rightarrow V.$$

Therefore the coproduct models V . The construction is

$$\begin{array}{ccc}
 P_i & \longrightarrow & \coprod_i P_i \\
 & \searrow \bar{\beta}_i & \downarrow \Pi_i \bar{\beta}_i \\
 & & V \\
 & & \downarrow j_V \\
 & & \mathbf{Beh}^{\text{can}}.
 \end{array}$$

4. **The theory models itself.** The identity map

$$V \rightarrow V$$

exhibits V as a model of itself. By Lemma 3.2.2, the semantic map of a restricted canonical subobject is its inclusion

$$j_V : V \hookrightarrow \mathbf{Beh}^{\text{can}}.$$

Hence

$$\mathbf{Beh}(V) = V.$$

5. **Prove monotonicity and order-reflection.** If $V \leq W$, then every factorization through V is a factorization through W , so

$$\mathbf{Mod}(V) \subseteq \mathbf{Mod}(W).$$

Conversely, if

$$\mathbf{Mod}(V) \subseteq \mathbf{Mod}(W),$$

then $V \in \mathbf{Mod}(V)$, hence $V \in \mathbf{Mod}(W)$. Therefore the inclusion

$$V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

factors through

$$W \hookrightarrow \mathbf{Beh}^{\text{can}},$$

so $V \leq W$. This is the triangle

$$\begin{array}{ccc} V & \xrightarrow{j_V} & \mathbf{Beh}^{\text{can}} \\ & \searrow \text{dashed} & \nearrow j_W \\ & W & \end{array}$$

obtained from the fact that V is a model of W .

□

Closure under small coproducts includes the empty coproduct. Hence the empty replete full subcategory is not coequationally definable.

3.5.2 Joins of local theories

Let

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

be a replete full subcategory. Define

$$\mathcal{S}_{\mathcal{C}} := \{S \hookrightarrow \mathbf{Beh}^{\text{can}} \mid S = \mathbf{Beh}(P) \text{ for some } P \in \mathcal{C}\}.$$

Although $\mathcal{C}$ may be large, the collection

$$\mathcal{S}_{\mathcal{C}}$$

is a subclass of the set

$$\mathbf{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_0).$$

We therefore choose a small set of representatives for the subobjects in $\mathcal{S}_{\mathcal{C}}$ and form their join. Define

$$V_{\mathcal{C}} := \bigvee_{S \in \mathcal{S}_{\mathcal{C}}} S \hookrightarrow \mathbf{Beh}^{\text{can}},$$

where the displayed join is taken over the chosen small representative set.

Lemma 3.5.1 (Joins of local coequational theories). *Let*

$$(S_i \hookrightarrow \mathbf{Beh}^{\text{can}})_{i \in I}$$

be a small family of local coequational theories. Then their join

$$S := \bigvee_{i \in I} S_i$$

in

$$\mathbf{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_0)$$

is again a local coequational theory.

Proof.

1. **Represent the join.** In the Grothendieck topos $\mathcal{E}/Z$, the join is represented by the regular image of the coproduct map

$$\coprod_i S_i \rightarrow \mathbf{Beh}_0.$$

Write this image factorization as

$$\begin{array}{ccc} \coprod_i S_i & \xrightarrow{\quad} & \mathbf{Beh}_0 \\ & \searrow & \nearrow \\ & S & \end{array}$$

in $\mathcal{E}/Z$.

2. **Regard the coproduct as a Mealy object.** By Proposition 3.3.2, the coproduct

$$\coprod_i S_i$$

carries the componentwise Mealy structure. Since each

$$S_i \hookrightarrow \mathbf{Beh}^{\text{can}}$$

is strict, the induced coproduct map to $\mathbf{Beh}^{\text{can}}$ is strict. Strictness is tested after precomposition with the jointly epic coproduct injections:

$$\begin{array}{ccc} S_i & \xrightarrow{\quad} & \coprod_i S_i \\ & \searrow & \downarrow \text{dashed} \\ & & \mathbf{Beh}^{\text{can}}. \end{array}$$

3. **Take the regular image in the strict Mealy category.** By Proposition 3.3.1, the regular image of this strict map is created in $\mathcal{E}/Z$ and carries a Mealy structure. Hence

$$S \hookrightarrow \mathbf{Beh}^{\text{can}}$$

is a subobject in the strict Mealy category. The created image is

$$\begin{array}{ccc} \coprod_i S_i & \xrightarrow{\quad} & \mathbf{Beh}^{\text{can}} \\ & \searrow & \nearrow \\ & S & \end{array}$$

in $\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$.

4. **Conclude local coequationality.** By Proposition 3.2.3, being a subobject of the canonical behaviour object is precisely being a local coequational theory.

□

Lemma 3.5.2 (The representing theory belongs to the class). *Let*

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

be a replete full subcategory closed under regular-epimorphic quotients and small coproducts. Let

$$\mathcal{S}_{\mathcal{C}} := \{S \hookrightarrow \mathbf{Beh}^{\text{can}} \mid S = \text{Beh}(P) \text{ for some } P \in \mathcal{C}\},$$

and let

$$V_{\mathcal{C}} := \bigvee_{S \in \mathcal{S}_{\mathcal{C}}} S \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be formed using a chosen small representative set of the distinct semantic-image subobjects.

Then the restricted canonical Mealy morphism

$$V_{\mathcal{C}}$$

belongs to $\mathcal{C}$.

Proof. For every chosen representative

$$S \in \mathcal{S}_{\mathcal{C}},$$

choose an object

$$P_S \in \mathcal{C}$$

together with an identification

$$S = \text{Beh}(P_S)$$

as a subobject of

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

The semantic-image factorization of P_S gives a regular quotient

$$P_S \twoheadrightarrow \text{Beh}(P_S).$$

Since $\mathcal{C}$ is closed under regular-epimorphic quotients,

$$\text{Beh}(P_S) \in \mathcal{C}.$$

The chosen representative S is canonically isomorphic to this semantic-image object. Repleteness therefore gives

$$S \in \mathcal{C}.$$

Form the small coproduct

$$C_{\mathcal{C}} := \coprod_{S \in \mathcal{S}_{\mathcal{C}}} S.$$

Since every summand belongs to $\mathcal{C}$ and $\mathcal{C}$ is closed under small coproducts,

$$C_{\mathcal{C}} \in \mathcal{C}.$$

The inclusions

$$S \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

induce a strict semantic homomorphism

$$c_{\mathcal{C}} : C_{\mathcal{C}} \longrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}.$$

Its regular image in the carrier slice is the join of the component subobjects:

$$\text{Im}(c_{\mathcal{C}}) = \bigvee_{S \in \mathcal{S}_{\mathcal{C}}} S = V_{\mathcal{C}}.$$

By Proposition 3.3.1, this carrier image is created in the strict Mealy category. Thus there is a regular quotient

$$C_{\mathcal{C}} \twoheadrightarrow V_{\mathcal{C}}$$

fitting into

$$\begin{array}{ccc} \coprod_{S \in \mathcal{S}_{\mathcal{C}}} S & \xrightarrow{c_{\mathcal{C}}} & \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}} \\ & \searrow & \nearrow j_{V_{\mathcal{C}}} \\ & V_{\mathcal{C}} & \end{array}$$

in

$$\mathbf{Mly}_{\mathbf{A}, \mathbf{B}}^{\text{str}}.$$

Since $C_{\mathcal{C}} \in \mathcal{C}$ and $\mathcal{C}$ is closed under regular-epimorphic quotients, the chosen image representative belongs to $\mathcal{C}$ up to canonical isomorphism. Repleteness therefore gives

$$V_{\mathcal{C}} \in \mathcal{C}.$$

□

3.5.3 The theorem

Theorem 3.5.2 (Local coequational Birkhoff theorem). *Let*

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbf{A}, \mathbf{B}}^{\text{str}}$$

be a replete full subcategory. Then $\mathcal{C}$ is of the form

$$\mathcal{C} = \text{Mod}(V)$$

for a local coequational theory

$$V \hookrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}$$

if and only if $\mathcal{C}$ is closed under:

1. *strict homomorphic preimages;*
2. *regular-epimorphic quotients;*
3. *small coproducts.*

When these conditions hold, the defining theory is

$$V_{\mathcal{C}} = \bigvee_{P \in \mathcal{C}} \text{Beh}(P) \hookrightarrow \mathbf{Beh}^{\text{can}},$$

where the join is taken over a chosen small set of representatives for the resulting semantic-image subobjects.

Proof.

1. Prove necessity.

Suppose

$$\mathcal{C} = \text{Mod}(V)$$

for a local coequational theory V . Closure under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts is Theorem 3.5.1.

2. Construct the representing theory.

Conversely, suppose that $\mathcal{C}$ is closed under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts.

Define

$$\mathcal{S}_{\mathcal{C}} := \{S \hookrightarrow \mathbf{Beh}^{\text{can}} \mid S = \text{Beh}(P) \text{ for some } P \in \mathcal{C}\}.$$

Choose a small representative set of the distinct subobjects in $\mathcal{S}_{\mathcal{C}}$, and put

$$V_{\mathcal{C}} := \bigvee_{S \in \mathcal{S}_{\mathcal{C}}} S \hookrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}.$$

Every semantic image

$$\text{Beh}(P) \hookrightarrow \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}}$$

is a local coequational theory by Corollary 3.3.2. Hence Lemma 3.5.1 implies that

$$V_{\mathcal{C}}$$

is a local coequational theory.

3. Place the representing theory inside the class.

By Lemma 3.5.2,

$$V_{\mathcal{C}} \in \mathcal{C}.$$

4. Show that every object of $\mathcal{C}$ models the representing theory.

Let

$$P \in \mathcal{C}.$$

By construction,

$$\text{Beh}(P) \leq V_{\mathcal{C}}.$$

The semantic map factors through its regular image:

$$P \twoheadrightarrow \text{Beh}(P) \hookrightarrow \mathbf{Beh}^{\text{can}}.$$

Composing the semantic-image inclusion with

$$\text{Beh}(P) \hookrightarrow V_{\mathcal{C}}$$

gives a factorization of β_P through $V_{\mathcal{C}}$:

$$P \xrightarrow{\quad} \text{Beh}(P) \xrightarrow{\quad} V_{\mathcal{C}} \xrightarrow{j_{V_{\mathcal{C}}}} \mathbf{Beh}^{\text{can}}.$$

β_P

Hence

$$P \models V_{\mathcal{C}}.$$

Therefore

$$\mathcal{C} \subseteq \text{Mod}(V_{\mathcal{C}}).$$

5. Use the representing theory to prove the converse inclusion.

Let

$$X \in \text{Mod}(V_{\mathcal{C}}).$$

By definition, the semantic map of X admits a unique carrier factorization

$$\bar{\beta}_X : X \rightarrow V_{\mathcal{C}}$$

satisfying

$$j_{V_{\mathcal{C}}} \bar{\beta}_X = \beta_X.$$

This is the triangle

$$\begin{array}{ccc} X & \xrightarrow{\beta_X} & \mathbf{Beh}_{\mathbf{A}, \mathbf{B}}^{\text{can}} \\ & \searrow \bar{\beta}_X & \nearrow j_{V_{\mathcal{C}}} \\ & V_{\mathcal{C}} & \end{array}$$

By Lemma 3.2.1, the factorization

$$\bar{\beta}_X : X \rightarrow V_{\mathcal{C}}$$

is a strict semantic homomorphism.

Since

$$V_{\mathcal{C}} \in \mathcal{C}$$

and $\mathcal{C}$ is closed under strict homomorphic preimages, it follows immediately that

$$X \in \mathcal{C}.$$

Therefore

$$\text{Mod}(V_{\mathcal{C}}) \subseteq \mathcal{C}.$$

6. Conclude.

Combining the two inclusions gives

$$\mathcal{C} = \text{Mod}(V_{\mathcal{C}}).$$

□

Corollary 3.5.1 (Uniqueness of the defining local theory). *Let*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

be a local coequational theory. Then

$$V = \bigvee_{P \in \text{Mod}(V)} \text{Beh}(P)$$

as a subobject of the canonical behaviour machine.

Consequently, if a replete full subcategory

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

is coequationally definable, then its defining local coequational theory is unique and is

$$V_{\mathcal{C}} = \bigvee_{P \in \mathcal{C}} \text{Beh}(P).$$

Proof. If

$$P \in \text{Mod}(V),$$

then the semantic map of P factors through V . Therefore its semantic image satisfies

$$\text{Beh}(P) \leq V.$$

Taking the join over all models gives

$$\bigvee_{P \in \text{Mod}(V)} \text{Beh}(P) \leq V.$$

Conversely, the restricted canonical object V is itself a model of V , and

$$\text{Beh}(V) = V$$

by Lemma 3.2.2. Hence V occurs among the semantic images entering the join, so

$$V \leq \bigvee_{P \in \text{Mod}(V)} \text{Beh}(P).$$

The two inequalities give

$$V = \bigvee_{P \in \text{Mod}(V)} \text{Beh}(P).$$

Now suppose

$$\mathcal{C} = \text{Mod}(V).$$

Then

$$V = \bigvee_{P \in \mathcal{C}} \text{Beh}(P) = V_{\mathcal{C}}.$$

Thus the defining local coequational theory is uniquely determined by $\mathcal{C}$. □

3.6 Structured output and algebraic closure

3.6.1 Structured output categories

This section explains how additional closure operations on behaviours arise from algebraic structure on the output category. The categorical treatment of varieties of languages uses such algebraic closure conditions as part of the bridge between syntactic structure and semantic classes [6, 1, 22].

Definition 3.6.1 (Structured output category). *Let $\mathbb{T}$ be a single-sorted finite-product theory. A $\mathbb{T}$ -structured output category is a $\mathbb{T}$ -algebra object in internal categories, equivalently a finite-product-preserving functor*

$$\mathbb{T} \longrightarrow \mathbf{Cat}_{\text{int}}(\mathcal{E}).$$

We write

$$\mathbb{B}$$

for the internal category assigned to the generic object of $\mathbb{T}$. Thus each operation of arity n in $\mathbb{T}$ determines an internal functor

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B},$$

and the equations of $\mathbb{T}$ hold internally.

3.6.2 Pointwise operations on behaviours

For $n > 0$, put

$$\mathbf{Beh}_0^{[n]} := \underbrace{\mathbf{Beh}_0 \times_{A_0} \cdots \times_{A_0} \mathbf{Beh}_0}_n.$$

A generalized element is a tuple

$$(H_1, \dots, H_n)$$

of residual behaviours

$$H_i : \mathbb{A}/v \rightarrow \mathbb{B}$$

over the same input object v . For $n = 0$, put

$$\mathbf{Beh}_0^{[0]} := A_0.$$

Let

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

be the internal functor corresponding to an n -ary operation. For $n > 0$, write

$$\mathbf{Beh}_{\theta,0}^{(n)}$$

for the underlying object

$$\mathbf{Beh}_0^{[n]}$$

equipped with the structure map

$$z_\theta : \mathbf{Beh}_0^{[n]} \rightarrow Z = A_0 \times B_0$$

defined by

$$z_\theta = \langle p_{\mathbf{Beh}} \pi_1, \theta_0 \langle q_{\mathbf{Beh}} \pi_1, \dots, q_{\mathbf{Beh}} \pi_n \rangle \rangle.$$

Thus

$$z_\theta(H_1, \dots, H_n) = \left(p_{\mathbf{Beh}}(H_1), \theta_0(q_{\mathbf{Beh}}H_1, \dots, q_{\mathbf{Beh}}H_n) \right).$$

The structure map is characterized by the product cone

$$\begin{array}{ccccc} & & \mathbf{Beh}_{\theta,0}^{(n)} & & \\ & \swarrow p_{\mathbf{Beh}}\pi_1 & \downarrow z_\theta & \searrow \theta_0\langle q_{\mathbf{Beh}}\pi_1, \dots, q_{\mathbf{Beh}}\pi_n \rangle & \\ A_0 & \xleftarrow{\pi_{A_0}} & Z = A_0 \times B_0 & \xrightarrow{\pi_{B_0}} & B_0. \end{array}$$

For $n = 0$, if

$$\theta : 1 \rightarrow \mathbb{B}$$

selects the object

$$b_\theta : 1 \rightarrow B_0,$$

then

$$\mathbf{Beh}_{\theta,0}^{(0)} = A_0$$

is regarded as an object of $\mathcal{E}/Z$ by the structure map

$$\langle 1_{A_0}, b_\theta! \rangle : A_0 \rightarrow A_0 \times B_0.$$

In generalized-element notation,

$$v \mapsto (v, b_\theta).$$

Notice that

$$\mathbf{Beh}_{\theta,0}^{(n)}$$

is not, in general, the n -fold product of $\mathbf{Beh}_0$ in the slice

$$\mathcal{E}/Z.$$

Its underlying object is first formed as the fibre product

$$\mathbf{Beh}_0 \times_{A_0} \cdots \times_{A_0} \mathbf{Beh}_0$$

over the common input coordinate. It is then regarded as an object of $\mathcal{E}/Z$ using the output coordinate induced by

$$\theta_0 : B_0^n \rightarrow B_0.$$

This distinction is essential when closure under $\mathbb{T}$ -operations is expressed as a factorization in $\mathcal{E}/Z$.

Definition 3.6.2 (Pointwise structured behaviour). *Let*

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

be an operation of $\mathbb{T}$. The induced map

$$\theta_{\mathbf{Beh}} : \mathbf{Beh}_{\theta,0}^{(n)} \rightarrow \mathbf{Beh}_0$$

in $\mathcal{E}/Z$ is defined by

$$\theta_{\mathbf{Beh}}(H_1, \dots, H_n) = \theta \circ \langle H_1, \dots, H_n \rangle$$

for $n > 0$. For $n = 0$, the nullary operation

$$\theta : 1 \rightarrow \mathbb{B}$$

induces the constant residual behaviour

$$\mathbb{A}/v \rightarrow 1 \xrightarrow{\theta} \mathbb{B}$$

over each $v \in A_0$.

The construction is summarized by

$$\mathbb{A}/v \xrightarrow{\langle H_1, \dots, H_n \rangle} \mathbb{B}^n \xrightarrow{\theta} \mathbb{B}.$$

$$\theta_{\mathbf{Beh}(H_i)_i}$$

On anchor objects and residual arrows this is the pair of diagrams

$$\begin{array}{ccc} 1_v & \xrightarrow{\quad} & (H_1(1_v), \dots, H_n(1_v)) \\ \downarrow & & \downarrow \theta_0 \\ \theta_{\mathbf{Beh}(H_i)_i}(1_v) & \xlongequal{\quad} & \theta_0(H_1(1_v), \dots, H_n(1_v)) \end{array}$$

and

$$\begin{array}{ccc} r \circ a & \xrightarrow{\quad} & r \\ \downarrow & & \downarrow \\ (H_i(r \circ a))_i & \xrightarrow{(H_i(r \circ a \rightarrow r))_i} & (H_i(r))_i \\ \downarrow \theta_0 & & \downarrow \theta_0 \\ \theta(H_i(r \circ a))_i & \xrightarrow{\theta_1(H_i(r \circ a \rightarrow r))_i} & \theta(H_i(r))_i. \end{array}$$

For a nullary operation $\theta : 1 \rightarrow \mathbb{B}$, write

$$\theta()$$

for the Mealy morphism with carrier A_0 , input projection 1_{A_0} , constant output b_θ , transition

$$\delta_{\theta()}(a, v) = s_A(a),$$

and identity output arrow at b_θ .

Lemma 3.6.1 (Structured operations on strict Mealy morphisms). *For every operation*

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B},$$

there is an induced n -ary functor

$$\theta : (\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}})^n \rightarrow \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

for $n > 0$, with the evident nullary object for $n = 0$. The residual semantics is compatible with this operation:

$$\beta_{\theta(P_i)_i} = \theta_{\mathbf{Beh}}(\beta_{P_i})_i.$$

Proof.

1. **Construct the carrier.** For $n > 0$, the carrier is

$$P_\theta := P_1 \times_{A_0} \cdots \times_{A_0} P_n.$$

Its input projection is the common projection to A_0 , and its output projection is

$$q_\theta(x_1, \dots, x_n) = \theta_0(q_1 x_1, \dots, q_n x_n).$$

This is the pullback-and-output diagram

$$\begin{array}{ccccc} & & (q_i \pi_i)_i & & \\ & \nearrow \pi_i & & \searrow q_i & \\ P_\theta & & P_i & & B_0 \\ p_\theta \downarrow & \lrcorner & \downarrow p_i & \cdots & \\ A_0 & \equiv & A_0 & & \end{array}$$

together with

$$q_\theta = \theta_0(q_i \pi_i)_i.$$

For $n = 0$, the carrier is A_0 , with input projection 1_{A_0} and output projection

$$A_0 \rightarrow 1 \xrightarrow{b_\theta} B_0.$$

2. **Identify the transition domain.** For $n > 0$, there is a canonical pullback identification

$$A_1 \times_{A_0} (P_1 \times_{A_0} \cdots \times_{A_0} P_n) \cong (A_1 \times_{A_0} P_1) \times_{A_1} \cdots \times_{A_1} (A_1 \times_{A_0} P_n).$$

Under this identification, a transition datum consists of one input arrow a and compatible states $(x_i)_i$. The compatibility is expressed by

$$\begin{array}{ccc} A_1 \times_{A_0} P_\theta & \longrightarrow & P_\theta \\ \downarrow & \lrcorner & \downarrow \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

3. **Define transition and output.** For $n > 0$, define

$$\delta_\theta(a, (x_i)_i) = (\delta_i(a, x_i))_i.$$

The output arrows

$$\omega_i(a, x_i) : q_i(\delta_i(a, x_i)) \rightarrow q_i(x_i)$$

form an arrow in $\mathbb{B}^n$. Applying the internal functor

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

gives a well-typed arrow

$$\theta_1(\omega_i(a, x_i))_i : \theta_0(q_i \delta_i(a, x_i))_i \rightarrow \theta_0(q_i x_i)_i.$$

Define

$$\omega_\theta(a, (x_i)_i) = \theta_1(\omega_i(a, x_i))_i.$$

The construction is summarized by

$$\begin{array}{ccc} \theta_0(q_i \delta_i(a, x_i))_i & \xrightarrow{\theta_1(\omega_i(a, x_i))_i} & \theta_0(q_i x_i)_i \\ \parallel & & \parallel \\ q_\theta \delta_\theta(a, (x_i)_i) & \xrightarrow{\omega_\theta(a, (x_i)_i)} & q_\theta (x_i)_i. \end{array}$$

For $n = 0$, the transition is

$$\delta_\theta(a, v) = s_A(a),$$

and the output map is the composite

$$A_1 \xrightarrow{!} 1 \xrightarrow{b_\theta} B_0 \xrightarrow{e_B} B_1.$$

In generalized-element notation,

$$\omega_\theta(a, v) = e_B(b_\theta).$$

4. Define the operation on strict maps. If

$$f_i : P_i \rightarrow Q_i$$

are strict semantic homomorphisms, define

$$\theta(f_i)_i : P_1 \times_{A_0} \cdots \times_{A_0} P_n \rightarrow Q_1 \times_{A_0} \cdots \times_{A_0} Q_n$$

componentwise. Strictness follows from the componentwise transition equations and from functoriality of

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}.$$

The output square is

$$\begin{array}{ccc} A_1 \times_{A_0} P_\theta & \xrightarrow{1 \times \theta(f_i)} & A_1 \times_{A_0} Q_\theta \\ \omega_\theta^P \downarrow & & \downarrow \omega_\theta^Q \\ B_1 & \xlongequal{\quad} & B_1. \end{array}$$

Identities and composition are inherited from the product in the carrier slice.

5. Verify the Mealy laws structurally. The transition laws are inherited componentwise. The output laws follow from functoriality of

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}.$$

In the nullary case the result follows from the identity laws in $\mathbb{B}$.

6. **Compare residual behaviours.** For a residual object

$$r : u \rightarrow v,$$

one has

$$q_\theta(\delta_\theta(r, (x_i)_i)) = \theta_0(q_i(\delta_i(r, x_i)))_i,$$

which is the object value of

$$\theta_{\mathbf{Beh}}(\beta_{P_i}(x_i))_i.$$

The same calculation on residual arrows gives

$$\omega_\theta(a, \delta_\theta(r, (x_i)_i)) = \theta_1(\omega_i(a, \delta_i(r, x_i)))_i.$$

Therefore

$$\beta_{\theta(P_i)_i} = \theta_{\mathbf{Beh}}(\beta_{P_i})_i.$$

□

Lemma 3.6.2 (Structured operations preserve derivatives and canonical output). *For every operation*

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

and every arrow

$$a : u \rightarrow v$$

of $\mathbb{A}$,

$$\partial_a(\theta_{\mathbf{Beh}}(H_i))_i = \theta_{\mathbf{Beh}}(\partial_a H_i)_i$$

and

$$\omega_{\mathbf{Beh}}(a, \theta_{\mathbf{Beh}}(H_i))_i = \theta_1(\omega_{\mathbf{Beh}}(a, H_i))_i.$$

Proof.

1. **Use precomposition for derivatives.** Since

$$\partial_a H = H \circ a_*,$$

one has

$$\begin{aligned} \partial_a(\theta \circ \langle H_i \rangle_i) &= (\theta \circ \langle H_i \rangle_i) \circ a_* \\ &= \theta \circ \langle H_i \circ a_* \rangle_i \\ &= \theta_{\mathbf{Beh}}(\partial_a H_i)_i. \end{aligned}$$

The commutation is

$$\begin{array}{ccc} \mathbb{A}/u & \xrightarrow{a_*} & \mathbb{A}/v \\ & \searrow \langle \partial_a H_i \rangle_i & \downarrow \langle H_i \rangle_i \\ & & \mathbb{B}^n \\ & & \downarrow \theta \\ & & \mathbb{B}. \end{array}$$

2. **Use functoriality for canonical output.** The canonical output is the value on the residual morphism

$$a \rightarrow 1_v.$$

Applying

$$\theta \circ \langle H_i \rangle_i$$

gives

$$\theta_1(H_i(a, 1_v))_i = \theta_1(\omega_{\mathbf{Beh}}(a, H_i))_i.$$

Diagrammatically,

$$\begin{array}{ccc} (H_i(a))_i & \xrightarrow{(H_i(a \rightarrow 1_v))_i} & (H_i(1_v))_i \\ \theta_0 \downarrow & & \downarrow \theta_0 \\ \theta(H_i(a))_i & \xrightarrow{\theta_1(H_i(a \rightarrow 1_v))_i} & \theta(H_i(1_v))_i. \end{array}$$

3. **Handle nullary operations.** In the nullary case both claims say that a constant functor remains constant under residual reindexing and sends residual morphisms to the identity arrow selected by the nullary functor.

□

3.6.3 $\mathbb{T}$ -coequational theories

Definition 3.6.3 ($\mathbb{T}$ -coequational theory). *Let $\mathbb{T}$ be a single-sorted finite-product theory, and let $\mathbb{B}$ be a $\mathbb{T}$ -structured output category.*

*A **local $\mathbb{T}$ -coequational theory** is a local coequational theory*

$$j_V : V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

that is closed under the pointwise operation induced by every morphism

$$\theta : n \rightarrow 1$$

of $\mathbb{T}$.

For $n > 0$, put

$$V^{[n]} := \underbrace{V \times_{A_0} \cdots \times_{A_0} V}_n.$$

This is the n -fold product of

$$p_V : V \rightarrow A_0$$

in the slice $\mathcal{E}/A_0$.

For an operation

$$\theta : n \rightarrow 1$$

whose interpretation is the internal functor

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B},$$

regard $V^{[n]}$ as an object of

$$\mathcal{E}/Z, \quad Z = A_0 \times B_0,$$

by the structure map

$$z_{\theta,V} := \langle p_V \pi_1, \theta_0 \langle q_V \pi_1, \dots, q_V \pi_n \rangle \rangle.$$

The resulting object of $\mathcal{E}/Z$ is denoted

$$V_\theta^{(n)}.$$

Thus, in generalized-element notation,

$$z_{\theta,V}(H_1, \dots, H_n) = \left(p_V(H_1), \theta_0(q_V(H_1), \dots, q_V(H_n)) \right).$$

For a nullary theory operation

$$\theta : 0 \rightarrow 1,$$

we use the same symbol

$$\theta : 1 \rightarrow \mathbb{B}$$

for its interpretation in the $\mathbb{T}$ -structured internal category. Let

$$b_\theta : 1 \rightarrow B_0$$

be its selected output object. Put

$$V_\theta^{(0)} := A_0,$$

regarded over Z by

$$\langle 1_{A_0}, b_\theta! \rangle : A_0 \rightarrow A_0 \times B_0.$$

Closure under a positive-arity operation

$$\theta : n \rightarrow 1$$

means that the pointwise behaviour operation restricts to V : there is a unique map

$$\theta_V : V_\theta^{(n)} \rightarrow V$$

in $\mathcal{E}/Z$ satisfying

$$j_V \theta_V = \theta_{\mathbf{Beh}} j_V^{[n]},$$

where

$$j_V^{[n]} := j_V \times_{A_0} \dots \times_{A_0} j_V : V^{[n]} \rightarrow \mathbf{Beh}_0^{[n]}.$$

Equivalently, after forgetting the B_0 -coordinate,

$$\theta_V : V^{[n]} \rightarrow V$$

is a map in $\mathcal{E}/A_0$ satisfying

$$q_V \theta_V = \theta_0 \langle q_V \pi_1, \dots, q_V \pi_n \rangle.$$

In the nullary case, closure means that there is a unique map

$$\theta_V : A_0 \rightarrow V$$

satisfying

$$j_V \theta_V = \theta_{\mathbf{Beh}},$$

equivalently

$$p_V \theta_V = 1_{A_0}, \quad q_V \theta_V = b_\theta!.$$

The proposition below shows that the resulting operations

$$\theta_V : V^{[n]} \rightarrow V$$

form a $\mathbb{T}$ -algebra structure on

$$p_V : V \rightarrow A_0$$

in the slice $\mathcal{E}/A_0$, and that the output map preserves this structure.

The closure condition is displayed by

$$\begin{array}{ccc} V_\theta^{(n)} & \xrightarrow{\theta_{\mathbf{Beh}}} & \mathbf{Beh}_0 \\ & \searrow \theta_V & \nearrow j_V \\ & V & \end{array} \quad \text{in } \mathcal{E}/Z.$$

The compatibility between algebraic closure and derivative closure is summarized by the cube-like diagram

$$\begin{array}{ccc} A_1 \times_{A_0} V_\theta^{(n)} & \xrightarrow{1 \times \theta_V} & A_1 \times_{A_0} V \\ \delta_{V_\theta^{(n)}} \downarrow & \circlearrowleft & \downarrow \delta_V \\ V_\theta^{(n)} & \xrightarrow{\theta_V} & V \\ \downarrow & & \downarrow j_V \\ \mathbf{Beh}_{\theta,0}^{(n)} & \xrightarrow{\theta_{\mathbf{Beh}}} & \mathbf{Beh}_0. \end{array}$$

Proposition 3.6.1 (Inherited structured operations). *Let*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

be a local $\mathbb{T}$ -coequational theory, and let

$$\theta : n \rightarrow 1$$

be a morphism of the finite-product theory $\mathbb{T}$, interpreted as an internal functor

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}.$$

For $n > 0$, the object

$$V_\theta^{(n)}$$

carries the Mealy structure of

$$\theta(V, \dots, V).$$

Explicitly,

$$\delta_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)) = (\partial_a H_1, \dots, \partial_a H_n),$$

and

$$\omega_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)) = \theta_1(\omega_V(a, H_1), \dots, \omega_V(a, H_n)).$$

The closure factorization

$$\theta_V : V_\theta^{(n)} \rightarrow V$$

is a strict semantic homomorphism. The analogous assertion holds for nullary operations.

After forgetting the B_0 -coordinate, the family

$$\theta_V : V^{[n]} \rightarrow V$$

preserves projections and substitution. Consequently it defines a $\mathbb{T}$ -algebra structure on

$$p_V : V \rightarrow A_0$$

in the slice $\mathcal{E}/A_0$.

Moreover,

$$\langle p_V, q_V \rangle : V \rightarrow A_0 \times B_0$$

is a homomorphism in $\mathcal{E}/A_0$ from this $\mathbb{T}$ -algebra to the fibrewise $\mathbb{T}$ -algebra

$$A_0 \times B_0 \rightarrow A_0$$

induced by the $\mathbb{T}$ -algebra structure on B_0 .

Proof. Suppose first that $n > 0$.

By Lemma 3.6.1, the fibre product

$$V^{[n]} = V \times_{A_0} \cdots \times_{A_0} V$$

carries the Mealy structure

$$\theta(V, \dots, V)$$

when equipped with the θ -induced output coordinate. Thus its carrier over Z is $V_\theta^{(n)}$, its transition is

$$\delta_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)) = (\partial_a H_1, \dots, \partial_a H_n),$$

and its output is

$$\omega_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)) = \theta_1(\omega_V(a, H_1), \dots, \omega_V(a, H_n)).$$

Since V is closed under θ , there is a unique map

$$\theta_V : V_\theta^{(n)} \rightarrow V$$

in $\mathcal{E}/Z$ satisfying

$$j_V \theta_V = \theta_{\mathbf{Beh} j_V^{[n]}}.$$

We prove that θ_V preserves transition. Let

$$a : u \rightarrow v$$

and let

$$(H_1, \dots, H_n) \in V^{[n]}$$

lie over v . After composing with j_V ,

$$\begin{aligned}
 j_V \theta_V \delta_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)) &= \theta_{\mathbf{Beh}}(j_V \partial_a H_1, \dots, j_V \partial_a H_n) \\
 &= \theta_{\mathbf{Beh}}(\partial_a(j_V H_1), \dots, \partial_a(j_V H_n)) \\
 &= \partial_a(\theta_{\mathbf{Beh}}(j_V H_1, \dots, j_V H_n)) \\
 &= \partial_a(j_V \theta_V(H_1, \dots, H_n)) \\
 &= j_V \delta_V(a, \theta_V(H_1, \dots, H_n)).
 \end{aligned}$$

Here the third equality is Lemma 3.6.2. Since j_V is monic,

$$\theta_V \delta_{V_\theta^{(n)}} = \delta_V \widehat{\theta_V}.$$

The map θ_V also preserves output. Indeed,

$$\begin{aligned}
 \omega_V(a, \theta_V(H_1, \dots, H_n)) &= \omega_{\mathbf{Beh}}(a, j_V \theta_V(H_1, \dots, H_n)) \\
 &= \omega_{\mathbf{Beh}}(a, \theta_{\mathbf{Beh}}(j_V H_1, \dots, j_V H_n)) \\
 &= \theta_1(\omega_{\mathbf{Beh}}(a, j_V H_1), \dots, \omega_{\mathbf{Beh}}(a, j_V H_n)) \\
 &= \theta_1(\omega_V(a, H_1), \dots, \omega_V(a, H_n)) \\
 &= \omega_{V_\theta^{(n)}}(a, (H_1, \dots, H_n)).
 \end{aligned}$$

Thus θ_V is a strict semantic homomorphism.

Now let

$$\theta : 0 \rightarrow 1$$

be nullary. Then

$$V_\theta^{(0)} = A_0$$

carries the nullary Mealy morphism $\theta()$. Closure gives a map

$$\theta_V : A_0 \rightarrow V$$

satisfying

$$j_V \theta_V = \theta_{\mathbf{Beh}}.$$

The nullary cases of Lemma 3.6.2 show, by the same monicity argument, that θ_V preserves transition and output. Hence it is strict.

We next verify that the underlying maps in $\mathcal{E}/A_0$ preserve the finite-product-theory structure.

Let

$$\pi_i : n \rightarrow 1$$

be the i -th product projection in $\mathbb{T}$. The corresponding pointwise operation on behaviours is the ordinary projection

$$\pi_{i, \mathbf{Beh}} : \mathbf{Beh}_0^{[n]} \rightarrow \mathbf{Beh}_0.$$

The ordinary projection

$$\pi_i : V^{[n]} \rightarrow V$$

satisfies

$$j_V \pi_i = \pi_{i, \mathbf{Beh}} j_V^{[n]}.$$

By uniqueness of factorization through the monomorphism j_V ,

$$\pi_{i, V} = \pi_i.$$

Hence the restricted operations preserve the distinguished projections.

Let

$$\theta : k \rightarrow 1$$

and

$$\varphi_i : n \rightarrow 1 \quad (1 \leq i \leq k)$$

be morphisms in $\mathbb{T}$, and put

$$\psi := \theta \circ \langle \varphi_1, \dots, \varphi_k \rangle : n \rightarrow 1.$$

In $\mathcal{E}/A_0$, the restricted operations determine a pairing

$$\langle \varphi_{1, V}, \dots, \varphi_{k, V} \rangle : V^{[n]} \rightarrow V^{[k]}.$$

Composing with j_V , one obtains

$$\begin{aligned} j_V \theta_V \langle \varphi_{1, V}, \dots, \varphi_{k, V} \rangle &= \theta_{\mathbf{Beh}} j_V^{[k]} \langle \varphi_{1, V}, \dots, \varphi_{k, V} \rangle \\ &= \theta_{\mathbf{Beh}} \langle j_V \varphi_{1, V}, \dots, j_V \varphi_{k, V} \rangle \\ &= \theta_{\mathbf{Beh}} \langle \varphi_{1, \mathbf{Beh}} j_V^{[n]}, \dots, \varphi_{k, \mathbf{Beh}} j_V^{[n]} \rangle \\ &= \theta_{\mathbf{Beh}} \langle \varphi_{1, \mathbf{Beh}}, \dots, \varphi_{k, \mathbf{Beh}} \rangle j_V^{[n]} \\ &= \psi_{\mathbf{Beh}} j_V^{[n]} \\ &= j_V \psi_V. \end{aligned}$$

Since j_V is monic,

$$\theta_V \langle \varphi_{1, V}, \dots, \varphi_{k, V} \rangle = \psi_V.$$

Thus the restricted operations preserve projections and substitution. These are precisely the structural requirements for an interpretation of the single-sorted finite-product theory $\mathbb{T}$. Therefore they define a $\mathbb{T}$ -algebra structure on

$$p_V : V \rightarrow A_0$$

in $\mathcal{E}/A_0$.

Finally, because

$$\theta_V : V_\theta^{(n)} \rightarrow V$$

is a map in $\mathcal{E}/Z$, one has, for every positive-arity operation,

$$q_V \theta_V = \theta_0 \langle q_V \pi_1, \dots, q_V \pi_n \rangle.$$

For a nullary operation,

$$q_V \theta_V = b_\theta!.$$

Hence

$$\langle p_V, q_V \rangle : V \rightarrow A_0 \times B_0$$

preserves every operation of $\mathbb{T}$ in the slice $\mathcal{E}/A_0$. It is therefore a homomorphism to the fibrewise $\mathbb{T}$ -algebra

$$A_0 \times B_0 \rightarrow A_0.$$

□

3.6.4 Semantic images and structured operations

Lemma 3.6.3 (Semantic images under structured operations). *Let*

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

be an operation of $\mathbb{T}$, and let

$$P_1, \dots, P_n$$

be Mealy morphisms. For $n > 0$, the semantic quotient maps

$$P_i \twoheadrightarrow \text{Beh}(P_i)$$

induce a regular epimorphism

$$\theta(P_1, \dots, P_n) \twoheadrightarrow \text{Beh}(P_1) \times_{A_0} \cdots \times_{A_0} \text{Beh}(P_n),$$

where the codomain is regarded over

$$Z = A_0 \times B_0$$

by the θ -induced output coordinate. Moreover,

$$\text{Im}(\theta_{\mathbf{Beh}} : \text{Beh}(P_1) \times_{A_0} \cdots \times_{A_0} \text{Beh}(P_n) \rightarrow \mathbf{Beh}_0) = \text{Beh}(\theta(P_1, \dots, P_n)).$$

For $n = 0$, the image of

$$A_0 \rightarrow \mathbf{Beh}_0$$

induced by the constant residual behaviour is

$$\text{Beh}(\theta()).$$

Proof.

1. **Form the product of semantic quotients.** For $n > 0$, the semantic quotients

$$P_i \twoheadrightarrow \text{Beh}(P_i)$$

are regular epimorphisms in the carrier slice $\mathcal{E}/Z$, hence in particular their underlying maps over A_0 are regular epimorphisms. Since regular epimorphisms are stable under pullback and finite products in slices of a Grothendieck topos, their fibre product over A_0 is a regular epimorphism:

$$P_1 \times_{A_0} \cdots \times_{A_0} P_n \twoheadrightarrow \text{Beh}(P_1) \times_{A_0} \cdots \times_{A_0} \text{Beh}(P_n).$$

The source is the carrier of

$$\theta(P_1, \dots, P_n),$$

and the target is regarded as an object of $\mathcal{E}/Z$ by the θ -induced coordinate

$$(H_1, \dots, H_n) \mapsto (p_{\mathbf{Beh}}(H_1), \theta_0(q_{\mathbf{Beh}} H_1, \dots, q_{\mathbf{Beh}} H_n)).$$

With this coordinate, the displayed map is a regular epimorphism in $\mathcal{E}/Z$.

2. **Compare the semantic composites.** By Lemma 3.6.1,

$$\beta_{\theta(P_i)_i} = \theta_{\mathbf{Beh}}(\beta_{P_i})_i.$$

Hence the semantic map of

$$\theta(P_1, \dots, P_n)$$

is the composite

$$\theta(P_i)_i \xrightarrow{\quad} \prod_{A_0} \text{Beh}(P_i) \xrightarrow{\theta_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

$\beta_{\theta(P_i)_i}$

3. **Compare regular images.** Precomposition with a regular epimorphism does not change the regular image. Therefore the image of

$$\theta_{\mathbf{Beh}} : \prod_{A_0} \text{Beh}(P_i) \rightarrow \mathbf{Beh}_0$$

is exactly the semantic image of

$$\theta(P_1, \dots, P_n).$$

Hence

$$\text{Im}(\theta_{\mathbf{Beh}}) = \text{Beh}(\theta(P_1, \dots, P_n)).$$

4. **Handle nullary operations.** For $n = 0$, the semantic map of $\theta()$ is the map

$$A_0 \rightarrow \mathbf{Beh}_0$$

assigning to v the constant residual behaviour

$$\mathbb{A}/v \rightarrow 1 \xrightarrow{\theta} \mathbb{B}.$$

Its regular image is, by definition,

$$\text{Beh}(\theta()).$$

□

3.6.5 $\mathbb{T}$ -coequational closure

Theorem 3.6.1 ($\mathbb{T}$ -coequational closure). *Let*

$$V \hookrightarrow \mathbf{Beh}^{\text{can}}$$

be a local $\mathbb{T}$ -coequational theory. Then

$$\text{Mod}(V)$$

is closed under strict homomorphic preimages, regular-epimorphic quotients, small coproducts, and the operations induced by $\mathbb{T}$.

Proof.

1. **Use local closure.** Closure under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts is Theorem 3.5.1.

2. **Use semantic compatibility.** Let

$$P_1, \dots, P_n \in \mathbf{Mod}(V).$$

Choose strict factorizations

$$\bar{\beta}_{P_i} : P_i \rightarrow V$$

with

$$j_V \bar{\beta}_{P_i} = \beta_{P_i}.$$

These induce a map from the carrier of

$$\theta(P_1, \dots, P_n)$$

to $V_\theta^{(n)}$:

$$\begin{array}{ccc} P_1 \times_{A_0} \dots \times_{A_0} P_n & \xrightarrow{(\bar{\beta}_{P_i})_i} & V_\theta^{(n)} \\ & \searrow (\beta_{P_i})_i & \downarrow \\ & & \mathbf{Beh}_{\theta,0}^{(n)}. \end{array}$$

3. **Use algebraic closure of V .** Since V is closed under θ , there is a map

$$\theta_V : V_\theta^{(n)} \rightarrow V$$

with

$$j_V \theta_V = \theta_{\mathbf{Beh}}|_{V_\theta^{(n)}}.$$

By Lemma 3.6.1,

$$\beta_{\theta(P_i)_i} = \theta_{\mathbf{Beh}}(\beta_{P_i})_i.$$

Hence

$$\beta_{\theta(P_i)_i}$$

factors through V , so

$$\theta(P_1, \dots, P_n) \in \mathbf{Mod}(V).$$

The factorization is

$$\theta(P_i)_i \xrightarrow{(\bar{\beta}_{P_i})_i} V_\theta^{(n)} \xrightarrow{\theta_V} V \xrightarrow{j_V} \mathbf{Beh}^{\text{can}}.$$

$\beta_{\theta(P_i)_i}$

4. **Handle nullary operations.** The nullary case is the same argument using

$$V_\theta^{(0)} = A_0$$

and the closure map

$$A_0 \rightarrow V.$$

□

3.6.6 The structured Birkhoff theorem

Theorem 3.6.2 ($\mathbb{T}$ -coequational Birkhoff theorem). *Let $\mathbb{B}$ be a $\mathbb{T}$ -structured output category, and let*

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

be a replete full subcategory. Then $\mathcal{C}$ is of the form

$$\mathcal{C} = \mathbf{Mod}(V)$$

for a local $\mathbb{T}$ -coequational theory

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

if and only if $\mathcal{C}$ is closed under:

1. *strict homomorphic preimages;*
2. *regular-epimorphic quotients;*
3. *small coproducts;*
4. *the operations induced by $\mathbb{T}$.*

Proof.

1. Prove necessity.

Suppose

$$\mathcal{C} = \mathbf{Mod}(V)$$

for a local $\mathbb{T}$ -coequational theory V .

Closure under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts follows from Theorem 3.5.1. Closure under the operations induced by $\mathbb{T}$ follows from Theorem 3.6.1.

2. Construct the underlying local theory.

Conversely, suppose that $\mathcal{C}$ has the four displayed closure properties. By the local coequational Birkhoff theorem,

$$V_{\mathcal{C}} := \bigvee_{P \in \mathcal{C}} \mathbf{Beh}(P) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

is a local coequational theory and

$$\mathcal{C} = \mathbf{Mod}(V_{\mathcal{C}}).$$

Moreover, Lemma 3.5.2 gives

$$V_{\mathcal{C}} \in \mathcal{C}.$$

It remains to prove that $V_{\mathcal{C}}$ is closed under every pointwise operation induced by $\mathbb{T}$.

3. Treat positive-arity operations.

Let

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

be an operation with $n > 0$.

Since

$$V_C \in \mathcal{C}$$

and $\mathcal{C}$ is closed under the operation induced by θ ,

$$\theta(V_C, \dots, V_C) \in \mathcal{C}.$$

Using

$$\mathcal{C} = \text{Mod}(V_C),$$

this object models V_C . Hence its semantic map admits a unique factorization

$$\bar{\beta}_\theta : \theta(V_C, \dots, V_C) \longrightarrow V_C$$

satisfying

$$j_{V_C} \bar{\beta}_\theta = \beta_{\theta(V_C, \dots, V_C)}.$$

By Lemma 3.2.1, $\bar{\beta}_\theta$ is a strict semantic homomorphism.

The carrier of

$$\theta(V_C, \dots, V_C)$$

is

$$(V_C)_\theta^{(n)}.$$

By Lemma 3.6.1,

$$\begin{aligned} \beta_{\theta(V_C, \dots, V_C)} &= \theta_{\mathbf{Beh}}(\beta_{V_C}, \dots, \beta_{V_C}) \\ &= \theta_{\mathbf{Beh}} j_{V_C}^{[n]}. \end{aligned}$$

The second equality uses Lemma 3.2.2:

$$\beta_{V_C} = j_{V_C}.$$

Therefore

$$j_{V_C} \bar{\beta}_\theta = \theta_{\mathbf{Beh}} j_{V_C}^{[n]}.$$

Define

$$\theta_{V_C} := \bar{\beta}_\theta : (V_C)_\theta^{(n)} \longrightarrow V_C.$$

Then

$$j_{V_C} \theta_{V_C} = \theta_{\mathbf{Beh}} j_{V_C}^{[n]},$$

which is exactly the required closure factorization.

The construction is displayed by

$$\begin{array}{ccc} (V_C)_\theta^{(n)} & \xrightarrow{\theta_{\mathbf{Beh}} j_{V_C}^{[n]}} & \mathbf{Beh}_0 \\ & \searrow \theta_{V_C} \quad \nearrow j_{V_C} & \\ & V_C & \end{array}$$

4. **Treat nullary operations.**

Let

$$\theta : 0 \rightarrow 1$$

be a nullary operation, interpreted as an internal functor

$$\theta : 1 \rightarrow \mathbb{B}.$$

Closure of $\mathcal{C}$ under this nullary operation gives

$$\theta() \in \mathcal{C}.$$

Therefore

$$\theta() \models V_{\mathcal{C}}.$$

Its semantic map admits a unique strict model factorization

$$\theta_{V_{\mathcal{C}}} : A_0 \longrightarrow V_{\mathcal{C}}$$

satisfying

$$j_{V_{\mathcal{C}}} \theta_{V_{\mathcal{C}}} = \beta_{\theta()}.$$

By the nullary case of Lemma 3.6.1, the semantic map of $\theta()$ is the constant residual-behaviour map

$$\theta_{\mathbf{Beh}} : A_0 \rightarrow \mathbf{Beh}_0.$$

Hence

$$j_{V_{\mathcal{C}}} \theta_{V_{\mathcal{C}}} = \theta_{\mathbf{Beh}}.$$

This is precisely the required nullary closure factorization.

5. **Conclude.**

Thus $V_{\mathcal{C}}$ is closed under every pointwise operation induced by $\mathbb{T}$. Since it is already a local coequational theory, it is a local $\mathbb{T}$ -coequational theory.

The underlying local Birkhoff theorem gives

$$\mathcal{C} = \mathbf{Mod}(V_{\mathcal{C}}),$$

completing the proof. □

Corollary 3.6.1 (Uniqueness of the defining local $\mathbb{T}$ -coequational theory). *Let $\mathbb{B}$ be a $\mathbb{T}$ -structured output category, and let*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

be a local $\mathbb{T}$ -coequational theory. Then

$$V = \bigvee_{P \in \mathbf{Mod}(V)} \mathbf{Beh}(P)$$

as a subobject of the canonical behaviour machine.

Consequently, if a replete full subcategory

$$\mathcal{C} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}$$

is definable by a local $\mathbb{T}$ -coequational theory, then that theory is unique and is

$$V_{\mathcal{C}} = \bigvee_{P \in \mathcal{C}} \text{Beh}(P).$$

Proof. Forgetting the additional $\mathbb{T}$ -closure, a local $\mathbb{T}$ -coequational theory is a local coequational theory. Therefore Corollary 3.5.1 gives

$$V = \bigvee_{P \in \text{Mod}(V)} \text{Beh}(P).$$

Now suppose

$$\mathcal{C} = \text{Mod}(V)$$

for a local $\mathbb{T}$ -coequational theory V . Then the underlying local coequational subobject is necessarily

$$V = \bigvee_{P \in \mathcal{C}} \text{Beh}(P).$$

It remains only to observe that the $\mathbb{T}$ -structure on this subobject is not additional independent data. For every operation

$$\theta : n \rightarrow 1$$

of $\mathbb{T}$, the induced map

$$\theta_V : V_{\theta}^{(n)} \rightarrow V$$

is uniquely determined by the factorization equation

$$j_V \theta_V = \theta_{\text{Beh} j_V^{[n]}},$$

because j_V is monic.

Hence both the underlying subobject and all its restricted $\mathbb{T}$ -operations are uniquely determined by $\mathcal{C}$. Therefore the defining local $\mathbb{T}$ -coequational theory is unique. $\square$

3.7 Global input reindexing

3.7.1 Reindexing residual behaviours along input functors

This section explains how local coequational theories vary with the input category. This is the categorical source of closure under inverse morphic preimage in the classical one-object setting, a closure operation appearing in Eilenberg-type correspondences for varieties of languages [6, 1].

Fix $\mathbb{B}$, and, when desired, a finite-product theory $\mathbb{T}$ for which $\mathbb{B}$ is a $\mathbb{T}$ -structured output category. Let

$$\mathbf{Inp}$$

be a chosen category of internal input categories and internal functors. For every

$$\mathbb{A} \in \mathbf{Inp},$$

there is a canonical behaviour object

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

and a strict Mealy category

$$\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

We write

$$Z_{\mathbb{A}} := A_0 \times B_0.$$

Let

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

be an internal functor. Let

$$\mathbb{R}_{\mathbb{A}'}$$

be the vertical residual category over A'_0 , and let

$$F_0^* \mathbb{R}_{\mathbb{A}}$$

be the pullback to A'_0 of the vertical residual category over A_0 .

Definition 3.7.1 (Internal residual-slice functor). *Application of F to residual arrows and residual triangles defines an internal functor over A'_0*

$$F_{/} : \mathbb{R}_{\mathbb{A}'} \longrightarrow F_0^* \mathbb{R}_{\mathbb{A}}.$$

On residual objects it is

$$r' : u' \rightarrow v' \quad \longmapsto \quad F_1(r') : F_0(u') \rightarrow F_0(v').$$

On residual arrows represented by a triangle

$$w' \xrightarrow{a'} u' \xrightarrow{r'} v',$$

it sends

$$r' \circ a' \rightarrow r'$$

to

$$F_1(r') \circ F_1(a') \longrightarrow F_1(r').$$

Functoriality follows from

$$F_1(1_{v'}) = 1_{F_0(v')}$$

and

$$F_1(r' \circ a') = F_1(r') \circ F_1(a').$$

Precomposition with $F_{/}$ defines an internal functor over A'_0

$$(F_{/})^* : F_0^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{vert}} \longrightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}}^{\text{vert}}.$$

Its object map is

$$F^{\sharp} : A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \longrightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}.$$

In generalized-element notation,

$$F^{\sharp}(v', H) = H \circ F_{/v'},$$

where

$$F_{/v'} : \mathbb{A}'/v' \rightarrow \mathbb{A}/F_0(v')$$

is the fibre of $F_{/}$ at v' .

The domain of $F^\sharp$ is regarded over

$$Z_{\mathbb{A}'} = A'_0 \times B_0$$

by

$$(v', H) \mapsto (v', q_{\mathbf{Beh}} H).$$

Since

$$F_{/v'}(1_{v'}) = 1_{F_0(v')},$$

one has

$$\begin{aligned} q_{\mathbf{Beh}}(F^\sharp(v', H)) &= (H \circ F_{/v'})(1_{v'}) \\ &= H(1_{F_0(v')}) \\ &= q_{\mathbf{Beh}}(H). \end{aligned}$$

Thus $F^\sharp$ is a map over $Z_{\mathbb{A}'}$.

Lemma 3.7.1 (Reindexing preserves canonical structure). *With the input-pullback Mealy structure of Definition 3.7.2 on*

$$A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0},$$

the map $F^\sharp$ is a strict semantic homomorphism

$$F^\sharp : F^* \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} \longrightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}}^{\text{can}}.$$

Explicitly, for every

$$a' : u' \rightarrow v'$$

and every behaviour

$$H : \mathbb{A}/F_0(v') \rightarrow \mathbb{B},$$

one has

$$\partial_{a'}(F^\sharp(v', H)) = F^\sharp(u', \partial_{F_1(a')} H),$$

and

$$\omega_{\mathbf{Beh}_{\mathbb{A}', \mathbb{B}}}(a', F^\sharp(v', H)) = \omega_{\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}}(F_1(a'), H).$$

If $\mathbb{B}$ is $\mathbb{T}$ -structured, then $F^\sharp$ also preserves all pointwise $\mathbb{T}$ -operations.

Proof. For

$$a' : u' \rightarrow v',$$

the residual-slice functors satisfy the strict equality

$$F_{/v'} \circ a'_* = (F_1 a')_* \circ F_{/u'}.$$

Indeed, both sides send a residual object

$$r' : w' \rightarrow u'$$

to

$$F_1(a' \circ r') = F_1(a') \circ F_1(r'),$$

and act identically on residual triangles.

Therefore

$$\begin{aligned}\partial_{a'}(F^\sharp(v', H)) &= (H \circ F_{/v'}) \circ a'_* \\ &= H \circ (F_1 a')_* \circ F_{/u'} \\ &= (\partial_{F_1 a'} H) \circ F_{/u'} \\ &= F^\sharp(u', \partial_{F_1 a'} H).\end{aligned}$$

This is the transition-preservation equation.

The canonical output of $F^\sharp(v', H)$ along a' is the arrow assigned by

$$H \circ F_{/v'}$$

to the residual morphism

$$a' \rightarrow 1_{v'}$$

in $\mathbb{A}'/v'$. The functor $F_{/v'}$ sends this morphism to

$$F_1(a') \rightarrow 1_{F_0(v')}$$

in $\mathbb{A}/F_0(v')$. Hence

$$\begin{aligned}\omega_{\mathbf{Beh}_{\mathbb{A}', \mathbb{B}}}(a', F^\sharp(v', H)) &= (H \circ F_{/v'})(a' \rightarrow 1_{v'}) \\ &= H(F_1(a') \rightarrow 1_{F_0(v')}) \\ &= \omega_{\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}}(F_1(a'), H).\end{aligned}$$

This is the output-preservation equation.

Suppose now that $\mathbb{B}$ is $\mathbb{T}$ -structured and

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

is an operation. Since $F^\sharp$ is defined by precomposition and $\theta_{\mathbf{Beh}}$ is defined by postcomposition,

$$\begin{aligned}F^\sharp(v', \theta_{\mathbf{Beh}}(H_i)_i) &= (\theta \circ \langle H_i \rangle_i) \circ F_{/v'} \\ &= \theta \circ \langle H_i \circ F_{/v'} \rangle_i \\ &= \theta_{\mathbf{Beh}}(F^\sharp(v', H_i))_i.\end{aligned}$$

Thus $F^\sharp$ preserves the pointwise structured operations. □

3.7.2 Input pullback of Mealy morphisms

Definition 3.7.2 (Input pullback). *Let*

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be a Mealy morphism. Define

$$F^*P : \mathbb{A}' \rightsquigarrow \mathbb{B}$$

as follows. The carrier is

$$F^*P := A'_0 \times_{F_0, A_0, p_P} P.$$

A state is a pair

$$(v', x)$$

with

$$F_0(v') = p_P(x).$$

The input projection is

$$p_{F^*P}(v', x) = v',$$

and the output projection is

$$q_{F^*P}(v', x) = q_P(x).$$

For an arrow

$$a' : u' \rightarrow v',$$

define

$$\delta_{F^*P}(a', (v', x)) = (u', \delta_P(F_1 a', x)),$$

and

$$\omega_{F^*P}(a', (v', x)) = \omega_P(F_1 a', x).$$

The carrier is the pullback

$$\begin{array}{ccc} F^*P & \xrightarrow{\pi_P} & P \\ \downarrow & \lrcorner & \downarrow p_P \\ A'_0 & \xrightarrow{F_0} & A_0. \end{array}$$

The transition is induced by the square

$$\begin{array}{ccc} A'_1 \times_{A'_0} F^*P & \xrightarrow{F_1 \times \pi_P} & A_1 \times_{A_0} P \\ \delta_{F^*P} \downarrow & & \downarrow \delta_P \\ F^*P & \xrightarrow{\pi_P} & P, \end{array}$$

together with the input-object component u' .

For a strict semantic homomorphism

$$f : P \rightarrow Q$$

define

$$F^*f : F^*P \rightarrow F^*Q$$

by

$$F^*f(v', x) = (v', fx).$$

This assignment is functorial.

Proposition 3.7.1 (Input pullback and semantics). *The data above define a Mealy morphism*

$$F^*P : \mathbb{A}' \rightsquigarrow \mathbb{B}.$$

Moreover,

$$F^* : \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \rightarrow \mathbf{Mly}_{\mathbb{A}', \mathbb{B}}^{\text{str}}$$

is a functor, and the semantic map satisfies

$$\beta_{F^*P} = F^\sharp \circ \langle p_{F^*P}, \beta_P \pi_P \rangle.$$

In generalized elements,

$$\beta_{F^*P}(v', x) = F^\sharp(v', \beta_P(x)).$$

Proof.

1. **Check typing.**

If

$$a' : u' \rightarrow v',$$

then

$$F_1(a') : F_0(u') \rightarrow F_0(v').$$

Hence $\delta_P(F_1 a', x)$ is defined whenever

$$p_P(x) = F_0(v'),$$

and satisfies

$$p_P(\delta_P(F_1 a', x)) = F_0(u').$$

Therefore

$$(u', \delta_P(F_1 a', x))$$

lies in

$$A'_0 \times_{F_0, A_0, p_P} P.$$

The output

$$\omega_P(F_1 a', x)$$

has source

$$q_P(\delta_P(F_1 a', x))$$

and target

$$q_P(x),$$

so it has the required typing for F^*P .

2. **Verify the Mealy laws.**

The transition unit law follows from preservation of identities by F :

$$\begin{aligned} \delta_{F^*P}(1_{v'}, (v', x)) &= (v', \delta_P(F_1(1_{v'}), x)) \\ &= (v', \delta_P(1_{F_0(v')}, x)) \\ &= (v', x). \end{aligned}$$

Let

$$a' : u' \rightarrow v', \quad b' : v' \rightarrow w'.$$

Then

$$\begin{aligned} \delta_{F^*P}(b' \circ a', (w', x)) &= (u', \delta_P(F_1(b' \circ a'), x)) \\ &= (u', \delta_P(F_1 b' \circ F_1 a', x)) \\ &= (u', \delta_P(F_1 a', \delta_P(F_1 b', x))) \\ &= \delta_{F^*P}(a', \delta_{F^*P}(b', (w', x))). \end{aligned}$$

The output unit law is

$$\begin{aligned} \omega_{F^*P}(1_{v'}, (v', x)) &= \omega_P(1_{F_0(v')}, x) \\ &= 1_{q_P(x)}. \end{aligned}$$

The output multiplication law is

$$\begin{aligned}\omega_{F^*P}(b' \circ a', (w', x)) &= \omega_P(F_1 b' \circ F_1 a', x) \\ &= \omega_P(F_1 b', x) \circ \omega_P(F_1 a', \delta_P(F_1 b', x)) \\ &= \omega_{F^*P}(b', (w', x)) \circ \omega_{F^*P}(a', \delta_{F^*P}(b', (w', x))).\end{aligned}$$

Hence the displayed data define a Mealy morphism

$$F^*P : \mathbb{A}' \rightsquigarrow \mathbb{B}.$$

3. Check the action on strict semantic homomorphisms.

Let

$$f : P \rightarrow Q$$

be strict. Define

$$F^*f(v', x) = (v', fx).$$

This map lies over A'_0 and B_0 .

Transition preservation follows from

$$\begin{aligned}F^*f(\delta_{F^*P}(a', (v', x))) &= F^*f(u', \delta_P(F_1 a', x)) \\ &= (u', f\delta_P(F_1 a', x)) \\ &= (u', \delta_Q(F_1 a', fx)) \\ &= \delta_{F^*Q}(a', (v', fx)).\end{aligned}$$

Output preservation follows from

$$\omega_P(F_1 a', x) = \omega_Q(F_1 a', fx).$$

Identities and composition are inherited from $\mathcal{E}$, so

$$F^* : \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \rightarrow \mathbf{Mly}_{\mathbb{A}', \mathbb{B}}^{\text{str}}$$

is a functor.

4. Identify residual semantics.

Let

$$(v', x) \in F^*P.$$

For a residual object

$$r' : u' \rightarrow v'$$

of $\mathbb{A}'/v'$, execution in F^*P gives

$$\delta_{F^*P}(r', (v', x)) = (u', \delta_P(F_1 r', x)).$$

Its output anchor is therefore

$$q_P(\delta_P(F_1 r', x)) = \beta_P(x)(F_1 r').$$

For a residual triangle

$$w' \xrightarrow{a'} u' \xrightarrow{r'} v',$$

the corresponding residual output is

$$\omega_P(F_1 a', \delta_P(F_1 r', x)),$$

which is the arrow assigned by $\beta_P(x)$ to the image under $F_{/v'}$ of the residual triangle.

Thus the residual behaviour of (v', x) is the residual behaviour of x precomposed with

$$F_{/v'} : \mathbb{A}' / v' \rightarrow \mathbb{A} / F_0(v').$$

Hence

$$\beta_{F^*P}(v', x) = F^\sharp(v', \beta_P(x)).$$

5. Display the correctly typed semantic factorization.

The semantic identity is the commutativity of

$$\begin{array}{ccc} F^*P \xrightarrow{\langle p_{F^*P}, \beta_P \pi_P \rangle} A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} & & \\ \searrow \beta_{F^*P} & & \downarrow F^\sharp \\ & & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}. \end{array}$$

Therefore

$$\beta_{F^*P} = F^\sharp \circ \langle p_{F^*P}, \beta_P \pi_P \rangle.$$

□

Proposition 3.7.2 (Coherence of input pullback and residual reindexing). *Let*

$$1_{\mathbb{A}} : \mathbb{A} \rightarrow \mathbb{A}$$

be the identity internal functor. There are canonical natural isomorphisms

$$(1_{\mathbb{A}})^* \cong 1_{\mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}},$$

and, after the canonical identification of its pulled-back domain,

$$(1_{\mathbb{A}})^\sharp = 1_{\mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}}.$$

Let

$$\mathbb{A}'' \xrightarrow{G} \mathbb{A}' \xrightarrow{F} \mathbb{A}$$

be composable internal functors. There is a canonical natural isomorphism

$$(F \circ G)^* \cong G^* F^*.$$

On residual behaviours, after the canonical pullback identification

$$\begin{aligned} A''_0 \times_{F_0 G_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \\ \cong A''_0 \times_{G_0, A'_0} (A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}), \end{aligned}$$

one has

$$(F \circ G)^\sharp = G^\sharp \circ (1_{A_0''} \times_{A_0'} F^\sharp).$$

These identity and composition isomorphisms satisfy the usual unit and associativity coherence conditions. They are compatible with residual semantics: under the canonical isomorphism

$$(F \circ G)^* P \cong G^* F^* P,$$

the two induced semantic maps to

$$\mathbf{Beh}_{\mathbb{A}'', \mathbb{B}, 0}$$

agree.

Proof.

1. Identity coherence for input pullback.

The carrier of

$$(1_{\mathbb{A}})^* P$$

is

$$A_0 \times_{1_{A_0}, A_0, p_P} P.$$

The second pullback projection gives the canonical isomorphism

$$A_0 \times_{A_0} P \xrightarrow{\cong} P.$$

Under this isomorphism, the pulled-back transition and output maps are exactly those of P , because

$$(1_{\mathbb{A}})_1 = 1_{A_1}.$$

These isomorphisms are natural in strict semantic homomorphisms, giving

$$(1_{\mathbb{A}})^* \cong 1.$$

2. Composition coherence for input pullback.

The carrier of $(F \circ G)^* P$ is

$$A_0'' \times_{F_0 G_0, A_0, p_P} P.$$

The carrier of $G^* F^* P$ is

$$A_0'' \times_{G_0, A_0'} (A_0' \times_{F_0, A_0, p_P} P).$$

The associativity universal property of pullbacks gives a canonical isomorphism between these two carriers.

On an input arrow

$$a'' : u'' \rightarrow v'',$$

both transition structures apply the transition of P to

$$F_1 G_1(a'') = (F \circ G)_1(a'').$$

Their outputs are likewise both

$$\omega_P(F_1 G_1(a''), x).$$

Hence the carrier isomorphism is a strict Mealy isomorphism, natural in P :

$$(F \circ G)^* \cong G^* F^*.$$

3. Identity coherence for residual reindexing.

For every $v \in A_0$, the residual-slice functor induced by the identity is

$$(1_{\mathbb{A}})_{/v} = 1_{\mathbb{A}/v}.$$

Therefore

$$(1_{\mathbb{A}})^{\sharp}(v, H) = H \circ 1_{\mathbb{A}/v} = H.$$

After identifying

$$A_0 \times_{A_0} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \cong \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0},$$

one has

$$(1_{\mathbb{A}})^{\sharp} = 1_{\mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}}.$$

4. Composition coherence for residual reindexing.

Let

$$v'' \in A_0''.$$

The residual-slice functor of the composite satisfies

$$(F \circ G)_{/v''} = F_{/G_0(v'')} \circ G_{/v''}.$$

This equality holds on residual objects because

$$(F \circ G)_1(r'') = F_1(G_1(r'')),$$

and it holds on residual arrows by functoriality of F and G .

Hence, for

$$\begin{aligned} H : \mathbb{A}/F_0G_0(v'') &\rightarrow \mathbb{B}, \\ (F \circ G)^{\sharp}(v'', H) &= H \circ (F \circ G)_{/v''} \\ &= H \circ F_{/G_0(v'')} \circ G_{/v''} \\ &= G^{\sharp}(v'', F^{\sharp}(G_0(v''), H)). \end{aligned}$$

This is the asserted composition equation.

5. Verify coherence and semantic compatibility.

The unit and associativity coherence conditions for F^* are the canonical coherence conditions of iterated pullbacks. The corresponding conditions for $F^{\sharp}$ follow from strict associativity and unitality of functor composition.

Finally, Proposition 3.7.1 identifies residual semantics by precomposition with the relevant residual-slice functor. Since

$$(F \circ G)_{/v''} = F_{/G_0(v'')} \circ G_{/v''},$$

the semantic map obtained directly from $(F \circ G)^*P$ agrees with the semantic map obtained through G^*F^*P under the canonical pullback isomorphism.

□

Lemma 3.7.2 (Input pullback and semantic images). *Let*

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

be an internal functor and let

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

be a Mealy morphism.

Write the semantic image factorization of P as

$$P \xrightarrow{e_P} \text{Beh}(P) \xrightarrow{m_P} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}.$$

Define

$$F_P^\sharp : A'_0 \times_{F_0, A_0, p_{\text{Beh}(P)}} \text{Beh}(P) \rightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

to be the restriction

$$F_P^\sharp := F^\sharp \circ (1_{A'_0} \times m_P).$$

Then

$$\text{Im}(F_P^\sharp) = \text{Beh}(F^*P)$$

as subobjects of

$$\mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

in the slice

$$\mathcal{E}/(A'_0 \times B_0).$$

Proof. Put

$$Z_{\mathbb{A}} := A_0 \times B_0, \quad Z_{\mathbb{A}'} := A'_0 \times B_0,$$

and let

$$\overline{F} := F_0 \times 1_{B_0} : Z_{\mathbb{A}'} \rightarrow Z_{\mathbb{A}}.$$

The semantic map of P has regular image factorization

$$\begin{array}{ccc} P & \xrightarrow{\beta_P} & \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \\ & \searrow e_P & \nearrow m_P \\ & \text{Beh}(P) & \end{array}$$

in $\mathcal{E}/Z_{\mathbb{A}}$.

Pulling this factorization back along

$$\overline{F} : Z_{\mathbb{A}'} \rightarrow Z_{\mathbb{A}}$$

gives a regular epimorphism

$$\overline{F}^* e_P : \overline{F}^* P \twoheadrightarrow \overline{F}^* \text{Beh}(P)$$

in $\mathcal{E}/Z_{\mathbb{A}'}$.

There are canonical pullback isomorphisms

$$\begin{aligned} \overline{F}^* P &= Z_{\mathbb{A}'} \times_{Z_{\mathbb{A}}} P \\ &\cong A'_0 \times_{F_0, A_0, p_P} P \\ &= F^* P, \end{aligned}$$

and

$$\begin{aligned}\overline{F}^* \text{Beh}(P) &= Z_{\mathbb{A}'} \times_{Z_{\mathbb{A}}} \text{Beh}(P) \\ &\cong A'_0 \times_{F_0, A_0, p_{\text{Beh}(P)}} \text{Beh}(P).\end{aligned}$$

The B_0 -coordinate in each pullback is forced by the output projection of the corresponding state, which gives these canonical identifications.

Under these identifications, write

$$\tilde{e}_P : F^*P \twoheadrightarrow A'_0 \times_{F_0, A_0, p_{\text{Beh}(P)}} \text{Beh}(P)$$

for the pullback of e_P . In generalized-element notation,

$$\tilde{e}_P(v', x) = (v', e_P x).$$

Since regular epimorphisms are stable under pullback, $\tilde{e}_P$ is a regular epimorphism in

$$\mathcal{E}/Z_{\mathbb{A}'}.$$

By Proposition 3.7.1,

$$\beta_{F^*P} = F^\sharp \circ \langle p_{F^*P}, \beta_P \pi_P \rangle.$$

Since

$$\beta_P = m_P e_P,$$

the map into the pullback behaviour object factors as

$$\langle p_{F^*P}, \beta_P \pi_P \rangle = (1_{A'_0} \times m_P) \tilde{e}_P.$$

Consequently,

$$\begin{aligned}\beta_{F^*P} &= F^\sharp (1_{A'_0} \times m_P) \tilde{e}_P \\ &= F_P^\sharp \tilde{e}_P.\end{aligned}$$

Thus the semantic map of F^*P is the composite

$$F^*P \xrightarrow{\tilde{e}_P} A'_0 \times_{F_0, A_0, p_{\text{Beh}(P)}} \text{Beh}(P) \xrightarrow{F_P^\sharp} \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}.$$

Precomposition with a regular epimorphism does not change a regular image. Therefore

$$\text{Im}(\beta_{F^*P}) = \text{Im}(F_P^\sharp).$$

By definition,

$$\text{Im}(\beta_{F^*P}) = \text{Beh}(F^*P).$$

Hence

$$\text{Im}(F_P^\sharp) = \text{Beh}(F^*P)$$

as subobjects of

$$\mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

in $\mathcal{E}/Z_{\mathbb{A}'}$. □

3.8 Global coequational theories

3.8.1 Global theories and global models

Definition 3.8.1 (Global coequational theory). A **global coequational theory** consists of a local coequational theory

$$V_{\mathbb{A}} \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

for each $\mathbb{A} \in \text{Inp}$, such that for every internal functor

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

the reindexing map

$$F^\sharp : A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \rightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

sends

$$A'_0 \times_{F_0, A_0, p_{V_{\mathbb{A}}}} V_{\mathbb{A}}$$

into

$$V_{\mathbb{A}'}$$

If $\mathbb{B}$ is $\mathbb{T}$ -structured, a **global $\mathbb{T}$ -coequational theory** is such a family for which each $V_{\mathbb{A}}$ is a local $\mathbb{T}$ -coequational theory.

The global reindexing condition is displayed by

$$\begin{array}{ccc} A'_0 \times_{A_0} V_{\mathbb{A}} & \xrightarrow{F^\sharp|_{V_{\mathbb{A}}}} & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0} \\ & \searrow & \nearrow \\ & V_{\mathbb{A}'} & \end{array}$$

where the fibre product is taken along

$$F_0 : A'_0 \rightarrow A_0 \quad \text{and} \quad p_{V_{\mathbb{A}}} : V_{\mathbb{A}} \rightarrow A_0.$$

Including the carrier slices, the condition is

$$\begin{array}{ccc} A'_0 \times_{A_0} V_{\mathbb{A}} & \xrightarrow{\quad} & V_{\mathbb{A}} \\ \downarrow & \lrcorner & \downarrow \\ A'_0 \times_{A_0} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} & \xrightarrow{F^\sharp} & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0} \\ \downarrow & \text{over } Z_{\mathbb{A}'} & \\ Z_{\mathbb{A}'} & & \end{array}$$

together with factorization through $V_{\mathbb{A}'}$.

Definition 3.8.2 (Global models). Let

$$V = (V_{\mathbb{A}})_{\mathbb{A} \in \text{Inp}}$$

be a global coequational theory. Define

$$\text{Mod}^g(V)_{\mathbb{A}} = \{P \in \text{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}} \mid \beta_P \text{ factors through } V_{\mathbb{A}}\}.$$

3.8.2 Global closure

Theorem 3.8.1 (Global coequational closure). *Let*

$$V = (V_{\mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

be a global coequational theory. Then

$$\mathbf{Mod}^g(V)$$

is closed under strict homomorphic preimages, regular-epimorphic quotients, small coproducts, and input pullback along functors in $\mathbf{Inp}$.

If V is a global $\mathbb{T}$ -coequational theory, then

$$\mathbf{Mod}^g(V)$$

is also closed under the operations induced by $\mathbb{T}$.

Proof.

1. **Apply local closure pointwise.** Closure under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts is Theorem 3.5.1 applied to each $V_{\mathbb{A}}$. In the structured case, closure under $\mathbb{T}$ -operations is Theorem 3.6.1 applied pointwise.
2. **Check input pullback.** Let

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

and let

$$P \in \mathbf{Mod}^g(V)_{\mathbb{A}}.$$

Choose a factorization

$$\bar{\beta}_P : P \rightarrow V_{\mathbb{A}}$$

of β_P . Proposition 3.7.1 gives

$$\beta_{F^*P} = F^\sharp \circ \langle p_{F^*P}, \beta_P \pi_P \rangle.$$

Since V is global, $F^\sharp$ sends the pullback of $V_{\mathbb{A}}$ into $V_{\mathbb{A}'}$. Therefore

$$\beta_{F^*P}$$

factors through $V_{\mathbb{A}'}$, so

$$F^*P \in \mathbf{Mod}^g(V)_{\mathbb{A}'}$$

The factorization is

$$\begin{array}{c}
 \xrightarrow{\beta_{F^*P}} \\
 F^*P \xrightarrow{\langle p_{F^*P}, \beta_P \pi_P \rangle} A'_0 \times_{A_0} V_{\mathbb{A}} \dashrightarrow V_{\mathbb{A}'} \hookrightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}.
 \end{array}$$

□

3.9 The global coequational Birkhoff theorem

3.9.1 The theorem

Theorem 3.9.1 (Global coequational Birkhoff theorem). *Let*

$$\mathcal{C} = (\mathcal{C}_{\mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

be a replete global family of full subcategories

$$\mathcal{C}_{\mathbb{A}} \subseteq \mathbf{Mly}_{\mathbb{A}, \mathbb{B}}^{\text{str}}.$$

Then $\mathcal{C}$ is of the form

$$\mathcal{C}_{\mathbb{A}} = \mathbf{Mod}^g(V)_{\mathbb{A}}$$

for a global coequational theory

$$V = (V_{\mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

if and only if each $\mathcal{C}_{\mathbb{A}}$ is closed under:

1. *strict homomorphic preimages;*
2. *regular-epimorphic quotients;*
3. *small coproducts;*

and the family $\mathcal{C}$ is closed under input pullback along functors in $\mathbf{Inp}$.

If $\mathbb{B}$ is $\mathbf{T}$ -structured, then global $\mathbf{T}$ -coequational theories correspond to such families that are also closed under the operations induced by $\mathbf{T}$.

Proof.

1. Prove necessity.

Suppose

$$\mathcal{C} = \mathbf{Mod}^g(V)$$

for a global coequational theory V . The required closure properties are Theorem 3.8.1.

2. Construct the local representing theories.

Conversely, suppose that each $\mathcal{C}_{\mathbb{A}}$ is closed under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts, and that the family $\mathcal{C}$ is closed under input pullback.

For every

$$\mathbb{A} \in \mathbf{Inp},$$

define

$$\mathcal{S}_{\mathcal{C}, \mathbb{A}} := \{S \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} \mid S = \mathbf{Beh}(P) \text{ for some } P \in \mathcal{C}_{\mathbb{A}}\}.$$

Choose a small representative set for the distinct subobjects in $\mathcal{S}_{\mathcal{C}, \mathbb{A}}$, and put

$$V_{\mathcal{C}, \mathbb{A}} := \bigvee_{S \in \mathcal{S}_{\mathcal{C}, \mathbb{A}}} S \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

By Theorem 3.5.2, $V_{\mathcal{C}, \mathbb{A}}$ is a local coequational theory and

$$\mathcal{C}_{\mathbb{A}} = \text{Mod}(V_{\mathcal{C}, \mathbb{A}}).$$

By Lemma 3.5.2, one also has

$$V_{\mathcal{C}, \mathbb{A}} \in \mathcal{C}_{\mathbb{A}}.$$

3. Use input-pullback closure on the representing theory.

Let

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

be an internal functor in Inp .

Since

$$V_{\mathcal{C}, \mathbb{A}} \in \mathcal{C}_{\mathbb{A}}$$

and the family $\mathcal{C}$ is closed under input pullback,

$$F^*V_{\mathcal{C}, \mathbb{A}} \in \mathcal{C}_{\mathbb{A}'}$$

Using

$$\mathcal{C}_{\mathbb{A}'} = \text{Mod}(V_{\mathcal{C}, \mathbb{A}'}),$$

the object $F^*V_{\mathcal{C}, \mathbb{A}}$ models $V_{\mathcal{C}, \mathbb{A}'}$. Hence there is a unique factorization

$$\varphi_F : F^*V_{\mathcal{C}, \mathbb{A}} \longrightarrow V_{\mathcal{C}, \mathbb{A}'}$$

satisfying

$$j_{V_{\mathcal{C}, \mathbb{A}'}} \varphi_F = \beta_{F^*V_{\mathcal{C}, \mathbb{A}}}.$$

By Lemma 3.2.1, φ_F is a strict semantic homomorphism.

4. Identify the model factorization with the required restriction of $F^\sharp$.

The carrier of the input pullback is

$$F^*V_{\mathcal{C}, \mathbb{A}} = A'_0 \times_{F_0, A_0, p_{V_{\mathcal{C}, \mathbb{A}}}} V_{\mathcal{C}, \mathbb{A}}.$$

Write

$$\pi_V : F^*V_{\mathcal{C}, \mathbb{A}} \rightarrow V_{\mathcal{C}, \mathbb{A}}$$

for its second pullback projection.

Proposition 3.7.1 gives

$$\beta_{F^*V_{\mathcal{C}, \mathbb{A}}} = F^\sharp \circ \langle p_{F^*V_{\mathcal{C}, \mathbb{A}}}, \beta_{V_{\mathcal{C}, \mathbb{A}}} \pi_V \rangle.$$

By Lemma 3.2.2,

$$\beta_{V_{\mathcal{C}, \mathbb{A}}} = j_{V_{\mathcal{C}, \mathbb{A}}}.$$

Therefore

$$\beta_{F^*V_{\mathcal{C}, \mathbb{A}}} = F^\sharp \circ \langle p_{F^*V_{\mathcal{C}, \mathbb{A}}}, j_{V_{\mathcal{C}, \mathbb{A}}} \pi_V \rangle.$$

The map

$$\langle p_{F^*V_{\mathcal{C}, \mathbb{A}}}, j_{V_{\mathcal{C}, \mathbb{A}}} \pi_V \rangle$$

is precisely the canonical pulled-back inclusion

$$A'_0 \times_{F_0, A_0, p_{V_{C, \mathbb{A}}}} V_{C, \mathbb{A}} \longrightarrow A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}.$$

Hence the factorization equation for φ_F becomes

$$j_{V_{C, \mathbb{A}'}} \varphi_F = F^\sharp \circ \langle p_{F^* V_{C, \mathbb{A}}}, j_{V_{C, \mathbb{A}}} \pi_V \rangle.$$

This is exactly the required global reindexing factorization:

$$\begin{array}{ccc} A'_0 \times_{F_0, A_0, p_{V_{C, \mathbb{A}}}} V_{C, \mathbb{A}} & \xrightarrow{F^\sharp \langle p, j_{V_{C, \mathbb{A}}} \pi_V \rangle} & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0} \\ & \searrow \varphi_F & \swarrow j_{V_{C, \mathbb{A}'}} \\ & V_{C, \mathbb{A}'} & \end{array}$$

Thus $F^\sharp$ sends the pullback of $V_{C, \mathbb{A}}$ into $V_{C, \mathbb{A}'}$.

Since F was arbitrary, the family

$$V_C := (V_{C, \mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

is a global coequational theory.

5. Handle the structured case.

Suppose now that $\mathbb{B}$ is $\mathbb{T}$ -structured and that every $\mathcal{C}_{\mathbb{A}}$ is also closed under the operations induced by $\mathbb{T}$.

By Theorem 3.6.2, applied separately to every input category $\mathbb{A}$, each

$$V_{C, \mathbb{A}}$$

is a local $\mathbb{T}$ -coequational theory. The preceding argument has already shown that these local theories satisfy the global reindexing condition. Therefore

$$V_C$$

is a global $\mathbb{T}$ -coequational theory.

6. Conclude.

For every

$$\mathbb{A} \in \mathbf{Inp},$$

the local Birkhoff theorem gives

$$\mathcal{C}_{\mathbb{A}} = \mathbf{Mod}(V_{C, \mathbb{A}}).$$

Since V_C is a global theory, this equality is precisely

$$\mathcal{C}_{\mathbb{A}} = \mathbf{Mod}^g(V_C)_{\mathbb{A}}.$$

Hence

$$\mathcal{C} = \mathbf{Mod}^g(V_C).$$

□

Corollary 3.9.1 (Uniqueness of the defining global theory). *Let*

$$V = (V_{\mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

be a global coequational theory. Then, for every

$$\mathbb{A} \in \mathbf{Inp},$$

one has

$$V_{\mathbb{A}} = \bigvee_{P \in \mathbf{Mod}^g(V)_{\mathbb{A}}} \mathbf{Beh}(P).$$

Consequently a globally coequationally definable family of model classes has a unique defining global coequational theory. The same statement holds for global $\mathbb{T}$ -coequational theories.

Proof. For each fixed input category $\mathbb{A}$,

$$\mathbf{Mod}^g(V)_{\mathbb{A}} = \mathbf{Mod}(V_{\mathbb{A}}).$$

Therefore Corollary 3.5.1 gives

$$V_{\mathbb{A}} = \bigvee_{P \in \mathbf{Mod}^g(V)_{\mathbb{A}}} \mathbf{Beh}(P).$$

Thus every component $V_{\mathbb{A}}$ is determined uniquely by the corresponding model class. Hence the entire global family V is uniquely determined.

The structured case follows pointwise in the same way. $\square$

3.10 One-object specialization

This section records the one-object form of the preceding constructions. Let the input category be determined by a monoid object M . In the purely one-object output case, let the output category be determined by a monoid object N . We keep the standing convention

$$b \circ a = ba.$$

Thus a Mealy morphism is a right M -object P , written

$$x \cdot m := \delta(m, x),$$

together with an N -valued output cocycle

$$\omega : M \times P \rightarrow N$$

satisfying

$$\omega(1, x) = 1, \quad \omega(mn, x) = \omega(m, x)\omega(n, x \cdot m).$$

The residual category of M is the right Cayley residual category. Its objects are elements $r \in M$, and a residual arrow labelled m has the form

$$rm \longrightarrow r.$$

A residual behaviour is therefore an N -valued cocycle

$$\lambda : M \times M \rightarrow N$$

satisfying

$$\lambda(r, 1) = 1, \quad \lambda(r, mn) = \lambda(r, m)\lambda(rm, n).$$

The canonical derivative is left translation in the residual-history variable:

$$(\partial_s \lambda)(r, m) = \lambda(sr, m),$$

and the canonical output on input s is

$$\lambda(1, s).$$

Hence a local coequational theory in the one-object monoid-valued case is a subobject

$$V \hookrightarrow \mathbf{Beh}_0$$

of residual cocycles closed under all derivatives:

$$\lambda \in V \implies \partial_s \lambda \in V \quad (s \in M).$$

A Mealy morphism P models V precisely when every residual cocycle generated by a state of P lies in V . The residual cocycle generated by $x \in P$ is

$$\lambda_x(r, m) = \omega(m, x \cdot r).$$

Thus

$$P \models V \iff \lambda_x \in V \text{ for every } x \in P.$$

The semantic image of P is the image of the state-to-behaviour map

$$P \rightarrow \mathbf{Beh}_0, \quad x \mapsto \lambda_x.$$

In **Set**, this is simply the set

$$\text{Beh}(P) = \{\lambda_x \mid x \in P\} \subseteq \mathbf{Beh}_0.$$

The semantic quotient

$$P \twoheadrightarrow \text{Beh}(P)$$

identifies precisely those states with the same residual cocycle:

$$x \sim y \iff \lambda_x = \lambda_y.$$

For a fixed one-object input monoid M , the local coequational Birkhoff theorem specializes as follows. Derivative-closed subobjects

$$V \hookrightarrow \mathbf{Beh}_0$$

correspond to replete full classes of strict Mealy morphisms closed under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts. The theory generated by such a class $\mathcal{C}$ is

$$V_{\mathcal{C}} = \bigvee_{P \in \mathcal{C}} \text{Beh}(P).$$

In **Set**, this join is the union of the residual behaviours generated by the objects of $\mathcal{C}$.

The Boolean language case is obtained by taking

$$\mathcal{E} = \mathbf{Set}, \quad M = \Sigma^*, \quad \mathbb{B} = 2_{\text{ch}},$$

where 2_{ch} is the chaotic category on the Boolean set 2. Since 2_{ch} has a unique arrow between any two Boolean values, the observable data of a state is its Boolean output anchor. A residual behaviour is therefore just a language

$$L : \Sigma^* \rightarrow 2.$$

The derivative convention becomes

$$(\partial_u L)(w) = L(uw).$$

Thus a local coequational theory over the alphabet Σ is a collection

$$V \subseteq 2^{\Sigma^*}$$

closed under left derivatives:

$$L \in V \implies \partial_u L \in V \quad (u \in \Sigma^*).$$

A Boolean recognizer P models V precisely when all residual languages generated by its states lie in V :

$$\{\beta_P(x) \mid x \in P\} \subseteq V.$$

Its semantic image is

$$\text{Beh}(P) = \{\beta_P(x) \mid x \in P\} \subseteq 2^{\Sigma^*},$$

and the semantic quotient is the usual Nerode-style quotient identifying states with the same residual language.

If the Boolean operations

$$\wedge, \vee, \neg, 0, 1$$

are included as structure on 2_{ch} , then a local Boolean coequational theory is a Boolean subalgebra of 2^{Σ^*} closed under residual derivatives. Finite recognizability is not part of this coequational closure condition by itself; it enters only through an additional finite-state restriction or by working inside the regular languages. Thus the derivative-closed Boolean structure is separated from the finite-state regularity condition, as in the classical distinction between all languages and regular languages [6, 8, 4].

The global input-reindexing construction specializes to inverse morphic preimage. A functor between one-object free-monoid input categories is a monoid homomorphism

$$h : \Gamma^* \rightarrow \Sigma^*.$$

For a language

$$L : \Sigma^* \rightarrow 2,$$

the reindexed behaviour is

$$h^{-1}L = L \circ h, \quad h^{-1}L(w) = L(h(w)).$$

Therefore a global Boolean coequational theory assigns to each alphabet Σ a derivative-closed collection

$$V_{\Sigma} \subseteq 2^{\Sigma^*}$$

such that

$$L \in V_{\Sigma} \implies h^{-1}L \in V_{\Gamma}$$

for every homomorphism

$$h : \Gamma^* \rightarrow \Sigma^*.$$

If the theory is Boolean, each V_Σ is also closed under the pointwise Boolean operations. The global coequational Birkhoff theorem then recovers the coequational inverse-image closure condition familiar from Eilenberg-type correspondences: the associated model classes are exactly the recognizers whose generated residual languages lie in the corresponding V_Σ .

3.11 Summary

This chapter has reformulated residual semantics in terms of coequational subobjects of the terminal behaviour Mealy morphism.

construction	role
$\mathbf{Beh}_{A,B}^{\text{can}}$	terminal residual semantic Mealy morphism
$V \hookrightarrow \mathbf{Beh}^{\text{can}}$	local coequational theory
$\beta_P : P \rightarrow \mathbf{Beh}^{\text{can}}$	unique semantic map to the terminal object
$\text{Beh}(P)$	regular semantic image of a Mealy morphism
$\text{Mod}(V)$	Mealy morphisms whose semantics land in V
$V_C = \bigvee_{P \in C} \text{Beh}(P)$	coequational theory generated by a model class
$\theta_{\mathbf{Beh}}$	pointwise structured operation on behaviours
$F^\#$	input reindexing of residual behaviours
F^*P	input pullback of a Mealy morphism

The local coequational Birkhoff theorem identifies local coequational theories with replete full classes of strict Mealy morphisms closed under strict homomorphic preimages, regular-epimorphic quotients, and small coproducts. The structured version adds closure under pointwise algebraic operations induced by a $\mathbb{T}$ -structured output category. The global version adds stability under input pullback, equivalently under residual behaviour reindexing.

The next chapter extracts the algebraic side of this coequational semantics. For a local coequational theory V , the restricted canonical object on V determines a residual transition–output action of

$$\mathbb{A}^{\text{op}},$$

reflecting the target-to-source processing convention. The regular image of this action is the transition–output quotient associated to V . For a specification K , applying this construction to the derivative theory

$$\text{Der}_0(K)$$

produces the syntactic transition–output category.

Chapter 4

Operational Equations, Syntactic Transition–Output Quotients, and Algebraic Eilenberg Duality

4.1 Overview and standing assumptions

4.1.1 Orientation

The preceding chapters developed residual semantics at the level of states and behaviours. Every Mealy morphism

$$P : \mathbb{A} \rightsquigarrow \mathbb{B}$$

has a unique strict semantic homomorphism

$$\beta_P : P \rightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B}}^{\text{can}},$$

and its regular image

$$\text{Beh}(P) \hookrightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B}}^{\text{can}}$$

is the local coequational theory of behaviours realized by P . The semantic image factorization is

$$\begin{array}{ccc} P & \xrightarrow{\eta_P} & \text{Beh}(P) \\ & \searrow \beta_P & \downarrow m_P \\ & & \mathbf{Beh}_{\mathbb{A},\mathbb{B}}^{\text{can}}. \end{array}$$

For a behaviour specification

$$k : K \rightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B},0},$$

the residual evaluation map

$$d_K : D_K \rightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B},0}$$

has regular image

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_{\mathbb{A},\mathbb{B},0},$$

which is the local coequational theory freely generated by the specified behaviours:

$$\begin{array}{ccc} D_K & \longrightarrow & \text{Der}_0(K) \\ & \searrow d_K & \downarrow \\ & & \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}. \end{array}$$

The present chapter develops the corresponding algebra of input operations. Its central idea is that a Mealy structure on a carrier

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

is a representation of the opposite input category

$$\mathbb{A}^{\text{op}}$$

by fibrewise output-comparison Kleisli operations on P . An input arrow

$$a : u \rightarrow v$$

acts on a state $x \in P_v$ by

$$x \mapsto (\delta(a, x), \omega(a, x)),$$

where

$$\delta(a, x) \in P_u, \quad \omega(a, x) : q(\delta(a, x)) \rightarrow q(x).$$

The action is displayed by

$$P_v \overset{x \mapsto (\delta(a, x), \omega(a, x))}{\dashrightarrow} P_u,$$

where the dashed arrow denotes a Kleisli map rather than an ordinary map in $\mathcal{E}/B_0$.

For a local coequational theory

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

this gives an internal representation

$$\chi_V : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(V).$$

Its kernel congruence

$$R_V := \ker(\chi_{V,1})$$

consists of the parallel input arrows having the same transition-output effect on every behaviour in V . Its regular image

$$\text{Op}_{\mathbb{A}, \mathbb{B}}(V) := \text{RegIm}(\chi_V)$$

is the corresponding faithful transition-output quotient:

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_V} & \text{KlEnd}_{\mathbb{B}}(V) \\ & \searrow q_V & \swarrow i_V \\ & \text{Op}_{\mathbb{A}, \mathbb{B}}(V) & \end{array}$$

Conversely, an identity-on-objects operation congruence

$$R \rightrightarrows A_1$$

determines a one-step equation predicate

$$\text{Eq}(R) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}.$$

The largest local coequational theory satisfying R is the derivative safety core

$$V_R := \text{Safe}_\partial(\text{Eq}(R)).$$

The resulting operational polarity is

$$V \leq V_R \iff R \leq R_V.$$

Diagrammatically,

$$\text{Loc}_{\mathbb{A}, \mathbb{B}} \begin{array}{c} \xrightarrow{V \mapsto R_V} \\ \xleftarrow{R \mapsto V_R} \end{array} \text{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}})^{\text{op}}$$

is a Galois connection. Its fixed points yield the local algebraic Eilenberg duality.

For a specification K , the syntactic transition-output category is

$$\text{SynTO}_{\mathbb{A}, \mathbb{B}}(K) := \text{Op}_{\mathbb{A}, \mathbb{B}}(\text{Der}_0(K)).$$

It is the faithful identity-on-objects quotient of $\mathbb{A}^{\text{op}}$ determined by all operational equations valid in the residual behaviours generated by K :

$$\mathbb{A}^{\text{op}} \twoheadrightarrow \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K) \hookrightarrow \text{KlEnd}_{\mathbb{B}}(\text{Der}_0(K)).$$

The final parts of the chapter globalize this correspondence along input functors and specialize it to monoids, stateless functors, Boolean languages, syntactic monoids, and the classical Eilenberg correspondence [6, 1, 23].

4.1.2 Standing assumptions and notation

Throughout this chapter we use the assumptions and notation of Chapters 2 and 3. Thus $\mathcal{E}$ is a Grothendieck topos with chosen pullbacks. In particular, $\mathcal{E}$ is Barr-exact and locally cartesian closed, regular epimorphisms are stable under pullback, internal equivalence relations are effective, and every slice has small limits, small colimits, and complete subobject lattices [12, 3, 10].

The input and output internal categories are

$$\mathbb{A} = (A_0, A_1, s_A, t_A, e_A, m_A), \quad \mathbb{B} = (B_0, B_1, s_B, t_B, e_B, m_B).$$

When $\mathbb{A}$ and $\mathbb{B}$ are fixed, write

$$Z := A_0 \times B_0.$$

A Mealy carrier is an object of

$$\mathcal{E}/Z,$$

displayed as a span

$$\begin{array}{ccc} & P & \\ p \swarrow & & \searrow q \\ A_0 & & B_0. \end{array}$$

Composition in $\mathbb{A}$ is written

$$m_A(a, b) = b \circ a$$

for composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w.$$

The target-to-source execution convention is

$$\delta(b \circ a, x) = \delta(a, \delta(b, x))$$

and

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

The comparison arrows form the triangle

$$\begin{array}{ccccc} & & \omega(b \circ a, x) & & \\ & \nearrow & & \searrow & \\ q(\delta(a, \delta(b, x))) & \xrightarrow{\omega(a, \delta(b, x))} & q(\delta(b, x)) & \xrightarrow{\omega(b, x)} & q(x). \end{array}$$

We regard A_1 , as the arrow object of $\mathbb{A}^{\text{op}}$, as an object over

$$A_0 \times A_0$$

by

$$(t_A, s_A) : A_1 \rightarrow A_0 \times A_0.$$

Thus an arrow

$$a : u \rightarrow v$$

of $\mathbb{A}$ is read as an arrow

$$a^{\text{op}} : v \rightarrow u$$

of $\mathbb{A}^{\text{op}}$.

We write

$$\mathbf{Loc}_{\mathbb{A}, \mathbb{B}}$$

for the complete lattice of local coequational theories

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

ordered by inclusion.

We write

$$\ker(f)$$

for the kernel pair, or kernel congruence, of a morphism f . Thus the notation does not refer to a pointed kernel.

Whenever local theories or operation congruences are written with $\cong$, this denotes a canonical isomorphism of chosen representatives. After passage to the corresponding subobject posets, these canonical isomorphisms are regarded as equalities.

4.2 Mealy morphisms as bicategorical representations

4.2.1 Internal categories as monads in spans

Recall the bicategory

$$\mathbf{Span}(\mathcal{E}).$$

Its objects are objects of $\mathcal{E}$, its 1-cells

$$X \rightsquigarrow Y$$

are spans

$$\begin{array}{ccc} & S & \\ \ell \swarrow & & \searrow r \\ X & & Y, \end{array}$$

and its 2-cells are maps of spans. Horizontal composition is defined by pullback.

An internal category

$$\mathbb{A}$$

is equivalently a monad on A_0 in $\mathbf{Span}(\mathcal{E})$, whose underlying 1-cell is

$$\begin{array}{ccc} & A_1 & \\ s_A \swarrow & & \searrow t_A \\ A_0 & & A_0. \end{array}$$

Likewise, $\mathbb{B}$ is a monad on B_0 .

A Mealy carrier

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

is a 1-cell

$$P : A_0 \rightsquigarrow B_0.$$

The composite

$$\mathbb{A} \odot P$$

has apex

$$A_1 \times_{t_A, A_0, p} P,$$

defined by

$$\begin{array}{ccc} A_1 \times_{A_0} P & \xrightarrow{\pi_P} & P \\ \pi_{A_1} \downarrow & \lrcorner & \downarrow p \\ A_1 & \xrightarrow{t_A} & A_0. \end{array}$$

The composite

$$P \odot \mathbb{B}$$

has apex

$$P \times_{q, B_0, s_B} B_1,$$

defined by

$$\begin{array}{ccc} P \times_{B_0} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_P \downarrow & \lrcorner & \downarrow s_B \\ P & \xrightarrow{q} & B_0. \end{array}$$

Definition 4.2.1 (Right-lax monad morphism structure). *A **right-lax monad morphism structure** from $\mathbb{A}$ to $\mathbb{B}$ on the span P is a 2-cell*

$$\alpha_P : \mathbb{A} \odot P \Longrightarrow P \odot \mathbb{B}$$

satisfying the following unit and multiplication axioms:

$$\alpha_P \circ (\eta_{\mathbb{A}} \odot 1_P) = 1_P \odot \eta_{\mathbb{B}},$$

and

$$\alpha_P \circ (\mu_{\mathbb{A}} \odot 1_P) = (1_P \odot \mu_{\mathbb{B}}) \circ (\alpha_P \odot 1_{\mathbb{B}}) \circ (1_{\mathbb{A}} \odot \alpha_P),$$

with the canonical associators and unitors of $\mathbf{Span}(\mathcal{E})$ understood.

The multiplication axiom compares the two paths in

$$\begin{array}{ccc} \mathbb{A} \odot \mathbb{A} \odot P & \xrightarrow{\mu_{\mathbb{A}} \odot 1_P} & \mathbb{A} \odot P \\ 1_{\mathbb{A}} \odot \alpha_P \downarrow & & \downarrow \alpha_P \\ \mathbb{A} \odot P \odot \mathbb{B} & \xrightarrow{\alpha_P \odot 1_{\mathbb{B}}} & P \odot \mathbb{B} \odot \mathbb{B} \\ & & \downarrow 1_P \odot \mu_{\mathbb{B}} \\ & & P \odot \mathbb{B}. \end{array}$$

Remark 4.2.1 (Terminology). *The term “right-lax” refers here to the direction*

$$\mathbb{A} \odot P \Longrightarrow P \odot \mathbb{B}$$

under the execution-order convention for horizontal composition used in this chapter. Under the opposite convention for bicategorical composition, the same structure is often called a colax monad morphism or monad opmorphism.

A map of spans

$$\alpha_P : A_1 \times_{t_A, A_0, p} P \longrightarrow P \times_{q, B_0, s_B} B_1$$

is equivalently a pair

$$\delta : A_1 \times_{t_A, A_0, p} P \rightarrow P, \quad \omega : A_1 \times_{t_A, A_0, p} P \rightarrow B_1$$

satisfying

$$p\delta = s_A \pi_{A_1},$$

$$s_B \omega = q\delta,$$

and

$$t_B \omega = q\pi_P.$$

The defining map is

$$\begin{array}{ccc} A_1 \times_{A_0} P & \xrightarrow{\langle \delta, \omega \rangle} & P \times_{B_0} B_1 \\ & \searrow \delta & \downarrow \pi_P \\ & & P, \end{array}$$

and the output typing equations form

$$\begin{array}{ccc} A_1 \times_{A_0} P & \xrightarrow{\omega} & B_1 \\ \delta \downarrow & \circlearrowleft & \downarrow s_B \\ P & \xrightarrow{q} & B_0 \end{array} \quad \begin{array}{ccc} A_1 \times_{A_0} P & \xrightarrow{\omega} & B_1 \\ \pi_P \downarrow & \circlearrowleft & \downarrow t_B \\ P & \xrightarrow{q} & B_0. \end{array}$$

Theorem 4.2.1 (Mealy structures as right-lax monad morphisms). *Let*

$$A_0 \xleftarrow{P} P \xrightarrow{q} B_0$$

be a carrier span. To give a Mealy structure on P is equivalent to giving a right-lax monad morphism structure

$$\alpha_P : \mathbb{A} \odot P \Longrightarrow P \odot \mathbb{B}.$$

Proof.

1. **Extract the transition-output pair.** A 2-cell α_P is equivalently a pair

$$(\delta, \omega)$$

with

$$\delta : A_1 \times_{A_0} P \rightarrow P, \quad \omega : A_1 \times_{A_0} P \rightarrow B_1,$$

satisfying the Mealy typing equations. It is represented by

$$A_1 \times_{A_0} P \xrightarrow{\langle \delta, \omega \rangle} P \times_{B_0} B_1.$$

2. **Read the unit axiom.** Compatibility with the units of $\mathbb{A}$ and $\mathbb{B}$ gives, for $x \in P_v$,

$$\delta(e_A v, x) = x, \quad \omega(e_A v, x) = e_B(qx).$$

The comparison is

$$\begin{array}{ccc} P & \xrightarrow{\langle e_{AP}, 1_P \rangle} & A_1 \times_{A_0} P \\ & \searrow \langle 1_P, e_{Bq} \rangle & \downarrow \langle \delta, \omega \rangle \\ & & P \times_{B_0} B_1. \end{array}$$

3. **Read the multiplication axiom.** For composable arrows

$$a : u \rightarrow v, \quad b : v \rightarrow w, \quad x \in P_w,$$

the multiplication axiom gives

$$\delta(b \circ a, x) = \delta(a, \delta(b, x))$$

and

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

The output equation is the commutativity of

$$q(\delta(a, \delta(b, x))) \xrightarrow{\omega(a, \delta(b, x))} q(\delta(b, x)) \xrightarrow{\omega(b, x)} q(x).$$

4. **Reconstruct the bicategorical structure.** Conversely, a transition-output pair satisfying the Mealy typing, unit, and multiplication laws defines a map of spans

$$\alpha_P : \mathbb{A} \odot P \Longrightarrow P \odot \mathbb{B}.$$

The displayed unit and multiplication equations are precisely the right-lax monad morphism axioms. Hence the two constructions are mutually inverse. $\square$

Remark 4.2.2 (Bicategorical role of the chapter). *The bicategorical formulation identifies the transition-output structure as a representation-type structure carried by the span P . The remaining constructions internalize the corresponding endomorphism object and take the faithful image of the induced representation.*

4.2.2 The output-comparison Kleisli monad

Right composition by the monad $\mathbb{B}$ induces the output-comparison monad

$$T_{\mathbb{B}} := (t_B)_!(s_B)^* : \mathcal{E}/B_0 \rightarrow \mathcal{E}/B_0.$$

For

$$q_X : X \rightarrow B_0,$$

the pullback part is

$$\begin{array}{ccc} X \times_{q_X, B_0, s_B} B_1 & \xrightarrow{\pi_{B_1}} & B_1 \\ \pi_X \downarrow & \lrcorner & \downarrow s_B \\ X & \xrightarrow{q_X} & B_0. \end{array}$$

Postcomposition with t_B gives

$$T_{\mathbb{B}}X = X \times_{q_X, B_0, s_B} B_1 \xrightarrow{t_B \pi_{B_1}} B_0.$$

A generalized element is a pair

$$(x, \alpha), \quad \alpha : q_X(x) \rightarrow b.$$

The unit is

$$\eta_X(x) = (x, e_B(q_X x)),$$

represented by

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T_{\mathbb{B}}X \\ & \searrow q_X & \downarrow t_B \pi_{B_1} \\ & & B_0. \end{array}$$

A generalized element of $T_{\mathbb{B}}^2 X$ is represented by

$$(x, \alpha, \beta), \quad \alpha : q_X x \rightarrow b, \quad \beta : b \rightarrow c,$$

and multiplication is

$$\mu_X(x, \alpha, \beta) = (x, \beta \circ \alpha).$$

The underlying comparison chain is

$$q_X x \xrightarrow{\alpha} b \xrightarrow{\beta} c. \\ \searrow \quad \quad \nearrow \\ \beta \circ \alpha$$

More generally, for any object S , base extension gives a monad

$$T_{\mathbb{B}}^S := (1_S \times t_B)!(1_S \times s_B)^*$$

on

$$\mathcal{E}/(S \times B_0).$$

Proposition 4.2.1 (Kleisli normal form). *Let*

$$q_X : X \rightarrow B_0, \quad q_Y : Y \rightarrow B_0$$

be objects of $\mathcal{E}/B_0$. A Kleisli map

$$X \rightsquigarrow Y$$

for $T_{\mathbb{B}}$ is equivalently a pair

$$F : X \rightarrow Y, \quad \alpha : X \rightarrow B_1$$

satisfying

$$s_B \alpha = q_Y F, \quad t_B \alpha = q_X.$$

Thus

$$\alpha_x : q_Y(Fx) \rightarrow q_X(x).$$

If

$$(F, \alpha) : X \rightsquigarrow Y, \quad (G, \gamma) : Y \rightsquigarrow Z,$$

then their Kleisli composite is

$$(G, \gamma) \odot (F, \alpha) = (GF, \alpha \circ_F \gamma),$$

where

$$(\alpha \circ_F \gamma)_x = \alpha_x \circ \gamma_{Fx}.$$

Proof.

1. **Unpack a Kleisli map.** A Kleisli map is a morphism

$$X \rightarrow T_{\mathbb{B}} Y = Y \times_{q_Y, B_0, s_B} B_1$$

in the slice $\mathcal{E}/B_0$. Hence it is a pair

$$(F, \alpha)$$

fitting into

$$\begin{array}{ccc} X & \xrightarrow{\langle F, \alpha \rangle} & Y \times_{B_0} B_1 \\ F \downarrow & & \downarrow \pi_Y \\ Y & \xlongequal{\quad} & Y. \end{array}$$

The pullback condition gives

$$s_B \alpha = q_Y F,$$

while the slice condition gives

$$t_B \alpha = q_X.$$

2. **Compute the comparison arrow.** For each generalized element x , the typing equations give

$$q_Y(Fx) \xrightarrow{\alpha_x} q_X(x).$$

3. **Compute Kleisli composition.** Given

$$(F, \alpha) : X \rightsquigarrow Y, \quad (G, \gamma) : Y \rightsquigarrow Z,$$

the comparison chain is

$$q_Z(GFx) \xrightarrow{\gamma_{Fx}} q_Y(Fx) \xrightarrow{\alpha_x} q_X(x).$$

Applying the multiplication of $T_{\mathbb{B}}$ composes these arrows, so the resulting comparison is

$$\alpha_x \circ \gamma_{Fx}.$$

The underlying state map is GF , proving the formula.

□

4.2.3 The internal Kleisli endomorphism category

Let

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

be a carrier. Put

$$A_0^{[2]} := A_0 \times A_0$$

with projections

$$\ell, r : A_0^{[2]} \rightarrow A_0.$$

A point

$$(v, u) \in A_0^{[2]}$$

is read as the source and target of an operation

$$v \rightarrow u.$$

Define the source-state and target-state families

$$P_{\text{src}} := A_0^{[2]} \times_{\ell, A_0, p} P$$

and

$$P_{\text{tgt}} := A_0^{[2]} \times_{r, A_0, P} P.$$

Write

$$\pi_{\text{src}} : P_{\text{src}} \rightarrow A_0^{[2]}, \quad \bar{\pi}_{\text{src}} : P_{\text{src}} \rightarrow P$$

and

$$\pi_{\text{tgt}} : P_{\text{tgt}} \rightarrow A_0^{[2]}, \quad \bar{\pi}_{\text{tgt}} : P_{\text{tgt}} \rightarrow P$$

for the pullback projections. The defining squares are

$$\begin{array}{ccc} P_{\text{src}} & \xrightarrow{\bar{\pi}_{\text{src}}} & P \\ \pi_{\text{src}} \downarrow & \lrcorner & \downarrow p \\ A_0^{[2]} & \xrightarrow{\ell} & A_0 \end{array} \quad \begin{array}{ccc} P_{\text{tgt}} & \xrightarrow{\bar{\pi}_{\text{tgt}}} & P \\ \pi_{\text{tgt}} \downarrow & \lrcorner & \downarrow p \\ A_0^{[2]} & \xrightarrow{r} & A_0. \end{array}$$

Over (v, u) , the two families are respectively P_v and P_u .

Regard P_{src} and P_{tgt} as objects of

$$\mathcal{E}/(A_0^{[2]} \times B_0)$$

by the structure maps

$$\hat{q}_{\text{src}} := \langle \pi_{\text{src}}, q\bar{\pi}_{\text{src}} \rangle : P_{\text{src}} \rightarrow A_0^{[2]} \times B_0$$

and

$$\hat{q}_{\text{tgt}} := \langle \pi_{\text{tgt}}, q\bar{\pi}_{\text{tgt}} \rangle : P_{\text{tgt}} \rightarrow A_0^{[2]} \times B_0.$$

Apply the base-extended output-comparison monad

$$T_{\mathbb{B}}^{[2]} := (1_{A_0^{[2]}} \times t_B)!(1_{A_0^{[2]}} \times s_B)^*$$

on

$$\mathcal{E}/(A_0^{[2]} \times B_0)$$

to P_{tgt} . Its underlying object is

$$T_{\mathbb{B}}^{[2]}(P_{\text{tgt}}) = P_{\text{tgt}} \times_{q\bar{\pi}_{\text{tgt}}, B_0, s_B} B_1.$$

Write

$$\pi_{P_{\text{tgt}}} : T_{\mathbb{B}}^{[2]}(P_{\text{tgt}}) \rightarrow P_{\text{tgt}}$$

and

$$\pi_{B_1} : T_{\mathbb{B}}^{[2]}(P_{\text{tgt}}) \rightarrow B_1$$

for its projections. Its structure map to $A_0^{[2]} \times B_0$ is

$$\hat{q}_T := \langle \pi_{\text{tgt}} \pi_{P_{\text{tgt}}}, t_B \pi_{B_1} \rangle.$$

A generalized element of $T_{\mathbb{B}}^{[2]}(P_{\text{tgt}})$ is written

$$(v, u, y, \alpha)$$

with

$$y \in P_u, \quad \alpha : q(y) \rightarrow b,$$

and it lies over

$$((v, u), b) \in A_0^{[2]} \times B_0.$$

Define

$$E_{[2]} := P_{\text{src}} \times_{\widehat{q}_{\text{src}}, A_0^{[2]} \times B_0, \widehat{q}_T} T_{\mathbb{B}}^{[2]}(P_{\text{tgt}}).$$

Write

$$\varepsilon : E_{[2]} \rightarrow P_{\text{src}}$$

for the first pullback projection. The defining square is

$$\begin{array}{ccc} E_{[2]} & \xrightarrow{\quad} & T_{\mathbb{B}}^{[2]}(P_{\text{tgt}}) \\ \varepsilon \downarrow & \lrcorner & \downarrow \widehat{q}_T \\ P_{\text{src}} & \xrightarrow[\widehat{q}_{\text{src}}]{} & A_0^{[2]} \times B_0. \end{array}$$

A generalized element of $E_{[2]}$ over

$$(v, u, x) \in P_{\text{src}}, \quad x \in P_v,$$

is a pair

$$y \in P_u, \quad \alpha : q(y) \rightarrow q(x).$$

Since $\mathcal{E}$ is locally cartesian closed, the pullback functor

$$\pi_{\text{src}}^* : \mathcal{E}/A_0^{[2]} \rightarrow \mathcal{E}/P_{\text{src}}$$

has a right adjoint

$$\Pi_{\pi_{\text{src}}}.$$

Define

$$\text{KlEnd}_{\mathbb{B}}(P)_1 := \Pi_{\pi_{\text{src}}}(\varepsilon : E_{[2]} \rightarrow P_{\text{src}}).$$

This is an object over $A_0^{[2]}$.

The counit of the dependent-product adjunction gives an evaluation map

$$\text{ev} : \text{KlEnd}_{\mathbb{B}}(P)_1 \times_{A_0^{[2]}} P_{\text{src}} \rightarrow E_{[2]}$$

over P_{src} , displayed by

$$\begin{array}{ccc} \text{KlEnd}_{\mathbb{B}}(P)_1 \times_{A_0^{[2]}} P_{\text{src}} & \xrightarrow{\text{ev}} & E_{[2]} \\ \downarrow & & \downarrow \varepsilon \\ P_{\text{src}} & \xlongequal{\quad\quad\quad} & P_{\text{src}}. \end{array}$$

A generalized element

$$\varphi \in \text{KlEnd}_{\mathbb{B}}(P)_1$$

over (v, u) therefore represents a family

$$x \in P_v \mapsto (\varphi_0(x), \alpha_x^\varphi),$$

where

$$\varphi_0(x) \in P_u$$

and

$$\alpha_x^\varphi : q(\varphi_0(x)) \rightarrow q(x).$$

Equivalently, φ represents a Kleisli map

$$P_v \rightsquigarrow P_u.$$

Proposition 4.2.2 (Internal Kleisli endomorphism category). *Let*

$$\pi_{\text{KlEnd}} : \text{KlEnd}_{\mathbb{B}}(P)_1 \rightarrow A_0 \times A_0$$

denote the structure map of the dependent product

$$\text{KlEnd}_{\mathbb{B}}(P)_1 = \Pi_{\pi_{\text{src}}}(\varepsilon : E_{[2]} \rightarrow P_{\text{src}}).$$

Then the object object A_0 and arrow object

$$\text{KlEnd}_{\mathbb{B}}(P)_1$$

form an internal category

$$\text{KlEnd}_{\mathbb{B}}(P).$$

Its source and target maps are

$$s_{\text{KlEnd}} := \ell\pi_{\text{KlEnd}}, \quad t_{\text{KlEnd}} := r\pi_{\text{KlEnd}}.$$

Identities are induced by the unit of $T_{\mathbb{B}}$, and composition is induced by parameterized Kleisli composition.

Proof.

1. **Define source and target.** By construction,

$$\text{KlEnd}_{\mathbb{B}}(P)_1 = \Pi_{\pi_{\text{src}}}(E_{[2]} \rightarrow P_{\text{src}})$$

is an object over

$$A_0 \times A_0$$

with structure map

$$\pi_{\text{KlEnd}} : \text{KlEnd}_{\mathbb{B}}(P)_1 \rightarrow A_0 \times A_0.$$

Define

$$s_{\text{KlEnd}} := \ell\pi_{\text{KlEnd}}, \quad t_{\text{KlEnd}} := r\pi_{\text{KlEnd}}.$$

Thus an operation lying over (v, u) is regarded as an internal arrow

$$v \rightarrow u$$

and represents a Kleisli map

$$P_v \rightsquigarrow P_u.$$

2. **Construct identities by transposition.** Regard A_0 as an object over $A_0 \times A_0$ through the diagonal

$$\Delta : A_0 \rightarrow A_0 \times A_0.$$

Over $v \in A_0$, consider the parameterized Kleisli identity

$$x \mapsto (x, e_B(qx)), \quad x \in P_v.$$

This defines a map from the pullback of P_{src} along Δ into the corresponding pullback of $E_{[2]}$, over the pulled-back source-state family.

Transposition through

$$\pi_{\text{src}}^* \dashv \Pi_{\pi_{\text{src}}}$$

therefore defines

$$e_{\text{KlEnd}} : A_0 \rightarrow \text{KlEnd}_{\mathbb{B}}(P)_1$$

over the diagonal:

$$\begin{array}{ccc} A_0 & \xrightarrow{e_{\text{KlEnd}}} & \text{KlEnd}_{\mathbb{B}}(P)_1 \\ & \searrow \Delta & \downarrow \pi_{\text{KlEnd}} \\ & & A_0 \times A_0. \end{array}$$

Consequently,

$$s_{\text{KlEnd}} e_{\text{KlEnd}} = 1_{A_0} = t_{\text{KlEnd}} e_{\text{KlEnd}}.$$

3. Form the object of composable operations. Put

$$\text{KlEnd}_{\mathbb{B}}(P)_2 := \text{KlEnd}_{\mathbb{B}}(P)_1 \times_{t_{\text{KlEnd}}, A_0, s_{\text{KlEnd}}} \text{KlEnd}_{\mathbb{B}}(P)_1.$$

A generalized element consists of operations

$$\varphi : P_v \rightsquigarrow P_u, \quad \psi : P_u \rightsquigarrow P_w.$$

Their composite must be an operation

$$P_v \rightsquigarrow P_w.$$

The composability square is

$$\begin{array}{ccc} \text{KlEnd}_{\mathbb{B}}(P)_2 & \xrightarrow{\pi_2} & \text{KlEnd}_{\mathbb{B}}(P)_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow s_{\text{KlEnd}} \\ \text{KlEnd}_{\mathbb{B}}(P)_1 & \xrightarrow{t_{\text{KlEnd}}} & A_0. \end{array}$$

4. Evaluate the two composable operations. Pulling back the universal evaluation map along the first projection from $\text{KlEnd}_{\mathbb{B}}(P)_2$ evaluates φ on a source state $x \in P_v$:

$$x \mapsto (\varphi_0(x), \alpha_x^\varphi),$$

where

$$\varphi_0(x) \in P_u$$

and

$$\alpha_x^\varphi : q(\varphi_0 x) \rightarrow q(x).$$

Pulling back the universal evaluation map along the second projection and substituting the state $\varphi_0(x)$ evaluates ψ :

$$\varphi_0(x) \mapsto (\psi_0(\varphi_0x), \alpha_{\varphi_0x}^\psi),$$

where

$$\alpha_{\varphi_0x}^\psi : q(\psi_0(\varphi_0x)) \rightarrow q(\varphi_0x).$$

Thus the two evaluations produce the composable comparison chain

$$q(\psi_0(\varphi_0x)) \xrightarrow{\alpha_{\varphi_0x}^\psi} q(\varphi_0x) \xrightarrow{\alpha_x^\varphi} q(x).$$

5. **Apply Kleisli multiplication.** Applying the multiplication of the output-comparison monad $T_{\mathbb{B}}$ composes the two comparison arrows. The resulting evaluated operation is

$$x \mapsto (\psi_0(\varphi_0x), \alpha_x^\varphi \circ \alpha_{\varphi_0x}^\psi).$$

Equivalently,

$$(\psi \odot \varphi)(x) = (\psi_0(\varphi_0x), \alpha_x^\varphi \circ \alpha_{\varphi_0x}^\psi).$$

This construction is natural in the parameter object $\mathbf{KlEnd}_{\mathbb{B}}(P)_2$ and defines a map from the outer source-state family to the corresponding outer operation-data object.

6. **Transpose the evaluated composite.** Transposition through the dependent-product adjunction for the outer source-state projection yields a unique map

$$m_{\mathbf{KlEnd}} : \mathbf{KlEnd}_{\mathbb{B}}(P)_2 \rightarrow \mathbf{KlEnd}_{\mathbb{B}}(P)_1$$

whose base map sends the outer endpoints

$$(v, u, w) \mapsto (v, w).$$

By construction, its evaluation is precisely parameterized Kleisli composition. In particular,

$$s_{\mathbf{KlEnd}} m_{\mathbf{KlEnd}} = s_{\mathbf{KlEnd}} \pi_1$$

and

$$t_{\mathbf{KlEnd}} m_{\mathbf{KlEnd}} = t_{\mathbf{KlEnd}} \pi_2.$$

7. **Verify associativity after evaluation.** Consider the two maps from the object of composable triples to $\mathbf{KlEnd}_{\mathbb{B}}(P)_1$ obtained from the two possible parenthesizations of $m_{\mathbf{KlEnd}}$.

Under dependent-product transposition, their evaluations are the two parenthesizations of the Kleisli composite of three operations. These evaluated maps are equal by associativity of Kleisli composition, equivalently by associativity of the multiplication of $T_{\mathbb{B}}$.

Dependent-product transposition is a bijection of hom-sets. Therefore equality of the evaluated transposes implies equality of the original maps. Hence $m_{\mathbf{KlEnd}}$ is associative.

8. **Verify the unit laws after evaluation.** The evaluations of

$$m_{\mathbf{KlEnd}}(e_{\mathbf{KlEnd}}, 1) \quad \text{and} \quad m_{\mathbf{KlEnd}}(1, e_{\mathbf{KlEnd}})$$

are respectively left and right Kleisli composition with the Kleisli identity

$$x \longmapsto (x, e_B(qx)).$$

They therefore agree with the universal evaluation map by the two unit laws of $T_{\mathbb{B}}$.

Bijectivity of dependent-product transposition again implies the corresponding equalities before evaluation. Thus the left and right internal-category unit laws hold.

Hence

$$\mathbf{KlEnd}_{\mathbb{B}}(P)$$

is an internal category.

□

Proposition 4.2.3 (Internal representation property). *Let*

$$u : U \rightarrow A_0 \times A_0$$

be an object of

$$\mathcal{E}/(A_0 \times A_0).$$

Put

$$P_{\text{src},U} := U \times_{A_0 \times A_0} P_{\text{src}},$$

and write

$$\pi_U : P_{\text{src},U} \rightarrow P_{\text{src}}$$

and

$$\pi_{\text{src},U} : P_{\text{src},U} \rightarrow U$$

for the pullback projections.

Define the operation-data object over $P_{\text{src},U}$ by

$$E_{[2],U} := P_{\text{src},U} \times_{P_{\text{src}}} E_{[2]},$$

where the pullback is taken along

$$\pi_U : P_{\text{src},U} \rightarrow P_{\text{src}}$$

and

$$\varepsilon : E_{[2]} \rightarrow P_{\text{src}}.$$

Thus there is a pullback square

$$\begin{array}{ccc} E_{[2],U} & \xrightarrow{\quad} & E_{[2]} \\ \varepsilon_U \downarrow & \lrcorner & \downarrow \varepsilon \\ P_{\text{src},U} & \xrightarrow{\pi_U} & P_{\text{src}}. \end{array}$$

Equivalently, there is a canonical isomorphism

$$E_{[2],U} \cong U \times_{A_0 \times A_0} E_{[2]}.$$

Maps

$$\varphi : U \rightarrow \mathbf{KlEnd}_{\mathbb{B}}(P)_1$$

over $A_0 \times A_0$ correspond naturally to sections

$$\tilde{\varphi} : P_{\text{src},U} \rightarrow E_{[2],U}$$

of ε_U , equivalently to maps

$$P_{\text{src},U} \rightarrow E_{[2]}$$

over P_{src} .

Proof. By definition,

$$\text{KlEnd}_{\mathbb{B}}(P)_1 = \Pi_{\pi_{\text{src}}}(\varepsilon : E_{[2]} \rightarrow P_{\text{src}}).$$

The dependent-product adjunction

$$\pi_{\text{src}}^* \dashv \Pi_{\pi_{\text{src}}}$$

gives a natural bijection

$$\begin{aligned} \text{Hom}_{\mathcal{E}/(A_0 \times A_0)}(U, \Pi_{\pi_{\text{src}}} E_{[2]}) \\ \cong \text{Hom}_{\mathcal{E}/P_{\text{src}}}(P_{\text{src}} \times_{A_0 \times A_0} U, E_{[2]}). \end{aligned}$$

The pullback

$$P_{\text{src}} \times_{A_0 \times A_0} U$$

is canonically $P_{\text{src},U}$. Hence maps

$$U \rightarrow \text{KlEnd}_{\mathbb{B}}(P)_1$$

over $A_0 \times A_0$ correspond naturally to maps

$$P_{\text{src},U} \rightarrow E_{[2]}$$

over P_{src} .

By the universal property of the pullback defining $E_{[2],U}$, such a map is equivalently a section

$$\tilde{\varphi} : P_{\text{src},U} \rightarrow E_{[2],U}$$

satisfying

$$\varepsilon_U \tilde{\varphi} = 1_{P_{\text{src},U}}.$$

Naturality follows from the naturality of the dependent-product adjunction and of pullback. $\square$

Lemma 4.2.1 (Epimorphic extensionality). *Let*

$$\varphi, \psi : U \rightarrow \text{KlEnd}_{\mathbb{B}}(P)_1$$

be maps over $A_0 \times A_0$, corresponding to evaluated maps

$$\tilde{\varphi}, \tilde{\psi} : P_{\text{src},U} \rightrightarrows E_{[2],U}.$$

Let

$$e : E \rightarrow P_{\text{src},U}$$

be an epimorphism. If

$$\tilde{\varphi}e = \tilde{\psi}e,$$

then

$$\varphi = \psi.$$

Proof.

1. **Cancel the epimorphism.** Since e is epic,

$$\tilde{\varphi}e = \tilde{\psi}e \implies \tilde{\varphi} = \tilde{\psi}.$$

This cancellation is represented by

$$\begin{array}{ccc} E & \xrightarrow{e} & P_{\text{src}, U} \\ & \searrow \tilde{\varphi}e = \tilde{\psi}e & \downarrow \tilde{\psi} \downarrow \tilde{\varphi} \\ & & E_{[2], U}. \end{array}$$

2. **Transpose the equality.** The dependent-product correspondence of Proposition 4.2.3 is bijective. Transposing

$$\tilde{\varphi} = \tilde{\psi}$$

therefore gives

$$\varphi = \psi.$$

□

4.2.4 Mealy actions as internal representations

Theorem 4.2.2 (Transition-output representation theorem). *Let*

$$A_0 \xleftarrow{p} P \xrightarrow{q} B_0$$

be a carrier. To give a Mealy structure on P is equivalent to giving an identity-on-objects internal functor

$$\chi_P : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P).$$

Under this correspondence, an arrow

$$a : u \rightarrow v$$

of $\mathbb{A}$, regarded as

$$a^{\text{op}} : v \rightarrow u,$$

is sent to the Kleisli map

$$P_v \rightsquigarrow P_u, \quad x \mapsto (\delta(a, x), \omega(a, x)).$$

Proof.

1. **Transpose the Mealy action.** Given a Mealy structure, the typing equations make

$$x \mapsto (\delta(a, x), \omega(a, x))$$

a Kleisli map

$$P_v \rightsquigarrow P_u$$

for every arrow

$$a : u \rightarrow v.$$

By the internal representation property, these operations transpose to an arrow map

$$\chi_{P,1} : A_1 \rightarrow \text{KlEnd}_{\mathbb{B}}(P)_1$$

over

$$\begin{array}{ccc} (t_A, s_A) : A_1 & \rightarrow & A_0 \times A_0 : \\ A_1 & \xrightarrow{\chi_{P,1}} & \text{KlEnd}_{\mathbb{B}}(P)_1 \\ & \searrow (t_A, s_A) & \downarrow (s_{\text{KlEnd}}, t_{\text{KlEnd}}) \\ & & A_0 \times A_0. \end{array}$$

Together with the identity object map 1_{A_0} , this is the underlying internal graph morphism

$$\mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P).$$

2. **Verify identities.** The Mealy unit equations

$$\delta(e_A v, x) = x, \quad \omega(e_A v, x) = e_B(qx)$$

say exactly that

$$\chi_P(e_A v)$$

is the Kleisli identity on P_v . Hence χ_P preserves identities.

3. **Fix the order of composable arrows.** Let

$$a : u \rightarrow v, \quad b : v \rightarrow w$$

be composable arrows of $\mathbb{A}$. In $\mathbb{A}^{\text{op}}$ they become

$$b^{\text{op}} : w \rightarrow v, \quad a^{\text{op}} : v \rightarrow u.$$

Thus the corresponding composable pair in the arrow object of $\mathbb{A}^{\text{op}}$ is represented, on the underlying arrow object A_1 , by the ordered pair

$$(b, a) \in A_1 \times_{s_A, A_0, t_A} A_1.$$

Its composite is

$$a^{\text{op}} \circ b^{\text{op}} = (b \circ a)^{\text{op}},$$

so the underlying arrow of $\mathbb{A}$ is $b \circ a$.

4. **Verify composition.** Functoriality requires

$$\chi_P(b \circ a) = \chi_P(a) \odot \chi_P(b).$$

With the ordered pair (b, a) made explicit, the functoriality square is

$$\begin{array}{ccc} A_1 \times_{s_A, A_0, t_A} A_1 & \xrightarrow{m_{\mathbb{A}^{\text{op}}}} & A_1 \\ \langle \chi_{P,1} \pi_1, \chi_{P,1} \pi_2 \rangle \downarrow & & \downarrow \chi_{P,1} \\ \text{KlEnd}_{\mathbb{B}}(P)_2 & \xrightarrow{m_{\text{KlEnd}}} & \text{KlEnd}_{\mathbb{B}}(P)_1. \end{array}$$

Here

$$m_{\mathbb{A}^{\text{op}}}(b, a) = b \circ a$$

on the underlying arrow object, while

$$m_{\text{KlEnd}}(\chi_P(b), \chi_P(a)) = \chi_P(a) \odot \chi_P(b).$$

Evaluation at $x \in P_w$ first applies $\chi_P(b)$:

$$x \mapsto (\delta(b, x), \omega(b, x)),$$

and then applies $\chi_P(a)$:

$$\delta(b, x) \mapsto (\delta(a, \delta(b, x)), \omega(a, \delta(b, x))).$$

Kleisli composition therefore gives

$$x \mapsto (\delta(a, \delta(b, x)), \omega(b, x) \circ \omega(a, \delta(b, x))).$$

Equality with the operation associated to $b \circ a$ is exactly the pair of Mealy multiplication equations

$$\delta(b \circ a, x) = \delta(a, \delta(b, x))$$

and

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

Hence χ_P preserves composition.

5. **Recover the Mealy maps.** Conversely, let

$$\chi_P : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P)$$

be an identity-on-objects internal functor. Evaluating its arrow map on the source-state family yields maps

$$\delta : A_1 \times_{t_A, A_0, P} P \rightarrow P, \quad \omega : A_1 \times_{t_A, A_0, P} P \rightarrow B_1.$$

The fact that the evaluated operation associated to $a : u \rightarrow v$ is a Kleisli map

$$P_v \rightsquigarrow P_u$$

gives the typing equations

$$p\delta = s_A\pi_{A_1},$$

$$s_B\omega = q\delta,$$

and

$$t_B\omega = q\pi_P.$$

Preservation of identities gives

$$\delta(e_A v, x) = x, \quad \omega(e_A v, x) = e_B(qx).$$

Preservation of composition, evaluated on the ordered composable pair (b, a) , gives

$$\delta(b \circ a, x) = \delta(a, \delta(b, x))$$

and

$$\omega(b \circ a, x) = \omega(b, x) \circ \omega(a, \delta(b, x)).$$

Thus (δ, ω) is a Mealy structure.

6. **Verify mutual inverse correspondence.** Starting from a Mealy structure, transposing its evaluated operations and then evaluating the transpose recovers the original transition and output maps by the counit of the dependent-product adjunction.

Conversely, starting from the internal functor χ_P , evaluating and then transposing recovers its original arrow map by the unit of the same adjunction. Hence the two constructions are mutually inverse.

□

Lemma 4.2.2 (Restriction of representations). *Let*

$$i : V \hookrightarrow W$$

be a strict monomorphism of Mealy morphisms over $(\mathbb{A}, \mathbb{B})$. Then the representation χ_V is the restriction of χ_W to the state fibres of V .

Proof.

1. **Restrict transitions.** Strictness gives

$$i\delta_V = \delta_W(1_{A_1} \times i),$$

represented by

$$\begin{array}{ccc} A_1 \times_{A_0} V & \xrightarrow{1 \times i} & A_1 \times_{A_0} W \\ \delta_V \downarrow & & \downarrow \delta_W \\ V & \xrightarrow{i} & W. \end{array}$$

2. **Restrict output comparisons.** Strictness also gives

$$\omega_V = \omega_W(1_{A_1} \times i).$$

Hence every input arrow acts on V by restricting its action on W , which is precisely the asserted restriction of representations.

□

4.3 Operational equations and behavioural satisfaction

4.3.1 Parallel pairs and operation congruences

Define the object of parallel arrows of $\mathbb{A}^{\text{op}}$ by

$$\text{Par}(\mathbb{A}) := A_1 \times_{A_0 \times A_0} A_1,$$

where each copy of A_1 is regarded over $A_0 \times A_0$ by

$$(t_A, s_A).$$

The defining pullback is

$$\begin{array}{ccc} \text{Par}(\mathbb{A}) & \xrightarrow{\pi_2} & A_1 \\ \pi_1 \downarrow & \lrcorner & \downarrow (t_A, s_A) \\ A_1 & \xrightarrow{(t_A, s_A)} & A_0 \times A_0. \end{array}$$

Definition 4.3.1 (Identity-on-objects operation congruence). An *identity-on-objects operation congruence* on $\mathbb{A}^{\text{op}}$ is an internal equivalence relation

$$\rho_0, \rho_1 : R \rightrightarrows A_1$$

in

$$\mathcal{E}/(A_0 \times A_0),$$

where A_1 is regarded over $A_0 \times A_0$ by

$$(t_A, s_A) : A_1 \rightarrow A_0 \times A_0,$$

and which is compatible with composition in $\mathbb{A}^{\text{op}}$.

We write

$$\text{Cong}_{\text{iio}}(\mathbb{A}^{\text{op}})$$

for the poset of operation congruences, ordered by inclusion.

Because ρ_0 and ρ_1 are morphisms over $A_0 \times A_0$, the relation R relates only parallel arrows.

No separate compatibility condition involving identity arrows is required. Indeed, reflexivity of the equivalence relation already gives

$$e_{\mathbb{A}^{\text{op}}}(v) R e_{\mathbb{A}^{\text{op}}}(v)$$

for every $v \in A_0$.

Compatibility with composition may be stated as follows. Let

$$R_2 := R \times_{t_{\mathbb{A}^{\text{op}}} \rho_0, A_0, s_{\mathbb{A}^{\text{op}}} \rho_0} R.$$

Since R relates parallel arrows, the same pullback is obtained by using ρ_1 in place of ρ_0 .

The two componentwise composition maps

$$R_2 \rightrightarrows A_1,$$

given internally by

$$(x R x', y R y') \longmapsto y \circ_{\mathbb{A}^{\text{op}}} x$$

and

$$(x R x', y R y') \longmapsto y' \circ_{\mathbb{A}^{\text{op}}} x',$$

must together factor through

$$R \hookrightarrow \text{Par}(\mathbb{A}).$$

Equivalently, whenever

$$x R x', \quad y R y'$$

and the corresponding pairs are composable, one has

$$y \circ_{\mathbb{A}^{\text{op}}} x R y' \circ_{\mathbb{A}^{\text{op}}} x'.$$

The poset

$$\text{Cong}_{\text{iio}}(\mathbb{A}^{\text{op}})$$

is a complete lattice. Meets are intersections of congruences. The join of a family is the intersection of all operation congruences containing the union of the given relations.

Since $\mathcal{E}/(A_0 \times A_0)$ is exact, every operation congruence is effective as a relation on A_1 . In Lemma 4.4.1, compatibility with identities and composition will equip the effective arrow quotient with an internal category structure.

4.3.2 Satisfaction of operational equations

Let

$$H \in \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

be a residual behaviour over the input object v . Let

$$a, a' : u \rightarrow v$$

be parallel arrows of $\mathbb{A}$.

Definition 4.3.2 (Transition-output satisfaction). *We write*

$$H \models_{\text{to}} a \equiv a'$$

when

$$\partial_a H = \partial_{a'} H$$

and

$$\omega_{\mathbf{Beh}}(a, H) = \omega_{\mathbf{Beh}}(a', H).$$

The second equality is taken in the common hom-object determined by the first equality of derivatives.

Equivalently, the two evaluated maps into the appropriate pullback of the operation-data object $E_{[2]}$ are equal. In this formulation, the transition and output components are compared simultaneously and all typing is automatic.

The satisfaction condition compares the two Kleisli operations

$$\{H\} \begin{array}{c} \xrightarrow{(\partial_a H, \omega_{\mathbf{Beh}}(a, H))} \\ \xrightarrow{(\partial_{a'} H, \omega_{\mathbf{Beh}}(a', H))} \end{array} \mathbf{Beh}_u.$$

Thus an operational equation asserts equality both of the transformed behaviour and of the output comparison produced at the current residual state.

For a local coequational theory

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}},$$

define a relation R_V internally by

$$a R_V a' \iff \forall H \in V_v, H \models_{\text{to}} a \equiv a'.$$

Theorem 4.3.1 (Valid equations form the kernel congruence). *For every local coequational theory V ,*

$$R_V = \ker(\chi_{V,1})$$

in

$$\mathcal{E}/(A_0 \times A_0).$$

In particular, R_V is an identity-on-objects operation congruence on $\mathbb{A}^{\text{op}}$.

Proof.

1. **Translate equality through evaluation.** By Proposition 4.2.3, equality

$$\chi_V(a) = \chi_V(a')$$

is equivalent to equality of the evaluated maps on the source-state family. The comparison is

$$V_v \begin{array}{c} \xrightarrow{\chi_V(a)} \\ \xrightarrow{\chi_V(a')} \end{array} V_u.$$

2. **Separate state and output components.** Equality of the evaluated Kleisli maps is exactly

$$\partial_a H = \partial_{a'} H$$

and

$$\omega_{\mathbf{Beh}}(a, H) = \omega_{\mathbf{Beh}}(a', H)$$

for every $H \in V_v$. Hence the kernel pair of $\chi_{V,1}$ is precisely R_V .

3. **Use functoriality.** Since χ_V is an internal functor, its kernel relation is compatible with identities and composition.

4. **Use exactness.** The slice $\mathcal{E}/(A_0 \times A_0)$ is exact, so the kernel relation is an effective equivalence relation. It is therefore an identity-on-objects operation congruence.

□

4.3.3 The one-step equation predicate

Let

$$R \in \mathbf{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}})$$

with projections

$$\rho_0, \rho_1 : R \rightrightarrows A_1.$$

Because these projections are morphisms over $A_0 \times A_0$, one has

$$(t_A, s_A)\rho_0 = (t_A, s_A)\rho_1.$$

Write

$$\tau_R : R \rightarrow A_0 \times A_0$$

for this common map:

$$\tau_R := (t_A, s_A)\rho_0 = (t_A, s_A)\rho_1.$$

Write also

$$\sigma_R := t_A \rho_0 = t_A \rho_1 : R \rightarrow A_0.$$

This is the common source of the represented arrows in $\mathbb{A}^{\text{op}}$.

Let

$$p_{\mathbf{Beh}} : \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \rightarrow A_0$$

be the input anchor of the canonical behaviour carrier. Form

$$X_R := R \times_{\sigma_R, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}.$$

Write

$$\pi_R^R : X_R \rightarrow R, \quad \pi_R^{\mathbf{Beh}} : X_R \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

for the pullback projections. Thus a generalized element of X_R is a triple

$$(a, a', H)$$

where

$$a R a'$$

and H lies over the common source of a^{op} and $(a')^{\text{op}}$.

Apply the construction of Subsection 4.2.3 to the canonical behaviour carrier. Write

$$\mathbf{Beh}_{\text{src}} := (A_0 \times A_0) \times_{\ell, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

and write

$$\varepsilon_{\mathbf{Beh}} : E_{[2]}^{\mathbf{Beh}} \rightarrow \mathbf{Beh}_{\text{src}}$$

for the corresponding operation-data projection.

There is a canonical map

$$\xi_R : X_R \rightarrow \mathbf{Beh}_{\text{src}}$$

defined by

$$\xi_R := \langle \tau_R \pi_R^R, \pi_R^{\mathbf{Beh}} \rangle.$$

Internally,

$$\xi_R(a, a', H) = ((t_A a, s_A a), H).$$

The pullback condition defining X_R ensures that this map is well typed, since

$$p_{\mathbf{Beh}}(H) = t_A a.$$

Define the operation-data object over X_R by

$$\mathcal{O}_R := X_R \times_{\xi_R, \mathbf{Beh}_{\text{src}}, \varepsilon_{\mathbf{Beh}}} E_{[2]}^{\mathbf{Beh}}.$$

The defining pullback is

$$\begin{array}{ccc} \mathcal{O}_R & \xrightarrow{\quad} & E_{[2]}^{\mathbf{Beh}} \\ \downarrow & \lrcorner & \downarrow \varepsilon_{\mathbf{Beh}} \\ X_R & \xrightarrow{\xi_R} & \mathbf{Beh}_{\text{src}}. \end{array}$$

A generalized element of $\mathcal{O}_R$ over

$$(a, a', H) \in X_R$$

is a pair

$$(K, \alpha)$$

where K lies over the common target

$$s_A a = s_A a'$$

and

$$\alpha : q_{\mathbf{Beh}}(K) \rightarrow q_{\mathbf{Beh}}(H).$$

Evaluation of the canonical representation along

$$\rho_0 \pi_R^R \quad \text{and} \quad \rho_1 \pi_R^R$$

gives two sections

$$\text{act}_0, \text{act}_1 : X_R \rightrightarrows \mathcal{O}_R.$$

Internally, these sections are

$$\text{act}_0(a, a', H) = (\partial_a H, \omega_{\mathbf{Beh}}(a, H))$$

and

$$\text{act}_1(a, a', H) = (\partial_{a'} H, \omega_{\mathbf{Beh}}(a', H)).$$

Let

$$e_R : E_R \hookrightarrow X_R$$

be their equalizer:

$$E_R \xleftarrow{e_R} X_R \xrightleftharpoons[\text{act}_1]{\text{act}_0} \mathcal{O}_R.$$

Regard X_R as an object of $\mathcal{E}/Z$ through

$$X_R \xrightarrow{\pi_R^{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \rightarrow Z.$$

Then

$$\pi_R^{\mathbf{Beh}} : X_R \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

is a morphism in $\mathcal{E}/Z$. Since $\mathcal{E}/Z$ is a topos, the pullback map on subobjects along $\pi_R^{\mathbf{Beh}}$ has a right adjoint

$$\forall_{\pi_R^{\mathbf{Beh}}}.$$

Definition 4.3.3 (One-step equation predicate). *The **one-step equation predicate***

$$\text{Eq}(R) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

is

$$\text{Eq}(R) := \forall_{\pi_R^{\mathbf{Beh}}} (E_R)$$

in the subobject lattice of $\mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$ in $\mathcal{E}/Z$.

Equivalently, in the internal language,

$$H \in \text{Eq}(R)$$

if and only if, for every pair

$$a, a' : u \rightarrow p_{\mathbf{Beh}}(H)$$

with

$$a R a',$$

one has

$$\partial_a H = \partial_{a'} H$$

and

$$\omega_{\mathbf{Beh}}(a, H) = \omega_{\mathbf{Beh}}(a', H).$$

Thus $H \in \text{Eq}(R)$ precisely when every equation in R is valid at the current residual state H .

The subobject

$$\text{Eq}(R)$$

need not be derivative-closed. An equation may hold at H while failing at a residual derivative of H . The invariant theory determined by R is therefore obtained by applying the derivative safety core.

4.3.4 The largest invariant theory satisfying a congruence

Recall from Chapter 3 that

$$\text{Safe}_\partial$$

is right adjoint to the inclusion of derivative-closed subobjects:

$$\text{Loc}_{\mathbb{A}, \mathbb{B}} \begin{array}{c} \xrightarrow{I_\partial} \\ \xleftarrow{\text{Safe}_\partial} \end{array} \text{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_0)$$

with

$$I_\partial \dashv \text{Safe}_\partial.$$

Thus, for every local coequational theory V and every subobject

$$C \hookrightarrow \mathbf{Beh}_0,$$

one has

$$V \leq \text{Safe}_\partial(C) \iff V \leq C.$$

Definition 4.3.4 (Theory of a congruence). *For an operation congruence R , define*

$$V_R := \text{Safe}_\partial(\text{Eq}(R)).$$

Thus V_R is the greatest derivative-closed subobject contained in the one-step model predicate $\text{Eq}(R)$:

$$\begin{array}{ccc} V_R & \hookrightarrow & \text{Eq}(R) \\ & \searrow & \downarrow \\ & & \mathbf{Beh}_0. \end{array}$$

Proposition 4.3.1 (Contextual form of V_R). *Reasoning in the internal language, a residual behaviour H lies in V_R if and only if, for every residual context*

$$r : v \rightarrow p_{\mathbf{Beh}}(H)$$

and every R -related pair

$$a, a' : u \rightarrow v,$$

one has

$$\partial_{r \circ a} H = \partial_{r \circ a'} H$$

and

$$\omega_{\mathbf{Beh}}(a, \partial_r H) = \omega_{\mathbf{Beh}}(a', \partial_r H).$$

The latter arrows are precisely the output comparisons assigned by H to the residual morphisms

$$r \circ a \rightarrow r, \quad r \circ a' \rightarrow r.$$

Proof.

1. **Expand safety-core membership.** Membership

$$H \in V_R$$

means that every derivative

$$\partial_r H$$

lies in $\text{Eq}(R)$.

2. **Apply the one-step equation predicate.** At the derivative $\partial_r H$, the relation $a R a'$ gives

$$\partial_a(\partial_r H) = \partial_{a'}(\partial_r H)$$

and

$$\omega_{\mathbf{Beh}}(a, \partial_r H) = \omega_{\mathbf{Beh}}(a', \partial_r H).$$

3. **Compose derivatives.** The derivative composition law gives

$$\partial_a(\partial_r H) = \partial_{r \circ a} H$$

and similarly for a' . The residual contexts fit into

$$u \xrightarrow[a']{a} v \xrightarrow{r} p_{\mathbf{Beh}}(H).$$

Substituting these identities yields the claimed contextual condition.

□

Theorem 4.3.2 (Operational soundness and completeness). *For every local coequational theory V and every operation congruence R ,*

$$V \leq V_R \iff R \leq R_V.$$

Proof.

1. **Use derivative invariance of V .** Since V is derivative-closed, the safety-core adjunction gives

$$V \leq V_R \iff V \leq \text{Eq}(R).$$

2. **Interpret the predicate inclusion.** The inclusion

$$V \leq \text{Eq}(R)$$

says that every equation in R is valid at every state of V .

3. **Identify the kernel congruence.** By Theorem 4.3.1, the equations valid at every state of V are precisely R_V . Therefore

$$V \leq \text{Eq}(R) \iff R \leq R_V.$$

□

Corollary 4.3.1 (Largest recognized theory). *The local coequational theory V_R is the largest local theory satisfying every operational equation in R .*

Corollary 4.3.2 (Join description). *One has*

$$V_R = \bigvee \{S \in \mathbf{Loc}_{\mathbb{A}, \mathbb{B}} \mid R \leq R_S\}.$$

Proof.

1. **Identify the indexed family.** By Theorem 4.3.2,

$$R \leq R_S \iff S \leq V_R.$$

2. **Take the join.** The displayed family is therefore exactly the family of local subtheories of V_R . Their join in the complete lattice $\mathbf{Loc}_{\mathbb{A}, \mathbb{B}}$ is V_R .

□

4.4 Faithful operational images

4.4.1 Identity-on-objects quotients and regular images

Definition 4.4.1 (Identity-on-objects regular epimorphism). *Let*

$$F : \mathbb{C} \rightarrow \mathbb{D}$$

*be an internal functor whose object map is the identity on a common object object $C_0 = D_0$. We call F an **identity-on-objects regular epimorphism** if its arrow map is a regular epimorphism in*

$$\mathcal{E}/(C_0 \times C_0).$$

Identity-on-objects monomorphisms are defined similarly.

Thus an identity-on-objects regular epimorphism is represented by

$$\begin{array}{ccc} C_1 & \xrightarrow{F_1} & D_1 \\ (s_C, t_C) \downarrow & & \downarrow (s_D, t_D) \\ C_0 \times C_0 & \xlongequal{\quad} & C_0 \times C_0. \end{array}$$

Lemma 4.4.1 (Quotient by an operation congruence). *Let*

$$R \rightrightarrows C_1$$

be an identity-on-objects operation congruence on an internal category $\mathbb{C}$ with object object C_0 . Let

$$q : C_1 \twoheadrightarrow Q_1$$

be the effective quotient of R in

$$\mathcal{E}/(C_0 \times C_0).$$

Then Q_1 , with object object C_0 , carries a unique internal category structure such that

$$q_R : \mathbb{C} \twoheadrightarrow \mathbb{C}/R$$

is an identity-on-objects internal functor.

Proof.

1. **Define source, target, and identities.** Since q is a morphism in $\mathcal{E}/(C_0 \times C_0)$, the structure map

$$Q_1 \rightarrow C_0 \times C_0$$

supplies source and target maps

$$s_Q, t_Q : Q_1 \rightrightarrows C_0.$$

Define

$$e_Q := qe_C.$$

The identity square is

$$\begin{array}{ccc} C_0 & \xrightarrow{e_C} & C_1 \\ & \searrow e_Q & \downarrow q \\ & & Q_1. \end{array}$$

2. **Construct the regular-epimorphic cover of composable pairs.** Put

$$C_2 := C_1 \times_{t_C, C_0, s_C} C_1, \quad Q_2 := Q_1 \times_{t_Q, C_0, s_Q} Q_1.$$

The induced map

$$q_2 : C_2 \rightarrow Q_2$$

factors as

$$C_1 \times_{C_0} C_1 \xrightarrow{q \times 1} Q_1 \times_{t_Q, C_0, s_Q} C_1 \xrightarrow{1 \times q} Q_1 \times_{C_0} Q_1.$$

Each arrow is a pullback of q , hence a regular epimorphism. Therefore q_2 is a regular epimorphism.

3. **Descend composition.** We claim that

$$qm_C : C_2 \rightarrow Q_1$$

descends through q_2 . Reasoning internally, if

$$q(a) = q(a'), \quad q(b) = q(b'),$$

then

$$a R a', \quad b R b'.$$

Compatibility of R with composition gives

$$m_C(a, b) R m_C(a', b'),$$

and hence

$$qm_C(a, b) = qm_C(a', b').$$

Thus qm_C coequalizes the kernel pair of q_2 . There is a unique map

$$m_Q : Q_2 \rightarrow Q_1$$

satisfying

$$m_Q q_2 = qm_C :$$

$$\begin{array}{ccc} C_2 & \xrightarrow{q_2} & Q_2 \\ m_C \downarrow & & \downarrow m_Q \\ C_1 & \xrightarrow{q} & Q_1. \end{array}$$

4. **Verify associativity.** Let C_3 and Q_3 be the corresponding objects of composable triples. The induced map

$$q_3 : C_3 \twoheadrightarrow Q_3$$

is a composite of pullbacks of q , hence a regular epimorphism. The two associativity composites on Q_3 become equal after precomposition with q_3 , because they become q applied to the two associative composites in $\mathbb{C}$:

$$\begin{array}{ccc} C_3 & \xrightarrow{q_3} & Q_3 \\ \downarrow \downarrow & & \downarrow \downarrow \\ C_1 & \xrightarrow{q} & Q_1. \end{array}$$

Since q_3 is epic, associativity holds in Q .

5. **Verify the unit laws.** The left and right unit equations in Q become the corresponding unit equations in $\mathbb{C}$ after precomposition with

$$q : C_1 \twoheadrightarrow Q_1.$$

Epimorphicity of q therefore proves the unit laws.

6. **Prove uniqueness.** The maps q , q_2 , and the higher-arity covers are epic. Consequently, source, target, identities, and composition are uniquely determined by the requirement that q_R be a functor.

□

The resulting quotient has the expected universal property.

Proposition 4.4.1 (Universal property of the operation quotient). *Let R be an operation congruence on $\mathbb{C}$. If*

$$F : \mathbb{C} \rightarrow \mathbb{D}$$

is an identity-on-objects internal functor whose arrow map coequalizes R , then there is a unique identity-on-objects internal functor

$$\bar{F} : \mathbb{C}/R \rightarrow \mathbb{D}$$

satisfying

$$\bar{F}q_R = F.$$

Proof.

1. **Descend the arrow map.** The quotient property in $\mathcal{E}/(C_0 \times C_0)$ gives a unique arrow map

$$\bar{F}_1 : Q_1 \rightarrow D_1$$

satisfying

$$\bar{F}_1 q = F_1 :$$

$$\begin{array}{ccc}
 C_1 & \xrightarrow{q} & Q_1 \\
 & \searrow F_1 & \downarrow \bar{F}_1 \\
 & & D_1.
 \end{array}$$

2. **Check identities.** One has

$$\begin{aligned}
 \bar{F}_1 e_Q &= \bar{F}_1 q e_C \\
 &= F_1 e_C \\
 &= e_D.
 \end{aligned}$$

3. **Check composition.** The two candidate composites

$$\bar{F}_1 m_Q, \quad m_D(\bar{F}_1 \times \bar{F}_1)$$

become equal after precomposition with the regular epimorphism

$$q_2 : C_2 \twoheadrightarrow Q_2,$$

because F preserves composition. Therefore they are equal.

4. **Prove uniqueness.** Any other factorization has the same arrow map after precomposition with the epimorphism q . Hence it equals $\bar{F}$.

□

Lemma 4.4.2 (Creation of identity-on-objects regular images). *Let*

$$F : \mathbb{C} \rightarrow \mathbb{D}$$

be an identity-on-objects internal functor. Factor its arrow map in

$$\mathcal{E}/(C_0 \times C_0)$$

as

$$C_1 \xrightarrow{r} I_1 \xrightarrow{m} D_1,$$

where r is a regular epimorphism and m is a monomorphism. Then I_1 , with object object C_0 , carries a unique internal category structure for which

$$\mathbb{C} \xrightarrow{r} I \xrightarrow{m} \mathbb{D}$$

are internal functors.

Moreover,

$$I \cong \mathbb{C} / \ker(F_1)$$

canonically as an identity-on-objects internal category.

Proof.

1. **Form the kernel congruence.** Since F is a functor,

$$\ker(F_1) \rightrightarrows C_1$$

is an operation congruence.

2. **Take its effective quotient.** By Lemma 4.4.1, the effective quotient

$$C_1 \twoheadrightarrow C_1 / \ker(F_1)$$

carries a unique internal category structure.

3. **Identify the quotient with the regular image.** Because m is monic,

$$\ker(F_1) = \ker(r).$$

Exactness of the slice identifies the quotient of C_1 by this kernel pair with the regular image I_1 . The comparison is

$$\begin{array}{ccccc} C_1 & \xrightarrow{r} & I_1 & \xleftarrow{m} & D_1 \\ & \searrow & \downarrow \cong & & \\ & & C_1 / \ker(F_1) & & \end{array}$$

4. **Transport the category structure.** Transporting the quotient category structure across the canonical isomorphism gives the required internal category structure on I_1 .
5. **Verify uniqueness.** The regular epimorphism r determines the identities and composition of I_1 , while the monomorphism m detects the required equations. Hence the structure is unique.

□

4.4.2 The transition-output image

Definition 4.4.2 (Transition-output image). *Let*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

*be a local coequational theory. Its **transition-output image** is the identity-on-objects regular image*

$$\text{Op}_{\mathbb{A}, \mathbb{B}}(V) := \text{RegIm}(\chi_V : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(V)).$$

The defining factorization is

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_V} & \text{KlEnd}_{\mathbb{B}}(V) \\ & \searrow q_V \quad \swarrow i_V & \\ & \text{Op}_{\mathbb{A}, \mathbb{B}}(V) & \end{array}$$

where the factorization is taken on arrow objects in

$$\mathcal{E} / (A_0 \times A_0).$$

Theorem 4.4.1 (Transition-output quotient theorem). *There is a canonical isomorphism of identity-on-objects internal categories*

$$\text{Op}_{\mathbb{A}, \mathbb{B}}(V) \cong \mathbb{A}^{\text{op}} / R_V.$$

Under this isomorphism, the regular-epimorphic part q_V corresponds to the effective quotient map

$$q_{R_V} : \mathbb{A}^{\text{op}} \twoheadrightarrow \mathbb{A}^{\text{op}} / R_V.$$

Proof.

1. **Identify the kernel congruence.** By Theorem 4.3.1,

$$R_V = \ker(\chi_{V,1}).$$

2. **Apply regular-image creation.** Lemma 4.4.2 identifies the regular image of χ_V with the quotient of $\mathbb{A}^{\text{op}}$ by this kernel congruence:

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\quad} & \mathbb{A}^{\text{op}}/R_V \\ & \searrow q_V & \downarrow \cong \\ & & \text{Op}_{\mathbb{A},\mathbb{B}}(V). \end{array}$$

□

4.4.3 Descent of operational equality

Proposition 4.4.2 (Operational descent along strict regular epimorphisms). *Let*

$$e : P \twoheadrightarrow Q$$

be a strict regular epimorphism of Mealy morphisms over $(\mathbb{A}, \mathbb{B})$. If two parallel input-arrow families induce the same transition-output operation on P , then they induce the same transition-output operation on Q .

Equivalently,

$$\ker(\chi_{P,1}) \leq \ker(\chi_{Q,1}).$$

Proof.

1. **Choose a generalized parallel pair.** Let

$$a, a' : U \rightarrow A_1$$

be parallel input-arrow families, and suppose their evaluated operations on P agree.

2. **Pull back the state quotient.** Pulling e back to the corresponding source-state families gives a regular epimorphism

$$e_U : P_{\text{src},U} \twoheadrightarrow Q_{\text{src},U},$$

in the pullback square

$$\begin{array}{ccc} P_{\text{src},U} & \xrightarrow{e_U} & Q_{\text{src},U} \\ \downarrow & \lrcorner & \downarrow \\ P & \xrightarrow{e} & Q. \end{array}$$

3. **Use strictness.** Strictness gives

$$e \delta_P(a, -) = \delta_Q(a, -) e_U$$

and

$$\omega_P(a, -) = \omega_Q(a, -) e_U,$$

and similarly for a' . The transition square is

$$\begin{array}{ccc} P_{\text{src},U} & \xrightarrow{e_U} & Q_{\text{src},U} \\ \delta_P(a,-) \downarrow & & \downarrow \delta_Q(a,-) \\ P & \xrightarrow{e} & Q. \end{array}$$

4. **Descend transition equality.** Since

$$\delta_P(a,-) = \delta_P(a',-),$$

one has

$$\begin{aligned} \delta_Q(a,-) e_U &= e \delta_P(a,-) \\ &= e \delta_P(a',-) \\ &= \delta_Q(a',-) e_U. \end{aligned}$$

Epimorphicity of e_U gives

$$\delta_Q(a,-) = \delta_Q(a',-).$$

5. **Descend output equality.** Similarly,

$$\begin{aligned} \omega_Q(a,-) e_U &= \omega_P(a,-) \\ &= \omega_P(a',-) \\ &= \omega_Q(a',-) e_U. \end{aligned}$$

Hence

$$\omega_Q(a,-) = \omega_Q(a',-).$$

6. **Conclude kernel inclusion.** The two induced operations on Q are equal, so every pair in $\ker(\chi_{P,1})$ lies in $\ker(\chi_{Q,1})$.

□

4.4.4 Recognition and faithful recognition

Definition 4.4.3 (Recognition by an operation quotient). *Let*

$$q_R : \mathbb{A}^{\text{op}} \rightarrow \mathbb{A}^{\text{op}}/R$$

*be an operation quotient. We say that q_R **recognizes** a local coequational theory V if there is an identity-on-objects internal functor*

$$\bar{\chi}_V : \mathbb{A}^{\text{op}}/R \rightarrow \text{KlEnd}_{\mathbb{B}}(V)$$

satisfying

$$\bar{\chi}_V q_R = \chi_V.$$

Recognition is the commutative triangle

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{q_R} & \mathbb{A}^{\text{op}}/R \\ & \searrow \chi_V & \downarrow \bar{\chi}_V \\ & & \text{KlEnd}_{\mathbb{B}}(V). \end{array}$$

Proposition 4.4.3 (Recognition criterion). *For an operation quotient q_R and a local theory V , the following are equivalent:*

1. q_R recognizes V ;

2.

$$R \leq R_V;$$

3.

$$V \leq V_R.$$

Proof.

1. **Translate factorization into kernel containment.** The representation χ_V descends through q_R exactly when its arrow map coequalizes R . This is equivalent to

$$R \leq \ker(\chi_{V,1}).$$

2. **Identify the valid-equation congruence.** By Theorem 4.3.1,

$$\ker(\chi_{V,1}) = R_V.$$

Hence recognition is equivalent to

$$R \leq R_V.$$

3. **Apply operational soundness and completeness.** Theorem 4.3.2 gives

$$R \leq R_V \iff V \leq V_R.$$

□

Definition 4.4.4 (Faithful recognition). *A quotient*

$$q_R : \mathbb{A}^{\text{op}} \twoheadrightarrow \mathbb{A}^{\text{op}}/R$$

faithfully recognizes V if it recognizes V and the induced functor

$$\bar{\chi}_V : \mathbb{A}^{\text{op}}/R \rightarrow \text{KlEnd}_{\mathbb{B}}(V)$$

is monic on arrows.

Theorem 4.4.2 (Universal faithful recognition). *Let*

$$q_R : \mathbb{A}^{\text{op}} \twoheadrightarrow \mathbb{A}^{\text{op}}/R$$

recognize a local theory V . Then there is a unique identity-on-objects regular epimorphism

$$h : \mathbb{A}^{\text{op}}/R \twoheadrightarrow \text{Op}_{\mathbb{A},\mathbb{B}}(V)$$

satisfying

$$hq_R = q_V$$

and

$$i_V h = \bar{\chi}_V.$$

Moreover, q_R faithfully recognizes V if and only if

$$R = R_V.$$

Consequently, $\text{Op}_{\mathbb{A},\mathbb{B}}(V)$ is, up to unique isomorphism, the unique faithful recognizing quotient of $\mathbb{A}^{\text{op}}$.

Proof.

1. **Compare the kernel congruences.** Recognition gives

$$R \leq R_V.$$

2. **Induce the quotient comparison.** The quotient by R_V factors uniquely through the quotient by R :

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{q_R} & \mathbb{A}^{\text{op}}/R \\ & \searrow q_{R_V} & \downarrow h \\ & & \mathbb{A}^{\text{op}}/R_V. \end{array}$$

3. **Show that the comparison is regular epic.** Factor h_1 as

$$h_1 = mr$$

with r regular epic and m monic. Since

$$h_1 q_R = q_{R_V}$$

is a regular epimorphism, the monomorphic factor m is an isomorphism. Hence h_1 is a regular epimorphism.

4. **Identify the target with the transition-output image.** Theorem 4.4.1 gives

$$\mathbb{A}^{\text{op}}/R_V \cong \text{Op}_{\mathbb{A}, \mathbb{B}}(V).$$

Under this identification, the quotient comparison satisfies

$$h q_R = q_V.$$

5. **Compare the represented actions.** The two functors

$$i_V h, \quad \bar{\chi}_V$$

agree after precomposition with the regular epimorphism q_R :

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{q_R} & \mathbb{A}^{\text{op}}/R \\ & \searrow \chi_V & \downarrow \bar{\chi}_V \\ & & \text{KlEnd}_{\mathbb{B}}(V). \end{array}$$

Hence

$$i_V h = \bar{\chi}_V.$$

6. **Characterize faithful recognition.** The induced representation factors as

$$\mathbb{A}^{\text{op}}/R \xrightarrow{h} \mathbb{A}^{\text{op}}/R_V \xleftarrow{i_V} \text{KlEnd}_{\mathbb{B}}(V).$$

Since i_V is monic, the composite is monic exactly when h is monic. Since h is already regular epic, this occurs exactly when h is an isomorphism.

7. **Translate isomorphism into equality of congruences.** The quotient comparison h is an isomorphism exactly when

$$\ker(q_R) = \ker(q_{R_V}),$$

equivalently

$$R = R_V.$$

Uniqueness follows from the epimorphicity of q_R .

□

Proposition 4.4.4 (Contravariance under inclusion). *If*

$$V \leq W$$

are local coequational theories, then

$$R_W \leq R_V$$

and there is a canonical identity-on-objects regular epimorphism

$$\text{Op}_{\mathbb{A}, \mathbb{B}}(W) \twoheadrightarrow \text{Op}_{\mathbb{A}, \mathbb{B}}(V).$$

Proof.

1. **Restrict valid equations.** Equality of operations on every state of W implies equality on the subobject V . Hence

$$R_W \leq R_V.$$

2. **Pass to effective quotients.** The inclusion of congruences induces

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \longrightarrow & \mathbb{A}^{\text{op}}/R_W \\ & \searrow & \downarrow \\ & & \mathbb{A}^{\text{op}}/R_V. \end{array}$$

3. **Identify the quotient categories.** Apply Theorem 4.4.1 to identify the source and target with

$$\text{Op}(W) \quad \text{and} \quad \text{Op}(V).$$

□

4.5 Local algebraic Eilenberg duality

4.5.1 Semantic closure operators

Definition 4.5.1 (Saturation and semantic closure). *For a local coequational theory V , define its operation saturation by*

$$c(V) := V_{R_V}.$$

We call V operation-saturated if

$$c(V) = V.$$

For an operation congruence R , define its **semantic closure** by

$$d(R) := R_{V_R}.$$

We call R **semantically closed** if

$$d(R) = R.$$

The two composites are

$$\text{Loc} \xrightarrow{V \mapsto R_V} \text{Cong}^{\text{op}} \xrightarrow{R \mapsto V_R} \text{Loc},$$

and

$$\text{Cong} \xrightarrow{R \mapsto V_R} \text{Loc}^{\text{op}} \xrightarrow{V \mapsto R_V} \text{Cong}.$$

Proposition 4.5.1 (Closure operators). *The assignments*

$$c : \text{Loc}_{\mathbb{A}, \mathbb{B}} \rightarrow \text{Loc}_{\mathbb{A}, \mathbb{B}}$$

and

$$d : \text{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}}) \rightarrow \text{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}})$$

are monotone, inflationary, and idempotent.

Proof.

1. **Establish antitonicity of the polarity.** One has

$$V \leq W \implies R_W \leq R_V,$$

and

$$R \leq S \implies V_S \leq V_R.$$

Hence both composites c and d are monotone.

2. **Prove inflationarity of c .** Apply operational soundness and completeness to the reflexive inequality

$$R_V \leq R_V.$$

This gives

$$V \leq V_{R_V} = c(V).$$

3. **Prove inflationarity of d .** Apply operational soundness and completeness to

$$V_R \leq V_R.$$

This gives

$$R \leq R_{V_R} = d(R).$$

4. **Prove idempotence of c .** Since

$$V \leq c(V),$$

antitonicity gives

$$R_{c(V)} \leq R_V.$$

Inflationarity of d , applied to R_V , gives

$$R_V \leq d(R_V) = R_{V_{R_V}} = R_{c(V)}.$$

Hence

$$R_{c(V)} = R_V,$$

and therefore

$$\begin{aligned} c^2(V) &= V_{R_{c(V)}} \\ &= V_{R_V} \\ &= c(V). \end{aligned}$$

5. **Prove idempotence of d .** Since

$$R \leq d(R),$$

antitonicity gives

$$V_{d(R)} \leq V_R.$$

Inflationarity of c , applied to V_R , gives

$$V_R \leq c(V_R) = V_{R_{V_R}} = V_{d(R)}.$$

Hence

$$V_{d(R)} = V_R,$$

and therefore

$$\begin{aligned} d^2(R) &= R_{V_{d(R)}} \\ &= R_{V_R} \\ &= d(R). \end{aligned}$$

□

Corollary 4.5.1 (Galois triangle identities). *For every local theory V and operation congruence R ,*

$$R_{V_{R_V}} = R_V$$

and

$$V_{R_{V_R}} = V_R.$$

In particular, every congruence of the form R_V is semantically closed, and every theory of the form V_R is operation-saturated.

4.5.2 Complete-lattice identities

Theorem 4.5.1 (Galois formulas for joins). *For every small family of local coequational theories $(V_i)_i$,*

$$R_{\bigvee_i V_i} = \bigwedge_i R_{V_i}.$$

For every small family of operation congruences $(R_j)_j$,

$$V_{\bigvee_j R_j} = \bigwedge_j V_{R_j}.$$

Proof.

1. **Test the first equality against an arbitrary congruence.** For every operation congruence R ,

$$\begin{aligned}
 R \leq R_{\bigvee_i V_i} &\iff \bigvee_i V_i \leq V_R \\
 &\iff \forall i, V_i \leq V_R \\
 &\iff \forall i, R \leq R_{V_i} \\
 &\iff R \leq \bigwedge_i R_{V_i}.
 \end{aligned}$$

Hence

$$R_{\bigvee_i V_i} = \bigwedge_i R_{V_i}.$$

2. **Test the second equality against an arbitrary theory.** For every local theory V ,

$$\begin{aligned}
 V \leq V_{\bigvee_j R_j} &\iff \bigvee_j R_j \leq R_V \\
 &\iff \forall j, R_j \leq R_V \\
 &\iff \forall j, V \leq V_{R_j} \\
 &\iff V \leq \bigwedge_j V_{R_j}.
 \end{aligned}$$

Hence

$$V_{\bigvee_j R_j} = \bigwedge_j V_{R_j}.$$

□

Corollary 4.5.2 (Fixed-point lattices). *Operation-saturated local theories form a complete lattice with*

$$\bigwedge_{\text{sat}} V_i = \bigwedge_i V_i$$

and

$$\bigvee_{\text{sat}} V_i = c \left(\bigvee_i V_i \right).$$

Semantically closed operation congruences form a complete lattice with

$$\bigwedge_{\text{cl}} R_j = \bigwedge_j R_j$$

and

$$\bigvee_{\text{cl}} R_j = d \left(\bigvee_j R_j \right).$$

Proof.

1. **Use the general fixed-point construction.** The fixed points of any closure operator on a complete lattice are closed under arbitrary ambient meets.
2. **Compute fixed-point joins.** The join of fixed points is obtained by taking the ambient join and applying the closure operator. Applying this general fact to c and d gives the displayed formulas.

□

4.5.3 The local algebraic Eilenberg theorem

Theorem 4.5.2 (Local algebraic Eilenberg duality). *The assignments*

$$V \mapsto R_V, \quad R \mapsto V_R$$

restrict to an order-reversing isomorphism of complete lattices between:

1. *operation-saturated local coequational theories over $(\mathbb{A}, \mathbb{B})$;*
2. *semantically closed identity-on-objects operation congruences on $\mathbb{A}^{\text{op}}$.*

Proof.

1. **Send saturated theories to closed congruences.** If V is operation-saturated, then

$$V = V_{R_V}.$$

By the Galois triangle identity,

$$R_{V_{R_V}} = R_V,$$

so R_V is semantically closed.

2. **Send closed congruences to saturated theories.** If R is semantically closed, then

$$R = R_{V_R}.$$

Again by the Galois triangle identity,

$$V_{R_{V_R}} = V_R,$$

so V_R is operation-saturated.

3. **Verify the two composites.** On fixed points,

$$V \mapsto R_V \mapsto V_{R_V} = V$$

and

$$R \mapsto V_R \mapsto R_{V_R} = R.$$

Thus the restrictions are mutually inverse.

4. **Verify order reversal and completeness.** Both assignments are antitone. The complete-lattice structures are those of Corollary 4.5.2.

□

The fixed-point duality is summarized by

$$\text{Loc}_{\mathbb{A}, \mathbb{B}}^{\text{sat}} \begin{array}{c} \xrightarrow{V \mapsto R_V} \\ \xleftarrow{R \mapsto V_R} \end{array} \text{Cong}_{\text{ioo}}^{\text{cl}}(\mathbb{A}^{\text{op}})^{\text{op}}$$

4.5.4 Subdirect operational representation

Let

$$\mathbf{Cat}_{A_0}^{\text{ioo}}(\mathcal{E})$$

denote the category of internal categories with object object A_0 and identity-on-objects internal functors. Products in this category are formed on arrow objects by products in

$$\mathcal{E}/(A_0 \times A_0).$$

Theorem 4.5.3 (Subdirect product theorem). *Let $(V_i)_i$ be a nonempty small family of local co-equational theories. There is a canonical identity-on-objects monomorphism*

$$\text{Op}\left(\bigvee_i V_i\right) \hookrightarrow \prod_i \text{Op}(V_i)$$

whose composites with the product projections are identity-on-objects regular epimorphisms. Thus

$$\text{Op}\left(\bigvee_i V_i\right)$$

is a subdirect product of the categories $\text{Op}(V_i)$.

Proof.

1. **Form the component quotient functors.** For every i , write

$$R_i := R_{V_i}$$

and let

$$q_i : \mathbb{A}^{\text{op}} \twoheadrightarrow \mathbb{A}^{\text{op}}/R_i$$

be the effective identity-on-objects quotient.

By Theorem 4.4.1,

$$\mathbb{A}^{\text{op}}/R_i \cong \text{Op}(V_i).$$

2. **Form the product representation.** Products in $\mathbf{Cat}_{A_0}^{\text{ioo}}(\mathcal{E})$ are formed on arrow objects by products in $\mathcal{E}/(A_0 \times A_0)$. Hence the quotient functors assemble into an identity-on-objects internal functor

$$q := \langle q_i \rangle_i : \mathbb{A}^{\text{op}} \longrightarrow \prod_i \mathbb{A}^{\text{op}}/R_i.$$

3. **Compute the kernel congruence of the product map.** On arrow objects, equality after applying q_1 is equality after applying every component $(q_i)_1$. Therefore, in the subobject lattice of

$$\text{Par}(\mathbb{A}),$$

one has

$$\begin{aligned} \ker(q_1) &= \bigwedge_i \ker((q_i)_1) \\ &= \bigwedge_i R_i \\ &= \bigwedge_i R_{V_i}. \end{aligned}$$

By Theorem 4.5.1,

$$\bigwedge_i R_{V_i} = R_{\bigvee_i V_i}.$$

Consequently,

$$\ker(q_1) = R_{\bigvee_i V_i}.$$

4. **Take the identity-on-objects regular image.** Factor q on arrow objects as

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{e} & I \\ & \searrow q & \downarrow m \\ & & \prod_i \mathbb{A}^{\text{op}}/R_i. \end{array}$$

By Lemma 4.4.2, the arrow image carries a unique internal category structure for which e and m are identity-on-objects internal functors.

Since m_1 is monic,

$$\ker(e_1) = \ker(q_1) = R_{\bigvee_i V_i}.$$

Exactness therefore gives a canonical isomorphism

$$I \cong \mathbb{A}^{\text{op}}/R_{\bigvee_i V_i}.$$

5. **Identify the regular image with the operational image.** Applying Theorem 4.4.1 to the local theory $\bigvee_i V_i$ gives

$$\mathbb{A}^{\text{op}}/R_{\bigvee_i V_i} \cong \text{Op} \left(\bigvee_i V_i \right).$$

Transporting the monomorphism m across this canonical isomorphism gives

$$\text{Op} \left(\bigvee_i V_i \right) \hookrightarrow \prod_i \text{Op}(V_i).$$

6. **Prove that every projection is regular epic.** Let

$$\pi_i : \prod_j \mathbb{A}^{\text{op}}/R_j \rightarrow \mathbb{A}^{\text{op}}/R_i$$

be the i -th product projection, and put

$$p_i := \pi_i m : I \rightarrow \mathbb{A}^{\text{op}}/R_i.$$

Since

$$p_i e = \pi_i m e = \pi_i q = q_i,$$

the composite $p_i e$ is regular epic on arrows.

Factor the arrow map of p_i as

$$(p_i)_1 = n_i r_i,$$

where r_i is regular epic and n_i is monic. Then

$$(q_i)_1 = n_i r_i e_1.$$

The composite $r_i e_1$ is regular epic, and $(q_i)_1$ is regular epic. Since a regular epimorphism is extremal, the monomorphism n_i through which $(q_i)_1$ factors is an isomorphism. Hence $(p_i)_1$ is a regular epimorphism.

Transporting across the canonical quotient-image isomorphisms shows that every projection

$$\text{Op} \left(\bigvee_i V_i \right) \twoheadrightarrow \text{Op}(V_i)$$

is an identity-on-objects regular epimorphism.

□

Corollary 4.5.3 (Operational image of a generated theory). *Let*

$$V = \bigvee_i V_i$$

for a nonempty small family. Then every operation represented faithfully on V is determined by its family of represented operations on the V_i , and every projection

$$\text{Op}(V) \twoheadrightarrow \text{Op}(V_i)$$

is a quotient.

4.6 Specifications and syntactic transition-output categories

4.6.1 Generated derivative theories

Let

$$k : K \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

be a behaviour specification. Recall that

$$d_K : D_K \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

is the residual evaluation map and that

$$\text{Der}_0(K) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

is its regular image:

$$\begin{array}{ccc} D_K & \xrightarrow{\eta_K} & \text{Der}_0(K) \\ & \searrow d_K & \downarrow j_K \\ & & \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \end{array}$$

Equivalently,

$$\text{Der}_0(K) = \text{Gen}_{\partial}(\text{Im}(k)).$$

Thus $\text{Der}_0(K)$ is the least local coequational theory containing the specified behaviours.

Proposition 4.6.1 (Formal derivative-orbit presentation). *Let*

$$k : K \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}$$

be a behaviour specification, and write

$$p_K := p_{\mathbf{Beh}} k : K \rightarrow A_0.$$

Define the formal derivative-orbit carrier

$$D_K := A_1 \times_{t_A, A_0, p_K} K.$$

A generalized element is written

$$(r, \kappa), \quad r : v \rightarrow p_K(\kappa).$$

Give D_K the input anchor

$$p_{D_K}(r, \kappa) := s_A(r) = v$$

and output anchor

$$q_{D_K}(r, \kappa) := k(\kappa)(r),$$

where $k(\kappa)(r)$ denotes the object of $\mathbb{B}$ assigned by the residual behaviour $k(\kappa)$ to the object

$$r : v \rightarrow p_K(\kappa)$$

of the residual slice.

Define the transition by

$$\delta_{D_K}(a, (r, \kappa)) := (r \circ a, \kappa)$$

for

$$a : u \rightarrow v, \quad r : v \rightarrow p_K(\kappa),$$

and define the output comparison by

$$\omega_{D_K}(a, (r, \kappa)) := k(\kappa)(r \circ a \rightarrow r).$$

Then D_K is a Mealy morphism, and the residual evaluation

$$d_K : D_K \rightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}, \quad d_K(r, \kappa) := \partial_r k(\kappa),$$

is a strict semantic homomorphism.

Consequently, its regular-image factorization

$$\begin{array}{ccc} D_K & \xrightarrow{\eta_K} & \text{Der}_0(K) \\ & \searrow d_K & \downarrow j_K \\ & & \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}} \end{array}$$

is a strict regular-image factorization of Mealy morphisms.

Proof.

1. **Verify the typing equations.** Let

$$a : u \rightarrow v, \quad r : v \rightarrow p_K(\kappa).$$

Then

$$r \circ a : u \rightarrow p_K(\kappa),$$

so

$$\delta_{D_K}(a, (r, \kappa)) = (r \circ a, \kappa)$$

lies over $u = s_A(a)$.

Moreover,

$$\omega_{D_K}(a, (r, \kappa)) = k(\kappa)(r \circ a \rightarrow r)$$

is an arrow

$$k(\kappa)(r \circ a) \rightarrow k(\kappa)(r),$$

hence

$$q_{D_K}(\delta_{D_K}(a, (r, \kappa))) \rightarrow q_{D_K}(r, \kappa).$$

2. **Verify the unit equations.** For the identity arrow $e_A(v)$,

$$\begin{aligned} \delta_{D_K}(e_A(v), (r, \kappa)) &= (r \circ e_A(v), \kappa) \\ &= (r, \kappa). \end{aligned}$$

Since $k(\kappa)$ is a functor on the residual slice,

$$k(\kappa)(r \rightarrow r) = e_B(k(\kappa)(r)).$$

Therefore

$$\omega_{D_K}(e_A(v), (r, \kappa)) = e_B(q_{D_K}(r, \kappa)).$$

3. **Verify transition multiplication.** Let

$$a : u \rightarrow v, \quad b : v \rightarrow w, \quad r : w \rightarrow p_K(\kappa).$$

Then

$$\begin{aligned} \delta_{D_K}(b \circ a, (r, \kappa)) &= (r \circ (b \circ a), \kappa) \\ &= ((r \circ b) \circ a, \kappa) \\ &= \delta_{D_K}(a, \delta_{D_K}(b, (r, \kappa))). \end{aligned}$$

4. **Verify output multiplication.** Functoriality of $k(\kappa)$ on the residual slice gives

$$\begin{aligned} k(\kappa)(r \circ (b \circ a) \rightarrow r) \\ = k(\kappa)(r \circ b \rightarrow r) \circ k(\kappa)((r \circ b) \circ a \rightarrow r \circ b). \end{aligned}$$

Hence

$$\begin{aligned} \omega_{D_K}(b \circ a, (r, \kappa)) &= \omega_{D_K}(b, (r, \kappa)) \\ &\quad \circ \omega_{D_K}(a, \delta_{D_K}(b, (r, \kappa))). \end{aligned}$$

Thus D_K satisfies the Mealy multiplication law.

5. **Verify strict preservation of transitions.** By the derivative composition law,

$$\begin{aligned} d_K(\delta_{D_K}(a, (r, \kappa))) &= d_K(r \circ a, \kappa) \\ &= \partial_{r \circ a} k(\kappa) \\ &= \partial_a(\partial_r k(\kappa)) \\ &= \delta_{\mathbf{Beh}}(a, d_K(r, \kappa)). \end{aligned}$$

6. **Verify strict preservation of outputs.** The canonical output comparison produced by acting with a on

$$\partial_r k(\kappa)$$

is the value of $k(\kappa)$ on the residual morphism

$$r \circ a \rightarrow r.$$

Therefore

$$\omega_{\mathbf{Beh}}(a, d_K(r, \kappa)) = k(\kappa)(r \circ a \rightarrow r) = \omega_{D_K}(a, (r, \kappa)).$$

Hence d_K is strict.

7. **Pass to the regular image.** Regular images of strict Mealy homomorphisms are created on their carrier maps by the regular-image construction established in Chapter 3. Therefore the carrier image of d_K inherits a unique Mealy structure for which

$$\eta_K : D_K \rightarrow \mathbf{Der}_0(K)$$

and

$$j_K : \mathbf{Der}_0(K) \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}$$

are strict.

□

Definition 4.6.1 (Syntactic operation congruence). *The **syntactic operation congruence** of K is*

$$R_K^{\text{syn}} := R_{\mathbf{Der}_0(K)}.$$

Definition 4.6.2 (Syntactic transition-output category). *The **syntactic transition-output category** of K is*

$$\text{SynTO}_{\mathbb{A}, \mathbb{B}}(K) := \text{Op}_{\mathbb{A}, \mathbb{B}}(\mathbf{Der}_0(K)).$$

We write

$$q_K^{\text{syn}} : \mathbb{A}^{\text{op}} \rightarrow \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K)$$

for the regular-epimorphic part of the image factorization of

$$\chi_{\mathbf{Der}_0(K)}.$$

By Theorem 4.4.1,

$$\text{SynTO}_{\mathbb{A}, \mathbb{B}}(K) \cong \mathbb{A}^{\text{op}} / R_K^{\text{syn}},$$

and under this isomorphism q_K^{syn} corresponds to $q_{R_K^{\text{syn}}}$. The defining diagram is

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_{\mathbf{Der}_0(K)}} & \mathbf{KlEnd}_{\mathbb{B}}(\mathbf{Der}_0(K)) \\ & \searrow q_K^{\text{syn}} & \swarrow \\ & \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K) & \end{array}$$

4.6.2 Regular presentations of operational equations

Theorem 4.6.1 (Regular-presentation detection). *Let*

$$e : G \twoheadrightarrow V$$

be a strict regular epimorphism of Mealy morphisms, with V a local coequational theory. Let

$$U \rightarrow \text{Par}(\mathbb{A})$$

classify a pair of parallel input-arrow families

$$a, a' : U \rightarrow A_1.$$

Pulling e back to the corresponding source-state families gives a regular epimorphism

$$e_U : G_{\text{src}, U} \twoheadrightarrow V_{\text{src}, U}.$$

Then the operations induced by a and a' on V are equal if and only if their evaluated transition and output components agree after precomposition with e_U .

Proof.

1. **Prove the forward implication.** Equality of the two operations on V implies equality after precomposition with any map, in particular with

$$e_U : G_{\text{src}, U} \twoheadrightarrow V_{\text{src}, U}.$$

2. **Represent the two operation families.** The parallel input-arrow families determine maps

$$U \rightrightarrows \text{KlEnd}_{\mathbb{B}}(V)_1.$$

Their evaluations fit into

$$\begin{array}{ccc} G_{\text{src}, U} & \xrightarrow{e_U} & V_{\text{src}, U} \\ & \searrow & \downarrow \downarrow \\ & & E_{[2], U}. \end{array}$$

3. **Apply epimorphic extensionality.** If the two evaluated maps agree after precomposition with e_U , Lemma 4.2.1 implies equality of the two maps

$$U \rightrightarrows \text{KlEnd}_{\mathbb{B}}(V)_1.$$

Hence the two operations on V are equal.

□

4.6.3 Context expansion

Reasoning in the internal language, let

$$\kappa \in K$$

and let

$$r : v \rightarrow p_K(\kappa)$$

be a residual context. For

$$a : u \rightarrow v,$$

the composite

$$r \circ a : u \rightarrow p_K(\kappa)$$

is an object of the residual slice. We write

$$k(\kappa)(a, r) := k(\kappa)(r \circ a \rightarrow r)$$

for the output comparison assigned to the corresponding residual morphism:

$$r \circ a \longrightarrow r \longrightarrow 1_{p_K(\kappa)}.$$

Theorem 4.6.2 (Context expansion). *Let*

$$a, a' : u \rightarrow v$$

be parallel arrows of $\mathbb{A}$. Then

$$a R_K^{\text{syn}} a'$$

if and only if, internally, for every

$$\kappa \in K$$

and every residual context

$$r : v \rightarrow p_K(\kappa),$$

one has

$$\partial_{r \circ a} k(\kappa) = \partial_{r \circ a'} k(\kappa)$$

and

$$k(\kappa)(a, r) = k(\kappa)(a', r).$$

Proof.

1. **Use the regular presentation of the derivative theory.** The derivative theory is presented by the strict regular epimorphism

$$D_K \twoheadrightarrow \text{Der}_0(K).$$

2. **Restrict to the relevant source fibre.** Pulling the presentation back to the fibre over v gives a regular-epimorphic cover of

$$\text{Der}_0(K)_v$$

by generated residual states

$$\partial_r k(\kappa).$$

The pullback is

$$\begin{array}{ccc} (D_K)_v & \twoheadrightarrow & \text{Der}_0(K)_v \\ \downarrow & \lrcorner & \downarrow \\ D_K & \twoheadrightarrow & \text{Der}_0(K). \end{array}$$

3. **Evaluate the action of a .** Acting by a on the generated residual state gives

$$\partial_a(\partial_r k(\kappa)) = \partial_{r \circ a} k(\kappa),$$

and the output comparison is

$$\omega_{\mathbf{Beh}}(a, \partial_r k(\kappa)) = k(\kappa)(r \circ a \rightarrow r).$$

4. **Evaluate the action of a' .** Similarly,

$$\partial_{a'}(\partial_r k(\kappa)) = \partial_{r \circ a'} k(\kappa)$$

and

$$\omega_{\mathbf{Beh}}(a', \partial_r k(\kappa)) = k(\kappa)(r \circ a' \rightarrow r).$$

5. **Apply regular-presentation detection.** Theorem 4.6.1 says that the operations induced by a and a' agree on $\text{Der}_0(K)$ exactly when the displayed transition and output components agree on this regular-epimorphic presentation. This is exactly the asserted contextual condition. □

4.6.4 The syntactic universal property

Definition 4.6.3 (Recognition of a specification). *An operation quotient*

$$q_R : \mathbb{A}^{\text{op}} \rightarrow \mathbb{A}^{\text{op}}/R$$

recognizes K if it recognizes the generated local theory

$$\text{Der}_0(K).$$

Theorem 4.6.3 (Syntactic transition-output universal property). *Let*

$$q_R : \mathbb{A}^{\text{op}} \rightarrow \mathbb{A}^{\text{op}}/R$$

be an operation quotient. The following are equivalent:

1. q_R recognizes K ;

2.

$$R \leq R_K^{\text{syn}};$$

3. *there is a unique identity-on-objects regular epimorphism*

$$h : \mathbb{A}^{\text{op}}/R \rightarrow \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K)$$

satisfying

$$hq_R = q_K^{\text{syn}}.$$

Moreover,

$$\text{SynTO}_{\mathbb{A}, \mathbb{B}}(K)$$

faithfully recognizes $\text{Der}_0(K)$.

Proof.

1. **Apply the recognition criterion.** Proposition 4.4.3, applied to

$$V = \text{Der}_0(K),$$

gives

$$q_R \text{ recognizes } K \iff R \leq R_{\text{Der}_0(K)} = R_K^{\text{syn}}.$$

2. **Apply universal faithful recognition.** Theorem 4.4.2 gives a unique regular-epimorphic comparison

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{q_R} & \mathbb{A}^{\text{op}}/R \\ & \searrow q_K^{\text{syn}} & \downarrow h \\ & & \text{SynTO}_{\mathbb{A},\mathbb{B}}(K). \end{array}$$

3. **Verify faithfulness of the syntactic quotient.** The quotient by

$$R_K^{\text{syn}} = \ker(\chi_{\text{Der}_0(K),1})$$

is the regular image of the residual representation. Its induced map into

$$\text{KlEnd}_{\mathbb{B}}(\text{Der}_0(K))$$

is therefore monic on arrows.

□

Corollary 4.6.1 (Coarsest recognizing quotient). *The syntactic transition-output category is the coarsest identity-on-objects quotient of $\mathbb{A}^{\text{op}}$ recognizing K , and the unique such quotient whose induced action on the generated derivative theory is faithful.*

Equivalently, it is terminal among recognizing quotients ordered by factorization: every recognizing quotient admits a unique identity-on-objects regular-epimorphic comparison onto $\text{SynTO}_{\mathbb{A},\mathbb{B}}(K)$.

4.6.5 Two-stage operational Myhill–Nerode theory

Definition 4.6.4 (Raw action image). *For a Mealy morphism P , define*

$$\text{ActIm}(P) := \text{RegIm}(\chi_P : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P)).$$

Define also

$$\text{Op}(P) := \text{Op}(\text{Beh}(P)).$$

The raw action image records the action on the original state object:

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_P} & \text{KlEnd}_{\mathbb{B}}(P) \\ & \searrow & \swarrow \\ & \text{ActIm}(P) & \end{array}$$

The semantic operation image records the action after states with identical residual behaviour have been identified.

Theorem 4.6.4 (Two-stage operational quotient). *Let P be a Mealy morphism realizing a specification K . Then there is a canonical chain of identity-on-objects regular epimorphisms*

$$\text{ActIm}(P) \twoheadrightarrow \text{Op}(P) \twoheadrightarrow \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K).$$

Proof.

1. **Define the raw action congruence.** Put

$$R_P^{\text{act}} := \ker(\chi_{P,1}).$$

By Lemma 4.4.2,

$$\text{ActIm}(P) \cong \mathbb{A}^{\text{op}} / R_P^{\text{act}}.$$

2. **Descend operational equality through state minimization.** The semantic quotient

$$\eta_P : P \twoheadrightarrow \text{Beh}(P)$$

is strict and regular epic. By Proposition 4.4.2,

$$R_P^{\text{act}} \leq R_{\text{Beh}(P)}.$$

3. **Form the first quotient comparison.** The preceding inclusion gives

$$\text{ActIm}(P) \longrightarrow \text{Op}(P).$$

4. **Use realization of the specification.** Since P realizes K ,

$$\text{Der}_0(K) \leq \text{Beh}(P).$$

Contravariance of valid equations gives

$$R_{\text{Beh}(P)} \leq R_{\text{Der}_0(K)}.$$

5. **Form the second quotient comparison.** Passing to effective quotients gives

$$\text{Op}(P) \longrightarrow \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K).$$

6. **Assemble the two stages.** The complete chain is

$$\mathbb{A}^{\text{op}} \longrightarrow \text{ActIm}(P) \longrightarrow \text{Op}(P) \longrightarrow \text{SynTO}(K).$$

□

Corollary 4.6.2 (Separated realizations). *If P is behaviourally separated, then*

$$\text{ActIm}(P) \cong \text{Op}(P).$$

Proof.

1. **Use monicity of the semantic map.** Behavioural separatedness means that

$$\beta_P : P \rightarrow \mathbf{Beh}^{\text{can}}$$

is monic.

2. **Inspect the semantic image factorization.** Write

$$\begin{array}{ccc} P & \xrightarrow{\eta_P} & \mathbf{Beh}(P) \\ & \searrow \beta_P & \downarrow m_P \\ & & \mathbf{Beh}^{\text{can}}. \end{array}$$

The composite

$$m_P \eta_P = \beta_P$$

is monic.

3. **Prove that η_P is monic.** Suppose

$$\eta_P g = \eta_P h.$$

Then

$$\begin{aligned} \beta_P g &= m_P \eta_P g \\ &= m_P \eta_P h \\ &= \beta_P h. \end{aligned}$$

Since β_P is monic,

$$g = h.$$

Thus η_P is monic.

4. **Conclude that η_P is an isomorphism.** Since η_P is both regular epic and monic, it is an isomorphism.
5. **Transport the operation representation.** Transporting the transition-output representation across this strict isomorphism identifies

$$\text{ActIm}(P) \cong \text{Op}(\mathbf{Beh}(P)) = \text{Op}(P).$$

□

Corollary 4.6.3 (Reachable realizations). *If P is reachable as a realization of K , then*

$$\text{Op}(P) \cong \text{SynTO}_{\mathbb{A}, \mathbb{B}}(K).$$

Proof.

1. **Use the reachability quotient.** Reachability gives a regular epimorphism

$$\rho_{P,K} : D_K \twoheadrightarrow P$$

satisfying

$$\begin{array}{ccc} & & \beta_P \rho_{P,K} = d_K : \\ D_K & \xrightarrow{\rho_{P,K}} & P \\ & \searrow d_K & \downarrow \beta_P \\ & & \mathbf{Beh}_0. \end{array}$$

2. **Compare regular images.** Precomposition with a regular epimorphism does not change a regular image. Hence

$$\begin{aligned}\mathrm{Im}(\beta_P) &= \mathrm{Im}(\beta_P \rho_{P,K}) \\ &= \mathrm{Im}(d_K).\end{aligned}$$

3. **Identify the behavioural theories.** Therefore

$$\mathrm{Beh}(P) \cong \mathrm{Der}_0(K)$$

as strict subobjects of the canonical behaviour object.

4. **Identify the operation images.** The strict isomorphism identifies their transition-output representations and hence their regular images:

$$\mathrm{Op}(P) \cong \mathrm{SynTO}(K).$$

□

Corollary 4.6.4 (Operational Myhill–Nerode theorem). *If P is reachable and behaviourally separated as a realization of K , then*

$$\mathrm{ActIm}(P) \cong \mathrm{SynTO}_{\mathbb{A},\mathbb{B}}(K).$$

Proof.

1. **Use separatedness.** Corollary 4.6.2 gives

$$\mathrm{ActIm}(P) \cong \mathrm{Op}(P).$$

2. **Use reachability.** Corollary 4.6.3 gives

$$\mathrm{Op}(P) \cong \mathrm{SynTO}(K).$$

3. **Compose the canonical isomorphisms.** Combining the two isomorphisms proves the result.

□

4.6.6 A finiteness criterion

Proposition 4.6.2 (Finiteness of syntactic operation categories). *Work in **Set**. Let V be a local coequational theory. Suppose:*

1. A_0 is finite;
2. every fibre V_v is finite;
3. every hom-set of $\mathbb{B}$ between output objects occurring in V is finite.

Then $\mathrm{Op}(V)$ is a finite category.

In particular, if $\mathrm{Der}_0(K)$ has finite fibres and the relevant output hom-sets are finite, then

$$\mathrm{SynTO}(K)$$

is finite.

Proof.

1. **Count state transformations.** A Kleisli operation

$$V_v \rightsquigarrow V_u$$

has an underlying function

$$V_v \rightarrow V_u.$$

Since both sets are finite, there are finitely many such functions.

2. **Count output comparisons.** For every $x \in V_v$, the operation also selects an arrow

$$q(Fx) \rightarrow q(x)$$

of $\mathbb{B}$. The relevant hom-sets are finite and V_v is finite, so there are finitely many possible comparison families.

3. **Count Kleisli hom-sets.** Hence every hom-set of

$$\mathbf{KlEnd}_{\mathbb{B}}(V)$$

is finite.

4. **Count the operational image.** The object set A_0 is finite, and $\mathbf{Op}(V)$ is an internal subcategory of $\mathbf{KlEnd}_{\mathbb{B}}(V)$. Therefore $\mathbf{Op}(V)$ is finite.

5. **Apply the result to a specification.** Taking

$$V = \mathbf{Der}_0(K)$$

gives the final statement. □

4.7 Structured output and automatic algebraic closure

4.7.1 Pointwise output operations

Let

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

be an internal functor. For Mealy morphisms

$$P_1, \dots, P_n : \mathbb{A} \rightsquigarrow \mathbb{B},$$

write

$$\theta(P_1, \dots, P_n)$$

for the carrier

$$P_1 \times_{A_0} \cdots \times_{A_0} P_n$$

with input anchor the common A_0 -anchor and output anchor

$$\theta_0 \circ \langle q_1, \dots, q_n \rangle.$$

Its transition is componentwise, and its output comparison is obtained by applying θ_1 componentwise.

The carrier map is

$$P_1 \times_{A_0} \cdots \times_{A_0} P_n \xrightarrow{\langle p, q_\theta \rangle} A_0 \times B_0,$$

where

$$q_\theta := \theta_0 \langle q_1, \dots, q_n \rangle.$$

Pointwise application defines a map of residual behaviour objects

$$\theta_{\mathbf{Beh}} : \underbrace{\mathbf{Beh}_0 \times_{A_0} \cdots \times_{A_0} \mathbf{Beh}_0}_{n \text{ factors}} \rightarrow \mathbf{Beh}_0.$$

Its input anchor is unchanged, and its output anchor is obtained by applying θ_0 to the n output anchors. The domain is regarded over $A_0 \times B_0$ using

$$1_{A_0} \times \theta_0 : A_0 \times B_0^n \rightarrow A_0 \times B_0.$$

For $n = 0$, the functor

$$\theta : \mathbb{B}^0 \rightarrow \mathbb{B}$$

selects a constant output object and its identity comparison. The associated carrier is A_0 with the induced map to $A_0 \times B_0$.

Proposition 4.7.1 (Pointwise operations intertwine residual actions). *Let*

$$H_1, \dots, H_n$$

be residual behaviours over the same input object. For every input arrow a ,

$$\partial_a \theta_{\mathbf{Beh}}(H_1, \dots, H_n) = \theta_{\mathbf{Beh}}(\partial_a H_1, \dots, \partial_a H_n),$$

and

$$\omega_{\mathbf{Beh}}(a, \theta_{\mathbf{Beh}}(H_1, \dots, H_n)) = \theta_1(\omega_{\mathbf{Beh}}(a, H_1), \dots, \omega_{\mathbf{Beh}}(a, H_n)).$$

Consequently,

$$\theta_{\mathbf{Beh}} : \theta(\mathbf{Beh}^{\text{can}}, \dots, \mathbf{Beh}^{\text{can}}) \rightarrow \mathbf{Beh}^{\text{can}}$$

is a strict semantic homomorphism.

Proof.

1. **Use pointwise residual functors.** Residual behaviours are internal functors on residual slices. Pointwise application of θ produces the composite

$$\mathbb{A}/v \xrightarrow{\langle H_1, \dots, H_n \rangle} \mathbb{B}^n \xrightarrow{\theta} \mathbb{B}.$$

2. **Commute with residual precomposition.** Derivation by a is precomposition with the residual-slice functor induced by a . Associativity of functor composition gives

$$\partial_a \theta_{\mathbf{Beh}}(H_1, \dots, H_n) = \theta_{\mathbf{Beh}}(\partial_a H_1, \dots, \partial_a H_n).$$

Diagrammatically,

$$\begin{array}{ccc} \mathbb{A}/u & \longrightarrow & \mathbb{A}/v \\ \langle \partial_a H_i \rangle_i \downarrow & & \downarrow \langle H_i \rangle_i \\ \mathbb{B}^n & \xlongequal{\quad} & \mathbb{B}^n \xrightarrow{\theta} \mathbb{B}. \end{array}$$

3. **Compute the canonical output comparison.** The canonical output is the value on the residual morphism

$$a \rightarrow 1_v.$$

Applying θ to the tuple of assigned arrows gives

$$\theta_1(\omega_{\mathbf{Beh}}(a, H_1), \dots, \omega_{\mathbf{Beh}}(a, H_n)).$$

The output diagram is

$$\theta_0(q(\partial_a H_i))_i \xrightarrow{\theta_1(\omega_{\mathbf{Beh}}(a, H_i))_i} \theta_0(q(H_i))_i.$$

4. **Conclude strictness.** The derivative and output equations are exactly the strict transition- and output-preservation equations for $\theta_{\mathbf{Beh}}$.
5. **Handle the nullary case.** For $n = 0$, the pointwise behaviour is constant on the object selected by θ , every derivative remains constant, and every residual morphism is sent to the selected identity arrow. The same strictness equations therefore hold.

□

4.7.2 Recognized theories are automatically structured

Theorem 4.7.1 (Automatic structured closure). *Let R be an operation congruence. For every internal functor*

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B},$$

pointwise application restricts to a strict map

$$\theta_{V_R} : \theta(V_R, \dots, V_R) \rightarrow V_R.$$

Thus V_R is closed under every pointwise finitary operation induced by an internal functor on $\mathbb{B}$.

Proof.

1. **Choose behaviours in the safety core.** Let

$$H_1, \dots, H_n \in V_R$$

and let r be an arbitrary residual context. Then

$$\partial_r H_i \in \text{Eq}(R)$$

for every i .

2. **Apply an operational equation componentwise.** If

$$a R a',$$

then for every i ,

$$\partial_a(\partial_r H_i) = \partial_{a'}(\partial_r H_i)$$

and

$$\omega_{\mathbf{Beh}}(a, \partial_r H_i) = \omega_{\mathbf{Beh}}(a', \partial_r H_i).$$

3. **Apply the pointwise output operation.** Proposition 4.7.1 gives

$$\begin{aligned}\partial_a \theta_{\mathbf{Beh}}(\partial_r H_i)_i &= \theta_{\mathbf{Beh}}(\partial_a \partial_r H_i)_i \\ &= \theta_{\mathbf{Beh}}(\partial_{a'} \partial_r H_i)_i \\ &= \partial_{a'} \theta_{\mathbf{Beh}}(\partial_r H_i)_i.\end{aligned}$$

It also gives equality of the corresponding output comparisons after applying θ_1 .

4. **Use derivative compatibility.** Since derivatives commute with pointwise θ ,

$$\partial_r \theta_{\mathbf{Beh}}(H_1, \dots, H_n) = \theta_{\mathbf{Beh}}(\partial_r H_1, \dots, \partial_r H_n).$$

Hence every derivative of

$$\theta_{\mathbf{Beh}}(H_1, \dots, H_n)$$

lies in $\text{Eq}(R)$.

5. **Conclude membership in the safety core.** Therefore

$$\theta_{\mathbf{Beh}}(H_1, \dots, H_n) \in V_R.$$

The factorization is

$$\begin{array}{ccc}\theta(V_R, \dots, V_R) & \xrightarrow{\theta_{\mathbf{Beh}}} & \mathbf{Beh}^{\text{can}} \\ & \searrow \theta_{V_R} & \uparrow \\ & & V_R.\end{array}$$

6. **Prove strictness.** The factorization is the restriction of the strict semantic homomorphism $\theta_{\mathbf{Beh}}$, so θ_{V_R} is strict.

7. **Handle nullary operations.** For $n = 0$, the pointwise behaviour is constant. Its derivatives are equal to itself, and every R -related pair acts by the same identity comparison. Hence the constant behaviour lies in V_R , giving a strict map

$$A_0 \rightarrow V_R.$$

□

Corollary 4.7.1 (Saturated theories inherit output structure). *If V is operation-saturated, then V is closed under every pointwise operation induced by an internal functor*

$$\mathbb{B}^n \rightarrow \mathbb{B}.$$

In particular, if $\mathbb{B}$ carries the operations of a finite-product theory $\mathbb{T}$, then every operation-saturated local theory is automatically a local $\mathbb{T}$ -coequational theory.

Proof.

1. **Represent the saturated theory by its equations.** Operation saturation gives

$$V = V_{R_V}.$$

2. **Apply automatic structured closure.** Theorem 4.7.1, applied to R_V , gives closure of $V_{R_V} = V$ under every pointwise operation induced by $\mathbb{B}$.

□

Remark 4.7.1 (Pointwise structure versus synchronized machine lifting). *The pointwise operations on the canonical behaviour object and their restrictions to V_R should be distinguished from the synchronized lifting of an output operation to arbitrary Mealy morphisms.*

On the canonical behaviour object, an internal functor

$$\theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

induces an actual pointwise operation

$$\theta_{\mathbf{Beh}} : \mathbf{Beh}_0 \times_{A_0} \cdots \times_{A_0} \mathbf{Beh}_0 \rightarrow \mathbf{Beh}_0.$$

When V_R is invariant under this operation, the factorization

$$\theta_{V_R} : V_R \times_{A_0} \cdots \times_{A_0} V_R \rightarrow V_R$$

is a genuine operation on the behavioural subobject. If the chosen operations on $\mathbb{B}$ interpret a finite-product theory $\mathbb{T}$, then the induced pointwise operations on $\mathbf{Beh}_0$ and on V_R satisfy the equations of $\mathbb{T}$, subject to the standing coherence convention for chosen finite products.

By contrast, for arbitrary Mealy morphisms

$$P_1, \dots, P_n : \mathbb{A} \rightsquigarrow \mathbb{B},$$

the synchronized lifting

$$\theta(P_1, \dots, P_n)$$

has carrier

$$P_1 \times_{A_0} \cdots \times_{A_0} P_n.$$

Even when θ is a projection operation on $\mathbb{B}$, the lifted machine retains this full synchronized product carrier rather than becoming literally one of the factors P_i .

Consequently, the synchronized lifting does not by itself make the entire category of Mealy morphisms into a strict algebra for $\mathbb{T}$. The genuine algebraic conclusion used here concerns the pointwise behavioural object and the invariant behavioural subobjects V_R , not a strict finite-product-theory structure on the whole Mealy category.

4.8 Globalization as an indexed operational polarity

4.8.1 Residual reindexing

Let

$$\mathbf{Inp}$$

be a chosen category of internal input categories and internal functors. Fix the output category $\mathbb{B}$.

Let

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

be an internal functor. Residual reindexing is the map

$$F^\sharp : A'_0 \times_{F_0, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0} \rightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

given by

$$F^\sharp(v', H) = H \circ F_{/v'}.$$

The residual-slice comparison is

$$\mathbb{A}'/v' \xrightarrow{F_{/v'}} \mathbb{A}/F_0(v') \xrightarrow{H} \mathbb{B}.$$

Write

$$\overline{F} := F_0 \times 1_{B_0} : A'_0 \times B_0 \rightarrow A_0 \times B_0.$$

Proposition 4.8.1 (Ambient computation of joins of local theories). *Let*

$$(V_i)_i$$

be a small family of local coequational theories in

$$\mathbf{Beh}_{\mathbb{A}, \mathbb{B}}^{\text{can}}.$$

Their join in

$$\mathbf{Loc}_{\mathbb{A}, \mathbb{B}}$$

is computed as their join in the ambient subobject lattice

$$\text{Sub}_{\mathcal{E}/Z}(\mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0}).$$

In particular, the ambient join

$$V := \bigvee_i V_i$$

is derivative-closed.

Proof. Let

$$\delta_{\mathbf{Beh}} : A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0 \rightarrow \mathbf{Beh}_0$$

be the derivative transition of the canonical behaviour Mealy morphism.

For a subobject

$$C \hookrightarrow \mathbf{Beh}_0,$$

write

$$D(C) := A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} C$$

for its pulled-back action domain, regarded as a subobject of

$$A_1 \times_{t_A, A_0, p_{\mathbf{Beh}}} \mathbf{Beh}_0.$$

Pullback on subobjects preserves small joins in a topos. Therefore

$$\begin{aligned} D(V) &= D\left(\bigvee_i V_i\right) \\ &= \bigvee_i D(V_i). \end{aligned}$$

Let

$$\exists_{\delta_{\mathbf{Beh}}}$$

denote the direct-image operation on subobjects along $\delta_{\mathbf{Beh}}$. Since it is a left adjoint, it preserves small joins. Hence

$$\begin{aligned} \exists_{\delta_{\mathbf{Beh}}} D(V) &= \exists_{\delta_{\mathbf{Beh}}} \left(\bigvee_i D(V_i) \right) \\ &= \bigvee_i \exists_{\delta_{\mathbf{Beh}}} D(V_i). \end{aligned}$$

Each V_i is derivative-closed, so

$$\exists_{\delta_{\mathbf{Beh}}} D(V_i) \leq V_i.$$

Consequently,

$$\begin{aligned} \exists_{\delta_{\mathbf{Beh}}} D(V) &\leq \bigvee_i V_i \\ &= V. \end{aligned}$$

This inequality says precisely that $\delta_{\mathbf{Beh}}$ restricts to a transition

$$A_1 \times_{A_0} V \rightarrow V.$$

Therefore V is derivative-closed and hence is a local coequational theory.

Since V is the least ambient subobject containing all V_i , it is also the least local theory containing all V_i . Thus it is their join in $\mathbf{Loc}_{\mathbb{A}, \mathbb{B}}$. $\square$

Definition 4.8.1 (Direct reindexing of local theories). *For a local theory*

$$V \hookrightarrow \mathbf{Beh}_{\mathbb{A}, \mathbb{B}, 0},$$

define

$$F^\bullet V := \text{Im} \left(\overline{F}^* V \xrightarrow{F^\sharp|_V} \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0} \right).$$

Equivalently,

$$F^\bullet V = \text{Beh}(F^* V)$$

as a subobject of $\mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$.

The defining regular image is

$$\begin{array}{ccc} \overline{F}^* V & \xrightarrow{\quad} & F^\bullet V \\ & \searrow F^\sharp|_V & \downarrow \\ & & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}. \end{array}$$

Proposition 4.8.2 (Basic properties of $F^\bullet$). *The assignment*

$$F^\bullet : \mathbf{Loc}_{\mathbb{A}, \mathbb{B}} \rightarrow \mathbf{Loc}_{\mathbb{A}', \mathbb{B}}$$

is monotone and preserves small joins.

For the identity input functor there is a canonical equality in the subobject lattice

$$(1_{\mathbb{A}})^{\bullet}V = V.$$

For composable input functors

$$\mathbb{A}'' \xrightarrow{G} \mathbb{A}' \xrightarrow{F} \mathbb{A},$$

there is a canonical coherent isomorphism of chosen representatives

$$G^{\bullet}(F^{\bullet}V) \cong (F \circ G)^{\bullet}V.$$

Proof.

1. **Identify $F^{\bullet}V$ as a semantic image.** The restricted residual-reindexing map

$$F^{\sharp}|_V : \overline{F}^*V \rightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}$$

is the strict semantic map of the input-pullback Mealy morphism F^*V . Therefore its regular image

$$F^{\bullet}V = \mathbf{Beh}(F^*V)$$

is a local coequational theory.

2. **Express $F^{\bullet}$ in subobject calculus.** On ambient subobject lattices,

$$F^{\bullet} = \exists_{F^{\sharp}} \circ \overline{F}^*.$$

Here

$$\overline{F}^*$$

is pullback along

$$\overline{F} = F_0 \times 1_{B_0},$$

and

$$\exists_{F^{\sharp}}$$

is the direct-image operation along the residual-reindexing map.

Both operations are monotone, so $F^{\bullet}$ is monotone.

3. **Prove preservation of small joins.** Let $(V_i)_i$ be a small family of local theories. By Proposition 4.8.1, their join in $\mathbf{Loc}_{\mathbb{A}, \mathbb{B}}$ is their ambient subobject join:

$$\bigvee_{\mathbf{Loc}} V_i = \bigvee_{\mathbf{Sub}} V_i.$$

Pullback on subobjects in a topos preserves small joins, and $\exists_{F^{\sharp}}$ preserves small joins because it is a left adjoint. Hence

$$\begin{aligned} F^{\bullet} \left(\bigvee_i V_i \right) &= \exists_{F^{\sharp}} \left(\overline{F}^* \left(\bigvee_i V_i \right) \right) \\ &= \exists_{F^{\sharp}} \left(\bigvee_i \overline{F}^* V_i \right) \\ &= \bigvee_i \exists_{F^{\sharp}} \left(\overline{F}^* V_i \right) \\ &= \bigvee_i F^{\bullet} V_i. \end{aligned}$$

The join on the right is again the ambient join and therefore, by Proposition 4.8.1, the join in $\text{Loc}_{\mathbb{A}', \mathbb{B}}$.

4. **Prove the identity law.** For the identity functor $1_{\mathbb{A}}$, the residual-slice comparison is canonically the identity. Hence

$$(1_{\mathbb{A}})^{\sharp}$$

is canonically the identity map on the behaviour object. Its restriction to V has image V , so

$$(1_{\mathbb{A}})^{\bullet}V = V$$

in the subobject lattice.

5. **Factor the first-stage semantic map.** Factor the semantic map of F^*V as

$$\begin{array}{ccc} F^*V & \xrightarrow{e_F} & F^{\bullet}V \\ & \searrow & \downarrow \\ & & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}}. \end{array}$$

The map e_F is a strict regular epimorphism of Mealy morphisms.

6. **Pull the first-stage factorization back along G .** Pullback stability of regular epimorphisms gives a strict regular epimorphism

$$G^*e_F : G^*F^*V \twoheadrightarrow G^*(F^{\bullet}V).$$

7. **Compare the second-stage semantic images.** The semantic map of

$$G^*(F^{\bullet}V)$$

precomposed with G^*e_F is the semantic map of G^*F^*V . Precomposition with a regular epimorphism does not change a regular image. Therefore

$$\text{Beh}(G^*(F^{\bullet}V)) \cong \text{Beh}(G^*F^*V).$$

By definition, the left-hand side is

$$G^{\bullet}(F^{\bullet}V).$$

8. **Use pullback coherence.** The chosen pullbacks give a canonical coherent strict isomorphism of input-pulled-back Mealy morphisms

$$G^*F^*V \cong (F \circ G)^*V.$$

Strictly isomorphic Mealy morphisms have canonically isomorphic semantic images. Hence

$$\begin{aligned} G^{\bullet}(F^{\bullet}V) &= \text{Beh}(G^*(F^{\bullet}V)) \\ &\cong \text{Beh}(G^*F^*V) \\ &\cong \text{Beh}((F \circ G)^*V) \\ &= (F \circ G)^{\bullet}V. \end{aligned}$$

The coherence of these isomorphisms follows from the coherence of iterated chosen pullbacks and the uniqueness of regular-image factorizations up to canonical isomorphism.

□

4.8.2 Operational inverse image

Definition 4.8.2 (Operational inverse image). *Let*

$$R \in \text{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}}).$$

*Define its **operational inverse image** along F by*

$$F^{\flat} R := R_{F^{\bullet}(V_R)}.$$

Thus

$$(\mathbb{A}')^{\text{op}} / F^{\flat} R$$

is the faithful syntactic quotient of the reindexed theory

$$F^{\bullet}(V_R) :$$

$$(\mathbb{A}')^{\text{op}} \longrightarrow (\mathbb{A}')^{\text{op}} / F^{\flat} R \hookrightarrow \text{KlEnd}_{\mathbb{B}}(F^{\bullet}(V_R)).$$

Proposition 4.8.3 (Monotonicity of $F^{\flat}$). *If*

$$R \leq S,$$

then

$$F^{\flat} R \leq F^{\flat} S.$$

Proof.

1. **Reverse the congruence inclusion.** Since $R \leq S$, antitonicity gives

$$V_S \leq V_R.$$

2. **Apply direct reindexing.** Monotonicity of $F^{\bullet}$ gives

$$F^{\bullet} V_S \leq F^{\bullet} V_R.$$

3. **Apply valid-equation contravariance.** Applying the antitone map $R_{(-)}$ gives

$$R_{F^{\bullet} V_R} \leq R_{F^{\bullet} V_S}.$$

By definition, this is

$$F^{\flat} R \leq F^{\flat} S.$$

□

4.8.3 Raw pullback and semantic completion

Let

$$R \hookrightarrow \text{Par}(\mathbb{A})$$

be an operation congruence. The arrow map F_1 induces

$$F_{\text{Par}} : \text{Par}(\mathbb{A}') \rightarrow \text{Par}(\mathbb{A}), \quad (a', b') \longmapsto (F_1 a', F_1 b').$$

The defining diagram is

$$\begin{array}{ccc} \text{Par}(\mathbb{A}') & \xrightarrow{F_{\text{Par}}} & \text{Par}(\mathbb{A}) \\ \downarrow & & \downarrow \\ A'_1 \times A'_1 & \xrightarrow{F_1 \times F_1} & A_1 \times A_1. \end{array}$$

Define the literal inverse-image relation

$$F^{-1}R := F_{\text{Par}}^* R.$$

Equivalently, $F^{-1}R$ is the kernel pair of

$$A'_1 \longrightarrow (A'_0 \times A'_0) \times_{A_0 \times A_0} (A_1/R)$$

induced by F_1 and the quotient map $A_1 \rightarrow A_1/R$.

Proposition 4.8.4 (Raw pullback is contained in operational inverse image). *The relation $F^{-1}R$ is an identity-on-objects operation congruence on $(\mathbb{A}')^{\text{op}}$, and*

$$F^{-1}R \leq F^b R.$$

Consequently,

$$d(F^{-1}R) \leq F^b R.$$

Proof.

1. **Pull back the effective congruence.** The relation $F^{-1}R$ is the pullback

$$\begin{array}{ccc} F^{-1}R & \longrightarrow & R \\ \downarrow & \lrcorner & \downarrow \\ \text{Par}(\mathbb{A}') & \xrightarrow{F_{\text{Par}}} & \text{Par}(\mathbb{A}). \end{array}$$

Effectiveness of equivalence relations is stable under pullback in a topos.

2. **Verify operational compatibility.** Compatibility with identities and composition follows from functoriality of F . Hence $F^{-1}R$ is an operation congruence.
3. **Choose a raw pulled-back equation.** Let

$$a', b' : u' \rightarrow v'$$

satisfy

$$F_1(a') R F_1(b').$$

4. **Use satisfaction of R by V_R .** Since V_R satisfies R , the arrows $F_1(a')$ and $F_1(b')$ induce the same operation on V_R .
5. **Pull the operation equality back along F .** Their input-pulled-back operations on F^*V_R are equal. The equality is represented by

$$F^*V_R \begin{array}{c} \xrightarrow{a'} \\ \xrightarrow{b'} \end{array} F^*V_R.$$

6. **Descend through the semantic image.** The semantic image factorization gives a strict regular epimorphism

$$F^*V_R \twoheadrightarrow F^\bullet(V_R).$$

Proposition 4.4.2 therefore implies that a' and b' induce the same operation on $F^\bullet(V_R)$.

7. **Conclude the first inclusion.** Thus

$$a' F^\flat R b',$$

and hence

$$F^{-1}R \leq F^\flat R.$$

8. **Apply semantic closure.** The congruence $F^\flat R$ is of the form R_W , so it is semantically closed. Monotonicity of d gives

$$\begin{aligned} d(F^{-1}R) &\leq d(F^\flat R) \\ &= F^\flat R. \end{aligned}$$

□

Thus

$$F^{-1}R \leq d(F^{-1}R) \leq F^\flat R.$$

The inequality may be strict because the reindexed semantic theory

$$F^\bullet(V_R)$$

can be strictly smaller than the largest local theory

$$V_{F^{-1}R}$$

satisfying the literal pullback congruence. Passing to the valid-equation congruence of the smaller theory can therefore create additional operational equations.

4.8.4 Lax functoriality

Proposition 4.8.5 (Lax composition of operational inverse image). *For every operation congruence R ,*

$$(1_{\mathbb{A}})^\flat R = R_{V_R} = d(R).$$

For composable input functors

$$\mathbb{A}'' \xrightarrow{G} \mathbb{A}' \xrightarrow{F} \mathbb{A},$$

one has

$$G^\flat(F^\flat R) \leq (F \circ G)^\flat R.$$

In particular, $F^\flat$ is strictly unital on semantically closed congruences.

Proof.

1. **Compute the identity case.** Since

$$(1_{\mathbb{A}})^{\bullet} V_R = V_R,$$

one has

$$(1_{\mathbb{A}})^{\flat} R = R_{V_R} = d(R).$$

2. **Introduce the first reindexed theory.** Put

$$W := F^{\bullet}(V_R).$$

Then

$$F^{\flat} R = R_W.$$

3. **Apply inflationary saturation.** One has

$$W \leq V_{R_W}.$$

4. **Reindex along G .** Monotonicity gives

$$G^{\bullet} W \leq G^{\bullet}(V_{R_W}).$$

5. **Reverse the inclusion by valid equations.** Contravariance of $R_{(-)}$ gives

$$R_{G^{\bullet}(V_{R_W})} \leq R_{G^{\bullet} W}.$$

The left-hand side is

$$G^{\flat}(F^{\flat} R).$$

6. **Use composition coherence of $F^{\bullet}$.** The right-hand side is canonically identified with

$$R_{(F \circ G)^{\bullet}(V_R)} = (F \circ G)^{\flat} R.$$

Thus

$$G^{\flat}(F^{\flat} R) \leq (F \circ G)^{\flat} R.$$

7. **Restrict to closed congruences.** If $R = d(R)$, then

$$(1_{\mathbb{A}})^{\flat} R = R,$$

proving strict unitality on semantically closed congruences.

□

4.8.5 Saturation-adjusted reindexing

Definition 4.8.3 (Saturated reindexing). *For a local theory V , define*

$$F^*V := V_{R_{F^\bullet V}} = c(F^\bullet V).$$

Thus F^*V is the least operation-saturated local theory containing the raw reindexing $F^\bullet V$:

$$F^\bullet V \hookrightarrow F^*V \hookrightarrow \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0}.$$

Proposition 4.8.6 (Compatibility with local duality). *If V is operation-saturated, then*

$$R_{F^*V} = F^\flat(R_V).$$

For every operation congruence R ,

$$V_{F^\flat R} = F^*(V_R).$$

Proof.

1. **Assume V is saturated.** Since

$$V = V_{R_V},$$

one has

$$\begin{aligned} F^\flat(R_V) &= R_{F^\bullet(V_{R_V})} \\ &= R_{F^\bullet V}. \end{aligned}$$

2. **Apply the triangle identity.** By definition,

$$F^*V = V_{R_{F^\bullet V}}.$$

Hence

$$\begin{aligned} R_{F^*V} &= R_{V_{R_{F^\bullet V}}} \\ &= R_{F^\bullet V} \\ &= F^\flat(R_V). \end{aligned}$$

3. **Compute the theory of an inverse image.** By definition,

$$\begin{aligned} V_{F^\flat R} &= V_{R_{F^\bullet(V_R)}} \\ &= c(F^\bullet(V_R)) \\ &= F^*(V_R). \end{aligned}$$

□

4.8.6 Global theories and global congruence systems

Definition 4.8.4 (Global coequational theory). *A **global coequational theory** is a family*

$$V = (V_{\mathbb{A}})_{\mathbb{A} \in \mathbf{Inp}}$$

where

$$V_{\mathbb{A}} \in \mathbf{Loc}_{\mathbb{A}, \mathbb{B}},$$

such that for every

$$F : \mathbb{A}' \rightarrow \mathbb{A}$$

one has

$$F^\bullet V_{\mathbb{A}} \leq V_{\mathbb{A}'}.$$

The globality condition is represented by

$$\begin{array}{ccc} \overline{F}^* V_{\mathbb{A}} & \xrightarrow{F^\sharp|_{V_{\mathbb{A}}}} & \mathbf{Beh}_{\mathbb{A}', \mathbb{B}, 0} \\ & \searrow & \uparrow \\ & & V_{\mathbb{A}'} \end{array}$$

If every $V_{\mathbb{A}'}$ is operation-saturated, the globality condition is equivalent to

$$F^* V_{\mathbb{A}} \leq V_{\mathbb{A}'}.$$

Definition 4.8.5 (Global operation congruence system). *A **global operation congruence system** is a family*

$$R = (R_{\mathbb{A}})_{\mathbb{A} \in \text{Inp}}$$

with

$$R_{\mathbb{A}} \in \text{Cong}_{\text{ioo}}(\mathbb{A}^{\text{op}})$$

such that, for every

$$F : \mathbb{A}' \rightarrow \mathbb{A},$$

one has

$$R_{\mathbb{A}'} \leq F^{\flat} R_{\mathbb{A}}.$$

The lax-composition inequality of Proposition 4.8.5 makes this condition coherent under composite input functors. Indeed, if

$$R_{\mathbb{A}''} \leq G^{\flat} R_{\mathbb{A}'}$$

and

$$R_{\mathbb{A}'} \leq F^{\flat} R_{\mathbb{A}},$$

then monotonicity of $G^{\flat}$ and lax composition give

$$R_{\mathbb{A}''} \hookrightarrow G^{\flat} R_{\mathbb{A}'} \hookrightarrow G^{\flat} (F^{\flat} R_{\mathbb{A}}) \hookrightarrow (F \circ G)^{\flat} R_{\mathbb{A}}.$$

A global theory is operation-saturated when every component is operation-saturated. A global congruence system is semantically closed when every component is semantically closed.

Proposition 4.8.7 (Saturated global theories induce closed congruence systems). *Let V be an operation-saturated global coequational theory. Then*

$$R_{\mathbb{A}}^V := R_{V_{\mathbb{A}}}$$

defines a semantically closed global operation congruence system.

Proof.

1. **Prove local semantic closure.** Each $R_{V_{\mathbb{A}}}$ is semantically closed by Corollary 4.5.1.
2. **Apply globality of V .** Let

$$F : \mathbb{A}' \rightarrow \mathbb{A}.$$

Globality gives

$$F^{\bullet} V_{\mathbb{A}} \leq V_{\mathbb{A}'}.$$

3. **Reverse the inclusion by valid equations.** Contravariance gives

$$R_{V_{\mathbb{A}'}} \leq R_{F^\bullet V_{\mathbb{A}}}.$$

4. **Use saturation of the source component.** Since

$$V_{\mathbb{A}} = V_{R_{V_{\mathbb{A}}}},$$

one has

$$R_{F^\bullet V_{\mathbb{A}}} = F^\flat R_{V_{\mathbb{A}}}.$$

5. **Conclude the global congruence inequality.** Therefore

$$\begin{aligned} R_{\mathbb{A}'}^V &= R_{V_{\mathbb{A}'}} \\ &\leq R_{F^\bullet V_{\mathbb{A}}} \\ &= F^\flat R_{V_{\mathbb{A}}} \\ &= F^\flat R_{\mathbb{A}}^V. \end{aligned}$$

Thus R^V is a global operation congruence system. □

Proposition 4.8.8 (Congruence systems induce saturated global theories). *Let R be a global operation congruence system. Then*

$$V_{\mathbb{A}}^R := V_{R_{\mathbb{A}}}$$

defines an operation-saturated global coequational theory.

Proof.

1. **Prove local saturation.** Each theory

$$V_{R_{\mathbb{A}}}$$

is operation-saturated by Corollary 4.5.1.

2. **Apply globality of R .** For

$$F : \mathbb{A}' \rightarrow \mathbb{A},$$

one has

$$R_{\mathbb{A}'} \leq F^\flat R_{\mathbb{A}}.$$

3. **Expand the operational inverse image.** By definition,

$$F^\flat R_{\mathbb{A}} = R_{F^\bullet(V_{R_{\mathbb{A}}})}.$$

4. **Apply operational soundness and completeness.** The inclusion

$$R_{\mathbb{A}'} \leq R_{F^\bullet(V_{R_{\mathbb{A}}})}$$

is equivalent to

$$F^\bullet(V_{R_{\mathbb{A}}}) \leq V_{R_{\mathbb{A}'}}.$$

This is precisely the globality condition for V^R .

□

Theorem 4.8.1 (Global algebraic Eilenberg duality). *The assignments*

$$V \longmapsto R^V, \quad R \longmapsto V^R$$

restrict to an order-reversing isomorphism between:

1. *operation-saturated global coequational theories;*
2. *semantically closed global operation congruence systems.*

Proof.

1. **Verify well-definedness from theories to congruences.** Proposition 4.8.7 shows that a saturated global theory determines a semantically closed global congruence system.
2. **Verify well-definedness from congruences to theories.** Proposition 4.8.8 shows that a closed global congruence system determines a saturated global theory.
3. **Verify the first composite pointwise.** For every input category $\mathbb{A}$,

$$V_{\mathbb{A}}^{R^V} = V_{R_{V_{\mathbb{A}}}} = V_{\mathbb{A}},$$

because $V_{\mathbb{A}}$ is saturated.

4. **Verify the second composite pointwise.** Similarly,

$$R_{\mathbb{A}}^{V^R} = R_{V_{R_{\mathbb{A}}}} = R_{\mathbb{A}},$$

because $R_{\mathbb{A}}$ is semantically closed.

5. **Conclude order reversal.** Both componentwise assignments are antitone, so the resulting global correspondence is an order-reversing isomorphism.

□

4.9 One-object and stateless specializations

4.9.1 One-object input categories

Let $\mathbb{A}$ be the one-object category determined by a monoid object M . With the standing convention

$$b \circ a = ba,$$

a Mealy transition is a right action

$$x \cdot m := \delta(m, x)$$

satisfying

$$x \cdot (mn) = (x \cdot m) \cdot n.$$

Suppose also that $\mathbb{B}$ is one-object, determined by a monoid object N . The output comparison is a cocycle

$$\omega : M \times P \rightarrow N$$

satisfying

$$\omega(1, x) = 1$$

and

$$\omega(mn, x) = \omega(m, x)\omega(n, x \cdot m).$$

The cocycle law is the one-object form of

$$q(x \cdot (mn)) \xrightarrow{\omega(n, x \cdot m)} q(x \cdot m) \xrightarrow{\omega(m, x)} q(x).$$

$\omega(mn, x)$

The transition-output representation is the monoid homomorphism

$$\chi_P : M^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P)$$

given by

$$m \mapsto (x \mapsto x \cdot m, \omega(m, x)).$$

4.9.2 Arrow monoids and execution monoids

For a one-object category $\mathbb{C}$, let

$$\text{ArrMon}(\mathbb{C})$$

denote its ordinary arrow monoid, whose multiplication is categorical composition:

$$x \cdot_{\text{Arr}} y := x \circ_{\mathbb{C}} y.$$

Define the **execution monoid** by

$$\text{ExecMon}(\mathbb{C}) := \text{ArrMon}(\mathbb{C})^{\text{op}}.$$

Equivalently, its multiplication is

$$x \star y := y \circ_{\mathbb{C}} x.$$

For the one-object category $\mathbb{A}_M$ determined by M under the standing convention,

$$\text{ArrMon}(\mathbb{A}_M^{\text{op}}) \cong M^{\text{op}},$$

whereas

$$\text{ExecMon}(\mathbb{A}_M^{\text{op}}) \cong M.$$

The execution monoid is therefore the monoid recording input operations in their temporal execution order.

A two-sided congruence on M^{op} is the same underlying equivalence relation as a two-sided congruence on M . Thus operation congruences may be regarded as ordinary two-sided monoid congruences when no multiplication-order ambiguity is present.

4.9.3 Residual cocycles

A residual behaviour is an N -valued cocycle

$$\lambda : M \times M \rightarrow N$$

satisfying

$$\lambda(r, 1) = 1$$

and

$$\lambda(r, mn) = \lambda(r, m)\lambda(rm, n).$$

The derivative is

$$(\partial_s \lambda)(r, m) = \lambda(sr, m),$$

and the canonical output on input s is

$$\lambda(1, s).$$

For a local theory V , the residual representation is

$$\chi_V : M^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(V),$$

with

$$\chi_V(s)(\lambda) = (\partial_s \lambda, \lambda(1, s)).$$

The action is

$$V \xrightarrow{\lambda \mapsto (\partial_s \lambda, \lambda(1, s))} V.$$

Thus

$$m R_V n$$

if and only if, for every $\lambda \in V$,

$$\partial_m \lambda = \partial_n \lambda$$

and

$$\lambda(1, m) = \lambda(1, n).$$

4.9.4 Stateless functors

Let

$$F : \mathbb{A} \rightarrow \mathbb{B}$$

be an internal functor. Its stateless Mealy carrier is

$$P_F := \left(A_0 \xleftarrow{1_{A_0}} A_0 \xrightarrow{F_0} B_0 \right),$$

displayed by

$$\begin{array}{ccc} & A_0 & \\ 1_{A_0} \swarrow & & \searrow F_0 \\ A_0 & & B_0. \end{array}$$

The fibre over every $v \in A_0$ is terminal.

A Kleisli operation

$$(P_F)_v \rightsquigarrow (P_F)_u$$

is therefore determined by a single arrow

$$F_0(u) \rightarrow F_0(v)$$

of $\mathbb{B}$.

Consequently,

$$\text{KlEnd}_{\mathbb{B}}(P_F)$$

is the identity-on-objects pullback of $\mathbb{B}^{\text{op}}$ along F_0 . Its arrow object consists internally of triples

$$(v, u, \alpha)$$

with

$$\alpha : F_0(u) \rightarrow F_0(v).$$

The pullback description is

$$\begin{array}{ccc} \text{KlEnd}_{\mathbb{B}}(P_F)_1 & \longrightarrow & B_1 \\ \downarrow & \lrcorner & \downarrow (t_B, s_B) \\ A_0 \times A_0 & \xrightarrow{F_0 \times F_0} & B_0 \times B_0. \end{array}$$

Definition 4.9.1 (Stateless action congruence). *Define*

$$R_F^{\text{act}} := \ker(\chi_{F,1}),$$

where

$$\chi_F : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P_F)$$

is the transition-output representation of the stateless Mealy morphism.

Proposition 4.9.1 (Stateless representation). *Under the preceding identification, the transition-output representation*

$$\chi_F : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(P_F)$$

is the identity-on-objects lift of

$$F^{\text{op}} : \mathbb{A}^{\text{op}} \rightarrow \mathbb{B}^{\text{op}}.$$

Hence

$$a R_F^{\text{act}} a' \iff F_1(a) = F_1(a')$$

for parallel input arrows a, a' .

Proof.

1. **Identify the state transition.** The state fibres are terminal, so the state transition

$$(P_F)_v \rightarrow (P_F)_u$$

is unique.

2. **Identify the output comparison.** For

$$a : u \rightarrow v,$$

the output comparison is

$$F_1(a) : F_0(u) \rightarrow F_0(v).$$

3. **Identify the representation.** The arrow map is therefore the lift

$$\begin{array}{ccc} A_1 & \xrightarrow{\chi_{F,1}} & \text{KlEnd}_{\mathbb{B}}(P_F)_1 \\ F_1 \downarrow & & \downarrow \\ B_1 & \xlongequal{\quad} & B_1. \end{array}$$

This is the identity-on-objects lift of F^{op} .

4. **Compute the kernel congruence.** Two parallel arrows induce the same operation exactly when their output arrows under F_1 are equal. Hence

$$a R_F^{\text{act}} a' \iff F_1(a) = F_1(a').$$

□

The residual behaviour of F at v is the functor

$$H_v^F : \mathbb{A}/v \rightarrow \mathbb{B}$$

defined by

$$H_v^F(r : u \rightarrow v) = F_0(u)$$

and

$$H_v^F(r \circ a \rightarrow r) = F_1(a).$$

Thus residual stateless semantics records every slice-restricted view of the functor F :

$$\begin{array}{ccc} \mathbb{A}/v & \xrightarrow{\quad} & \mathbb{A} \\ & \searrow H_v^F & \downarrow F \\ & & \mathbb{B}. \end{array}$$

4.10 Boolean languages and classical syntactic monoids

From this section onward, and throughout the comparison with classical language theory and the classical Eilenberg correspondence, we specialize to

$$\mathcal{E} = \mathbf{Set}.$$

Accordingly, the input and output categories, monoids, congruences, quotients, languages, Boolean algebras, syntactic monoids, and pseudovarieties appearing below are ordinary set-based structures. The internal constructions developed above specialize along the canonical identification of internal categories, internal functors, regular images, and effective equivalence relations in $\mathbf{Set}$ with their ordinary counterparts.

4.10.1 Boolean residual behaviour

Let

$$\mathbb{B} = 2_{\text{ch}}$$

be the chaotic category on the Boolean algebra 2. Between any two Boolean values there is a unique arrow, so output-comparison equality is determined entirely by equality of endpoints.

Let

$$\mathbb{A}_\Sigma$$

be the one-object category associated to the free monoid Σ^* . A residual behaviour is determined by a language

$$L : \Sigma^* \rightarrow 2.$$

The derivative convention is

$$(\partial_u L)(w) = L(uw).$$

We identify a language

$$L \in \mathbf{Beh}_{\mathbb{A}_\Sigma, 2_{\text{ch}}, 0}$$

with the singleton specification

$$L : 1 \rightarrow \mathbf{Beh}_{\mathbb{A}_\Sigma, 2_{\text{ch}}, 0}.$$

Accordingly, we write

$$\text{Der}_0(L) \quad \text{and} \quad \text{SynTO}(L)$$

for the constructions associated with this singleton specification.

For a monoid congruence R on Σ^* , write

$$\eta_R : \Sigma^* \rightarrow \Sigma^*/R$$

for the quotient morphism.

Proposition 4.10.1 (Languages recognized by a congruence). *A language $L \subseteq \Sigma^*$ lies in V_R if and only if it is recognized by the quotient*

$$\eta_R : \Sigma^* \twoheadrightarrow \Sigma^*/R.$$

Equivalently,

$$V_R = \{ \eta_R^{-1}(S) \mid S \subseteq \Sigma^*/R \}.$$

Proof.

1. **Assume recognition by the quotient.** Suppose

$$L = \eta_R^{-1}(S)$$

for some $S \subseteq \Sigma^*/R$.

2. **Apply two-sided compatibility of R .** If

$$u R v,$$

then for all contexts $r, s \in \Sigma^*$,

$$rus R rvs.$$

Therefore

$$L(rus) = L(rvs).$$

3. **Apply the contextual description of V_R .** In the chaotic output category, output comparisons are unique once their endpoints agree. The preceding equality therefore gives the contextual transition-output condition of Proposition 4.3.1. Hence

$$L \in V_R.$$

4. **Assume membership in V_R .** Conversely, suppose

$$L \in V_R.$$

Apply the contextual condition with the identity residual context and evaluate the derivative equality at the empty word. Then

$$u R v \implies L(u) = L(v).$$

5. **Factor through the quotient.** The language L is constant on R -classes, so it is the inverse image of a subset of Σ^*/R . Thus it is recognized by η_R .

□

4.10.2 Recovery of the syntactic monoid

Theorem 4.10.1 (Classical syntactic monoid recovery). *Let*

$$L \subseteq \Sigma^*$$

be a language. There is a canonical isomorphism

$$\text{ExecMon}(\text{SynTO}_{\mathbb{A}_{\Sigma}, 2_{\text{ch}}}(L)) \cong \text{Syn}(L),$$

where $\text{Syn}(L)$ is the classical syntactic monoid of L .

Equivalently,

$$\text{ArrMon}(\text{SynTO}_{\mathbb{A}_{\Sigma}, 2_{\text{ch}}}(L)) \cong \text{Syn}(L)^{\text{op}}.$$

If L is regular, these monoids are finite.

Proof.

1. **Describe the generated derivative states.** The derivative states generated by L are

$$\partial_r L, \quad r \in \Sigma^*.$$

2. **Compute the action of a word.** Acting by u sends

$$\partial_r L \mapsto \partial_u(\partial_r L) = \partial_{ru} L.$$

The action diagram is

$$\partial_r L \xrightarrow{\quad u \quad} \partial_{ru} L.$$

3. **Discard redundant output-comparison data.** Since 2_{ch} is chaotic, equality of output comparisons is automatic once the derivative states agree.

4. **Compute the syntactic congruence.** Therefore

$$u R_L^{\text{syn}} v$$

if and only if

$$\partial_{ru} L = \partial_{rv} L \quad \text{for every } r \in \Sigma^*.$$

5. **Evaluate the derivative equality.** Evaluating at $s \in \Sigma^*$ gives

$$L(rus) = L(rvs) \quad \text{for every } r, s \in \Sigma^*.$$

Hence

$$u R_L^{\text{syn}} v \iff \forall r, s, rus \in L \iff rvs \in L.$$

This is the classical syntactic congruence.

6. **Identify the categorical quotient.** Consequently,

$$\text{SynTO}(L) \cong \mathbb{A}_\Sigma^{\text{op}} / R_L^{\text{syn}}.$$

7. **Identify the ordinary arrow monoid.** The ordinary arrow monoid of the quotient category has opposite multiplication:

$$\text{ArrMon}(\text{SynTO}(L)) \cong \text{Syn}(L)^{\text{op}}.$$

8. **Pass to temporal execution order.** Taking the execution monoid reverses multiplication once more:

$$\text{ExecMon}(\text{SynTO}(L)) \cong \text{Syn}(L).$$

The comparison is

$$\Sigma^* \longrightarrow \text{Syn}(L) \xrightarrow{\cong} \text{ExecMon}(\text{SynTO}(L)).$$

9. **Prove finiteness for regular languages.** If L is regular, it has finitely many derivatives. Proposition 4.6.2 therefore applies, so the syntactic transition-output category and its arrow and execution monoids are finite.

□

4.10.3 Finite Boolean saturation and right quotients

For a language L and a word u , write

$$Lu^{-1} := \{w \in \Sigma^* \mid wu \in L\}$$

for the right quotient.

Theorem 4.10.2 (Finite Boolean saturation theorem). *Let V be a finite Boolean algebra of languages over Σ , closed under left derivatives. Then every language in V is regular, and the following are equivalent:*

1. V is operation-saturated:

$$V = V_{R_V};$$

2. *V is closed under right quotients.*

Proof.

1. **Establish regularity.** Let $L \in V$. Since V is closed under left derivatives, every left derivative

$$\partial_u L, \quad u \in \Sigma^*,$$

belongs to V . Because V is finite, L has only finitely many distinct left derivatives. Hence L is regular.

2. **Assume operation saturation.** Suppose

$$V = V_{R_V}.$$

By Proposition 4.10.1, every language $L \in V$ is recognized by the quotient

$$\eta_{R_V} : \Sigma^* \twoheadrightarrow \Sigma^*/R_V.$$

Thus there is a subset

$$S \subseteq \Sigma^*/R_V$$

such that

$$L = \eta_{R_V}^{-1}(S).$$

3. **Show that the same quotient recognizes right quotients.** Let $u \in \Sigma^*$. Define

$$Su^{-1} := \{c \in \Sigma^*/R_V \mid c\eta_{R_V}(u) \in S\}.$$

Then

$$\begin{aligned} w \in Lu^{-1} &\iff wu \in L \\ &\iff \eta_{R_V}(wu) \in S \\ &\iff \eta_{R_V}(w)\eta_{R_V}(u) \in S \\ &\iff \eta_{R_V}(w) \in Su^{-1}. \end{aligned}$$

Therefore

$$Lu^{-1} = \eta_{R_V}^{-1}(Su^{-1})$$

is recognized by the same quotient. Hence

$$Lu^{-1} \in V_{R_V} = V.$$

Notice that this implication does not use finiteness of V : every operation-saturated Boolean language theory is closed under right quotients.

4. **Assume closure under right quotients.** Conversely, suppose that V is closed under right quotients.

For words $x, w \in \Sigma^*$, the definition of R_V gives

$$x R_V w$$

if and only if

$$\partial_x L = \partial_w L \quad \text{for every } L \in V.$$

Equivalently,

$$L(xs) = L(ws) \quad \text{for every } L \in V \text{ and every } s \in \Sigma^*.$$

5. **Relate operational equivalence to membership profiles.** We claim that

$$x R_V w \iff \forall J \in V, J(x) = J(w).$$

For the forward implication, take the empty suffix $s = \varepsilon$. Then

$$J(x) = J(w)$$

for every $J \in V$.

Conversely, suppose

$$J(x) = J(w) \quad \text{for every } J \in V.$$

Let $L \in V$ and $s \in \Sigma^*$. Since V is closed under right quotients,

$$Ls^{-1} \in V.$$

Applying membership-profile equality to Ls^{-1} gives

$$\begin{aligned} L(xs) &= (Ls^{-1})(x) \\ &= (Ls^{-1})(w) \\ &= L(ws). \end{aligned}$$

Hence

$$\partial_x L = \partial_w L$$

for every $L \in V$, and therefore

$$x R_V w.$$

6. **Express each congruence class Booleanly.** Fix $w \in \Sigma^*$. By the membership-profile characterization,

$$x \in [w]_{R_V}$$

if and only if x and w have the same membership value in every $J \in V$. Therefore

$$[w]_{R_V} = \bigcap_{\substack{J \in V \\ w \in J}} J \cap \bigcap_{\substack{J \in V \\ w \notin J}} (\Sigma^* \setminus J).$$

Because V is finite and Boolean, both intersections are finite Boolean combinations of members of V . Hence

$$[w]_{R_V} \in V.$$

7. **Establish finite index.** The R_V -class of a word is determined by its membership profile

$$(J(w))_{J \in V} \in 2^V.$$

Since V is finite, there are only finitely many such profiles. Thus R_V has finite index.

8. **Recover every language recognized by the quotient.** Let

$$L \in V_{R_V}.$$

By Proposition 4.10.1, L is a union of R_V -classes. Since R_V has finite index, this is a finite union.

Every R_V -class belongs to V , and V is closed under finite unions. Therefore

$$L \in V.$$

Hence

$$V_{R_V} \leq V.$$

9. Use inflationarity. The general operational Galois connection gives

$$V \leq V_{R_V}.$$

Combining the two inclusions yields

$$V = V_{R_V}.$$

Thus V is operation-saturated.

□

Remark 4.10.1 (Regularity and operation saturation). *The fixed-point closure*

$$V \longmapsto V_{R_V}$$

is formed in the complete lattice of all local behavioural theories and need not preserve regularity when R_V has infinite index. Therefore the classical Eilenberg correspondence is not obtained by simply restricting the local fixed-point duality to arbitrary varieties of regular languages. Instead, it is recovered by taking the finite syntactic quotients of the individual regular languages in the variety and closing those finite monoids under products and divisors.

Remark 4.10.2 (Categorical origins of the classical closure conditions). *In the Boolean specialization:*

- *derivative closure is left-quotient closure;*
- *pointwise output structure gives Boolean closure;*
- *operation saturation gives right-quotient closure in the finite local setting;*
- *global input reindexing gives inverse morphic preimage.*

Thus the familiar closure conditions of language varieties arise from distinct categorical parts of the general construction.

4.10.4 Residual reindexing as inverse morphic preimage

Proposition 4.10.2 (Residual reindexing of languages). *Let*

$$h : \Gamma^* \rightarrow \Sigma^*$$

be a monoid homomorphism, regarded as an input functor

$$F_h : \mathbb{A}_\Gamma \rightarrow \mathbb{A}_\Sigma.$$

For every language

$$L \subseteq \Sigma^*,$$

one has

$$F_h^\sharp(L) = h^{-1}(L).$$

Consequently, for every local theory V over Σ ,

$$F_h^\bullet V = \text{Im} \left(V \longrightarrow \mathbf{Beh}_{\mathbb{A}_\Gamma, 2_{\text{ch}}, 0}, \quad L \longmapsto h^{-1}(L) \right).$$

Proof.

1. **Compute residual precomposition.** For every $w \in \Gamma^*$,

$$\begin{aligned} F_h^\sharp(L)(w) &= L(F_h(w)) \\ &= L(h(w)). \end{aligned}$$

2. **Identify the inverse image.** The characteristic function of $h^{-1}(L)$ is exactly

$$w \longmapsto L(h(w)).$$

Hence

$$F_h^\sharp(L) = h^{-1}(L).$$

3. **Pass to local theories.** By definition, $F_h^\bullet V$ is the regular image of the restriction of $F_h^\sharp$ to V . This is precisely the image of the map

$$L \longmapsto h^{-1}(L).$$

The factorization is

$$\begin{array}{ccc} V & \xrightarrow{\quad} & F_h^\bullet V \\ & \searrow L \mapsto h^{-1}(L) & \downarrow \\ & & 2^{\Gamma^*}. \end{array}$$

□

Thus the globality condition

$$F_h^\bullet V_\Sigma \leq V_\Gamma$$

is exactly closure under inverse morphic preimages.

4.11 Compatibility with the classical Eilenberg correspondence

4.11.1 Varieties of regular languages and pseudovarieties

Definition 4.11.1 (Global Boolean regular-language theory). A *global Boolean regular-language theory* is a family

$$\mathcal{V} = (\mathcal{V}_\Sigma)_\Sigma$$

indexed by finite alphabets, such that:

1. each $\mathcal{V}_\Sigma$ is a Boolean algebra of regular languages over Σ ;

2. each $\mathcal{V}_\Sigma$ is closed under left and right quotients;

3. for every monoid homomorphism

$$h : \Gamma^* \rightarrow \Sigma^*,$$

one has

$$h^{-1}(\mathcal{V}_\Sigma) \subseteq \mathcal{V}_\Gamma.$$

A **pseudovariety of finite monoids** is a nonempty, isomorphism-closed class of finite monoids closed under finite products and divisors. Equivalently, it is closed under finite products, submonoids, and quotient monoids. In particular, every pseudovariety contains the trivial monoid.

For a global Boolean regular-language theory $\mathcal{V}$, define

$$\mathbf{P}_\mathcal{V}$$

to be the pseudovariety generated by all classical syntactic monoids

$$\text{Syn}(L), \quad L \in \mathcal{V}_\Sigma,$$

as Σ varies. By Theorem 4.10.1, these are equivalently the execution monoids

$$\text{ExecMon}(\text{SynTO}(L)).$$

For a pseudovariety $\mathbf{P}$, define

$$\mathcal{V}_\Sigma^\mathbf{P} := \{L \subseteq \Sigma^* \mid \text{Syn}(L) \in \mathbf{P}\}.$$

4.11.2 Syntactic separation

Lemma 4.11.1 (Syntactic fibres). *Let $L \subseteq \Sigma^*$ be regular. Every fibre of its syntactic morphism*

$$\eta_L : \Sigma^* \twoheadrightarrow \text{Syn}(L)$$

is a Boolean combination of two-sided quotients

$$x^{-1}Ly^{-1} := \{w \in \Sigma^* \mid xwy \in L\}.$$

Proof.

1. **Use syntactic separation.** Two words lie in the same syntactic congruence class precisely when no two-sided context

$$x(-)y$$

separates them with respect to membership in L .

2. **Choose finitely many separating contexts.** Since $\text{Syn}(L)$ is finite, fix a syntactic class C . For every distinct class D , choose a context

$$x_D(-)y_D$$

separating C from D .

3. **Express the class as a Boolean combination.** For each D , the chosen quotient

$$x_D^{-1}Ly_D^{-1}$$

either contains C and excludes D , or its complement does. The finite intersection of these selected quotient conditions is exactly C .

4. **Conclude.** Hence each fibre of η_L is a Boolean combination of two-sided quotients of L .

□

4.11.3 Finite syntactic generation

Lemma 4.11.2 (Finite monoids from syntactic quotients). *Let M be a finite monoid generated by a finite alphabet Σ . Then M divides a finite product of syntactic monoids of languages recognized by M .*

Proof.

1. **Choose a finite generating morphism.** Since M is generated by the finite alphabet Σ , choose a surjective monoid morphism

$$h : \Sigma^* \twoheadrightarrow M.$$

2. **Form languages recognized by M .** For every subset $S \subseteq M$, define

$$L_S := h^{-1}(S).$$

The language L_S is recognized by M through the morphism h .

3. **Compare the kernel of h with the syntactic congruence.** Let

$$x, y \in \Sigma^*$$

satisfy

$$h(x) = h(y).$$

For every pair of two-sided contexts

$$p, q \in \Sigma^*,$$

one has

$$\begin{aligned} h(pqx) &= h(p)h(x)h(q) \\ &= h(p)h(y)h(q) \\ &= h(pyq). \end{aligned}$$

Consequently,

$$pxq \in L_S \iff h(pqx) \in S \iff h(pyq) \in S \iff pyq \in L_S.$$

Thus

$$x \equiv_{L_S} y,$$

where $\equiv_{L_S}$ is the classical syntactic congruence of L_S . Hence

$$\ker(h) \leq \equiv_{L_S}.$$

4. **Factor the syntactic morphism through M .** Let

$$\eta_{L_S} : \Sigma^* \twoheadrightarrow \text{Syn}(L_S)$$

be the syntactic morphism.

Since η_{L_S} coequalizes $\ker(h)$, there is a unique monoid morphism

$$\rho_S : M \rightarrow \text{Syn}(L_S)$$

satisfying

$$\rho_S h = \eta_{L_S}.$$

The factorization is

$$\begin{array}{ccc} \Sigma^* & \xrightarrow{h} & M \\ & \searrow \eta_{L_S} & \downarrow \rho_S \\ & & \text{Syn}(L_S). \end{array}$$

Because both h and η_{L_S} are surjective, ρ_S is surjective. Indeed, if $z \in \text{Syn}(L_S)$, choose $x \in \Sigma^*$ with $\eta_{L_S}(x) = z$. Then

$$z = \eta_{L_S}(x) = \rho_S(h(x)).$$

5. **Choose a finite separating family of subsets.** Since M is finite, the family of singleton subsets

$$\{\{m\} \mid m \in M\}$$

is finite and separates distinct elements of M .

For each $m \in M$, write

$$L_m := h^{-1}(\{m\})$$

and

$$\rho_m : M \twoheadrightarrow \text{Syn}(L_m)$$

for the preceding quotient morphism.

6. **Prove joint injectivity.** Form the product morphism

$$\rho := \langle \rho_m \rangle_{m \in M} : M \rightarrow \prod_{m \in M} \text{Syn}(L_m).$$

Suppose $a, b \in M$ with $a \neq b$. Choose words

$$x, y \in \Sigma^*$$

such that

$$h(x) = a, \quad h(y) = b.$$

Consider the singleton language

$$L_a = h^{-1}(\{a\}).$$

Then

$$x \in L_a, \quad y \notin L_a.$$

The empty two-sided context therefore separates x and y , so

$$\eta_{L_a}(x) \neq \eta_{L_a}(y).$$

Using

$$\eta_{L_a} = \rho_a h,$$

one obtains

$$\rho_a(a) \neq \rho_a(b).$$

Hence

$$\rho(a) \neq \rho(b).$$

Thus ρ is injective.

7. **Conclude divisibility.** The morphism ρ identifies M with a submonoid of the finite product

$$\prod_{m \in M} \text{Syn}(L_m).$$

Each language L_m is recognized by M . Therefore M divides a finite product of syntactic monoids of languages recognized by M .

□

4.11.4 Recognition through divisors of products

Lemma 4.11.3 (Recognition by products of syntactic monoids). *Let $\mathcal{V}$ be a global Boolean regular-language theory. Suppose $L \subseteq \Sigma^*$ is recognized by a finite monoid D that divides a finite product*

$$\prod_{i=1}^n \text{Syn}(L_i), \quad L_i \in \mathcal{V}_{\Sigma_i},$$

where $n \geq 0$, and where the case $n = 0$ is interpreted as the trivial monoid. Then

$$L \in \mathcal{V}_{\Sigma}.$$

Proof.

1. **Handle the empty-product case.** Suppose first that

$$n = 0.$$

Then

$$\prod_{i=1}^0 \text{Syn}(L_i)$$

is the trivial monoid. Every divisor of the trivial monoid is trivial, so D is the trivial monoid. A language recognized by the trivial monoid is either

$$\emptyset$$

or

$$\Sigma^*.$$

Since $\mathcal{V}_{\Sigma}$ is a Boolean algebra, both languages belong to $\mathcal{V}_{\Sigma}$. Hence the result holds when $n = 0$.

For the remainder of the proof, assume

$$n \geq 1.$$

2. **Choose divisor data.** Choose a submonoid

$$j : S \hookrightarrow \prod_{i=1}^n \text{Syn}(L_i)$$

and a surjective homomorphism

$$p : S \twoheadrightarrow D.$$

3. **Choose a recognizing morphism.** Let

$$\varphi : \Sigma^* \rightarrow D$$

recognize L . Thus there is a subset

$$T \subseteq D$$

such that

$$L = \varphi^{-1}(T).$$

4. **Lift through the divisor quotient.** Since Σ^* is free and p is surjective, choose, for each generator $a \in \Sigma$, an element

$$\tilde{a} \in S$$

such that

$$p(\tilde{a}) = \varphi(a).$$

These choices extend uniquely to a monoid homomorphism

$$\tilde{\varphi} : \Sigma^* \rightarrow S$$

satisfying

$$\begin{array}{ccc} \Sigma^* & \xrightarrow{\tilde{\varphi}} & S \\ & \searrow \varphi & \downarrow p \\ & & D. \end{array}$$

5. **Map into the product of syntactic monoids.** Put

$$\psi := j\tilde{\varphi} : \Sigma^* \rightarrow \prod_{i=1}^n \text{Syn}(L_i),$$

and write

$$\varphi_i : \Sigma^* \rightarrow \text{Syn}(L_i)$$

for its i -th component.

6. **Lift each component to a free monoid.** Since

$$\eta_{L_i} : \Sigma_i^* \twoheadrightarrow \text{Syn}(L_i)$$

is surjective and Σ^* is free, each φ_i lifts to a monoid homomorphism

$$h_i : \Sigma^* \rightarrow \Sigma_i^*$$

satisfying

$$\begin{array}{ccc} \Sigma^* & \xrightarrow{h_i} & \Sigma_i^* \\ & \searrow \varphi_i & \downarrow \eta_{L_i} \\ & & \text{Syn}(L_i). \end{array}$$

7. **Show that component fibres lie in the variety.** Fix

$$c \in \text{Syn}(L_i).$$

One has

$$\varphi_i^{-1}(c) = h_i^{-1}(\eta_{L_i}^{-1}(c)).$$

By Lemma 4.11.1, the syntactic fibre

$$\eta_{L_i}^{-1}(c)$$

is a Boolean combination of two-sided quotients of L_i . Since

$$L_i \in \mathcal{V}_{\Sigma_i}$$

and $\mathcal{V}$ is closed under left and right quotients, Boolean operations, and inverse morphic preimages, it follows that

$$\varphi_i^{-1}(c) \in \mathcal{V}_{\Sigma}.$$

8. **Show that product fibres lie in the variety.** For a tuple

$$c = (c_1, \dots, c_n) \in \prod_{i=1}^n \text{Syn}(L_i),$$

one has

$$\psi^{-1}(c) = \bigcap_{i=1}^n \varphi_i^{-1}(c_i).$$

Since $\mathcal{V}_{\Sigma}$ is closed under finite intersections, every fibre of ψ lies in $\mathcal{V}_{\Sigma}$.

9. **Restrict to the submonoid.** Since j is injective, for every $s \in S$,

$$\tilde{\varphi}^{-1}(s) = \psi^{-1}(j(s)).$$

Thus every fibre of $\tilde{\varphi}$ lies in $\mathcal{V}_{\Sigma}$.

10. **Descend to the quotient monoid.** For every $d \in D$,

$$\varphi^{-1}(d) = \bigcup_{s \in p^{-1}(d)} \tilde{\varphi}^{-1}(s).$$

The monoid S is finite because it is a submonoid of a finite product of finite monoids. Hence each fibre $p^{-1}(d)$ is finite, so

$$\varphi^{-1}(d) \in \mathcal{V}_{\Sigma}.$$

11. **Recover the recognized language.** Since

$$L = \varphi^{-1}(T),$$

one has

$$L = \bigcup_{d \in T} \varphi^{-1}(d).$$

The monoid D is finite, so this is a finite union of languages in $\mathcal{V}_{\Sigma}$. Therefore

$$L \in \mathcal{V}_{\Sigma}.$$

□

4.11.5 The classical correspondence

Lemma 4.11.4 (Normal form for generated pseudovarieties). *Let $\mathcal{G}$ be a class of finite monoids. The pseudovariety generated by $\mathcal{G}$ consists exactly of the finite monoids that divide a finite product*

$$G_1 \times \cdots \times G_n$$

with

$$G_i \in \mathcal{G}.$$

The case $n = 0$ is interpreted as the trivial monoid.

Proof. Let $D(\mathcal{G})$ be the class of finite monoids that divide finite products of members of $\mathcal{G}$.

The class $D(\mathcal{G})$ contains every member of $\mathcal{G}$, since each $G \in \mathcal{G}$ divides the one-factor product G . It also contains the trivial monoid, viewed as the empty product.

It is closed under finite products. Indeed, if M divides

$$G_1 \times \cdots \times G_m$$

and N divides

$$H_1 \times \cdots \times H_n,$$

then $M \times N$ divides

$$G_1 \times \cdots \times G_m \times H_1 \times \cdots \times H_n.$$

It is also closed under divisors, because divisibility of finite monoids is transitive. Hence $D(\mathcal{G})$ is a pseudovariety containing $\mathcal{G}$.

Conversely, every pseudovariety containing $\mathcal{G}$ contains every finite product of members of $\mathcal{G}$, and therefore contains every divisor of such a product. Thus every such pseudovariety contains $D(\mathcal{G})$.

It follows that $D(\mathcal{G})$ is precisely the pseudovariety generated by $\mathcal{G}$. $\square$

Theorem 4.11.1 (Compatibility with the classical Eilenberg theorem). *The assignments*

$$\mathcal{V} \longmapsto P_{\mathcal{V}}, \quad P \longmapsto \mathcal{V}^P$$

are mutually inverse and give the classical Eilenberg correspondence between varieties of regular languages and pseudovarieties of finite monoids.

Proof.

1. **Relate the categorical and classical syntactic objects.** By Theorem 4.10.1, for every regular language L ,

$$\text{ExecMon}(\text{SynTO}(L)) \cong \text{Syn}(L).$$

Thus the finite monoids used in the classical construction are exactly the execution monoids of the categorical syntactic transition-output objects.

2. **Verify that $P_{\mathcal{V}}$ is a pseudovariety.** By definition,

$$P_{\mathcal{V}}$$

is the pseudovariety generated by the finite monoids

$$\text{Syn}(L), \quad L \in \mathcal{V}_{\Sigma},$$

as Σ varies. Hence it is nonempty, isomorphism-closed, and closed under finite products and divisors.

3. **Verify regularity of $\mathcal{V}^P$.** Let

$$L \in \mathcal{V}_\Sigma^P.$$

Then

$$\text{Syn}(L) \in P.$$

Every member of a pseudovariety of finite monoids is finite, so $\text{Syn}(L)$ is finite. Therefore L is regular.

4. **Verify Boolean closure.** The empty and full languages are recognized by the trivial monoid, which belongs to every pseudovariety.

Let

$$L, K \in \mathcal{V}_\Sigma^P.$$

Their syntactic monoids

$$M := \text{Syn}(L), \quad N := \text{Syn}(K)$$

belong to P . Hence

$$M \times N \in P.$$

Every Boolean combination of L and K is recognized by $M \times N$. Its syntactic monoid therefore divides $M \times N$, and consequently belongs to P . Thus $\mathcal{V}_\Sigma^P$ is a Boolean algebra.

5. **Verify left- and right-quotient closure.** Let

$$L \in \mathcal{V}_\Sigma^P$$

and let $M = \text{Syn}(L)$. Every left or right quotient of L is recognized by M . The syntactic monoid of the quotient therefore divides M , and belongs to P . Hence $\mathcal{V}_\Sigma^P$ is closed under left and right quotients.

6. **Verify inverse-morphic-preimage closure.** Let

$$h : \Gamma^* \rightarrow \Sigma^*$$

be a monoid homomorphism and let

$$L \in \mathcal{V}_\Sigma^P.$$

If

$$\eta : \Sigma^* \rightarrow M$$

recognizes L , then

$$\eta h : \Gamma^* \rightarrow M$$

recognizes $h^{-1}(L)$. Taking

$$M = \text{Syn}(L) \in P,$$

the syntactic monoid of $h^{-1}(L)$ divides M , and hence belongs to P . Therefore

$$h^{-1}(L) \in \mathcal{V}_\Gamma^P.$$

7. **Conclude that $\mathcal{V}^P$ is a global Boolean regular-language theory.** The preceding steps establish regularity, Boolean closure, closure under both quotients, and closure under inverse morphic preimages. Thus

$$\mathcal{V}^P$$

is a global Boolean regular-language theory.

8. **Prove** $P_{\mathcal{V}^P} \subseteq P$. Every generator of

$$P_{\mathcal{V}^P}$$

has the form

$$\text{Syn}(L)$$

with

$$L \in \mathcal{V}_{\Sigma}^P.$$

By definition of $\mathcal{V}^P$,

$$\text{Syn}(L) \in P.$$

Since P is closed under finite products and divisors, it contains the pseudovariety generated by these monoids. Hence

$$P_{\mathcal{V}^P} \subseteq P.$$

9. **Prove** $P \subseteq P_{\mathcal{V}^P}$. Let

$$M \in P.$$

Since M is finite, it is generated by some finite alphabet. Lemma 4.11.2 gives languages

$$L_1, \dots, L_n$$

recognized by M such that M divides

$$\prod_{i=1}^n \text{Syn}(L_i).$$

Each $\text{Syn}(L_i)$ divides M , because M recognizes L_i . Since $M \in P$ and P is divisor-closed,

$$\text{Syn}(L_i) \in P.$$

Therefore

$$L_i \in \mathcal{V}^P.$$

The monoids $\text{Syn}(L_i)$ are consequently generators of

$$P_{\mathcal{V}^P}.$$

Since that class is closed under products and divisors, it contains M . Thus

$$P \subseteq P_{\mathcal{V}^P}.$$

10. **Conclude the monoid equality.** Combining the two preceding inclusions gives

$$P_{\mathcal{V}^P} = P.$$

11. **Prove** $\mathcal{V} \subseteq \mathcal{V}^{P_{\mathcal{V}}}$. Let

$$L \in \mathcal{V}_{\Sigma}.$$

Then

$$\text{Syn}(L)$$

is one of the generators used to define $P_{\mathcal{V}}$. Hence

$$\text{Syn}(L) \in P_{\mathcal{V}},$$

and therefore

$$L \in \mathcal{V}_{\Sigma}^{P_{\mathcal{V}}}.$$

12. **Prove $\mathcal{V}^{\mathcal{P}\mathcal{V}} \subseteq \mathcal{V}$.** Let

$$L \in \mathcal{V}_{\Sigma}^{\mathcal{P}\mathcal{V}}.$$

Then

$$\text{Syn}(L) \in \mathcal{P}\mathcal{V}.$$

By Lemma 4.11.4, there exist languages

$$L_i \in \mathcal{V}_{\Sigma_i}, \quad 1 \leq i \leq n,$$

such that

$$\text{Syn}(L)$$

divides the finite product

$$\prod_{i=1}^n \text{Syn}(L_i).$$

The language L is recognized by its own syntactic monoid. Therefore it is recognized by a finite monoid dividing the displayed product. Lemma 4.11.3 now gives

$$L \in \mathcal{V}_{\Sigma}.$$

Hence

$$\mathcal{V}^{\mathcal{P}\mathcal{V}} \subseteq \mathcal{V}.$$

13. **Conclude the language equality.** The two language inclusions give

$$\mathcal{V}^{\mathcal{P}\mathcal{V}} = \mathcal{V}.$$

14. **Conclude the correspondence.** The assignments

$$\mathcal{V} \longmapsto \mathcal{P}\mathcal{V}, \quad \mathcal{P} \longmapsto \mathcal{V}^{\mathcal{P}}$$

are mutually inverse. They are order-preserving in the ordinary inclusion orders, and therefore give the classical Eilenberg correspondence.

□

The classical correspondence may be summarized by

$$\left\{ \begin{array}{c} \text{varieties of} \\ \text{regular languages} \end{array} \right\} \begin{array}{c} \xrightarrow{\mathcal{V} \mapsto \mathcal{P}\mathcal{V}} \\ \xleftarrow{\mathcal{P} \mapsto \mathcal{V}^{\mathcal{P}}} \end{array} \left\{ \begin{array}{c} \text{pseudovarieties} \\ \text{of finite monoids} \end{array} \right\}$$

Remark 4.11.1 (Fixed-point duality and pseudovarieties). *The local fixed-point duality concerns a theory and the single possibly non-finite quotient determined by all of its operational equations. The classical Eilenberg correspondence packages the finite syntactic quotients of individual languages into a pseudovariety closed under products and divisors. The subdirect-product theorem and the syntactic universal property explain categorically why those algebraic closure operations arise.*

4.12 Synthesis: state minimization and operation minimization

Residual semantics gives two related minimization processes.

At the state level, the semantic image factorization

$$\begin{array}{ccc} P & \xrightarrow{\quad} & \text{Beh}(P) \\ & \searrow \beta_P & \downarrow \\ & & \mathbf{Beh}^{\text{can}} \end{array}$$

identifies states having the same residual behaviour. For a specification K , reachability identifies the relevant semantic image with

$$\text{Der}_0(K).$$

At the raw operation level, the image factorization

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_P} & \text{KlEnd}_{\mathbb{B}}(P) \\ & \searrow & \swarrow \\ & \text{ActIm}(P) & \end{array}$$

identifies input arrows having the same effect on the original states of P .

At the semantic operation level, the representation

$$\chi_{\text{Beh}(P)} : \mathbb{A}^{\text{op}} \rightarrow \text{KlEnd}_{\mathbb{B}}(\text{Beh}(P))$$

has image factorization

$$\begin{array}{ccc} \mathbb{A}^{\text{op}} & \xrightarrow{\chi_{\text{Beh}(P)}} & \text{KlEnd}_{\mathbb{B}}(\text{Beh}(P)) \\ & \searrow & \swarrow \\ & \text{Op}(P) & \end{array}$$

It identifies input arrows having the same effect after states with identical residual behaviour have been identified.

Let P be a realization of a specification K . For K , the corresponding faithful quotient is

$$\text{SynTO}_{\mathbb{A}, \mathbb{B}}(K).$$

The complete two-level reduction is

$$\begin{array}{lcl} \text{states} & P \xrightarrow{\quad} \text{Beh}(P) \hookrightarrow & \mathbf{Beh}^{\text{can}} \\ \\ \text{operations} & \mathbb{A}^{\text{op}} \xrightarrow{\quad} \text{ActIm}(P) \xrightarrow{\quad} \text{Op}(P) \xrightarrow{\quad} & \text{SynTO}(K). \end{array}$$

These constructions are summarized by

state semantics	operation semantics
β_P	χ_P
$\text{Beh}(P)$	$\text{ActIm}(P) \twoheadrightarrow \text{Op}(P)$
$\text{Der}_0(K)$	$\text{SynTO}(K)$
behavioural kernel	operation congruence
reachable and separated realization	faithful syntactic quotient

The operational satisfaction polarity

$$V \leq V_R \iff R \leq R_V$$

shows that behavioural theories and operational equations determine one another up to semantic completion:

$$V \longmapsto R_V \longmapsto V_{R_V} = c(V), \quad R \longmapsto V_R \longmapsto R_{V_R} = d(R).$$

Its fixed points yield the local and global algebraic Eilenberg dualities.

For a behaviour specification K , the syntactic transition-output category

$$\text{SynTO}(K)$$

is therefore simultaneously:

1. the regular image of the residual transition-output representation;
2. the quotient of $\mathbb{A}^{\text{op}}$ by the equations valid in every generated residual context;
3. the unique faithful quotient recognizing $\text{Der}_0(K)$;
4. the operation-level counterpart of the minimal residual realization.

These characterizations fit into

$$\begin{array}{ccccc}
 \mathbb{A}^{\text{op}} & \xrightarrow{q_K^{\text{syn}}} & \text{SynTO}(K) & \hookrightarrow & \text{KlEnd}_{\mathbb{B}}(\text{Der}_0(K)) \\
 & \searrow q_{R_K}^{\text{syn}} & \downarrow \cong & & \\
 & & \mathbb{A}^{\text{op}}/R_K^{\text{syn}} & &
 \end{array}$$

In the one-object Boolean specialization, the ordinary arrow monoid of this category is the opposite of the classical syntactic monoid, while its execution monoid is canonically the classical syntactic monoid itself:

$$\text{ExecMon}(\text{SynTO}(L)) \cong \text{Syn}(L).$$

Under finite-state recognizability and global closure, the resulting algebraic theory recovers Eilenberg's correspondence between varieties of regular languages and pseudovarieties of finite monoids.

Bibliography

- [1] Jiří Adámek et al. “Generalized Eilenberg Theorem: Varieties of Languages in a Category”. In: *ACM Transactions on Computational Logic* 20.1 (2019), pp. 1–47. DOI: [10.1145/3276771](https://doi.org/10.1145/3276771).
- [2] Jean Bénabou. “Introduction to Bicategories”. In: *Reports of the Midwest Category Seminar I*. Vol. 47. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1967, pp. 1–77. DOI: [10.1007/BFb0074299](https://doi.org/10.1007/BFb0074299).
- [3] Francis Borceux. *Handbook of Categorical Algebra 2: Categories and Structures*. Vol. 51. Encyclopedia of Mathematics and its Applications. Cambridge: Cambridge University Press, 1994. ISBN: 978-0521441797. DOI: [10.1017/CB09780511525865](https://doi.org/10.1017/CB09780511525865).
- [4] Janusz A. Brzozowski. “Derivatives of Regular Expressions”. In: *Journal of the ACM* 11.4 (1964), pp. 481–494. DOI: [10.1145/321239.321249](https://doi.org/10.1145/321239.321249).
- [5] Thomas Colcombet and Daniela Petrişan. “Automata Minimization: A Functorial Approach”. In: *Logical Methods in Computer Science* 16.1 (2020), 32:1–32:31. DOI: [10.23638/LMCS-16\(1:32\)2020](https://doi.org/10.23638/LMCS-16(1:32)2020).
- [6] Samuel Eilenberg. *Automata, Languages, and Machines*. Vol. A. Pure and Applied Mathematics. New York: Academic Press, 1974. ISBN: 978-0122340017.
- [7] Joseph A. Goguen. “Realization is Universal”. In: *Mathematical Systems Theory* 6.4 (1973), pp. 359–374. DOI: [10.1007/BF01706091](https://doi.org/10.1007/BF01706091).
- [8] John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman. *Introduction to Automata Theory, Languages, and Computation*. 3rd ed. Boston: Addison-Wesley, 2006. ISBN: 978-0321455369.
- [9] Victor Iwaniack. *Automata in Toposes, and General Myhill–Nerode Theorems*. 2023. arXiv: [2307.14855](https://arxiv.org/abs/2307.14855) [[cs.FL](#)].
- [10] Bart Jacobs. *Categorical Logic and Type Theory*. Vol. 141. Studies in Logic and the Foundations of Mathematics. Amsterdam: Elsevier, 1999. ISBN: 978-0444501707.
- [11] Bart Jacobs. *Introduction to Coalgebra: Towards Mathematics of States and Observation*. Cambridge: Cambridge University Press, 2016. ISBN: 9781316823187. DOI: [10.1017/CB09781316823187](https://doi.org/10.1017/CB09781316823187).
- [12] Peter T. Johnstone. *Sketches of an Elephant: A Topos Theory Compendium*. Vol. 43–44. Oxford Logic Guides. Volumes 1 and 2. Oxford: Clarendon Press, 2002.
- [13] Stephen Lack. “A 2-Categories Companion”. In: *Towards Higher Categories*. Ed. by John C. Baez and J. Peter May. Vol. 152. The IMA Volumes in Mathematics and its Applications. New York: Springer, 2010, pp. 105–191. DOI: [10.1007/978-1-4419-1524-5_4](https://doi.org/10.1007/978-1-4419-1524-5_4). arXiv: [math/0702535](https://arxiv.org/abs/math/0702535) [[math.CT](#)].

- [14] Adrian Toshir Miranda. “Internal Categories”. Masters by Research thesis. Sydney, Australia: Macquarie University, 2018. URL: https://figshare.mq.edu.au/articles/thesis/Internal_categories/19434626.
- [15] John Myhill. *Finite Automata and the Representation of Events*. Tech. rep. WADC Technical Report 57-624. Appeared in *Fundamental Concepts in the Theory of Systems*. Wright-Patterson Air Force Base, Ohio: Wright Air Development Center, 1957, pp. 112–137.
- [16] G. Naudé. “On the Adjoint Situations Between Behaviour and Realization”. In: *Quaestiones Mathematicae* 2 (1977), pp. 245–267. DOI: [10.1080/16073606.1977.9632546](https://doi.org/10.1080/16073606.1977.9632546).
- [17] G. Naudé. “Universal Realization”. In: *Journal of Computer and System Sciences* 19.3 (1979), pp. 277–289. DOI: [10.1016/0022-0000\(79\)90005-9](https://doi.org/10.1016/0022-0000(79)90005-9).
- [18] Anil Nerode. “Linear Automaton Transformations”. In: *Proceedings of the American Mathematical Society* 9.4 (1958), pp. 541–544. DOI: [10.1090/S0002-9939-1958-0135681-9](https://doi.org/10.1090/S0002-9939-1958-0135681-9).
- [19] Robert Paré. “Mealy Morphisms of Enriched Categories”. In: *Applied Categorical Structures* 20.3 (2012), pp. 251–273. DOI: [10.1007/s10485-010-9238-8](https://doi.org/10.1007/s10485-010-9238-8).
- [20] Jan J. M. M. Rutten. “Automata and Coinduction: An Exercise in Coalgebra”. In: *CONCUR’98: Concurrency Theory*. Vol. 1466. Lecture Notes in Computer Science. Springer, 1998, pp. 194–218. DOI: [10.1007/BFb0055624](https://doi.org/10.1007/BFb0055624).
- [21] Jan J. M. M. Rutten. “Universal Coalgebra: A Theory of Systems”. In: *Theoretical Computer Science* 249.1 (2000), pp. 3–80. DOI: [10.1016/S0304-3975\(00\)00056-6](https://doi.org/10.1016/S0304-3975(00)00056-6).
- [22] Julian Salamanca. “An Eilenberg-like Theorem for Algebras on a Monad”. In: *Electronic Proceedings in Theoretical Computer Science* 219 (2016), pp. 110–125. DOI: [10.4204/EPTCS.219.8](https://doi.org/10.4204/EPTCS.219.8).
- [23] Julian Salamanca. “Unveiling Eilenberg-Type Correspondences: Birkhoff’s Theorem for (Finite) Algebras + Duality”. In: *arXiv preprint arXiv:1702.02822* (2017). arXiv: [1702.02822](https://arxiv.org/abs/1702.02822) [cs.FL].
- [24] Ross Street. “The Formal Theory of Monads”. In: *Journal of Pure and Applied Algebra* 2.2 (1972), pp. 149–168. DOI: [10.1016/0022-4049\(72\)90019-9](https://doi.org/10.1016/0022-4049(72)90019-9).